# Technical Report EU-CATRINE WP5 & WP6: Testbed CloudRoots-Amazon22

Vincent S. de Feiter[1,*], Jordi Vilà-Gureau de Arellano[1]

[1]Meteorology and Air Quality Section, Wageningen University & Research, Wageningen, The Netherlands
[*]Contact: vincent.defeiter@wur.nl




**Abstract**

This technical report presents the (visual) results of the Amazon testbed based on the CloudRoots-Amazon22 campaign, which brings together two weeks of comprehensive field observations, large-eddy simulations that explicitly resolve turbulence and clouds, and simulations from ECMWF's Integrated Forecasting System (IFS) at three horizontal resolutions: 28 km, 9 km and 4.4 km. The Amazon case provides a proof of concept for the Carbon Atmospheric Tracer Research to Improve Numerics and Evaluation (CATRINE) testbed framework by demonstrating how these three sources can be used together to evaluate a global model. The report examines variables ranging from radiation and the surface energy balance to vertical profiles of temperature and humidity. It documents their temporal evolution and uses statistical comparisons to quantify model performance across resolutions. Together with the process-based scorecards developed in CATRINE, this analysis provides a framework for evaluating IFS systematically at other sites where comparable observations and large-eddy simulations are available. Applying the framework across sites would help identify which transport errors recur across environments and how sensitive they are to model resolution, supporting one of CATRINE's main goals of assessing uncertainties in atmospheric greenhouse gas transport.

# Contents


**1 Introduction** **3**

**2 Methodology** **3**
2.1 Testbed Focus: CloudRoots-Amazon22 . . . 3
2.1.1 Observational Data . . . 3
2.1.2 Modelling Tools . . . 3
2.2 Variable Groups . . . 4
2.3 Statistics . . . 5
2.4 Metrics . . . 6

**3 Results** **6**
3.1 Radiation . . . 6
3.1.1 $Q_{net}$ . . . 6
3.1.2 $SW_{net}$ . . . 12
3.1.3 $LW_{net}$ . . . 17
3.1.4 $SW_{in}$ . . . 22
3.1.5 $SW_{out}$ . . . 27
3.1.6 $LW_{in}$ . . . 32
3.1.7 $LW_{out}$ . . . 37
3.2 Energy Fluxes . . . 42
3.2.1 $EF$ . . . 42
3.2.2 $H$ . . . 47
3.2.3 $LE$ . . . 52
3.2.4 $NEE$ . . . 57
3.2.5 $Resp$ . . . 62
3.2.6 $GPP$ . . . 67

**4 Lower Troposphere** **72**
4.1 $z_i$ . . . 72
4.2 $\theta$ . . . 77
4.3 $q$ . . . 91
4.4 $CO_2$ . . . 105
4.5 $U$ . . . 117
4.6 $w_{dir}$ . . . 131
4.7 Cloud Cover . . . 145
4.7.1 Low Cloud Cover . . . 145
4.7.2 Medium Cloud Cover . . . 147
4.7.3 High Cloud Cover . . . 149
4.8 Metrics . . . 151
4.8.1 $DXR$ . . . 151

**Acknowledgements** **154**

**5 Data & Code Availability** **154**

# 1 Introduction

The Carbon Atmospheric Tracer Research to Improve Numerics and Evaluation (CATRINE) project is dedicated to improving the accuracy of atmospheric tracer transport in numerical weather and climatological models, which are essential for reliable emission monitoring. As part of this effort, dedicated **testbeds** have been developed to evaluate model performance under varying (meteorological) conditions. A central aim of these testbeds is to compare model outputs from the European Centre for Medium-Range Weather Forecasts (ECMWF) Integrated Forecasting System (IFS) with observational data and other modelling tools. Through these comparisons, the project seeks to identify the current status-quo and characterise **systematic errors**. Systematic errors refer to consistent and repeatable deviations between model results (or observations) and the true state of the atmosphere. These errors typically arise from shortcomings in model formulation, physical parameterisations, or observational techniques, and can lead to predictable biases, such as persistent underestimation of tracer transport efficiency. In this technical report we outline the key objectives, methodology, considerations and results from on of these testbeds over the Amazon rainforest.

# 2 Methodology

In this section we highlight the focus of the testbed, the CloudRoots-Amazon22 campaign, after which we discuss the general methodology to identify systematic errors: variable groups, statistics and metrics.

## 2.1 Testbed Focus: CloudRoots-Amazon22

The testbed is focused on the CloudRoots-Amazon22 campaign (Vilà-Guerau de Arellano et al., 2024). The campaign took place at the Amazon Tall Tower Observatory (ATTO) (Andreae et al., 2015) during the Amazonian dry season (9th to 18th August 2022, `https://cloudroots.wur.nl/`), a period typically characterised by the daytime presence of shallow-to-deep convective clouds, with influences from the larger-scale (e.g., horizontal advection) expected to be negligibly small. On a typical day, shallow convective clouds begin to develop from 10:00 local time (LT, UTC–4). These clouds either transition into deeper clouds near the end of the afternoon (15:00 LT) if conditions are favourable, or they dissipate following the setting sun. Consequently, this campaign is well suited to identifying systematic errors associated with the development and transition of clear-to-shallow-to-deep convective clouds, as well as the associated transport of $CO_2$. All details on the campaign can be found via Vilà-Guerau de Arellano et al. (2024).

### 2.1.1 Observational Data

From the CloudRoots-Amazon22 campaign, tall towers, radiosonde and aircraft observations are available to use. These observations include incoming and reflected radiation components, (turbulent) fluxes of energy (e.g., sensible heat ($H$)) and $CO_2$ (e.g.,net ecosystem exchange ($NEE$)) along the tall-towers, state variables (e.g., potential temperature ($\theta$) but also $CO_2$) along the tall-tower and in the lower troposphere through radiosondes and aircaft flights.

Data has been grouped (aggregated) based on the respective convective regime; *shallow convective* (6 days) and *shallow-to-deep convective* (4 days). Consequently, representative days (mean, with a standard deviation) can be used (Vilà-Guerau de Arellano et al., 2024).

### 2.1.2 Modelling Tools

The following modelling tools are used to compare to the observations:

- IFS CY49R1 simulations at 28, 9 and 4.4 km resolution (CATRINE specific). Experiment-IDs: ia61 (28 km), ia62 (9 km) and ia63 (4.4 km). To ensure a consistent comparison across horizontal resolutions, an area of approximately 100 km × 100 km centred on ATTO is selected from each simulation and the mean value is calculated. In this way, we ensure that approximately the same geographical area is analysed for each resolution for ATTO.

- Chemistry Land-surface Atmosphere Soil Slab (CLASS) model mixed-layer simulations (van Stratum et al., 2014). The simulation is only available for the diurnal cycle (6 - 18 LT) of the shallow convective regime, split into a simulation with (CLOUDS) and without (NO CLOUDS) shallow convective clouds.
- Dutch Atmospheric Large Eddy Simulation (DALES) (Heus et al., 2010) with respective domain sizes. More details can be found in de Feiter et al. (2025). The simulation is only available for the diurnal cycle (6 - 18 LT) of the shallow convective regime, split into a simulation with (CLOUDS) and without (NO CLOUDS) shallow convective clouds.

## 2.2 Variable Groups

To limit our scope for comparing observations and the various modelling tools, we distinguish **four** variable groups as listed by Table 1:

Table 1: Defined variable groups.

| **Variable Group** | **Variables** |
|---|---|
| Radiation | $Q_{net}$, $SW_{net}$, $LW_{net}$, $SW_{in}$, $SW_{out}$, $LW_{in}$, $LW_{out}$ |
| Energy fluxes | $EF$, $H$, $LE$ |
| Surface fluxes | $NEE$, $Resp$, $GPP$ |
| Lower troposphere | $\theta$, $q$, $U$, $w_{dir}$, $CO_2$, $z_i$... $u$, $v$, $w$<br>cloud cover, ... LCL, $M$, cloud base, cloud top |

The abbreviations in Tab 1 refer to:

Table 2: Abbreviations and their meaning.

| **Abbrevation** | **Variable** | **Unit** | **Abbrevation** | **Variable** | **Unit** |
|---|---|---|---|---|---|
| $Q_{net}$ | net radiation | W $m^{-2}$ | $\theta$ | potential temperature | K |
| $SW_{net}$ | net shortwave radiation | W $m^{-2}$ | $q$ | specific humidity | g $kg^{-1}$ |
| $LW_{net}$ | net longwave radiation | W $m^{-2}$ | $u$ | zonal wind | m $s^{-1}$ |
| $SW_{in}$ | incoming shortwave radiation | W $m^{-2}$ | $v$ | meridional wind | m $s^{-1}$ |
| $SW_{out}$ | outgoing shortwave radiation | W $m^{-2}$ | $w$ | vertical wind speed | m $s^{-1}$ |
| $LW_{in}$ | incoming longwave radiation | W $m^{-2}$ | $U$ | total wind speed | m $s^{-1}$ |
| $LW_{out}$ | outgoing longwave radiation | W $m^{-2}$ | $w_{dir}$ | wind direction | ° |
| $EF$ | evaporative fraction ($\frac{LE}{LE+H}$) | (0 - 1) | $CO_2$ | $CO_2$ mixing ratio | ppm |
| $H$ | sensible heat flux | W $m^{-2}$ | $z_i$ | boundary layer height | m |
| $LE$ | latent heat flux | W $m^{-2}$ | LCL | lifting condensation level | m |
| $NEE$ | net ecosystem exchange | $\mu$mol $m^{-2}$ $s^{-1}$ | cloud cover | cloud cover | (0 - 1) |
| $Resp$ | respiration | $\mu$mol $m^{-2}$ $s^{-1}$ | cloud base | cloud base height | m |
| $GPP$ | gross primary production | $\mu$mol $m^{-2}$ $s^{-1}$ | cloud top | cloud top height | m |
| | | | $M$ | mass flux | m $s^{-1}$ |

Next to visualising the daily or diurnal time series and vertical profiles of these variables, we compare different variables from different variable groups to gain insights in **cause-and-effect** relationships. **Please note**, for ICON-ART at this moment, only meteorological state variables, radiation components and surface energy fluxes are available.

## 2.3 Statistics

To quantify the systematic error, we utilise the following statistical quantities. These statistical quantities are assessed for diurnal ranges and the full campaign of ten days.

- **Pearson Correlation Coefficient ($r$)**. The $r$ is a statistical measure that describes the strength and direction of a linear relationship between model simulations and observations. The coefficient ranges from -1 (perfect negative correlation) to +1 (perfect positive correlation). A correlation close to 0 indicates no linear relationship.

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{1}$$

  Where: $x_i$, $y_i$ are the individual data points of respectively observations (reference) and model (forecast), $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ and $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ the means of the reference and forecast, with $n$ the sample size.

- **Root Mean squared Error (RMSE)**. The RMSE is a measure of the average magnitude of the error between model simulations and observations. It is calculated as the square root of the mean of the squared differences between predicted and observed values. RMSE gives a sense of how much error is typically present in model predictions, with higher values indicating larger discrepancies.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{2}$$

  Where: $y_i$, $\hat{y}_i$ are the individual data points of respectively observations (reference) and model (forecast), with $n$ the sample size.

- **Bias**. The bias is the average difference between model simulations and observations over a defined period, spatial domain, or set of cases. In this context, bias is considered a specific form of systematic error and is used as a quantitative indicator of persistent discrepancies.

$$\text{Bias} = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i) \tag{3}$$

  Where: $y_i$, $\hat{y}_i$ are the individual data points of respectively observations (reference) and model (forecast), with $n$ the sample size.

- **Absolute difference in diurnal range ($|DXR|$)**. The $|:DXR:|$ is the absolute difference between the diurnal range of variable $X$ in model simulations and in observations. The diurnal range refers to the difference between daily maximum and minimum values of $\phi$, capturing the amplitude of its daily cycle. The $|:DXR:|$ metric serves as an indicator of ability to reproduce observed daily variability. By focusing on the absolute difference, it quantifies the magnitude of deviation regardless of direction, making it particularly useful for assessing discrepancies in temporal variability patterns, independent of overall bias.

$$|\,DXR\,| = \left|(X_{obs}^{max} - X_{obs}^{min}) - (X_{model}^{max} - X_{model}^{min})\right| \tag{4}$$

  Where: $X_{obs}$ and $X_{model}$ refer to the data point individual data points of respectively observations (reference) and model (forecast), with superscript $max$ denoting the maximum daily value, and $min$ the minimum daily value.

## 2.4 Metrics

Next to the statistical quantities, we also define several (additional) metrics to quantify the systematic errors. These metrics quantities are assessed for diurnal ranges and the full campaign of ten days.

- **Diurnal Range ($DXR$).** The $DXR$ of a variable $\phi$ refers to the difference between its daily maximum ($X^{max}$) and minimum ($X^{min}$) values. It captures the amplitude of the daily cycle and reflects how much the variable fluctuates throughout the day.

$$DXR = (X^{max} - X^{min}) \tag{5}$$

# 3 Results

## 3.1 Radiation

### 3.1.1 $Q_{net}$

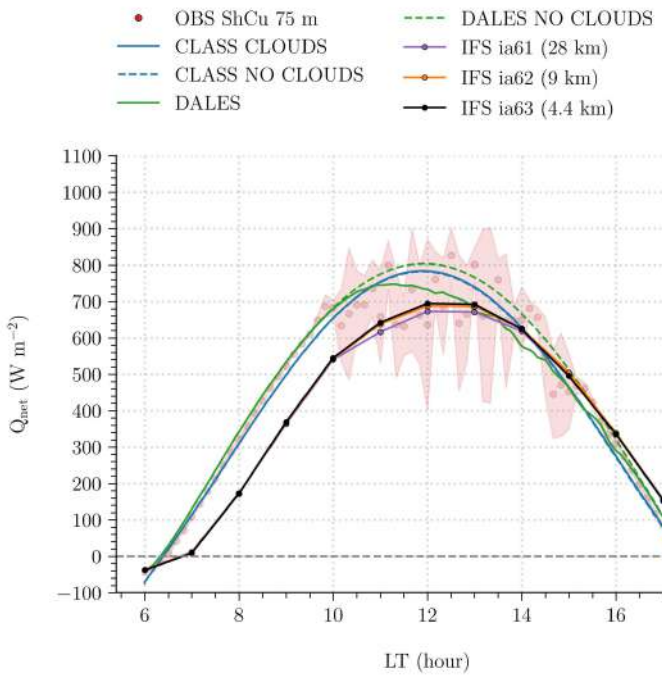



Figure 1: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 75 m) and simulated (CLASS, DALES and IFS) $Q_{net}$ for the Shallow Convective (ShCu) regime.

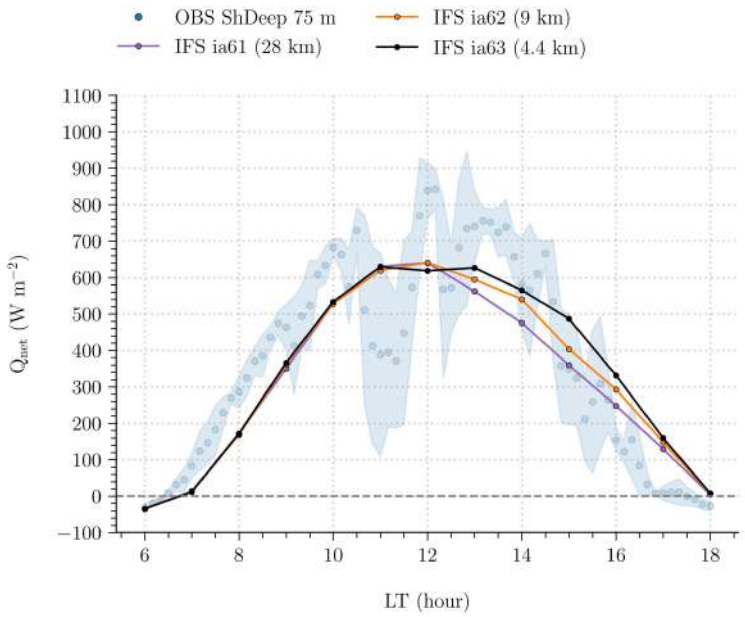



Figure 2: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 75 m) and simulated (IFS) $Q_{net}$ for the Shallow-to-Deep Convective (ShDeep) regime.

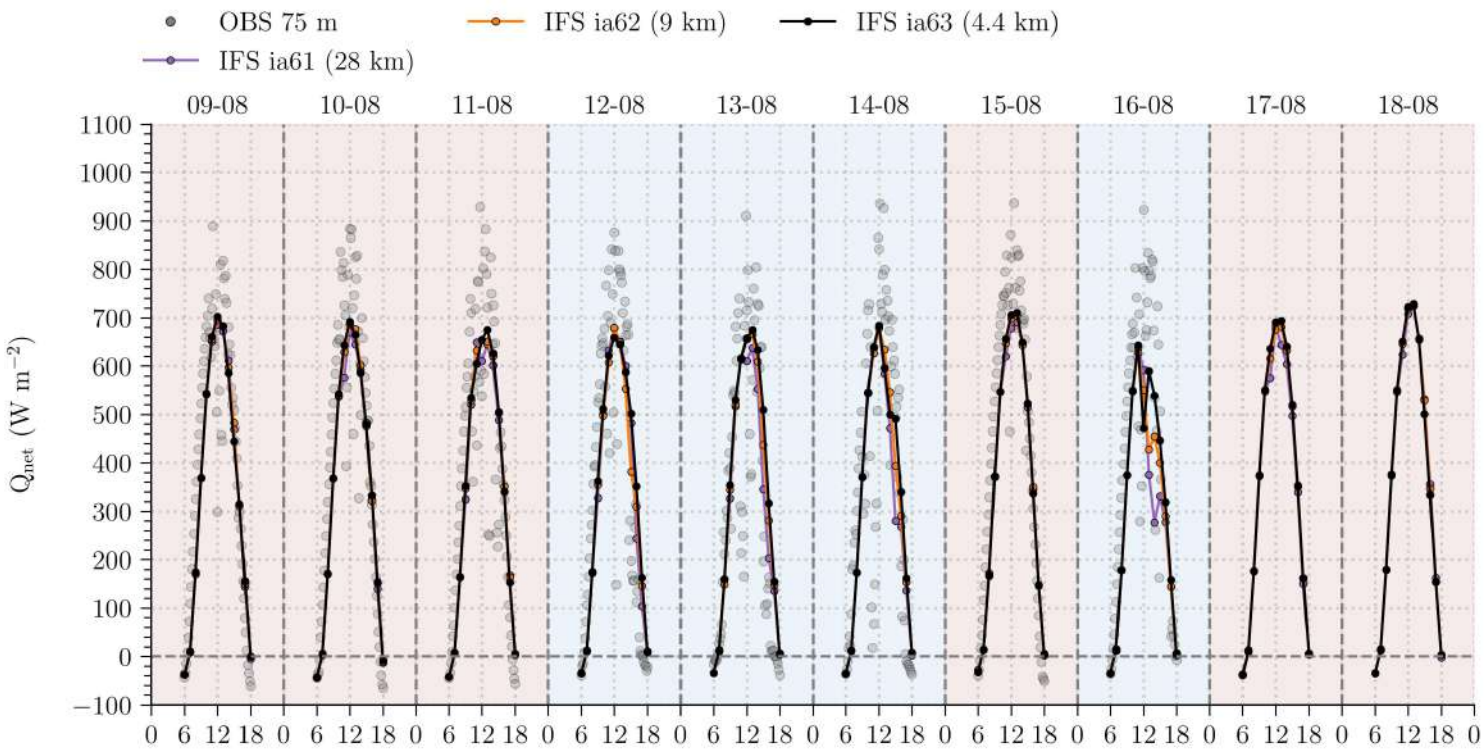



Figure 3: Diurnal cycle of observed (markers, at 75 m) and simulated (IFS) $Q_{net}$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).



Funded by the European Union

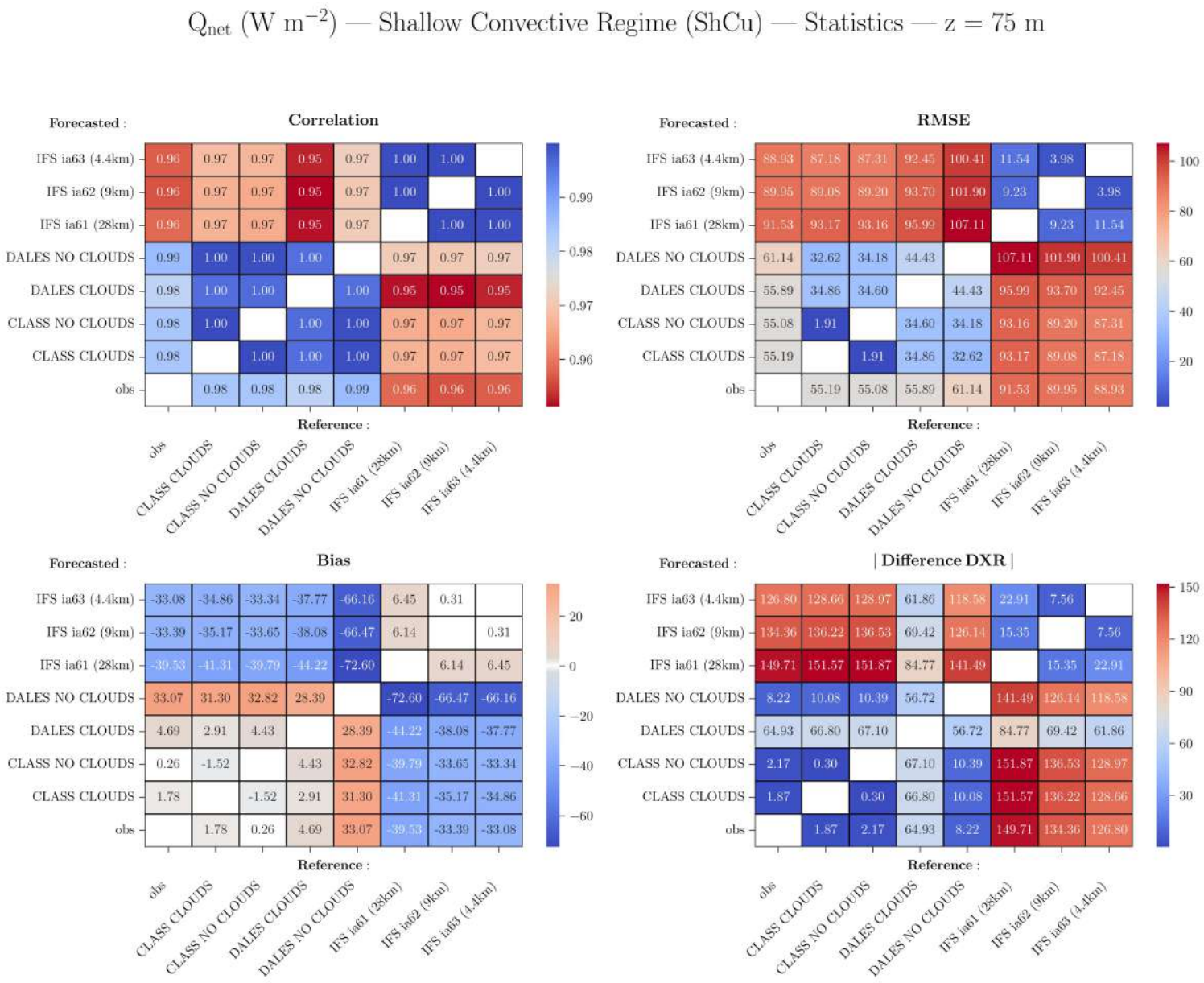



Figure 4: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (CLASS, DALES and IFS) $Q_{net}$ for the Shallow Convective (ShCu) regime.



Funded by the European Union

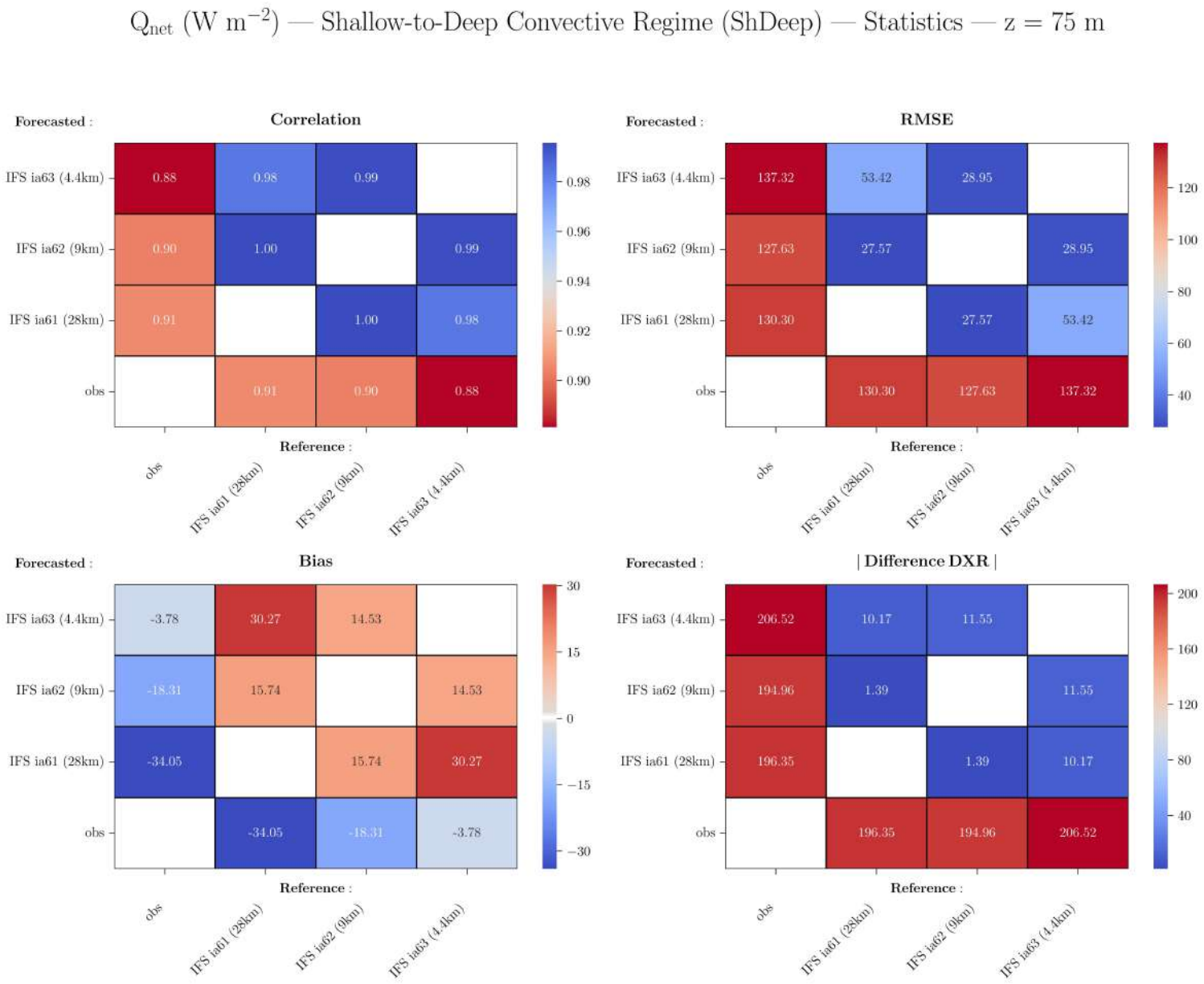



Figure 5: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (IFS) $Q_{net}$ for the Shallow-to-Deep Convective (ShDeep) regime.

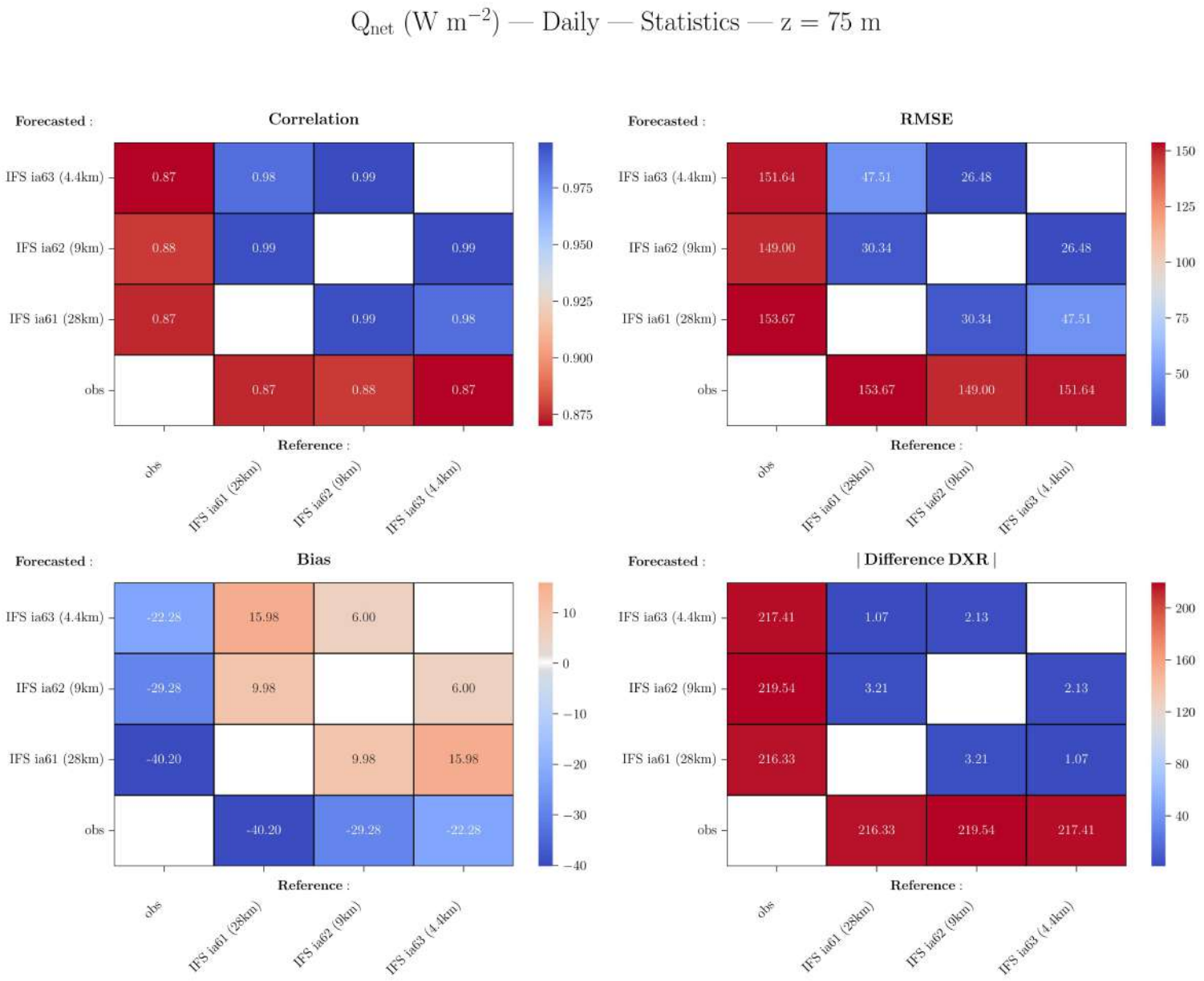



Figure 6: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (IFS) $Q_{net}$ for the full CloudRoots-Amazon22 campaign.



Funded by the European Union

### 3.1.2 $SW_{net}$

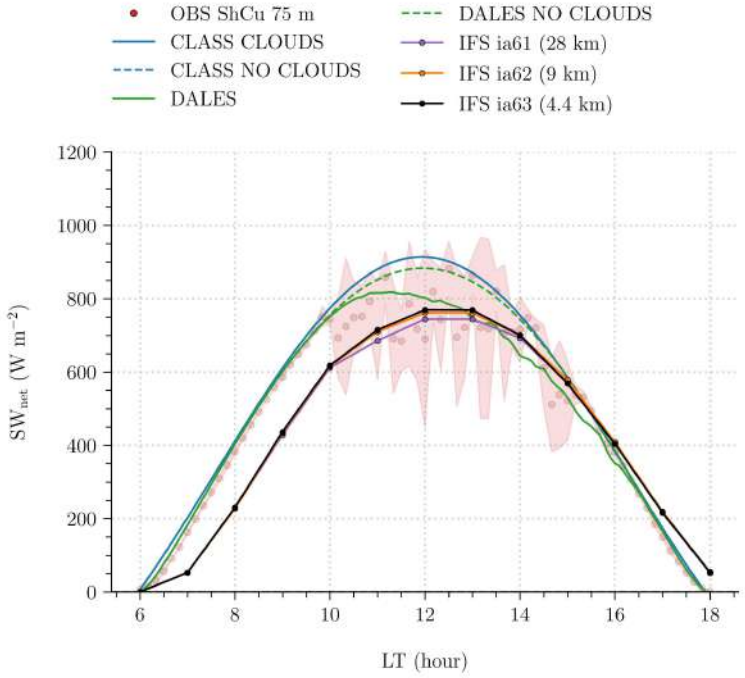



Figure 7: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 75 m) and simulated (CLASS, DALES and IFS) $SW_{net}$ for the Shallow Convective (ShCu) regime.

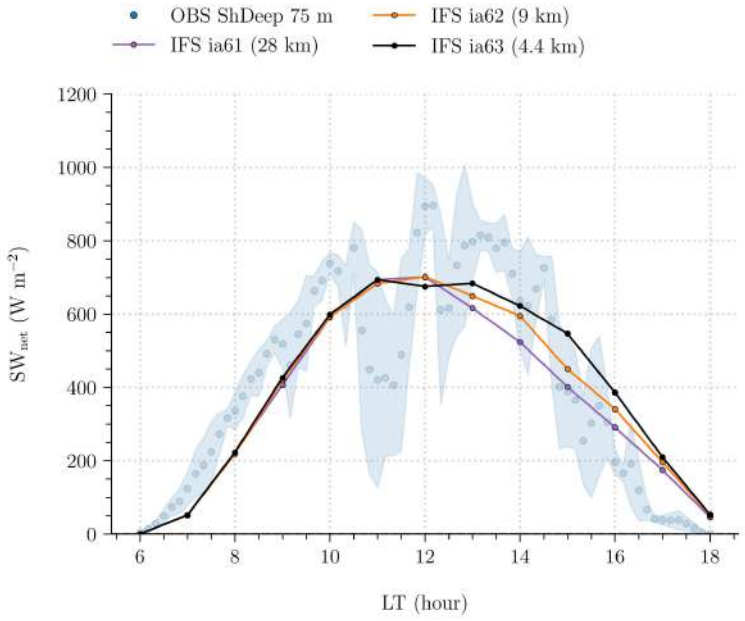



Figure 8: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 75 m) and simulated (IFS) $SW_{net}$ for the Shallow-to-Deep Convective (ShDeep) regime.

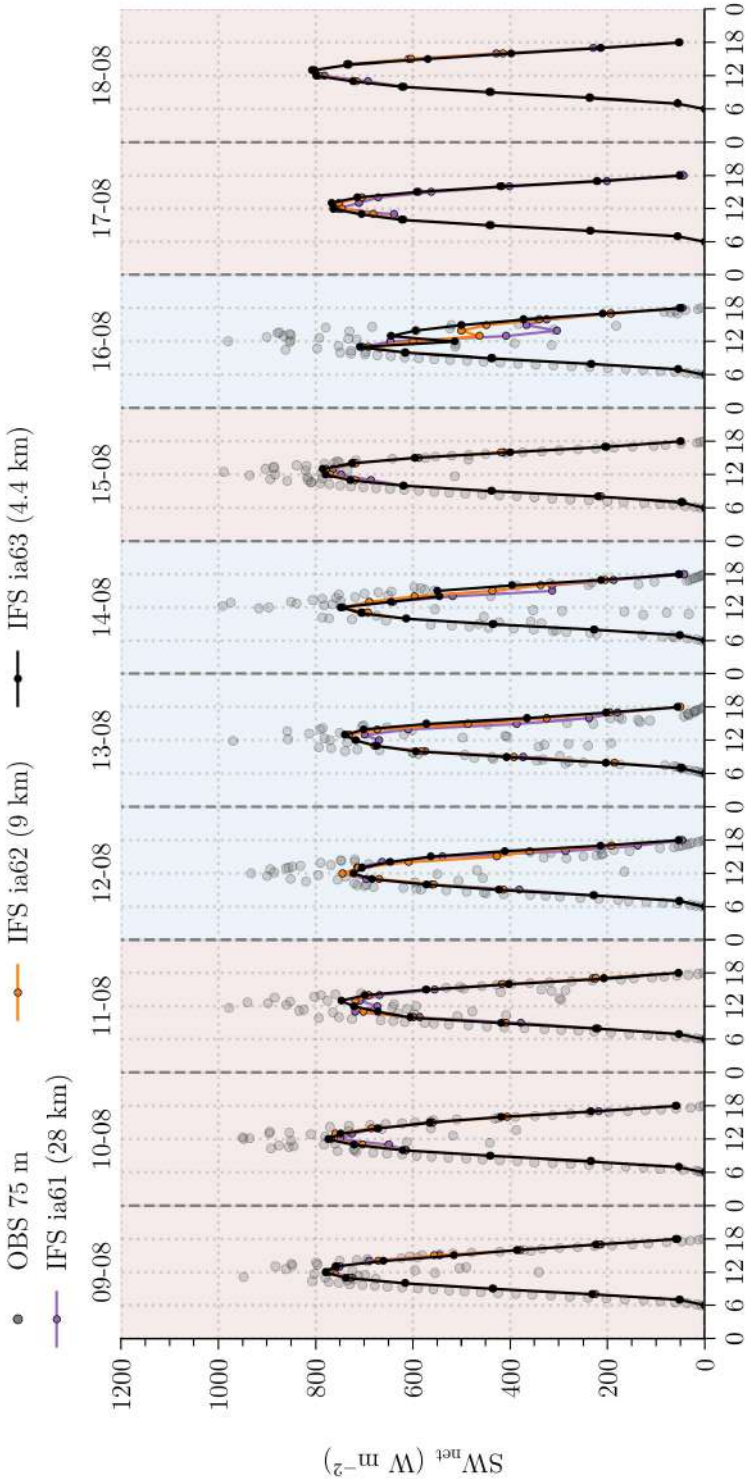



Figure 9: Diurnal cycle of observed (markers, at 75 m) and simulated (IFS) $SW_{net}$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).

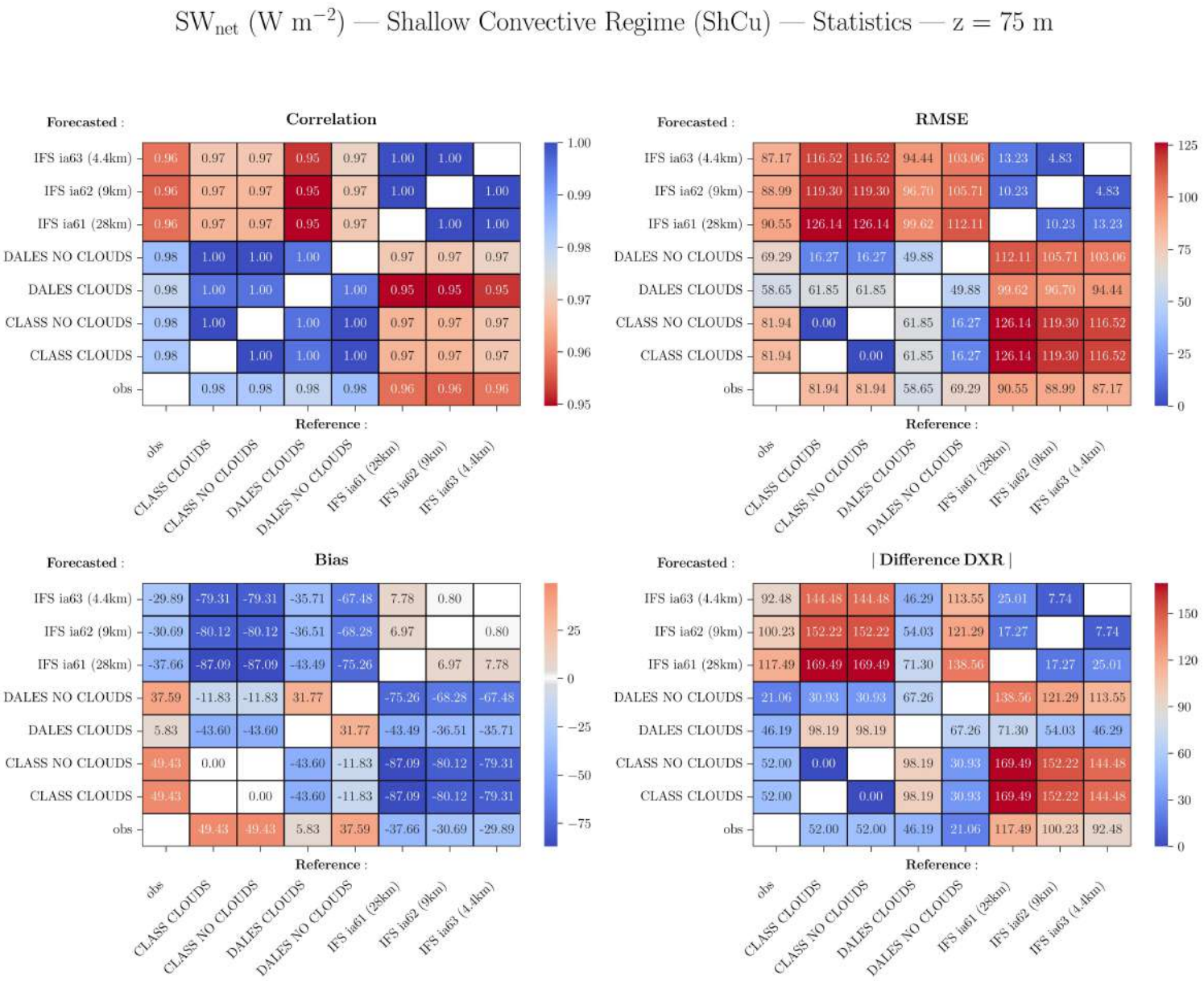



Figure 10: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (CLASS, DALES and IFS) $SW_{net}$ for the Shallow Convective (ShCu) regime.

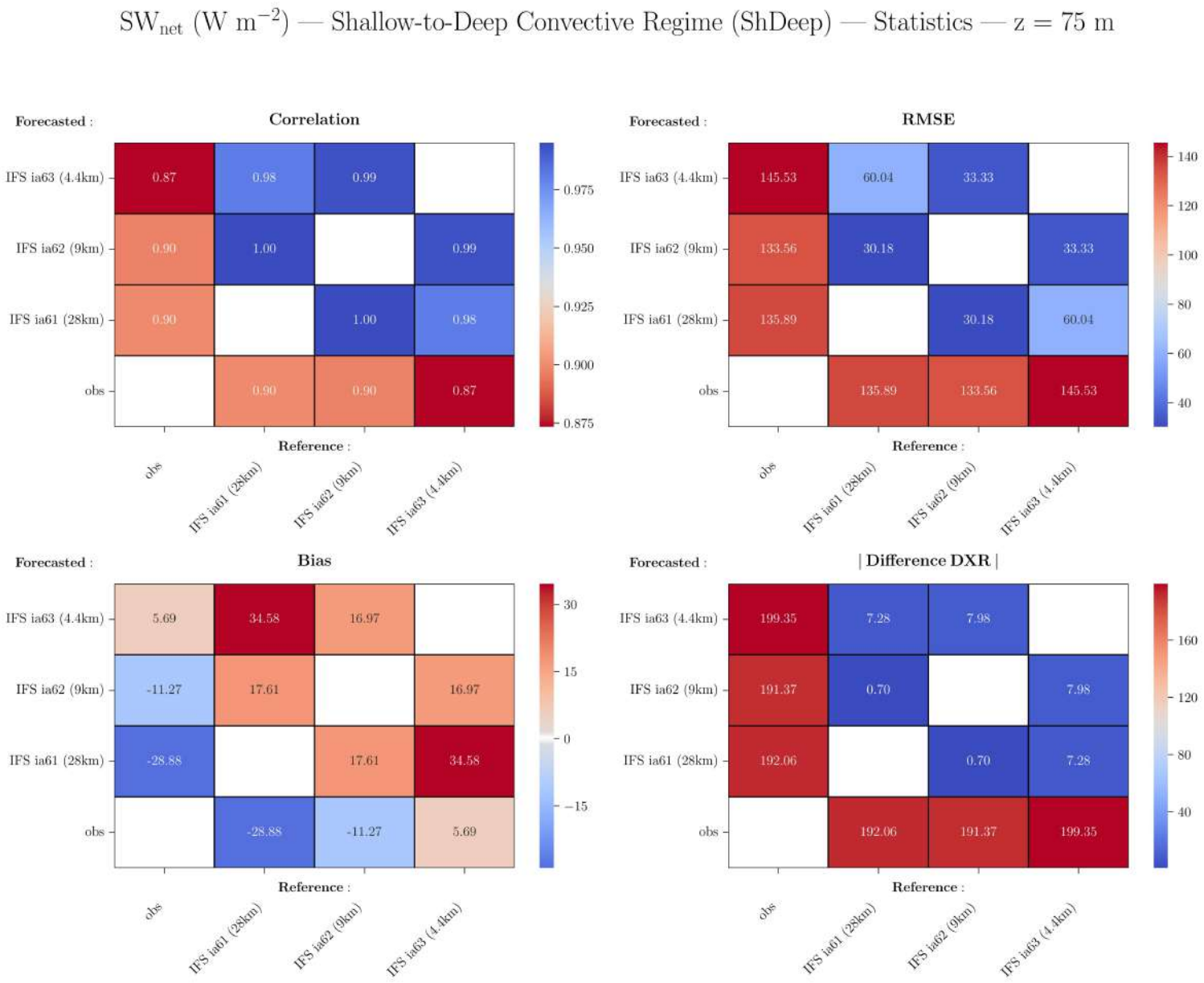



Figure 11: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (IFS) $SW_{net}$ for the Shallow-to-Deep Convective (ShDeep) regime.

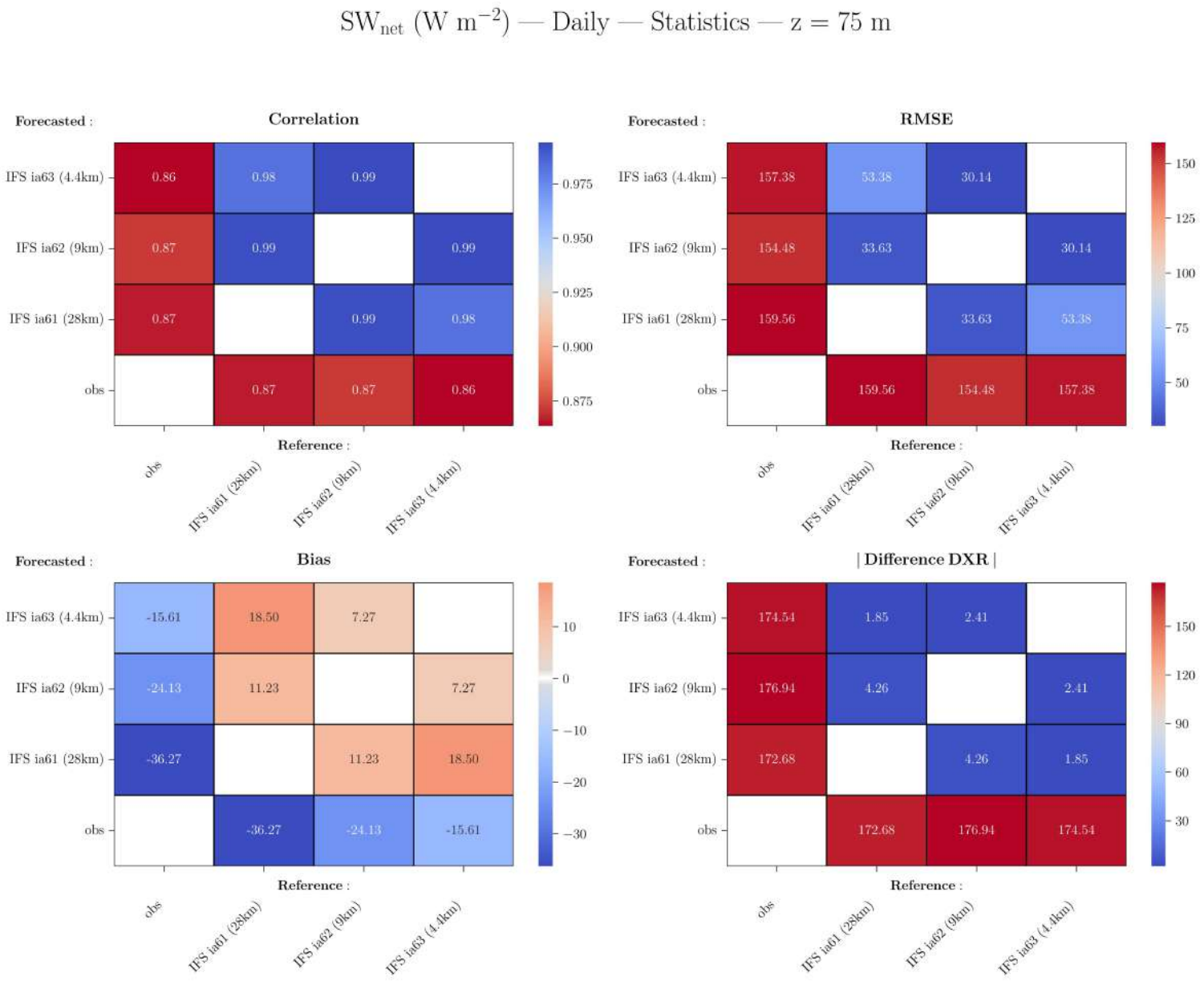



Figure 12: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (IFS) $SW_{net}$ for the full CloudRoots-Amazon22 campaign.



Funded by the European Union

### 3.1.3 $LW_{net}$

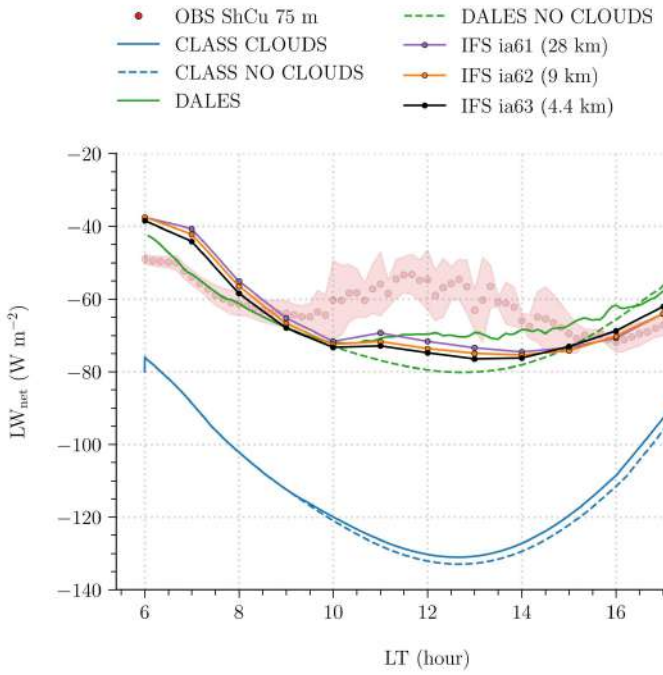



Figure 13: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 75 m) and simulated (CLASS, DALES and IFS) $LW_{net}$ for the Shallow Convective (ShCu) regime.

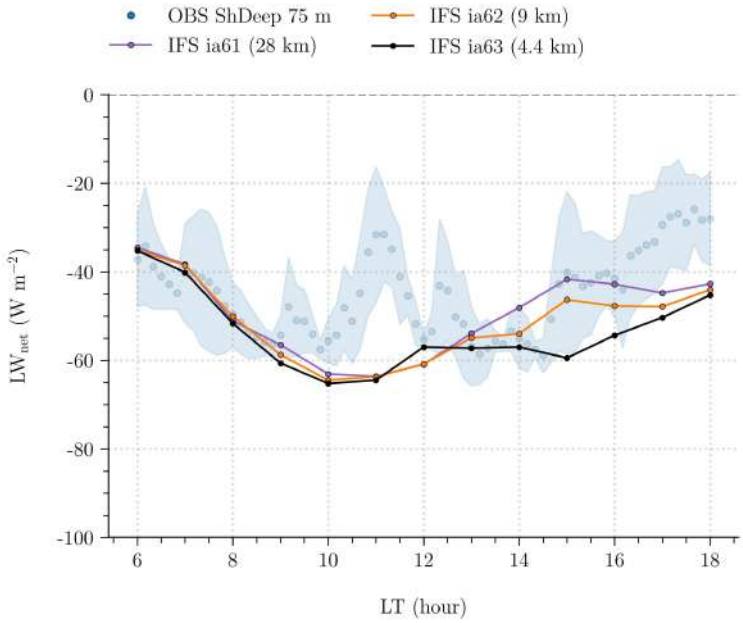



Figure 14: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 75 m) and simulated (IFS) $LW_{net}$ for the Shallow-to-Deep Convective (ShDeep) regime.

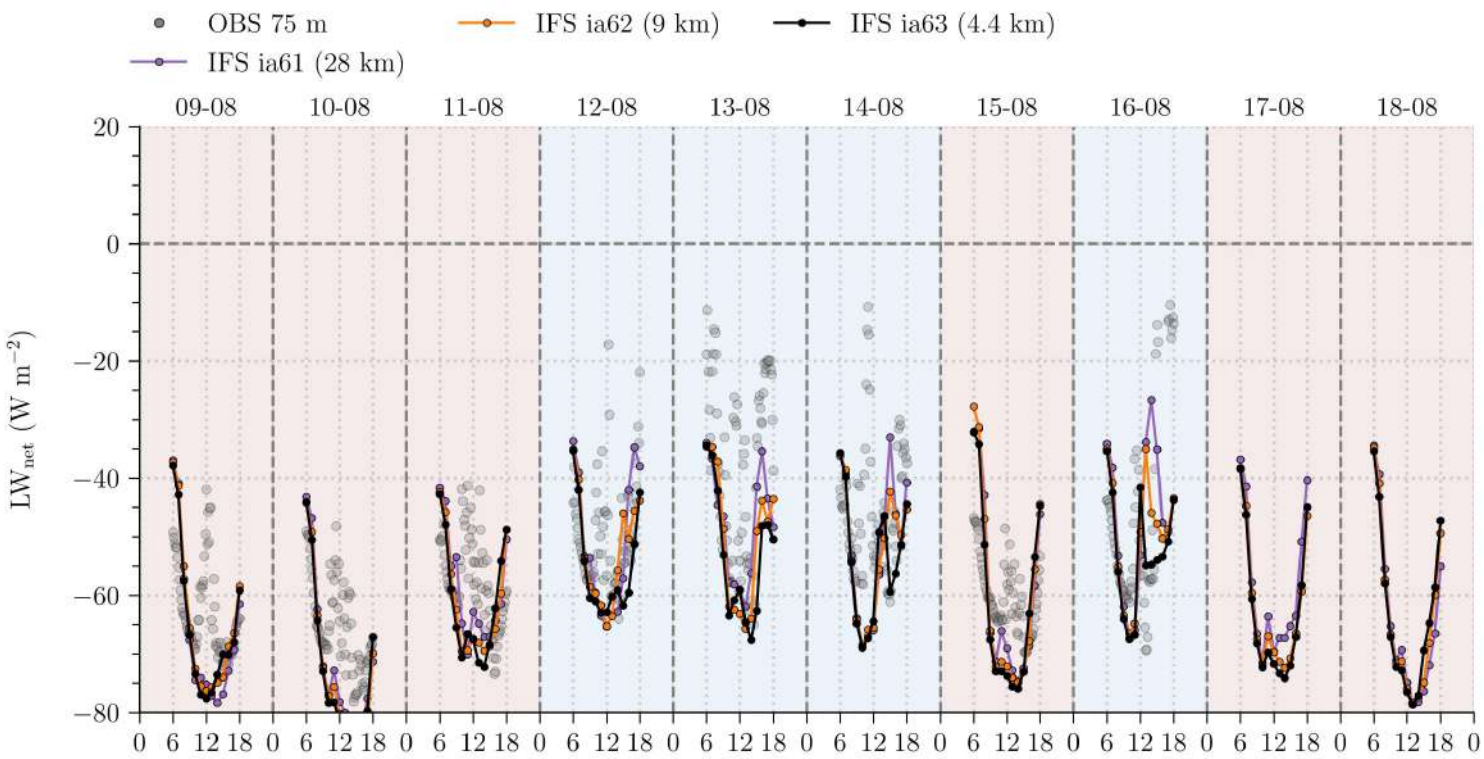



Figure 15: Diurnal cycle of observed (markers, at 75 m) and simulated (IFS) $LW_{net}$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).



Funded by the European Union

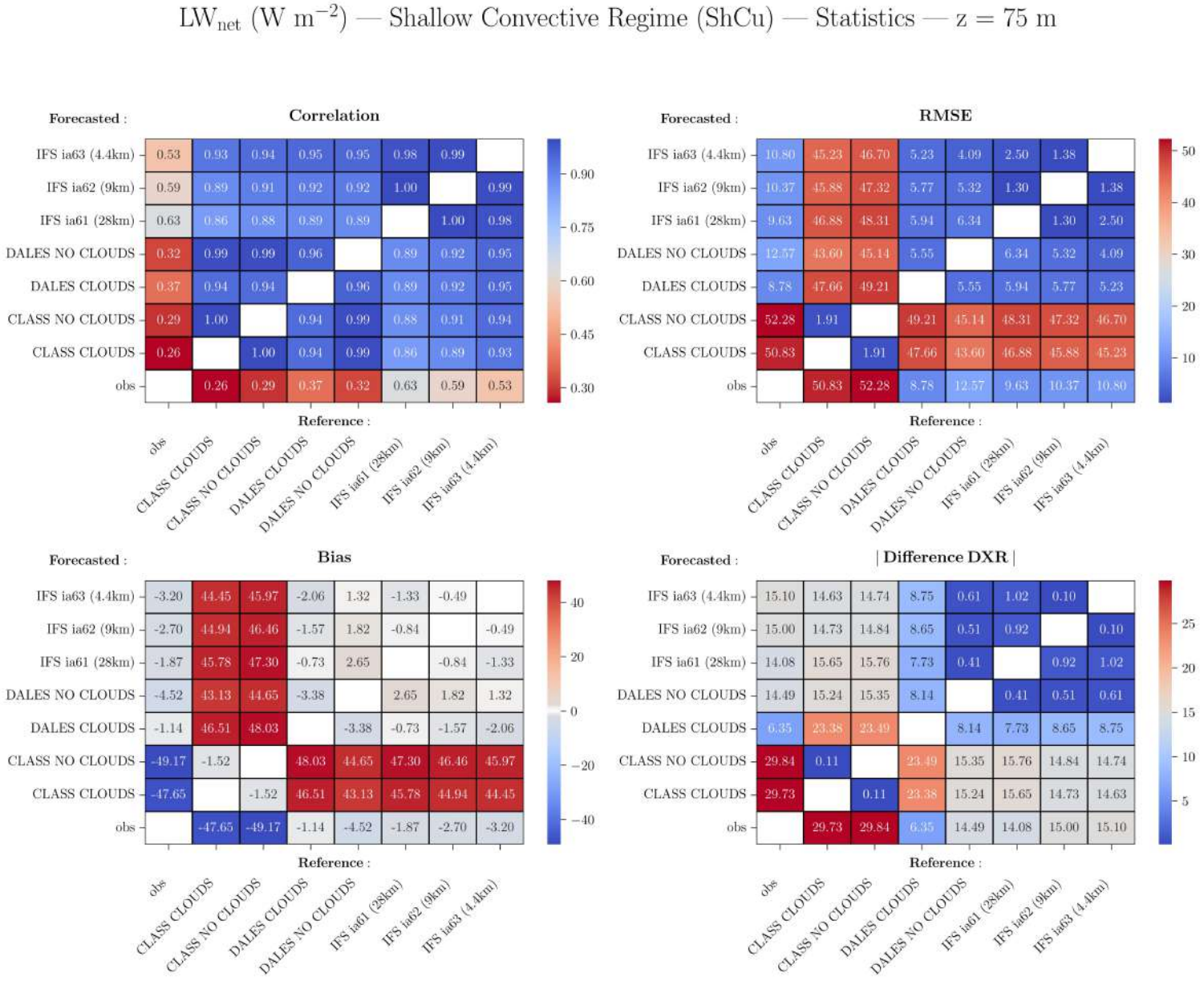



Figure 16: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (CLASS, DALES and IFS) LW$_{net}$ for the Shallow Convective (ShCu) regime.

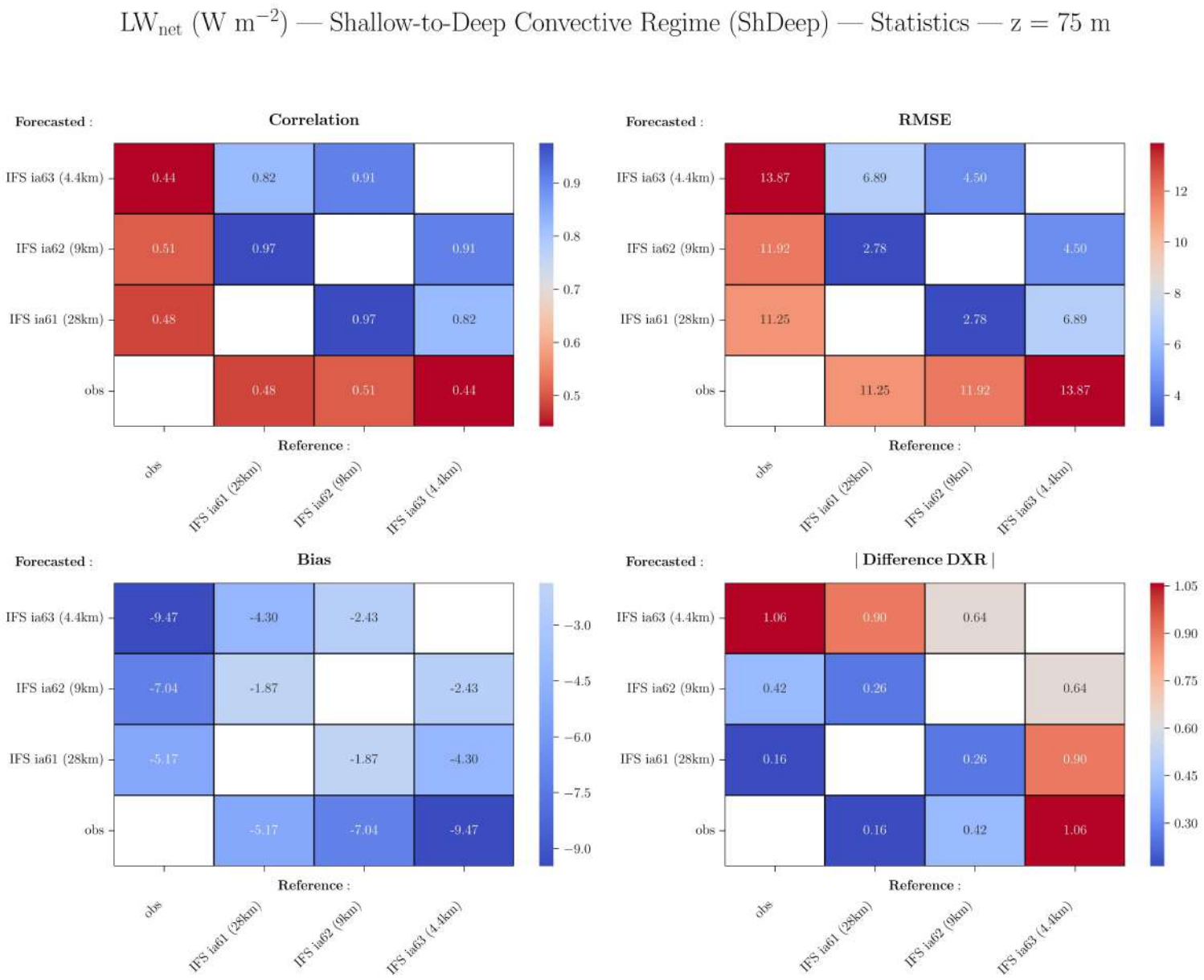



Figure 17: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (IFS) $LW_{net}$ for the Shallow-to-Deep Convective (ShDeep) regime.

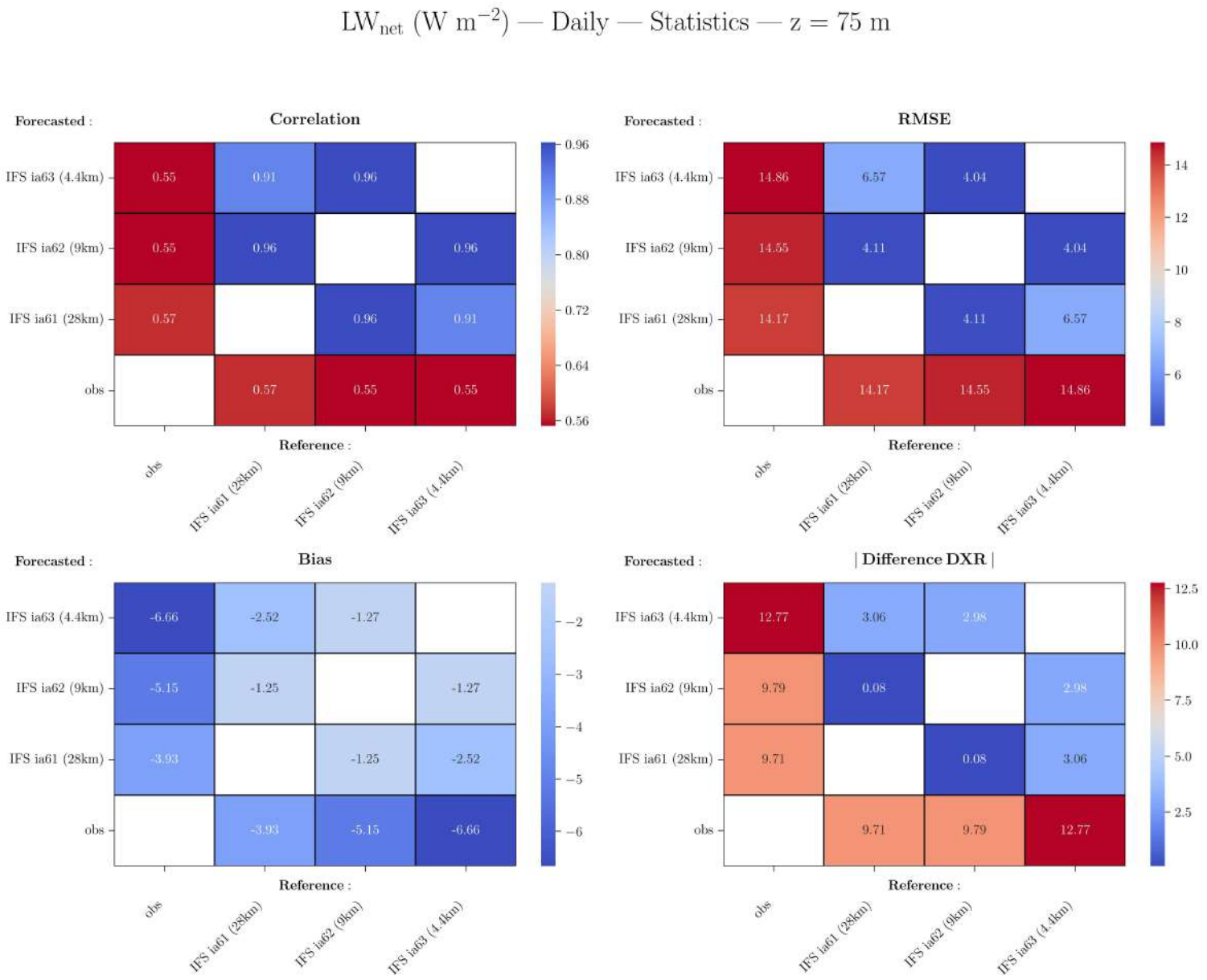



Figure 18: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (IFS) $LW_{net}$ for the full CloudRoots-Amazon22 campaign.



Funded by the European Union

### 3.1.4 $SW_{in}$

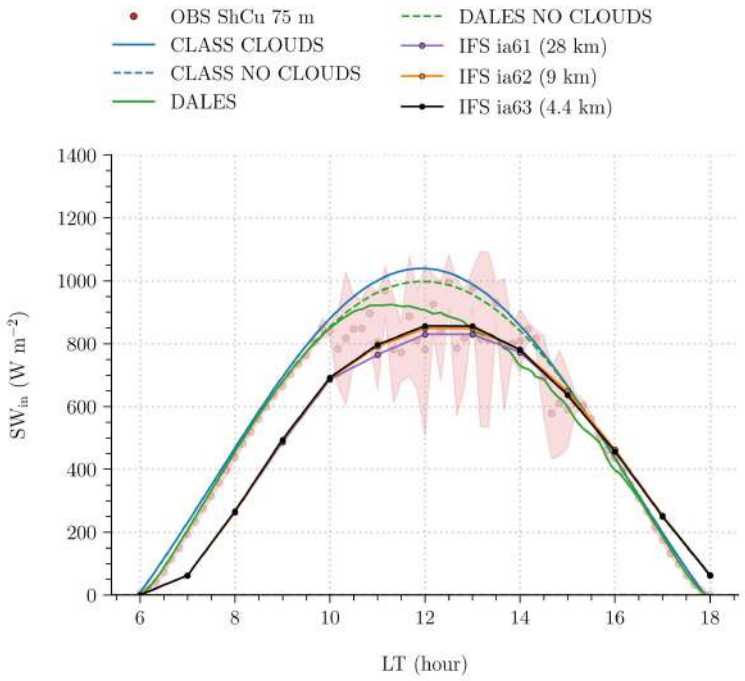



Figure 19: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 75 m) and simulated (CLASS, DALES and IFS) $SW_{in}$ for the Shallow Convective (ShCu) regime.

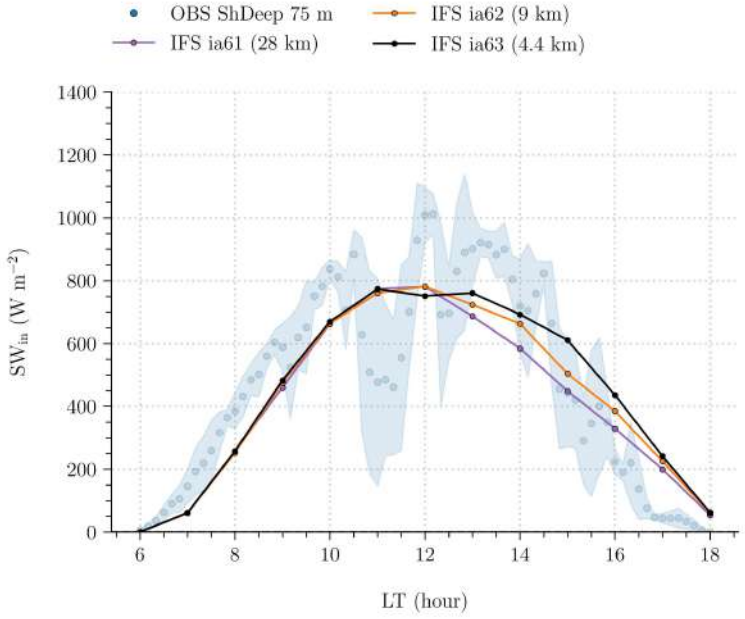



Figure 20: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 75 m) and simulated (IFS) $SW_{in}$ for the Shallow-to-Deep Convective (ShDeep) regime.

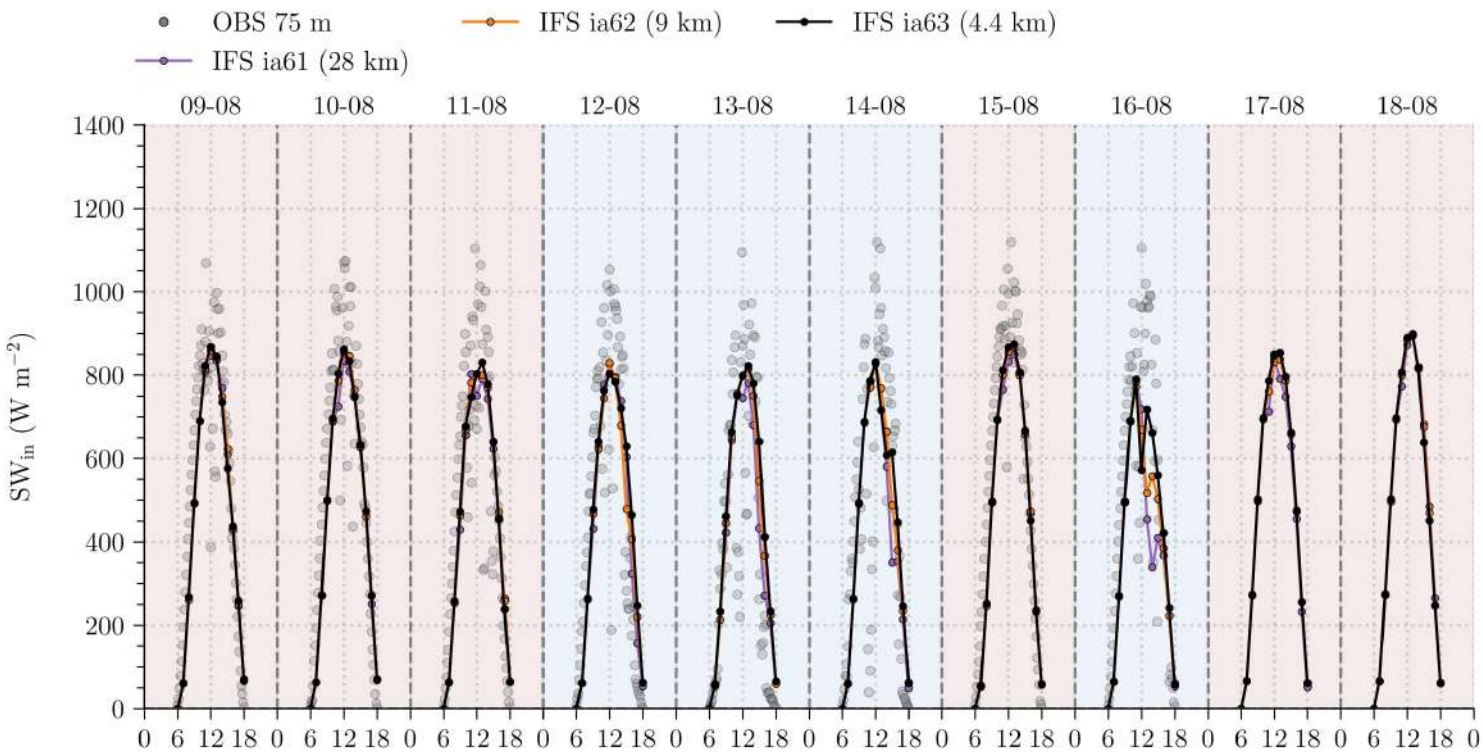



Figure 21: Diurnal cycle of observed (markers, at 75 m) and simulated (IFS) $SW_{in}$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).

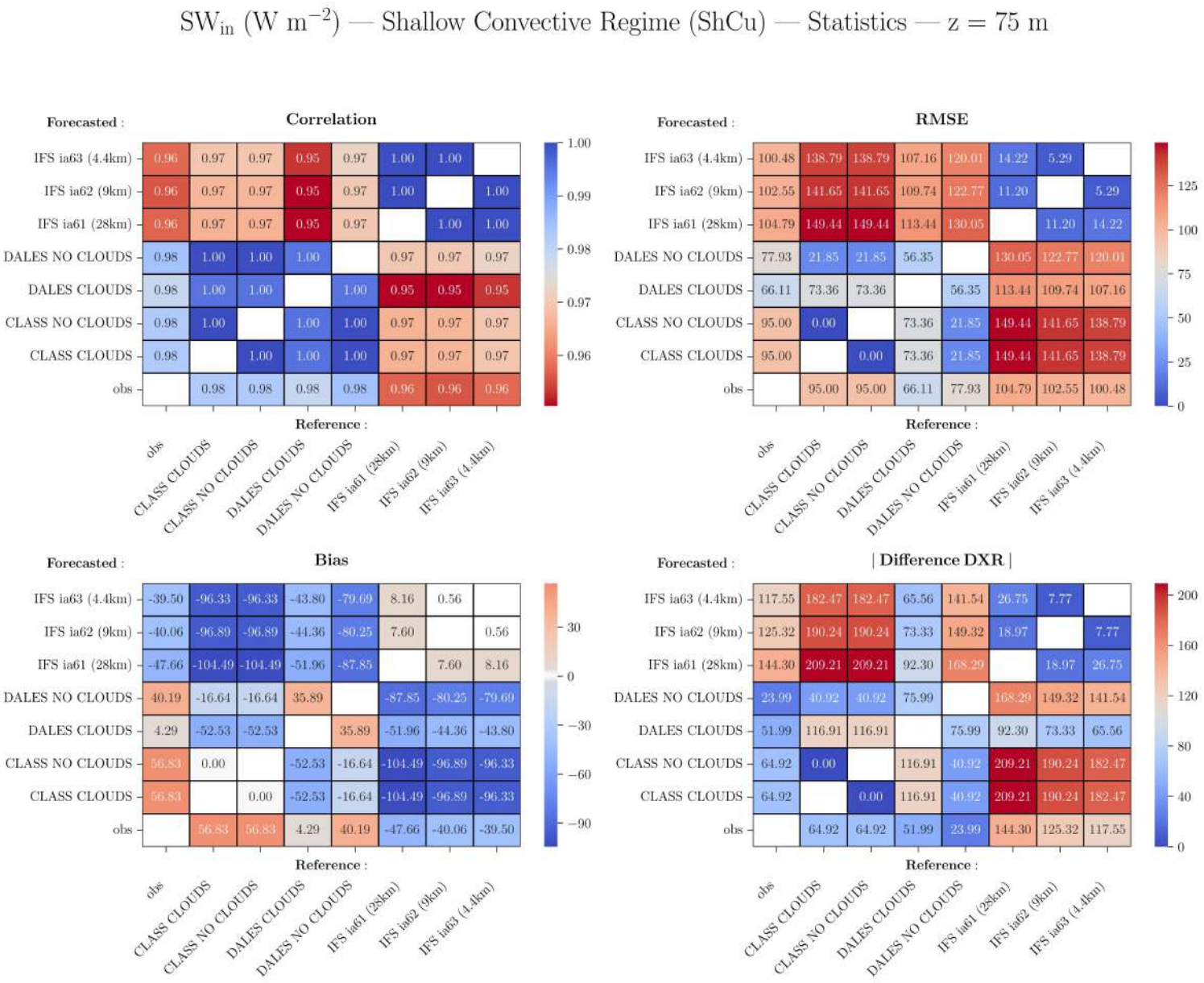



Figure 22: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (CLASS, DALES and IFS) $SW_{in}$ for the Shallow Convective (ShCu) regime.

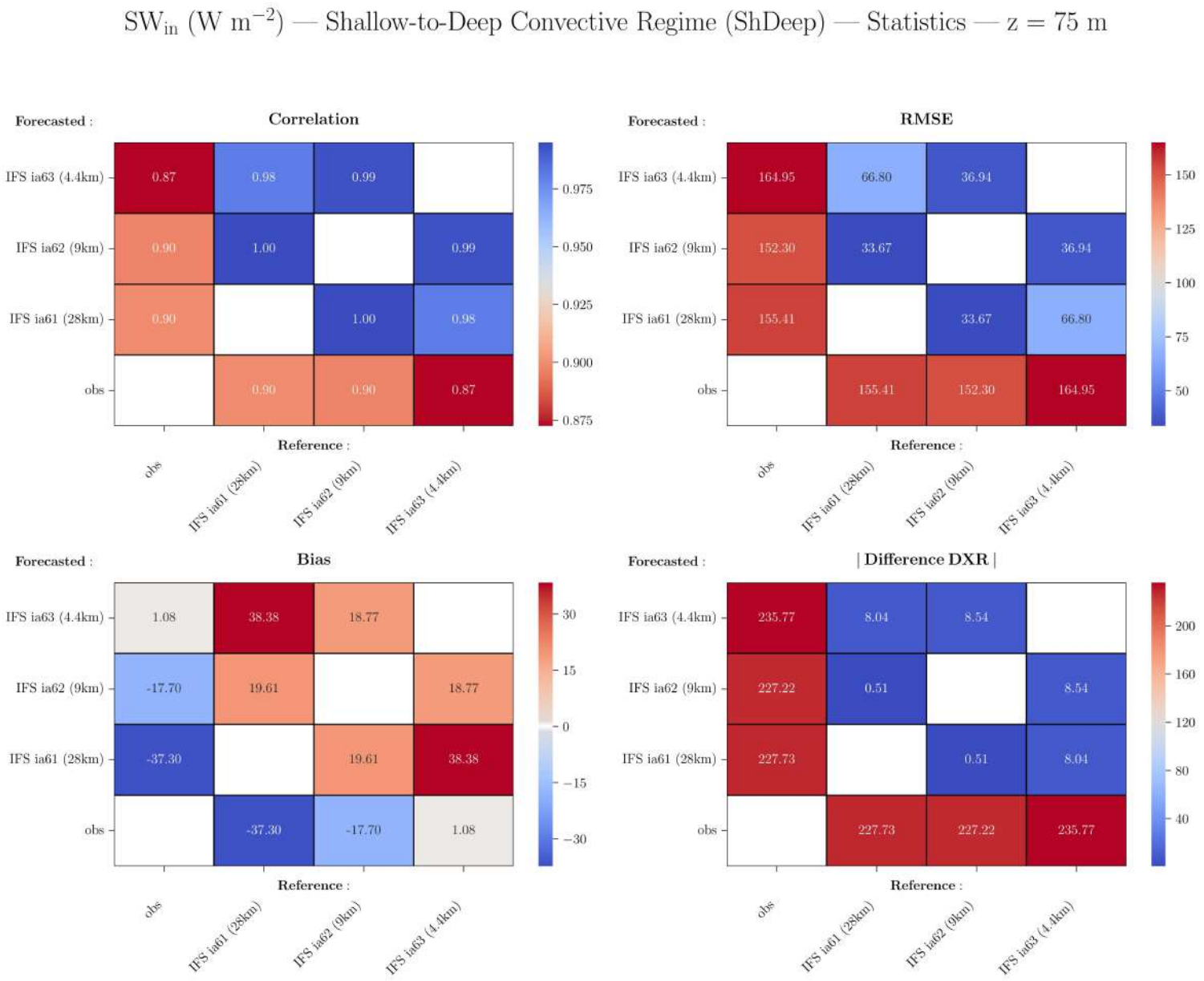



Figure 23: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (IFS) $SW_{in}$ for the Shallow-to-Deep Convective (ShDeep) regime.

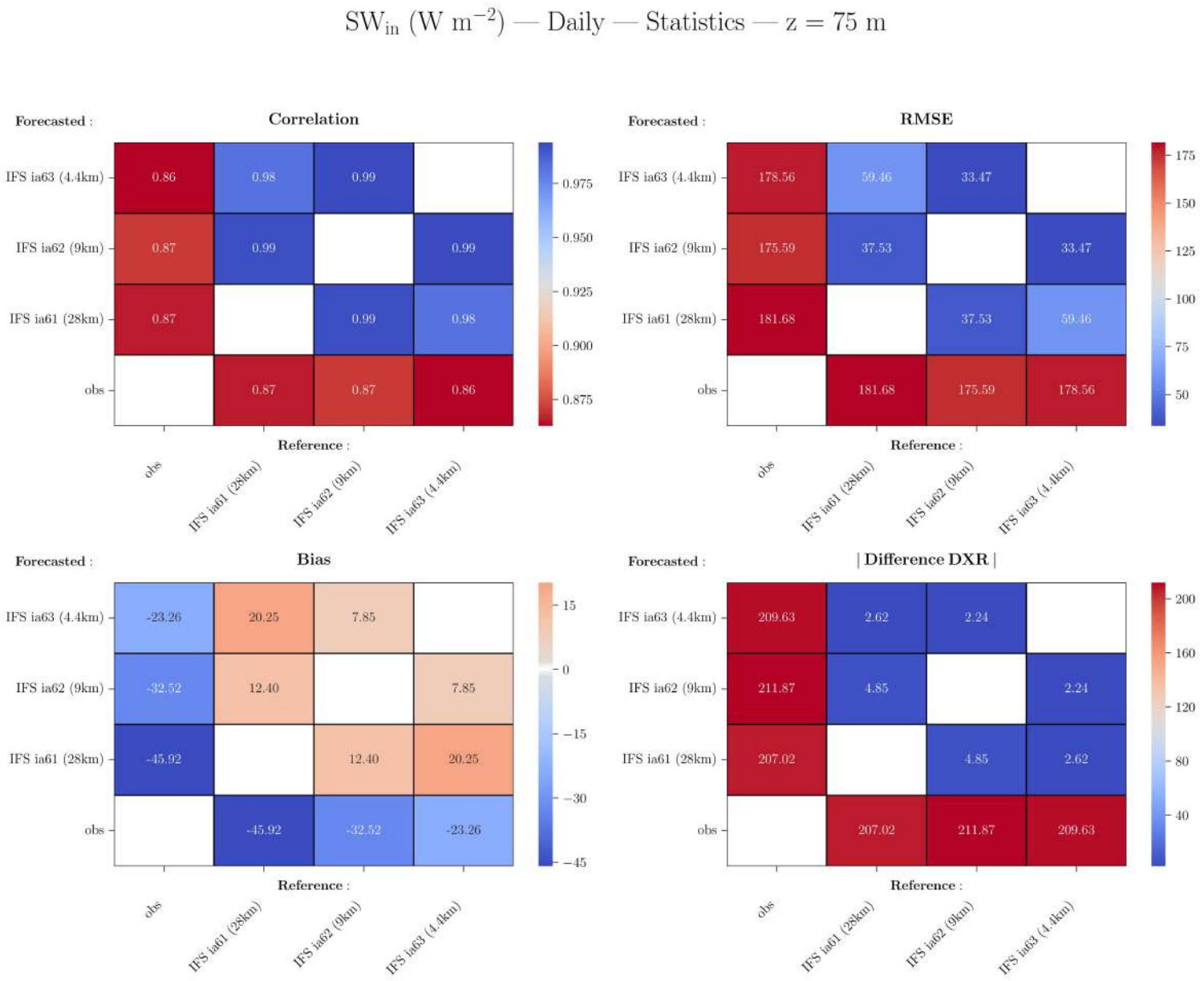



Figure 24: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (IFS) $SW_{in}$ for the full CloudRoots-Amazon22 campaign.



Funded by the European Union

### 3.1.5 $SW_{out}$

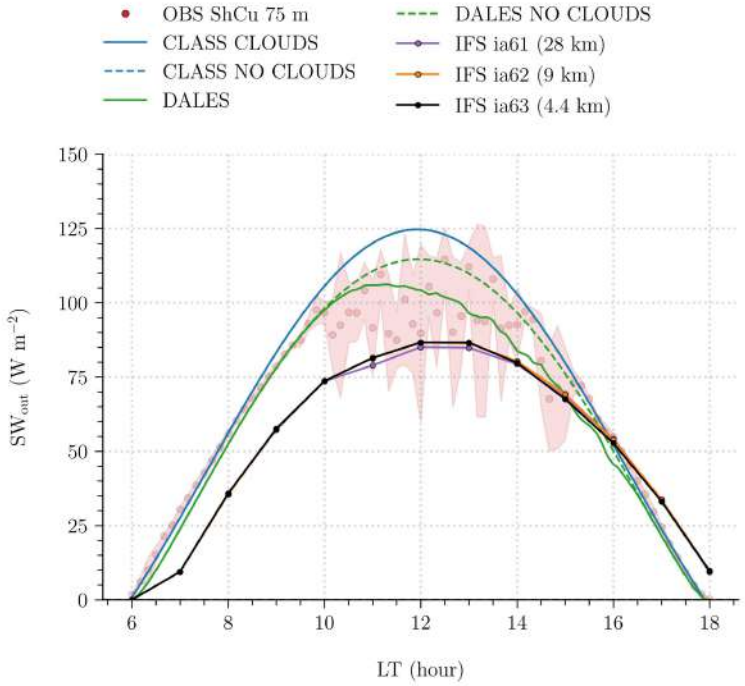



Figure 25: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 75 m) and simulated (CLASS, DALES and IFS) $SW_{out}$ for the Shallow Convective (ShCu) regime.

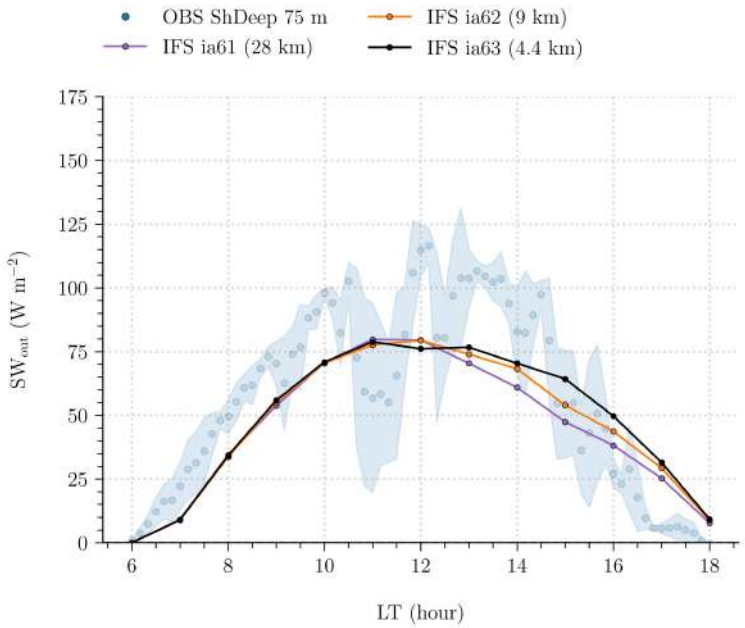



Figure 26: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 75 m) and simulated (IFS) $SW_{out}$ for the Shallow-to-Deep Convective (ShDeep) regime.

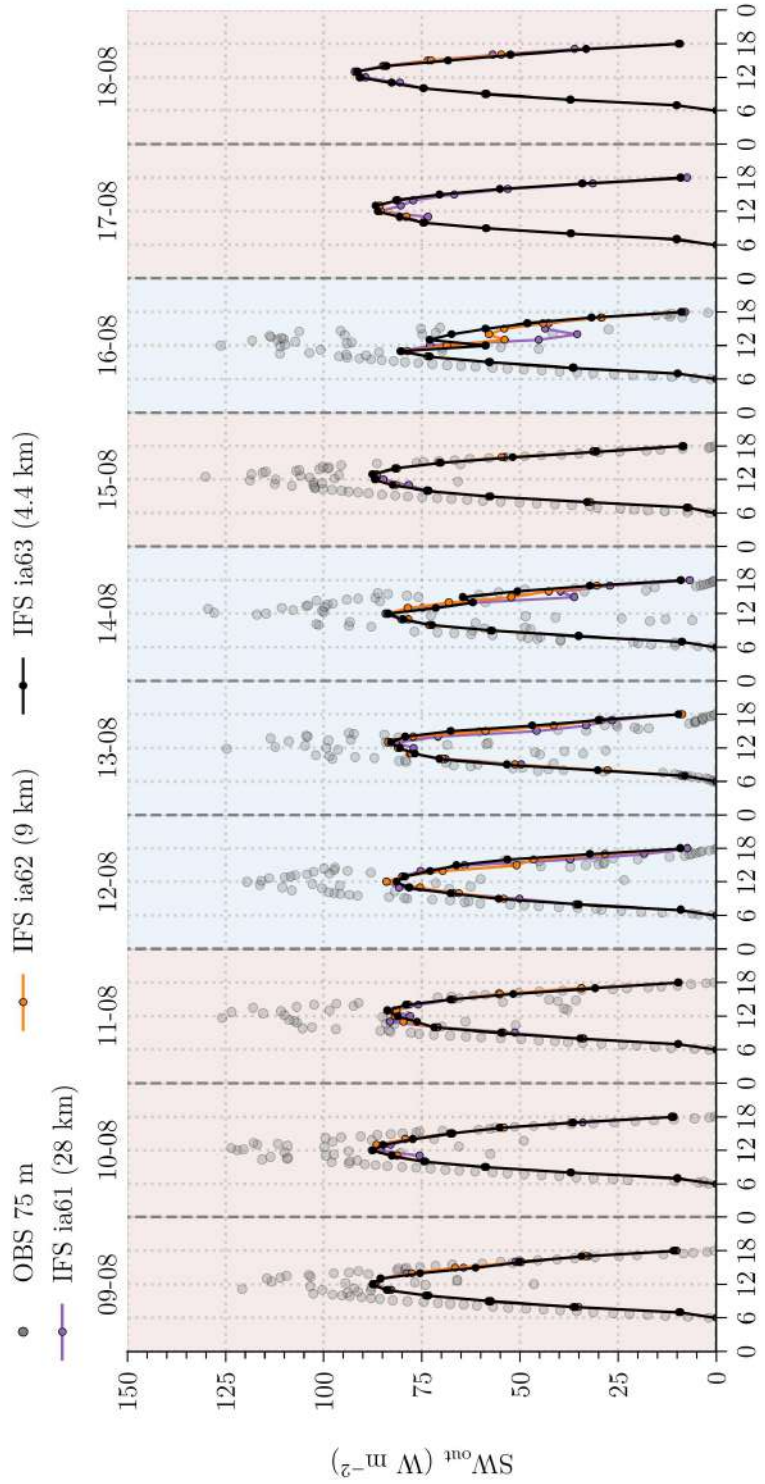



Figure 27: Diurnal cycle of observed (markers, at 75 m) and simulated (IFS) $SW_{out}$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).

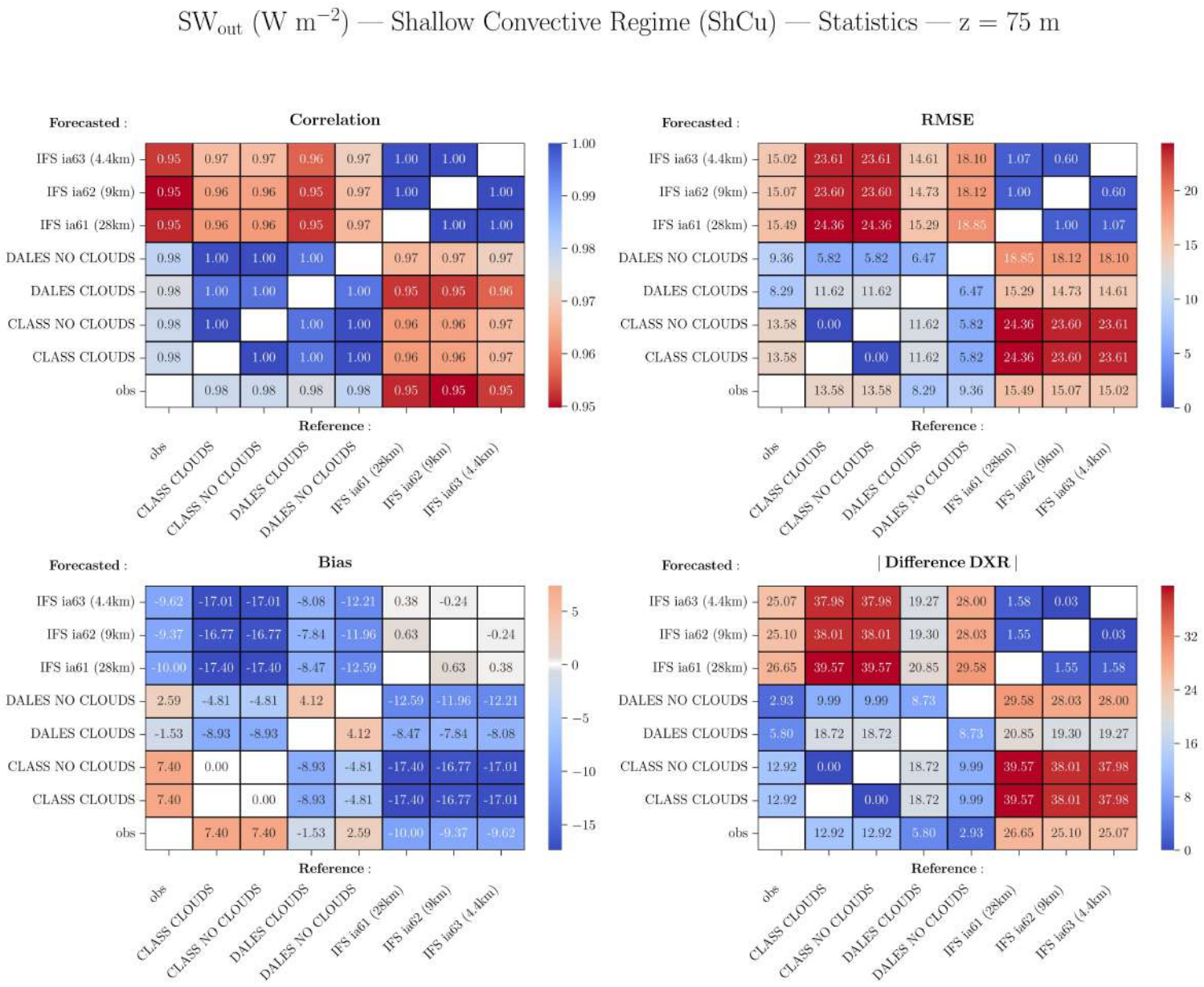



Figure 28: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (CLASS, DALES and IFS) $SW_{out}$ for the Shallow Convective (ShCu) regime.

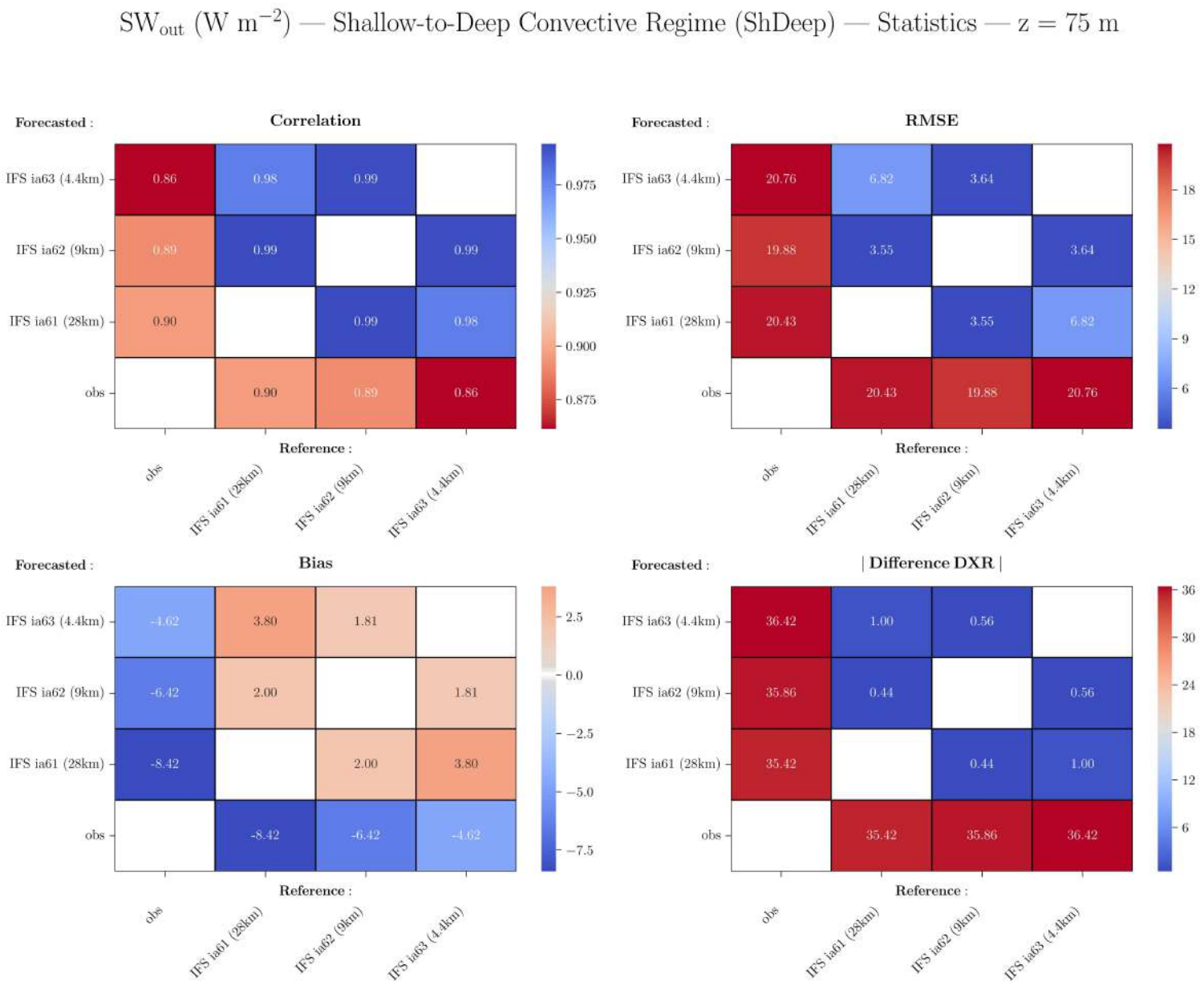



Figure 29: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (IFS) $SW_{out}$ for the Shallow-to-Deep Convective (ShDeep) regime.



Funded by the European Union

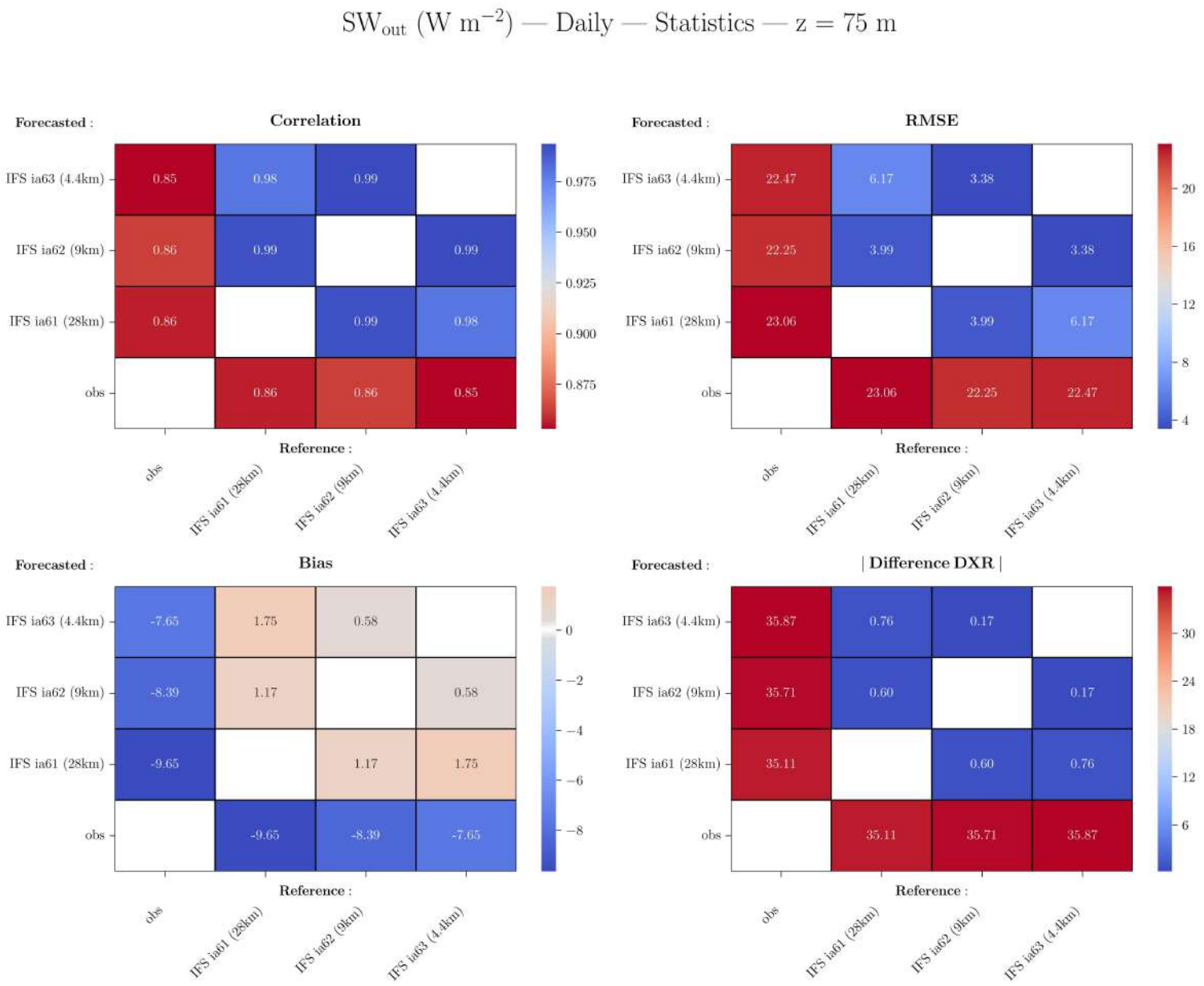



Figure 30: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (IFS) $SW_{out}$ for the full CloudRoots-Amazon22 campaign.



Funded by the European Union

### 3.1.6 $LW_{in}$

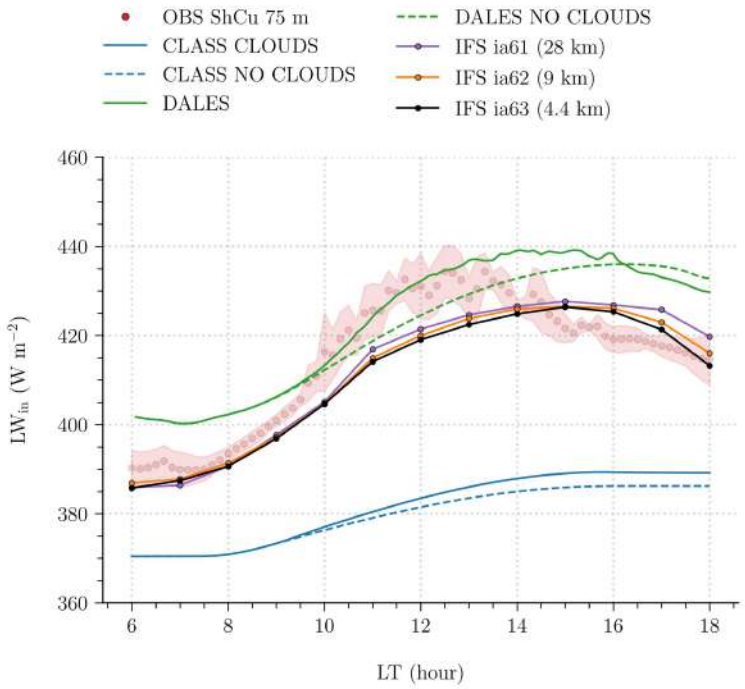



Figure 31: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 75 m) and simulated (CLASS, DALES and IFS) $LW_{in}$ for the Shallow Convective (ShCu) regime.

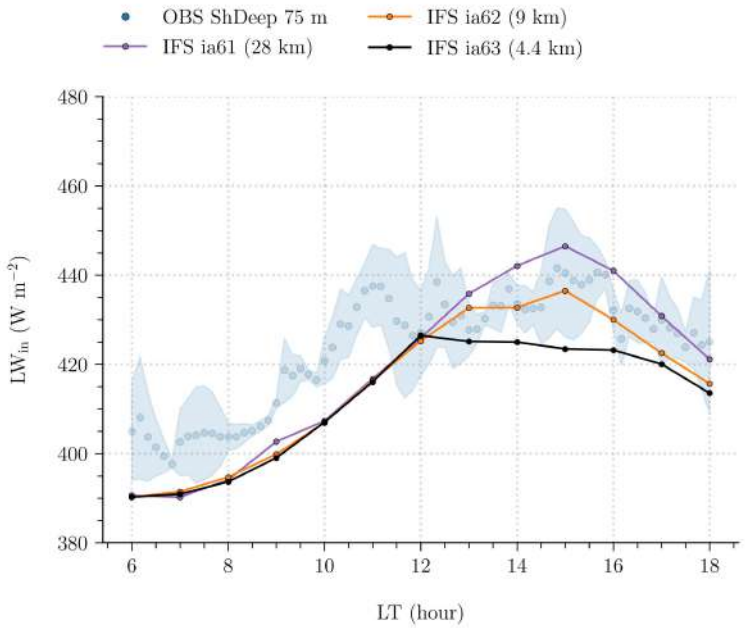



Figure 32: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 75 m) and simulated (IFS) $LW_{in}$ for the Shallow-to-Deep Convective (ShDeep) regime.

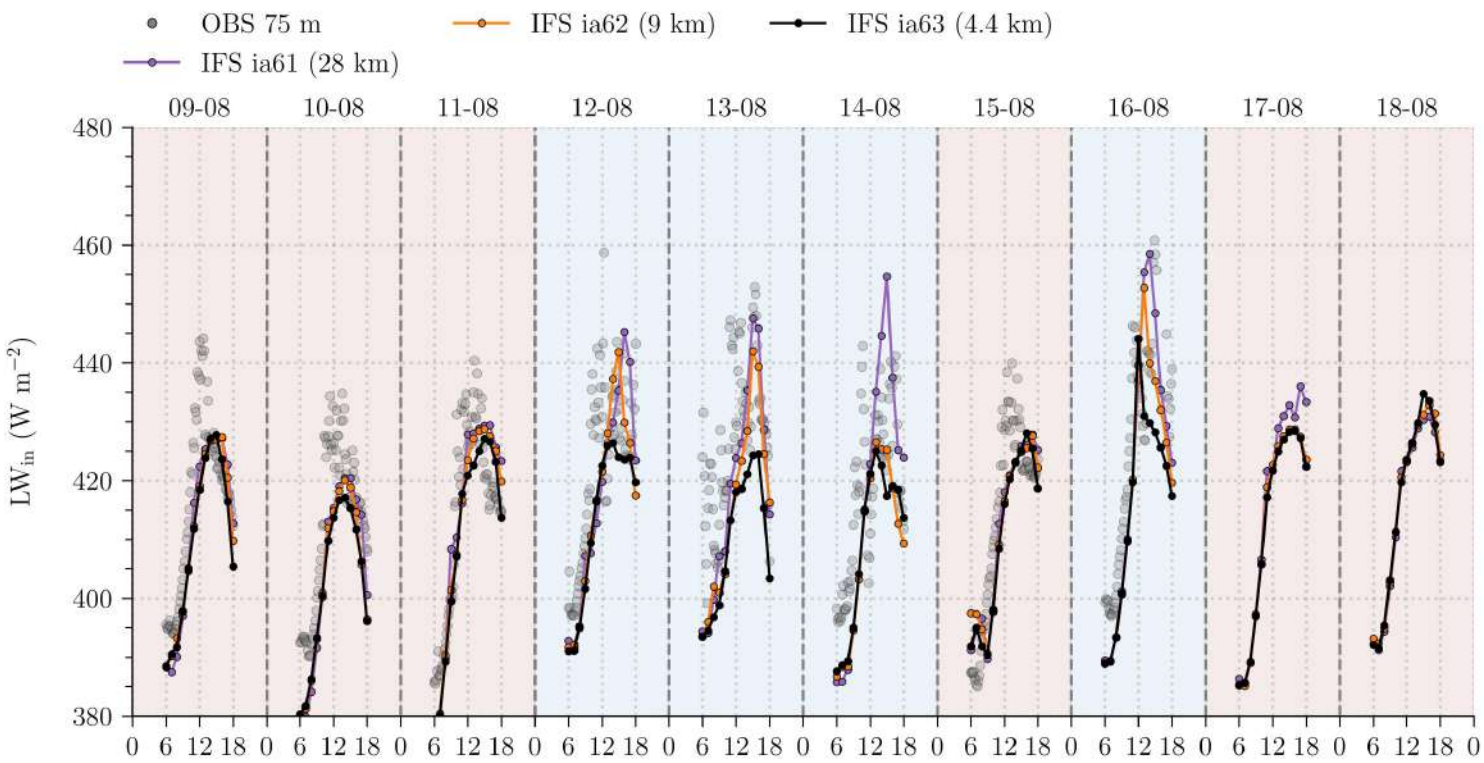



Figure 33: Diurnal cycle of observed (markers, at 75 m) and simulated (IFS) $LW_{in}$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).



Funded by the European Union

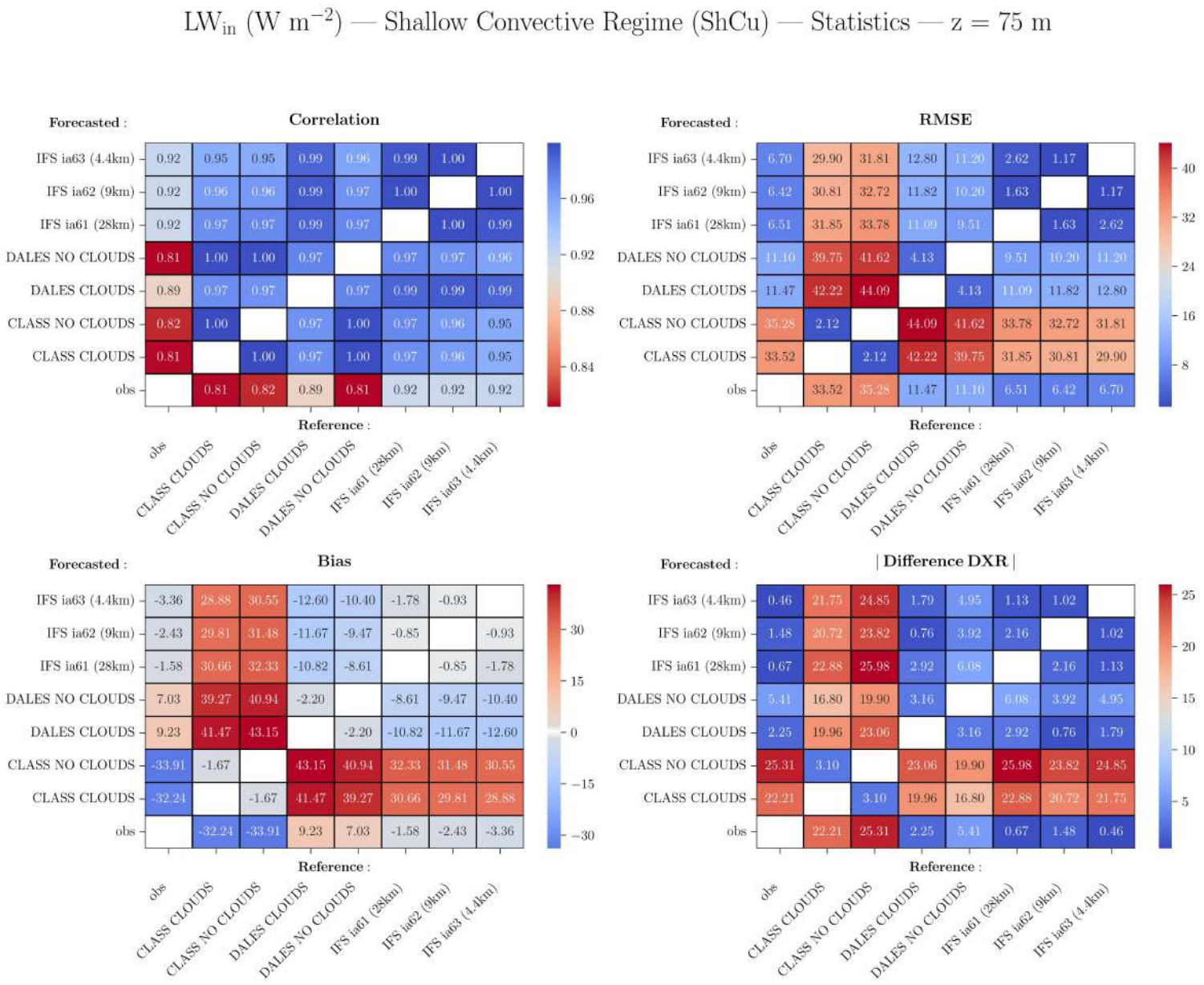



Figure 34: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (CLASS, DALES and IFS) $LW_{in}$ for the Shallow Convective (ShCu) regime.



Funded by the European Union

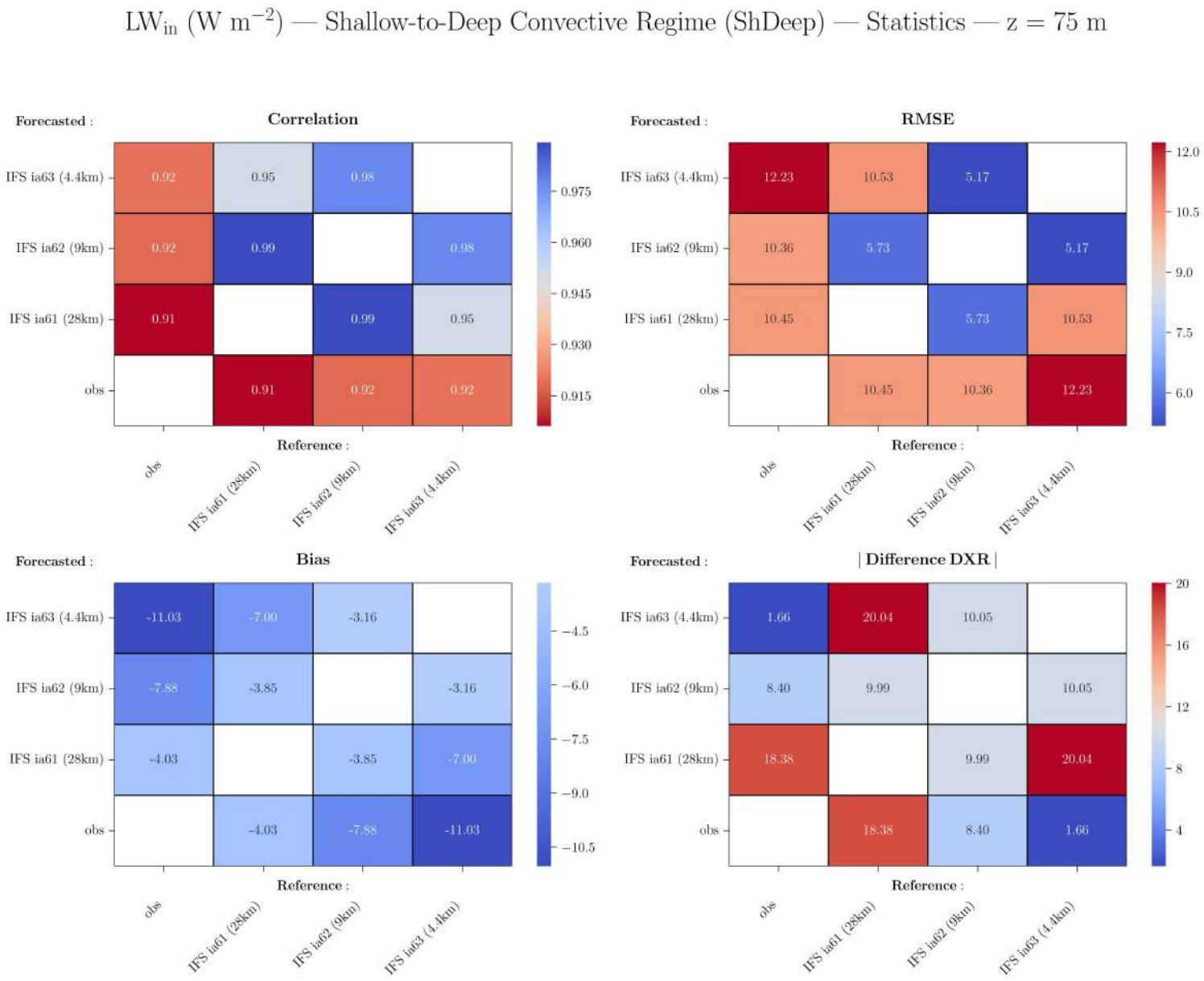



Figure 35: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (IFS) $LW_{in}$ for the Shallow-to-Deep Convective (ShDeep) regime.



Funded by the European Union

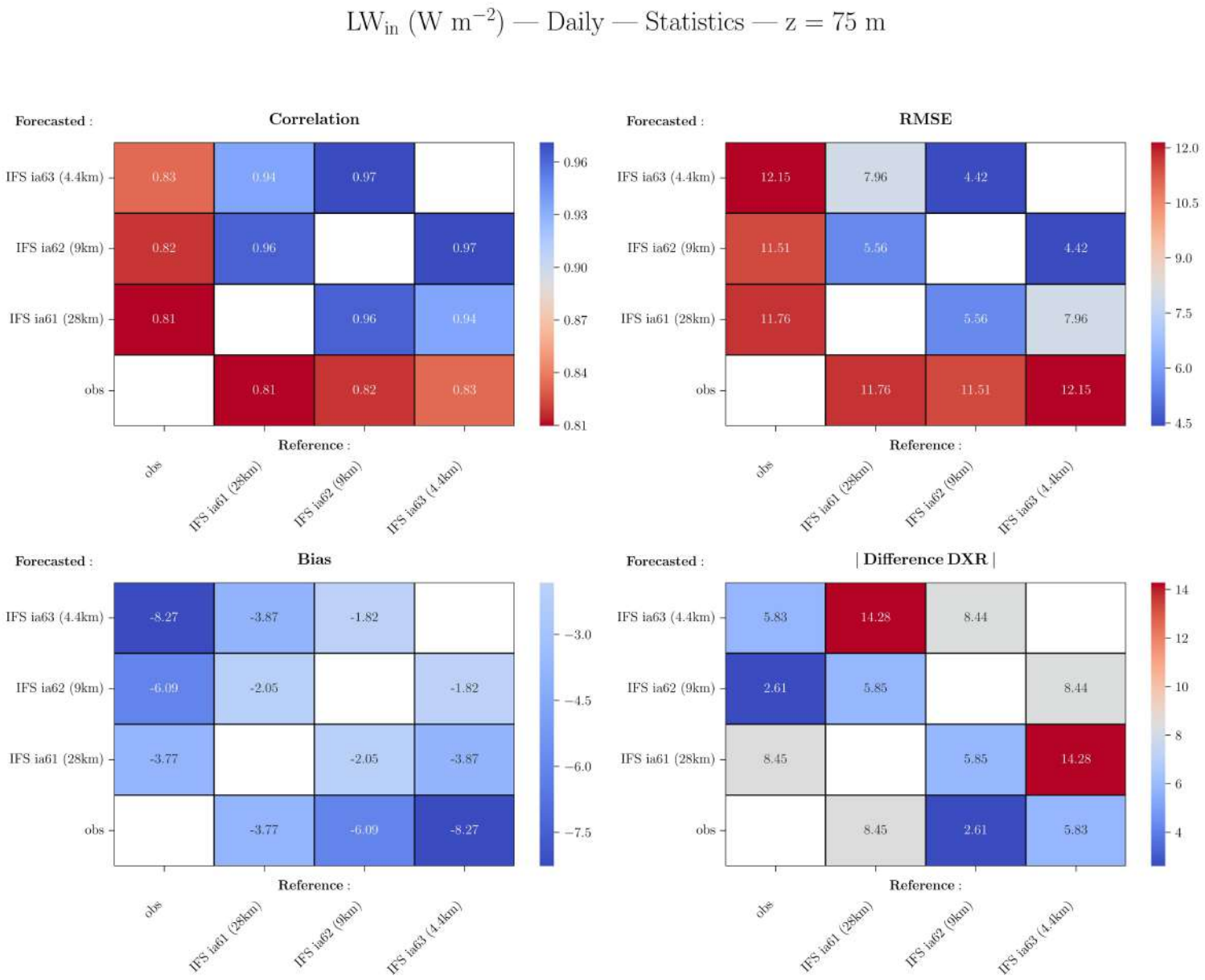



Figure 36: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (IFS) $LW_{in}$ for the full CloudRoots-Amazon22 campaign.



Funded by the European Union

### 3.1.7 $LW_{out}$

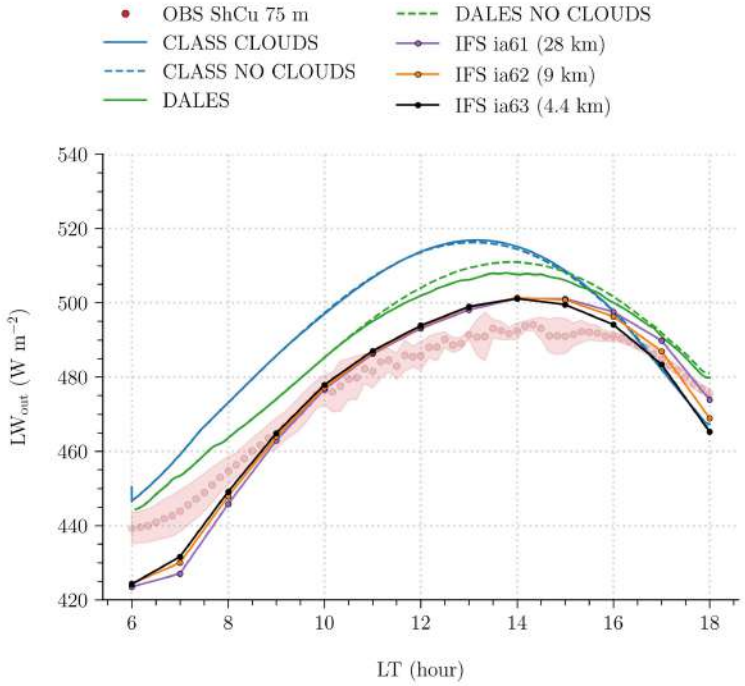



Figure 37: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 75 m) and simulated (CLASS, DALES and IFS) $LW_{out}$ for the Shallow Convective (ShCu) regime.

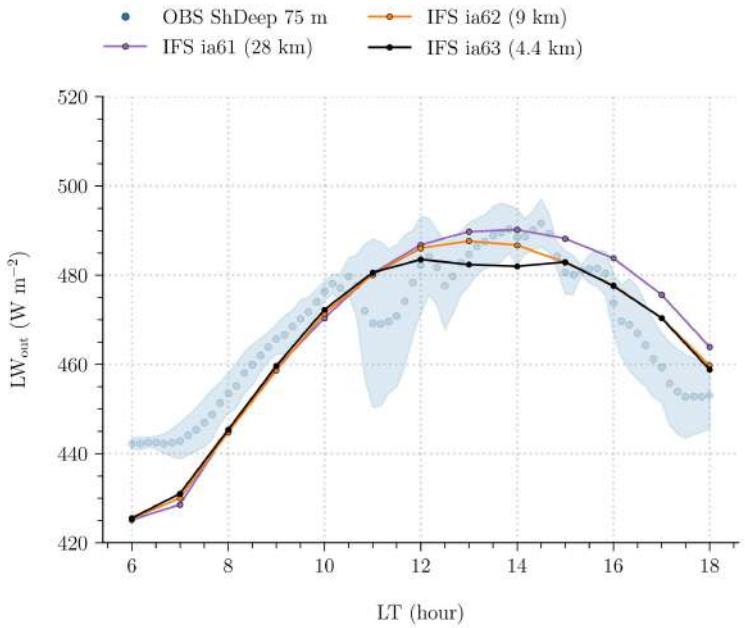



Figure 38: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 75 m) and simulated (IFS) $LW_{out}$ for the Shallow-to-Deep Convective (ShDeep) regime.

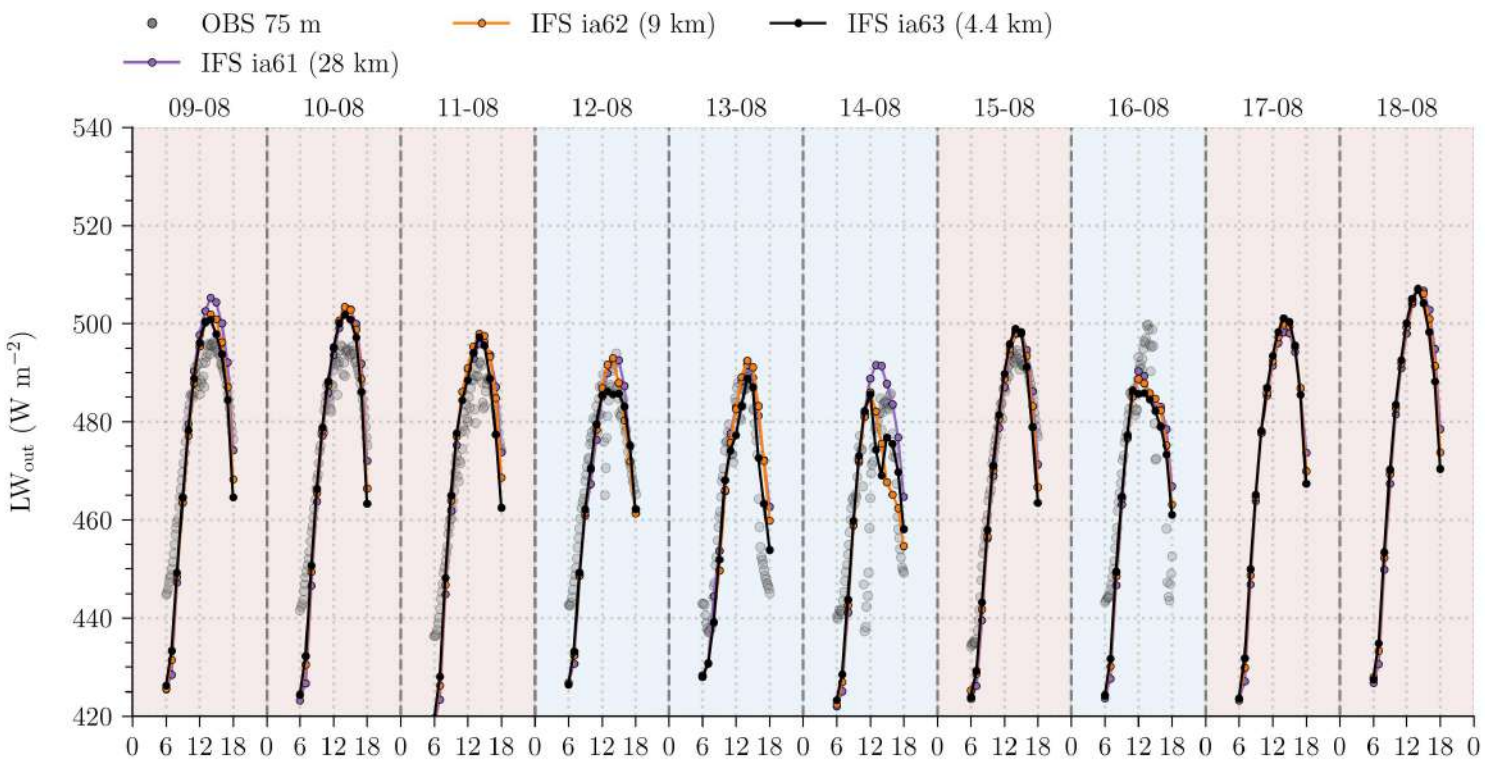



Figure 39: Diurnal cycle of observed (markers, at 75 m) and simulated (IFS) $LW_{out}$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).



Funded by the European Union

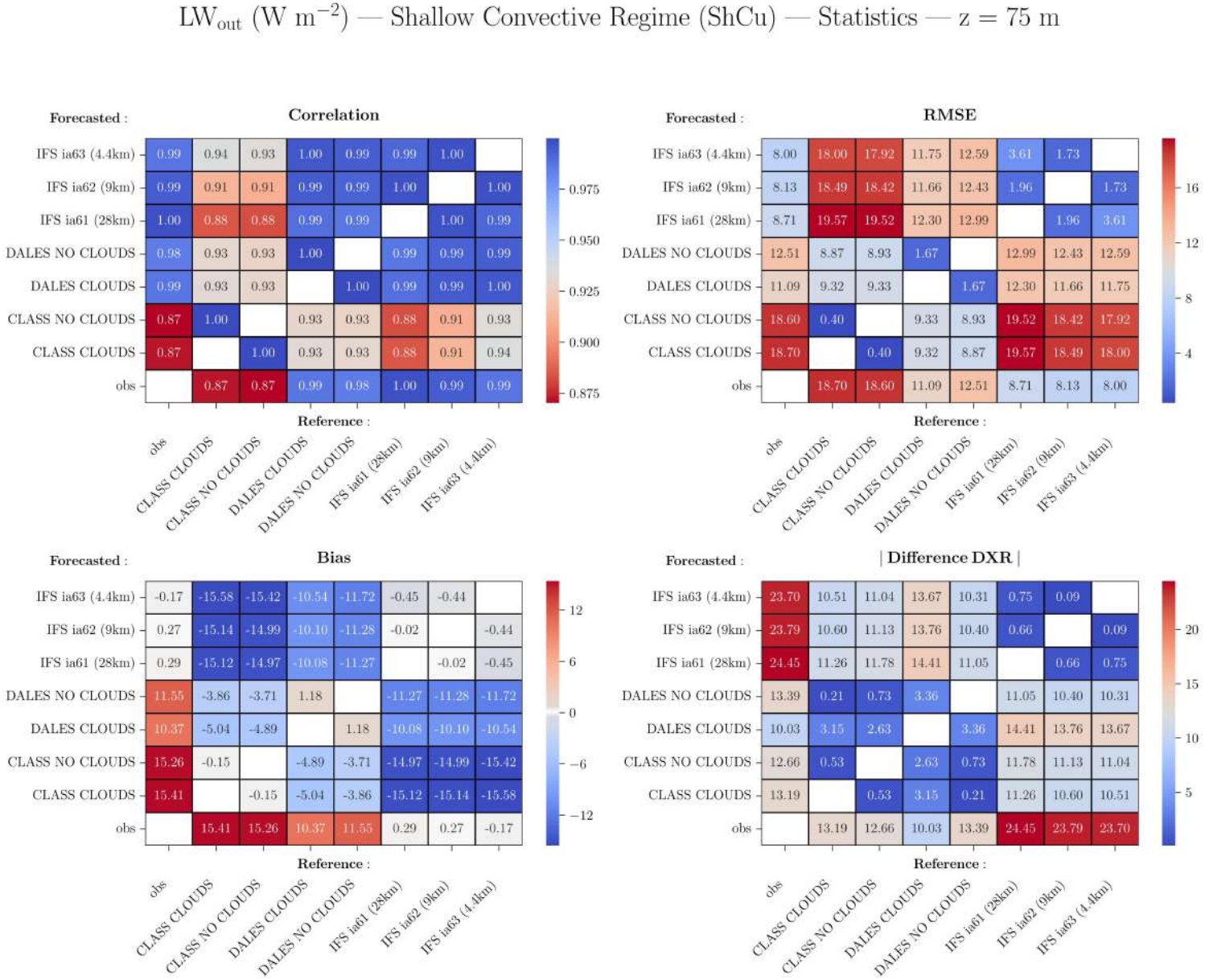



Figure 40: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (CLASS, DALES and IFS) LW$_{out}$ for the Shallow Convective (ShCu) regime.



Funded by the European Union

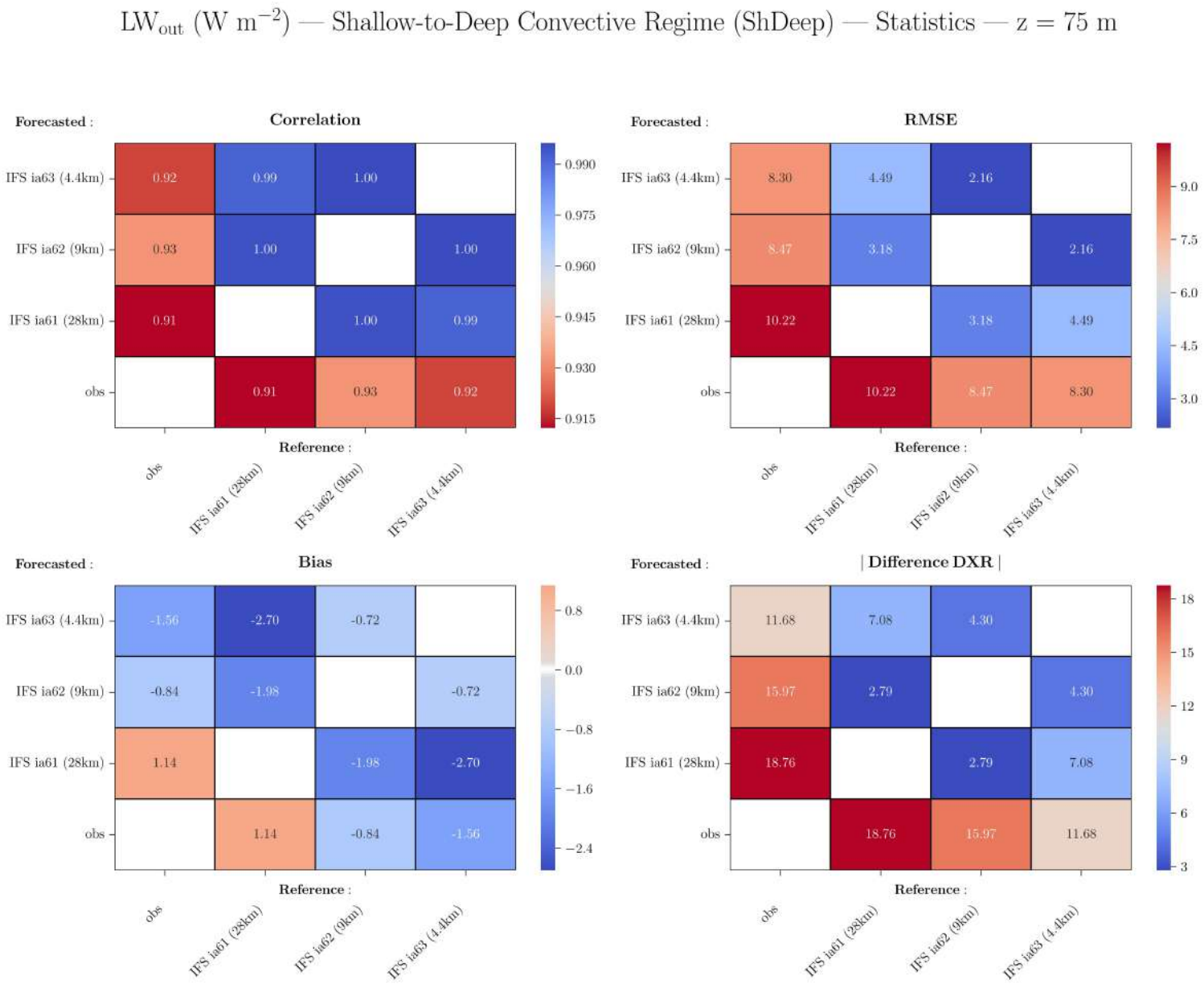



Figure 41: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (IFS) $LW_{out}$ for the Shallow-to-Deep Convective (ShDeep) regime.



Funded by the European Union

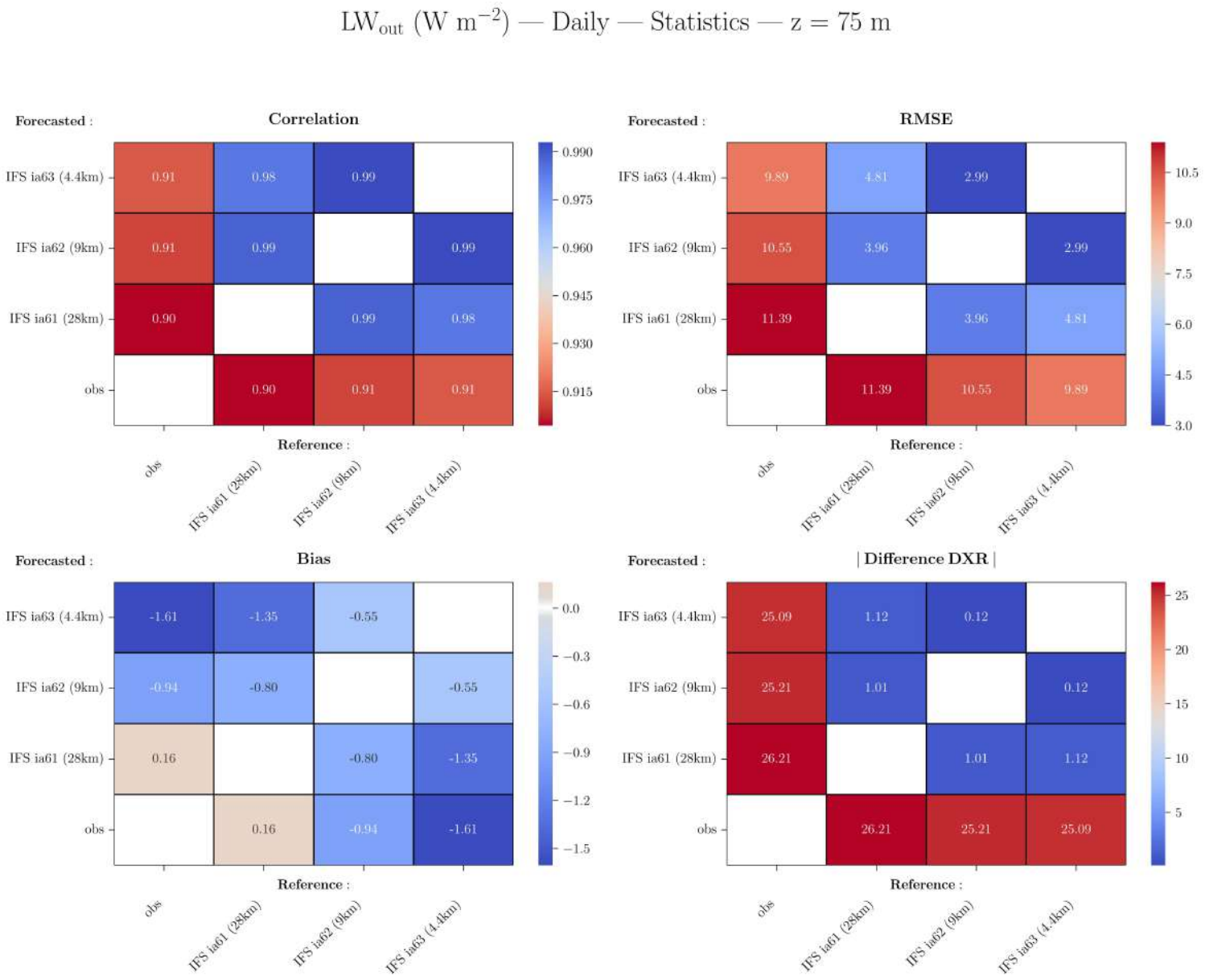



Figure 42: Statistical quantities of the diurnal cycle of observed (at 75 m) and simulated (IFS) $LW_{out}$ for the full CloudRoots-Amazon22 campaign.



Funded by the European Union

## 3.2 Energy Fluxes

### 3.2.1 *EF*

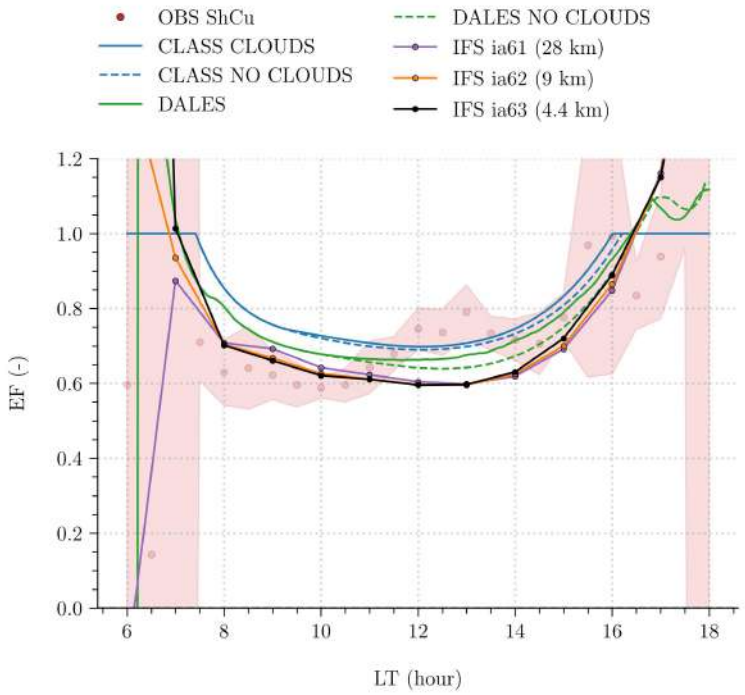



Figure 43: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (CLASS, DALES and IFS) *EF* for the Shallow Convective (ShCu) regime.

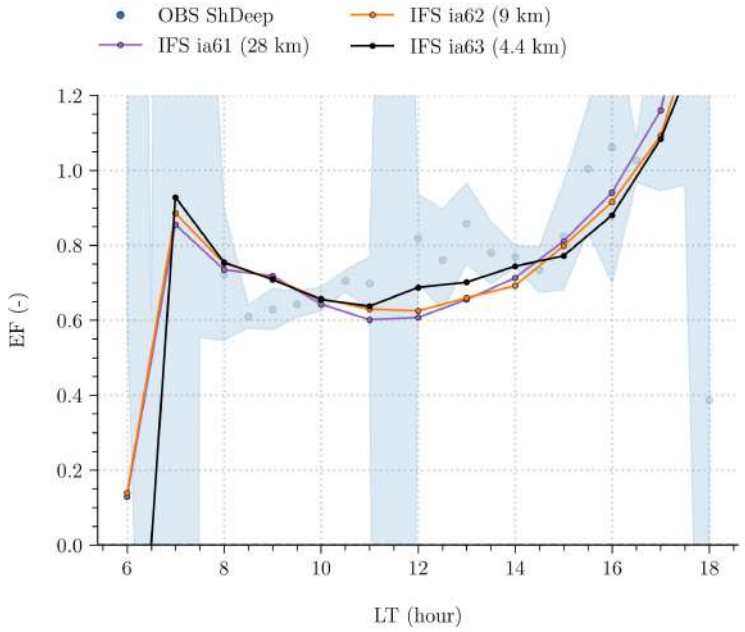



Figure 44: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (IFS) *EF* for the Shallow-to-Deep Convective (ShDeep) regime.

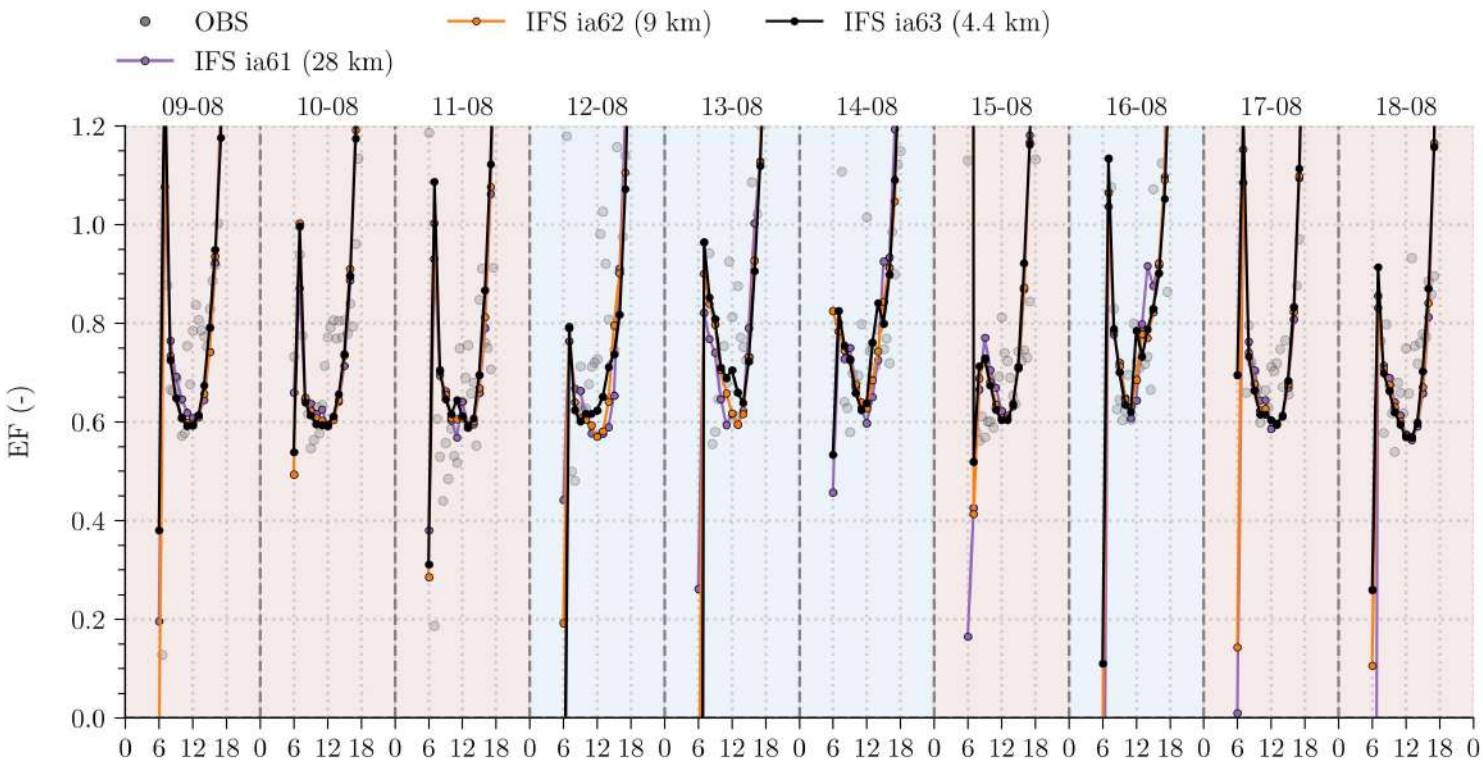



Figure 45: Diurnal cycle of observed (markers, at 50 m) and simulated (IFS) $EF$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).



Funded by the European Union

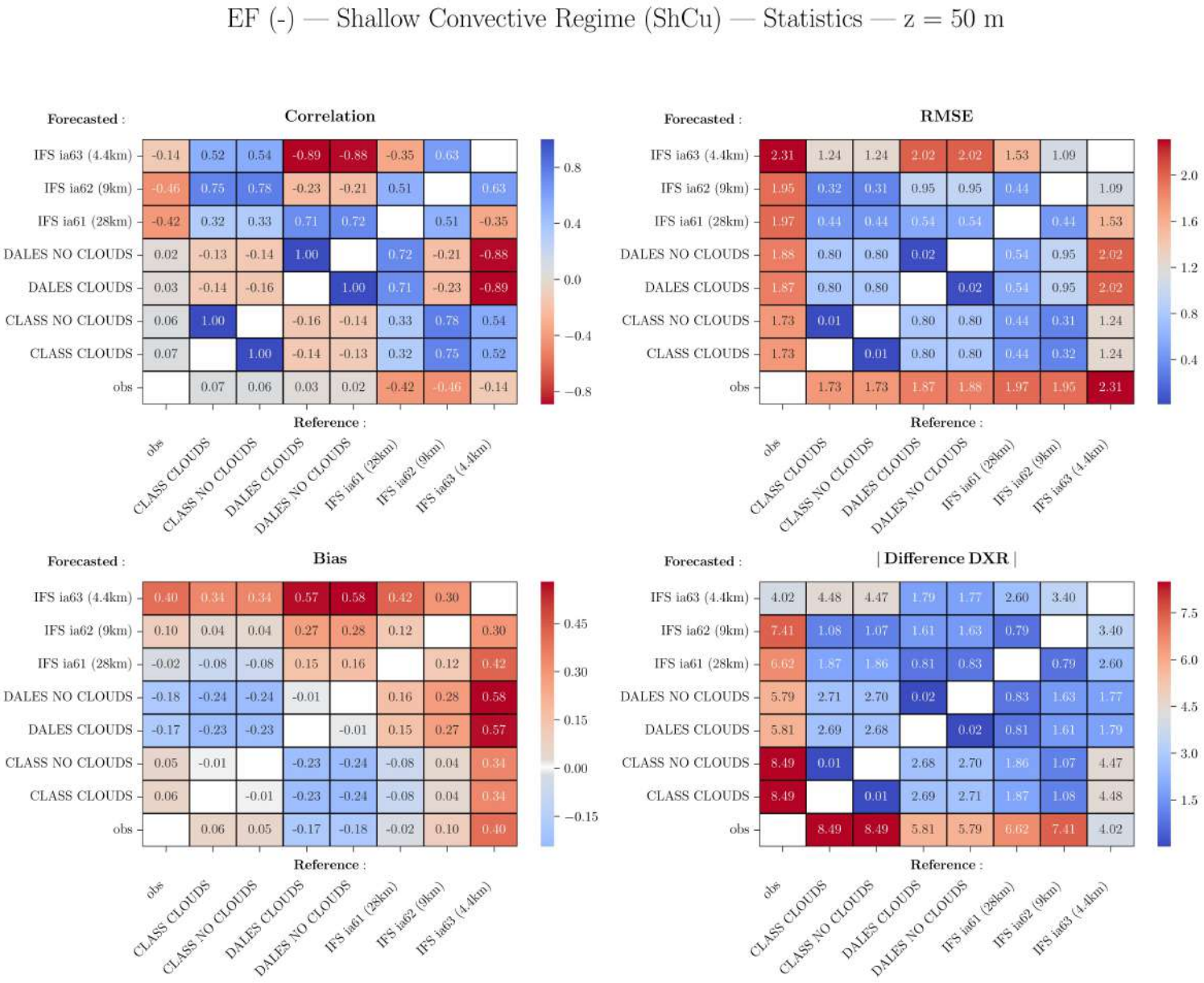



Figure 46: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (CLASS, DALES and IFS) $EF$ for the Shallow Convective (ShCu) regime.

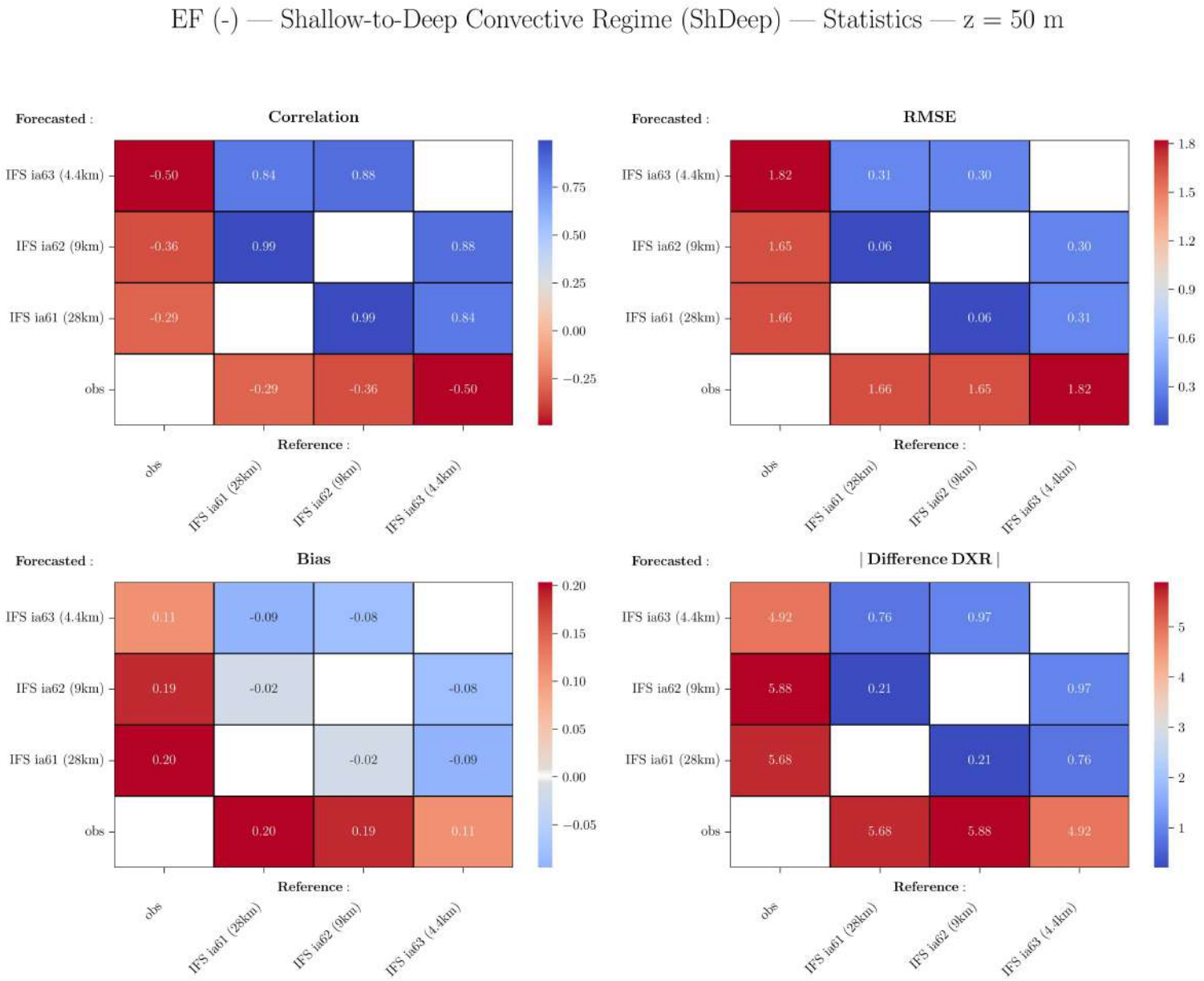



Figure 47: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) $Q_{net}$ for the Shallow-to-Deep Convective (ShDeep) regime.



Funded by the European Union

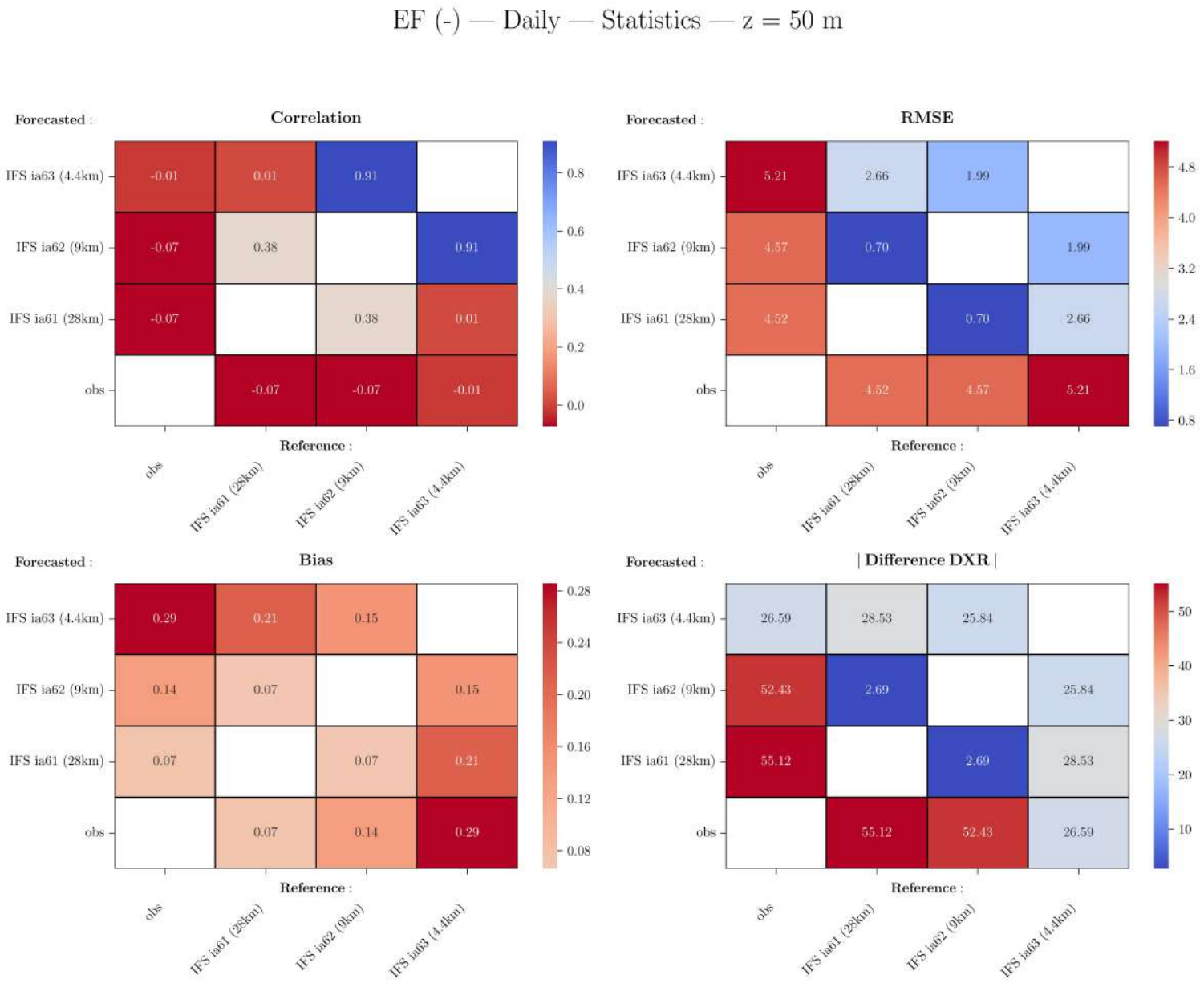



Figure 48: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) $EF$ for the full CloudRoots-Amazon22 campaign.



Funded by the European Union

### 3.2.2 *H*

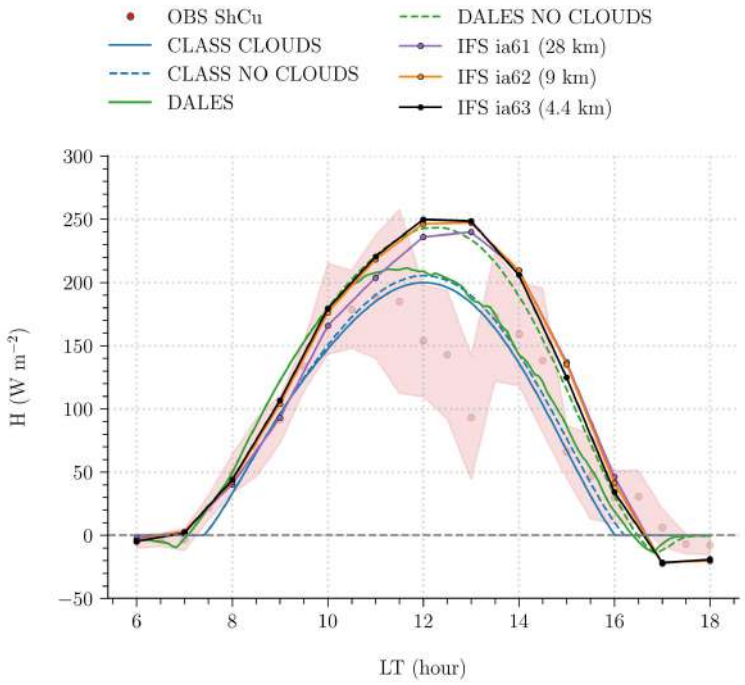



Figure 49: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (CLASS, DALES and IFS) $H$ for the Shallow Convective (ShCu) regime.

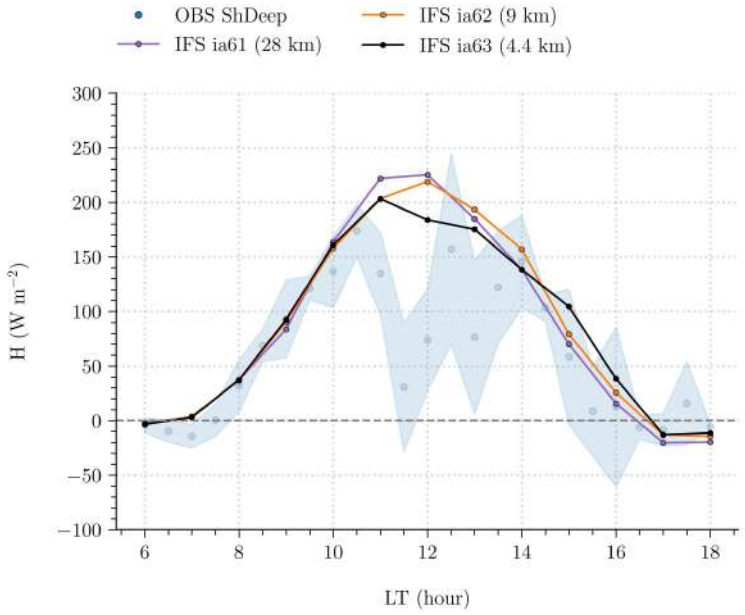



Figure 50: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (IFS) $H$ for the Shallow-to-Deep Convective (ShDeep) regime.

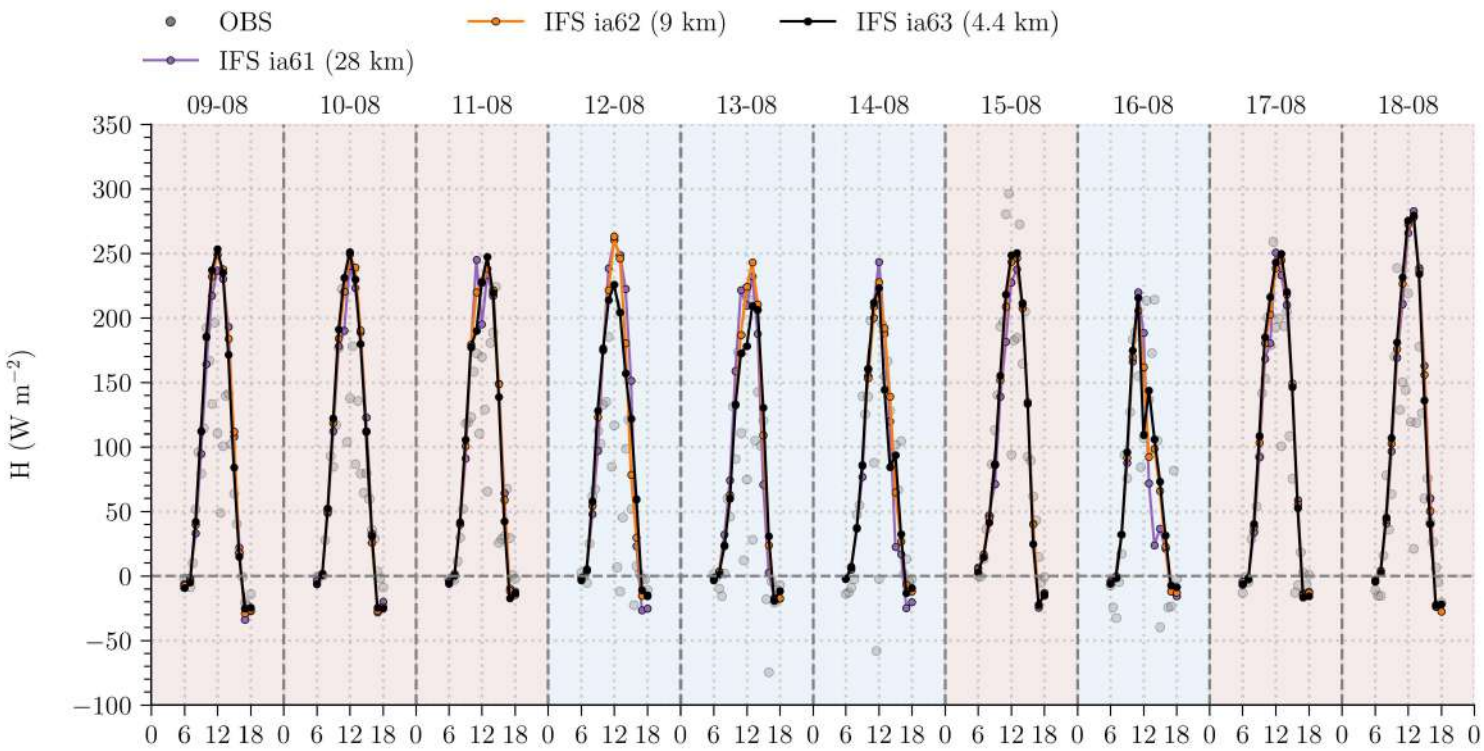



Figure 51: Diurnal cycle of observed (markers, at 50 m) and simulated (IFS) $H$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).



Funded by the European Union

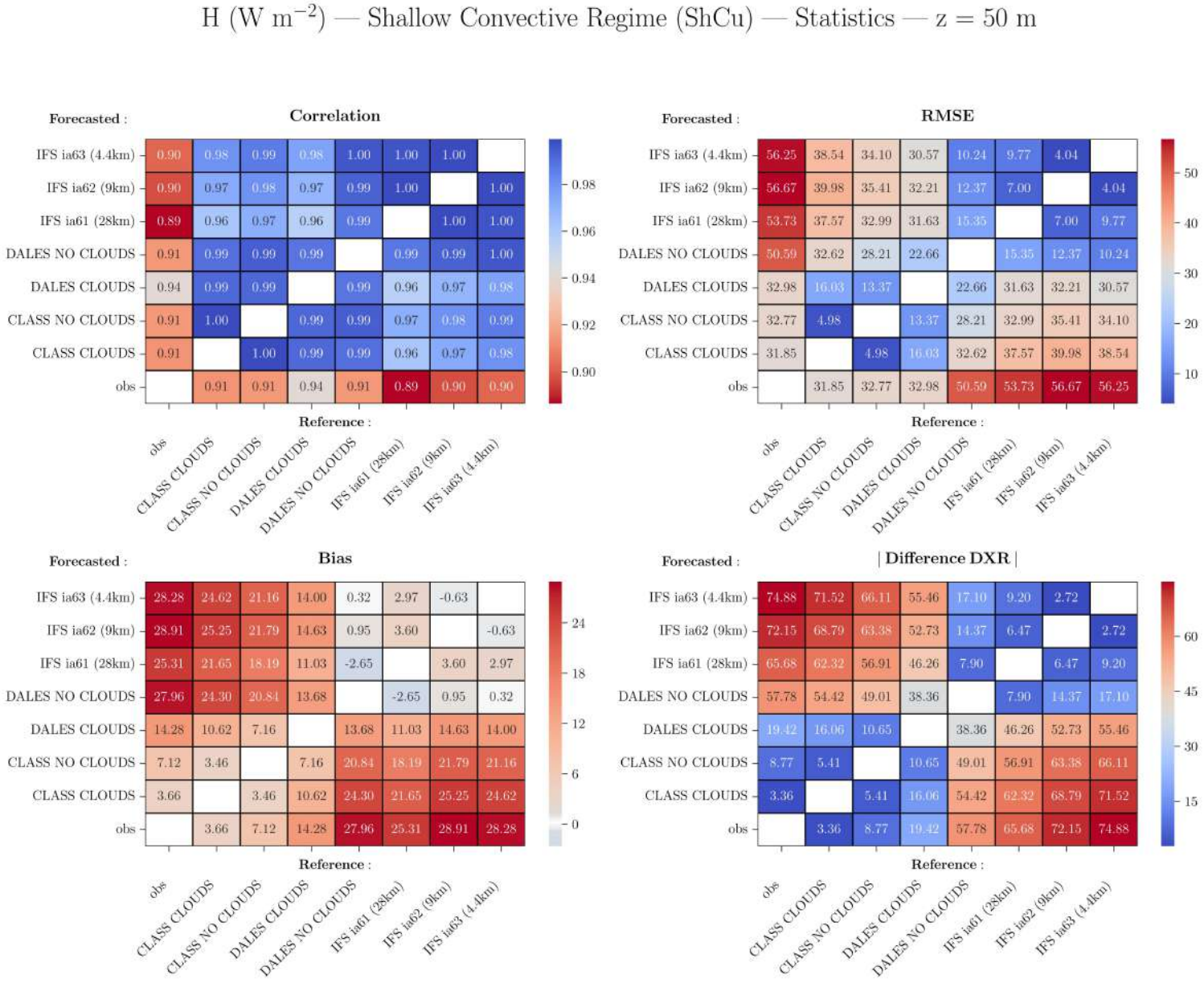



Figure 52: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (CLASS, DALES and IFS) $H$ for the Shallow Convective (ShCu) regime.

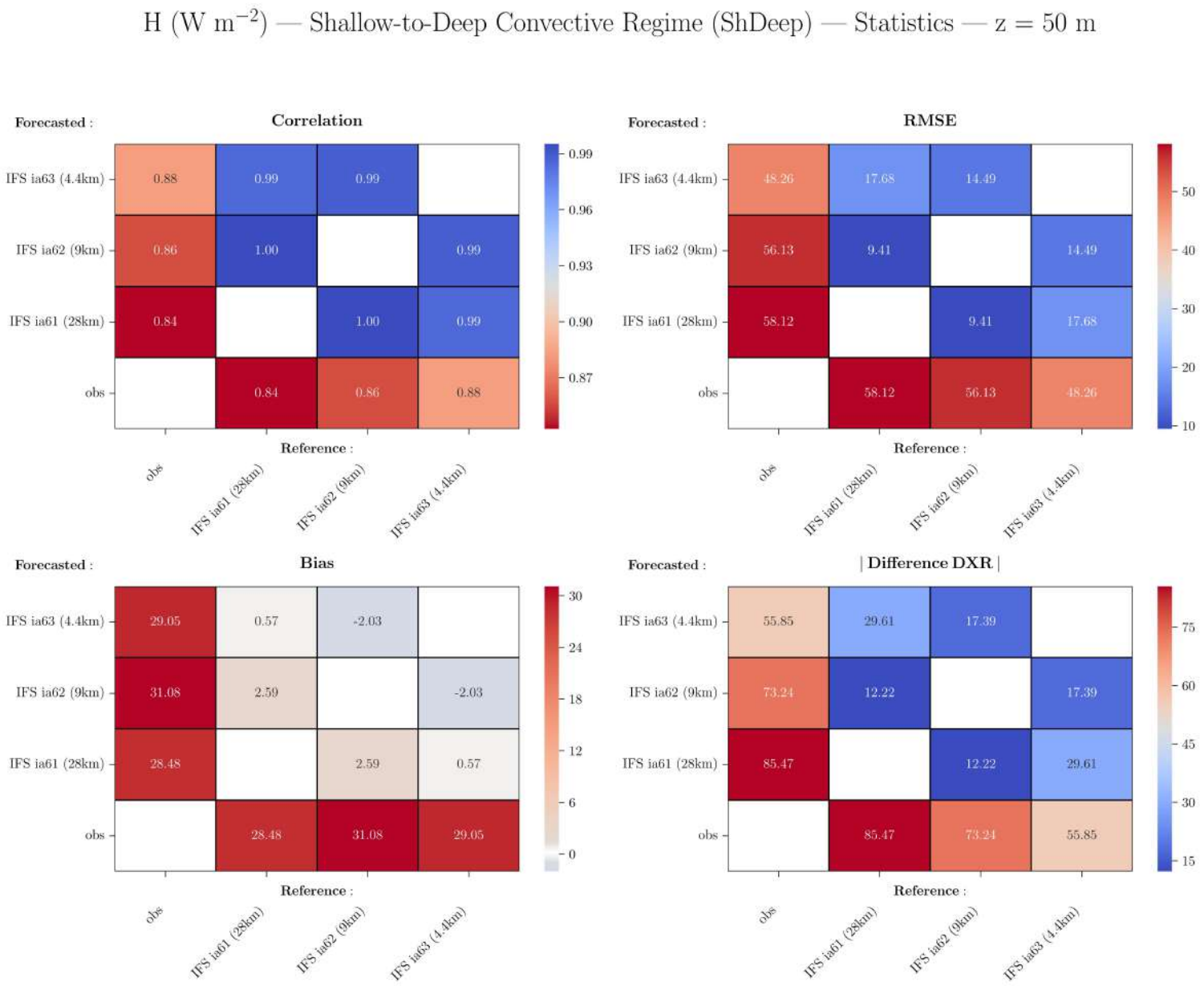



Figure 53: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) $Q_{net}$ for the Shallow-to-Deep Convective (ShDeep) regime.

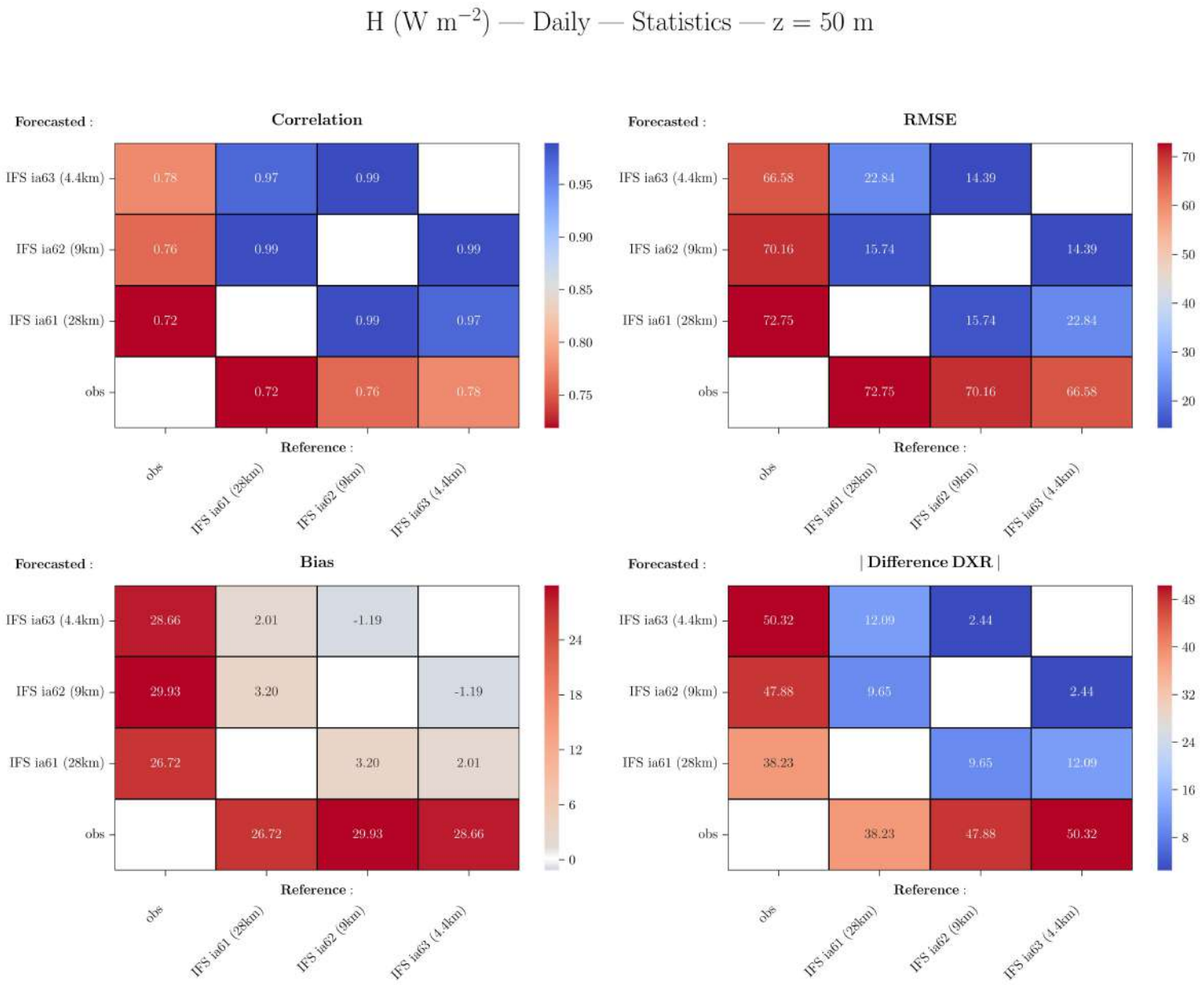



Figure 54: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) $H$ for the full CloudRoots-Amazon22 campaign.

### 3.2.3 *LE*

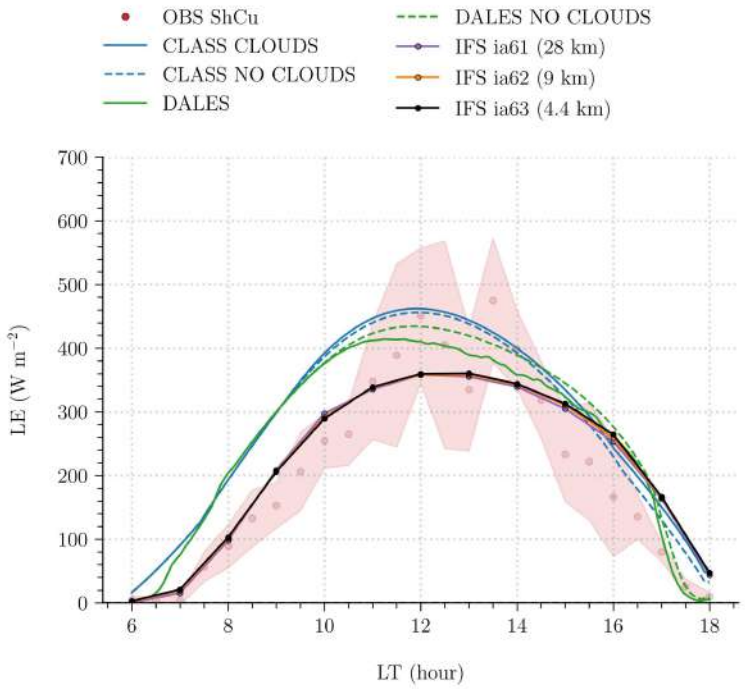



Figure 55: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (CLASS, DALES and IFS) *LE* for the Shallow Convective (ShCu) regime.

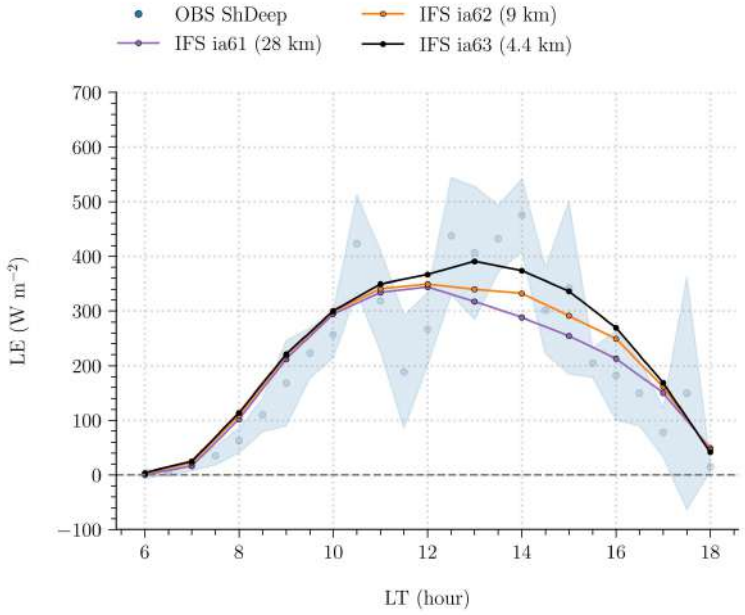



Figure 56: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (IFS) *LE* for the Shallow-to-Deep Convective (ShDeep) regime.

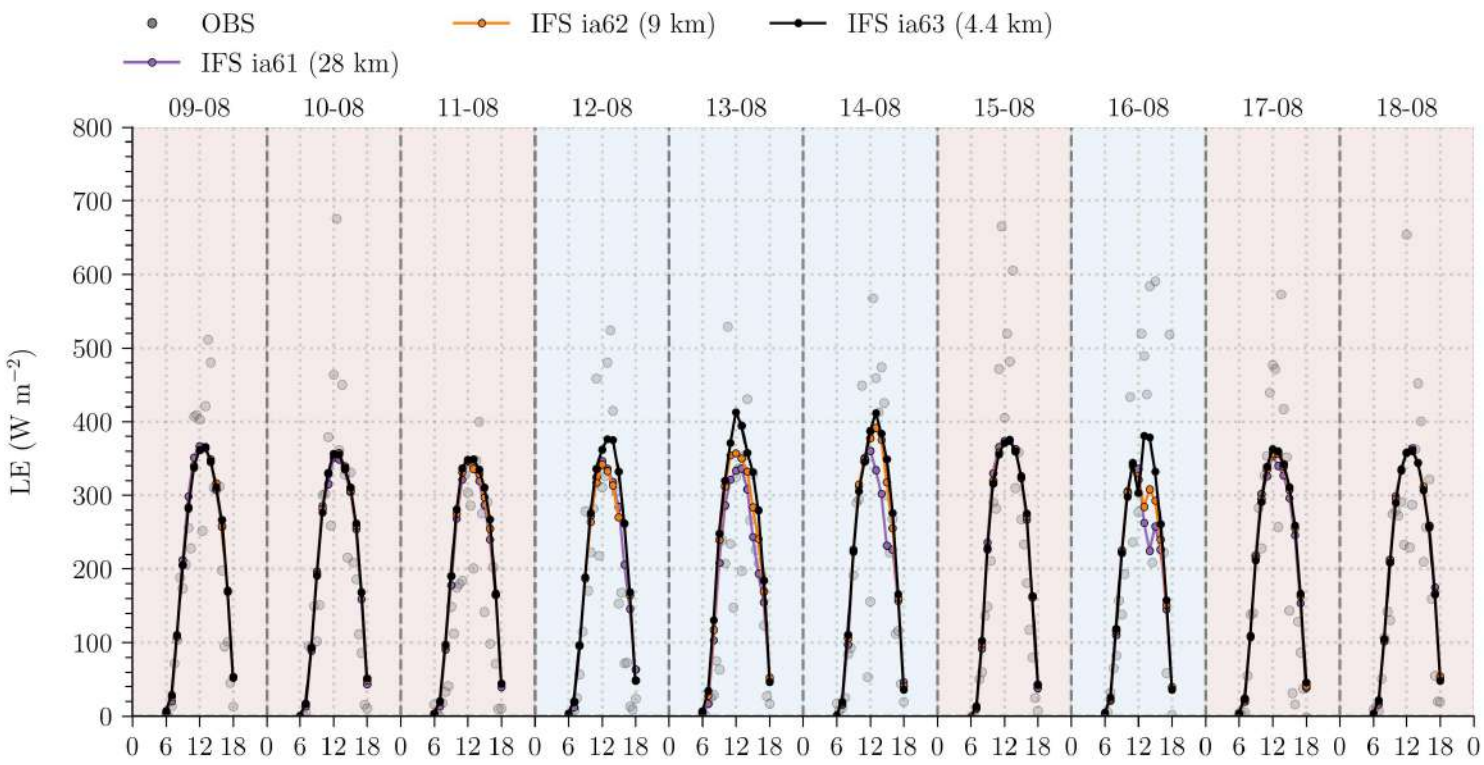



Figure 57: Diurnal cycle of observed (markers, at 50 m) and simulated (IFS) *LE* for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).



Funded by the European Union

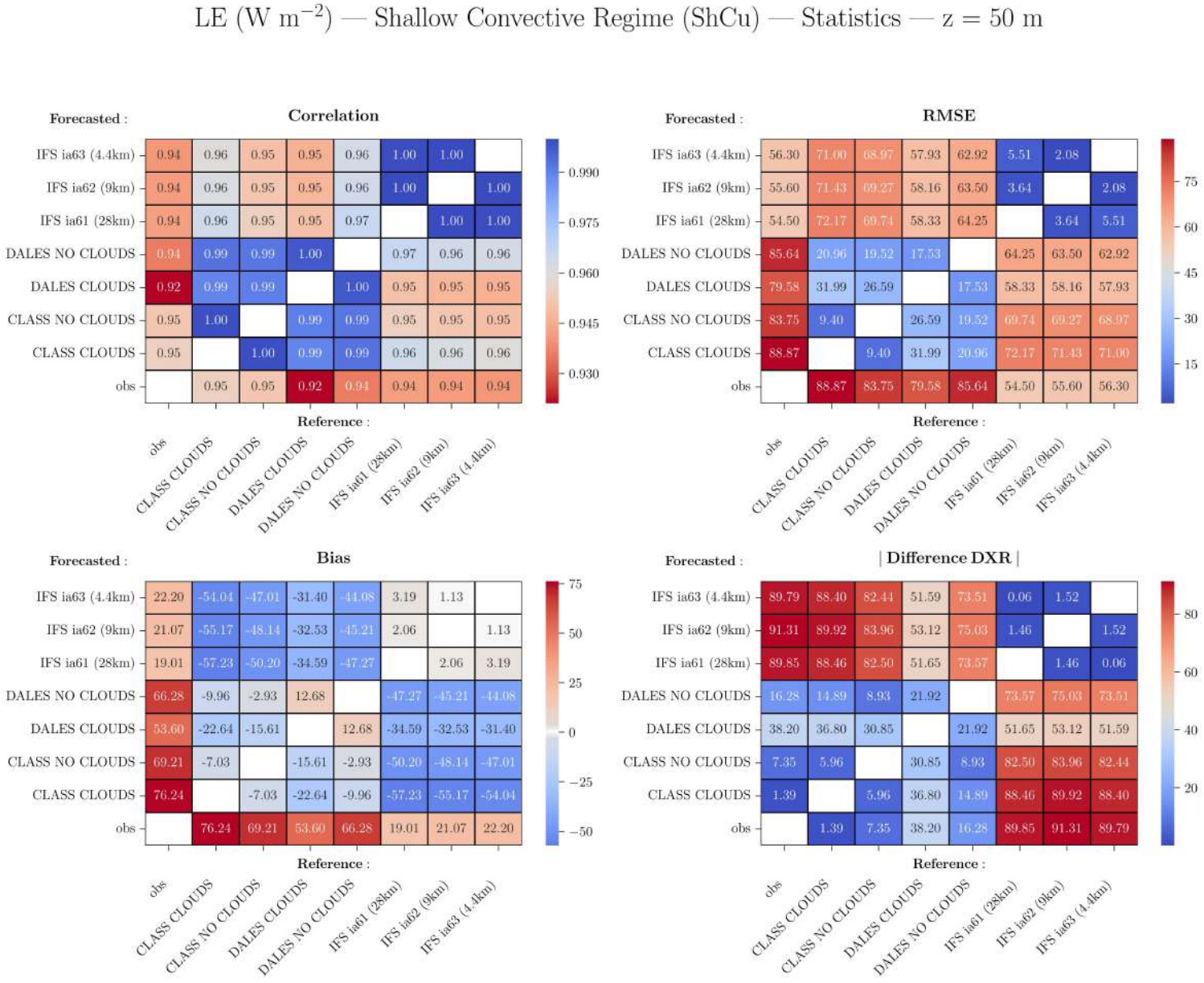



Figure 58: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (CLASS, DALES and IFS) $LE$ for the Shallow Convective (ShCu) regime.

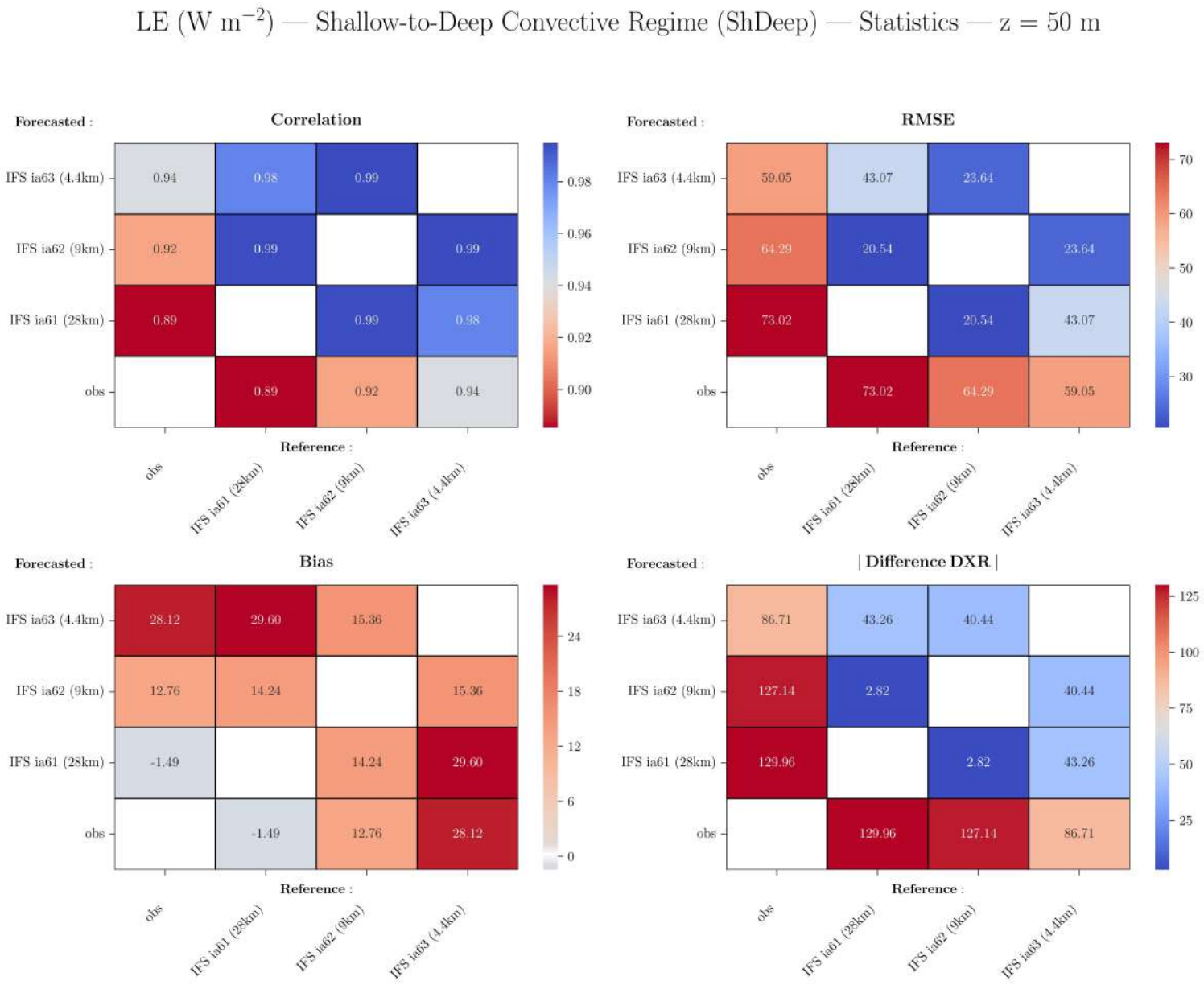



Figure 59: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) $Q_{net}$ for the Shallow-to-Deep Convective (ShDeep) regime.



Funded by the European Union

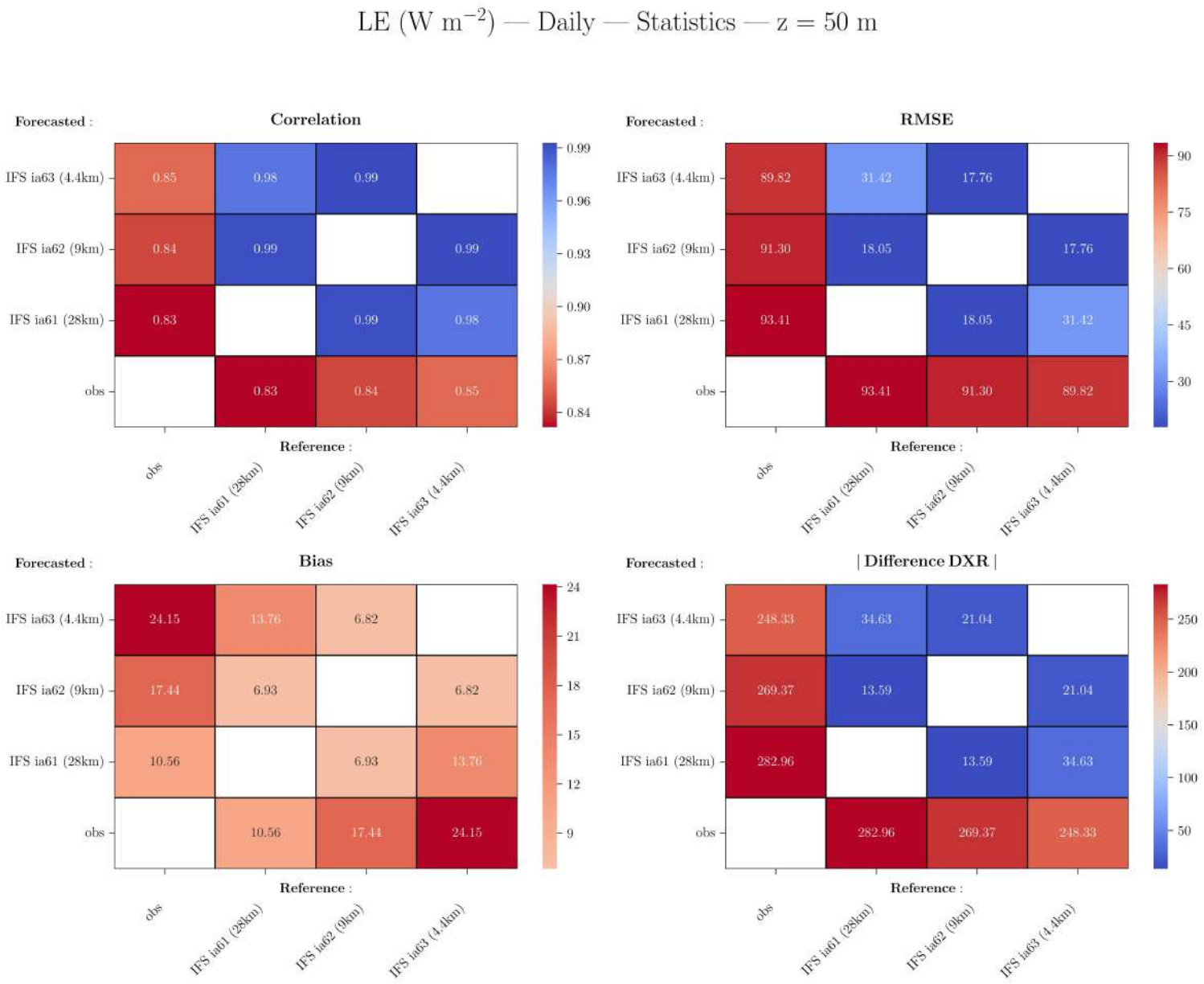



Figure 60: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) *LE* for the full CloudRoots-Amazon22 campaign.



Funded by the European Union

### 3.2.4 *NEE*

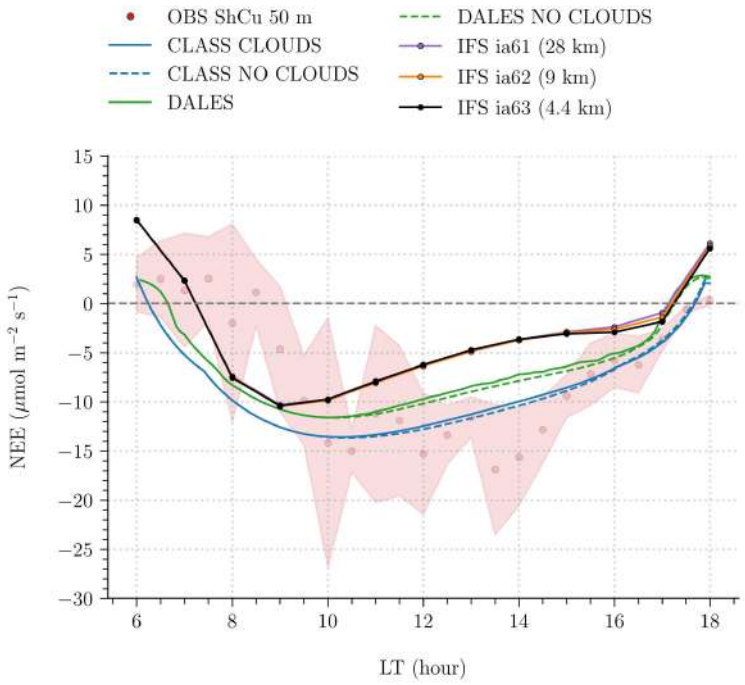



Figure 61: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (CLASS, DALES and IFS) *NEE* for the Shallow Convective (ShCu) regime.

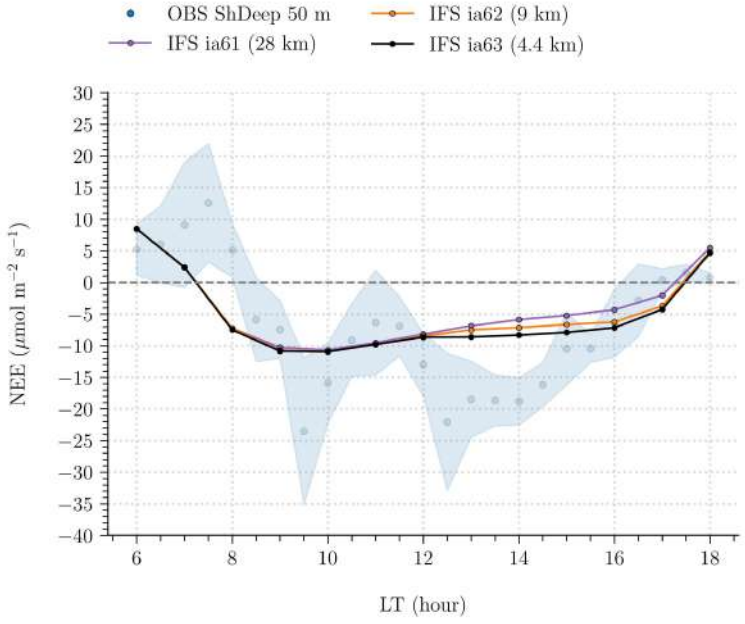



Figure 62: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (IFS) *NEE* for the Shallow-to-Deep Convective (ShDeep) regime.

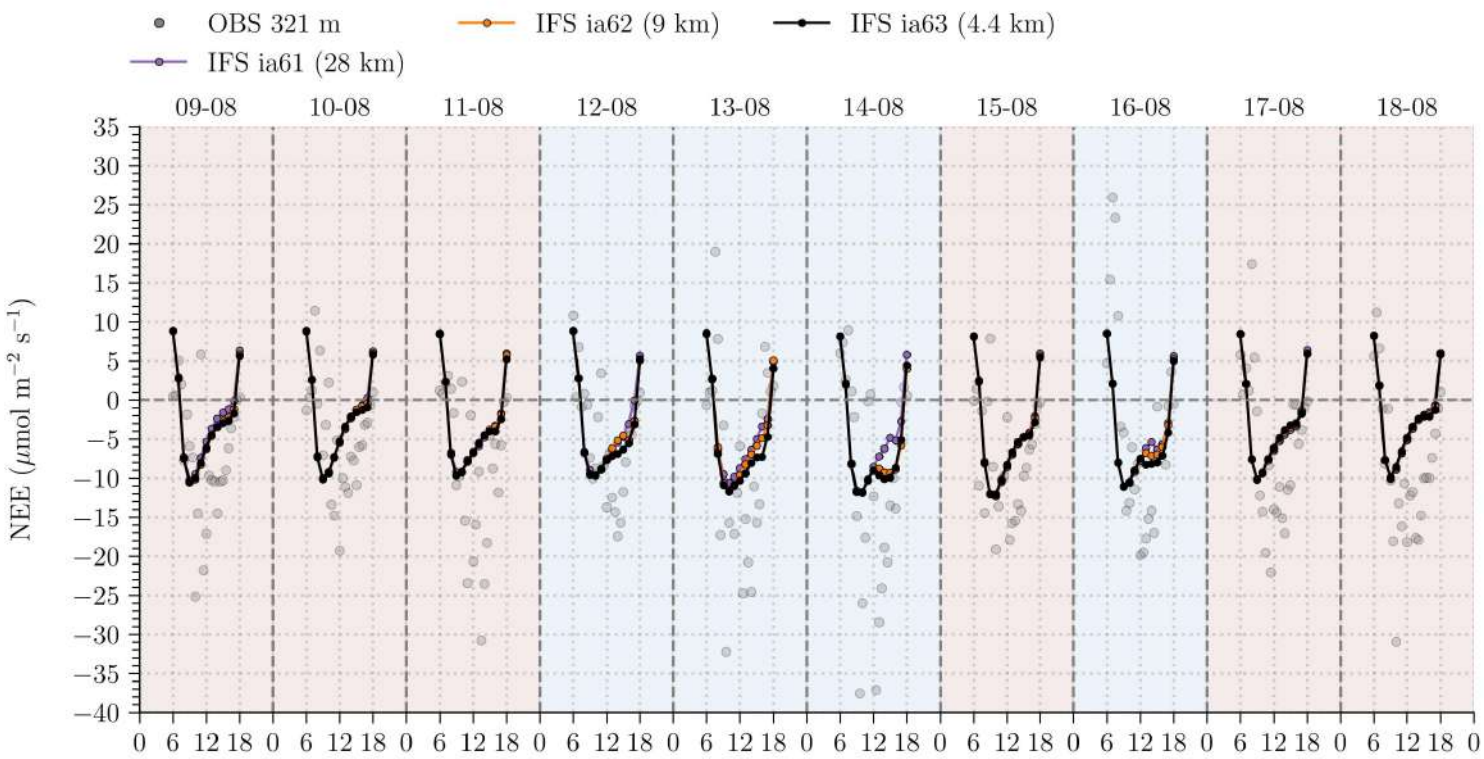



Figure 63: Diurnal cycle of observed (markers, at 50 m) and simulated (IFS) $NEE$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).



Funded by the European Union

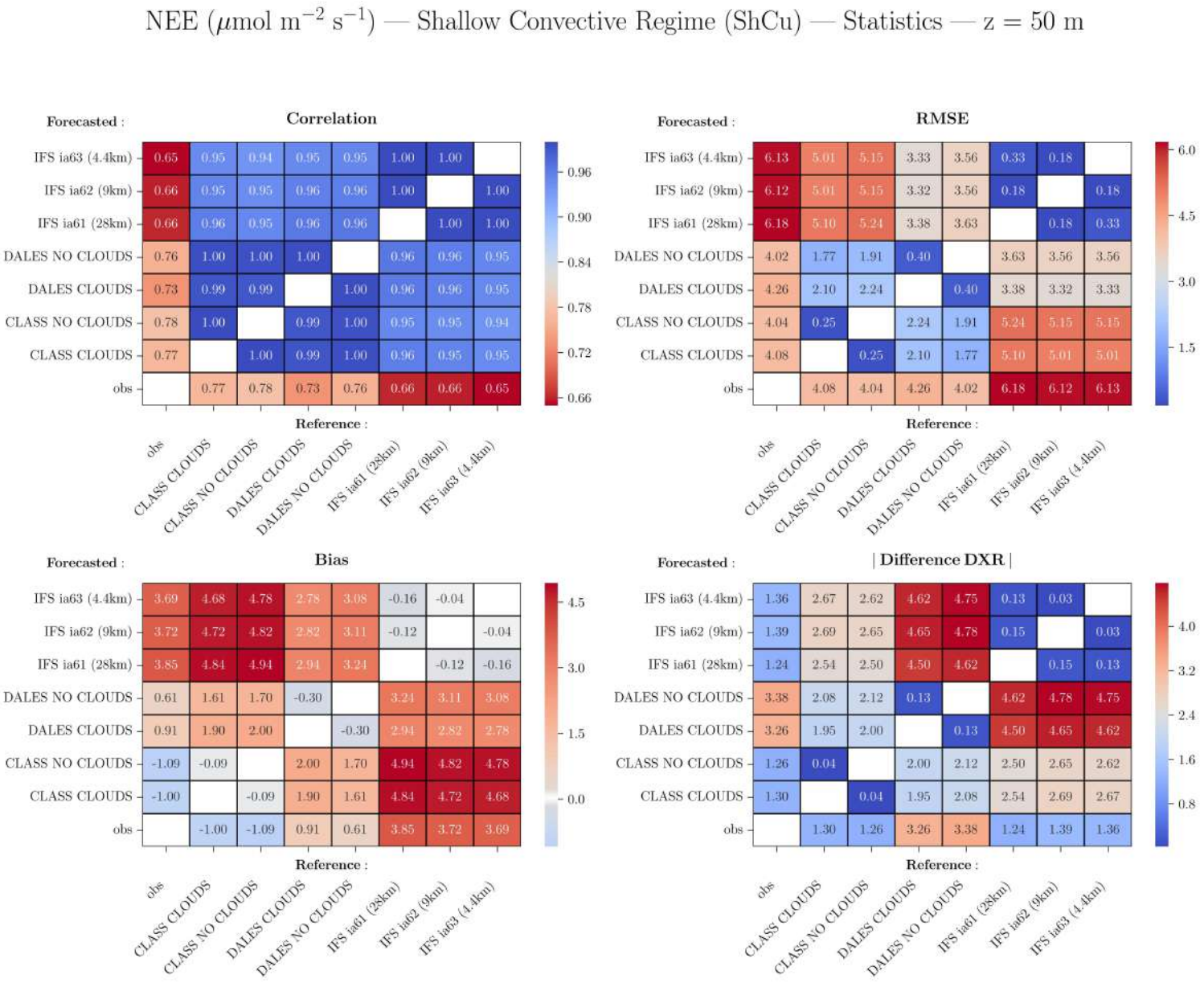



Figure 64: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (CLASS, DALES and IFS) $NEE$ for the Shallow Convective (ShCu) regime.



Funded by the European Union

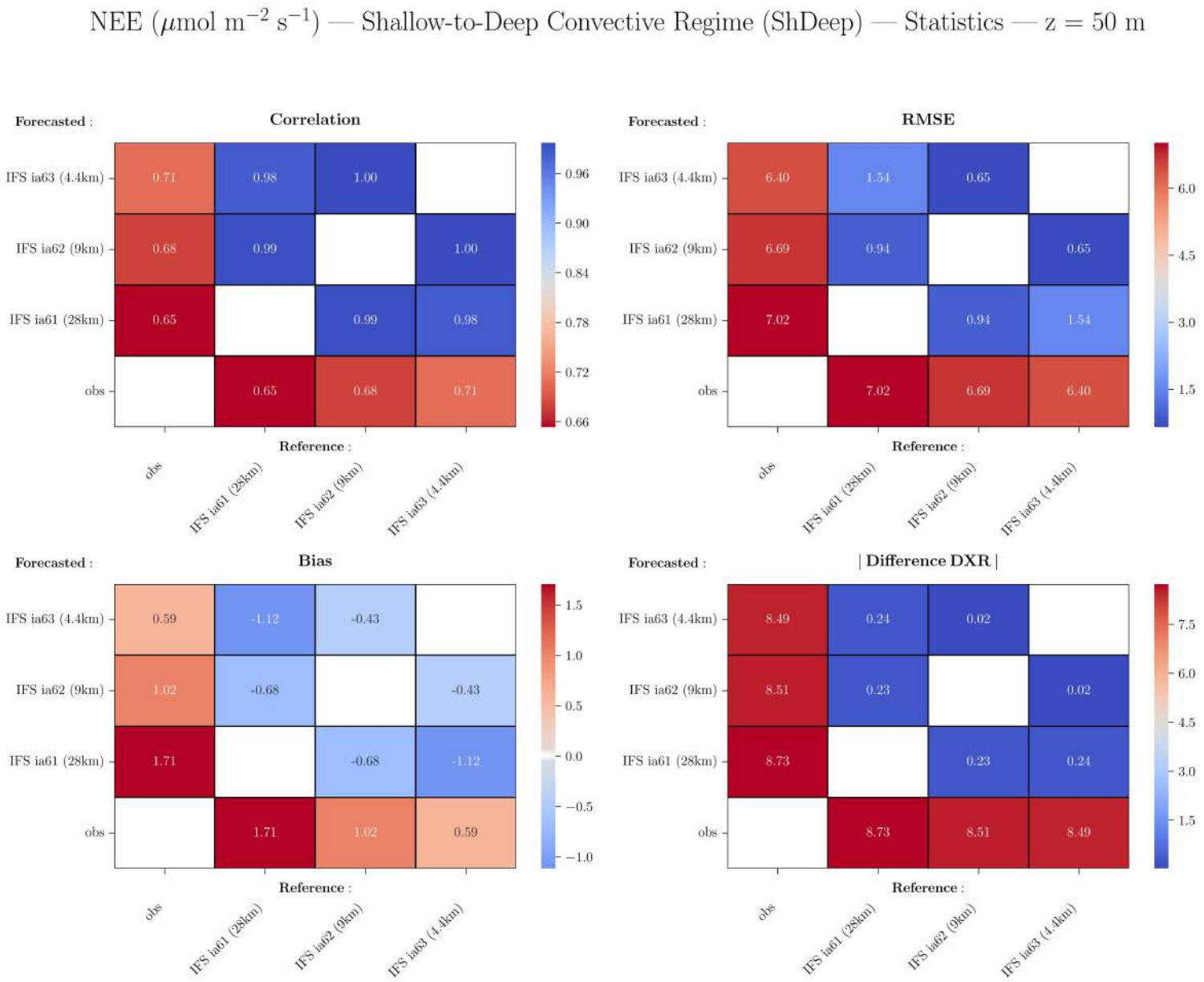



Figure 65: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) $NEE$ for the Shallow-to-Deep Convective (ShDeep) regime.



Funded by the European Union

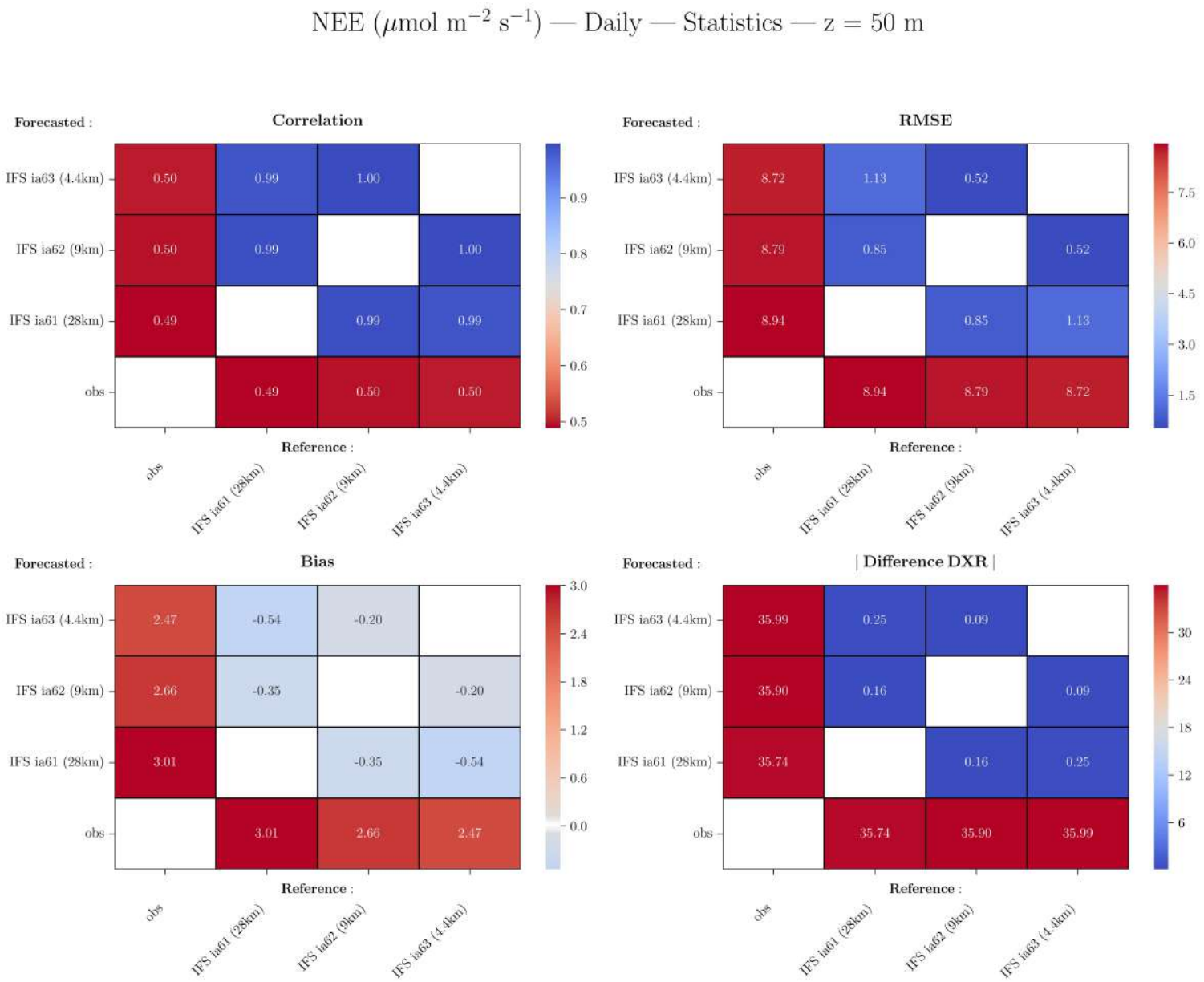



Figure 66: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) $NEE$ for the full CloudRoots-Amazon22 campaign.



Funded by the European Union

### 3.2.5 *Resp*

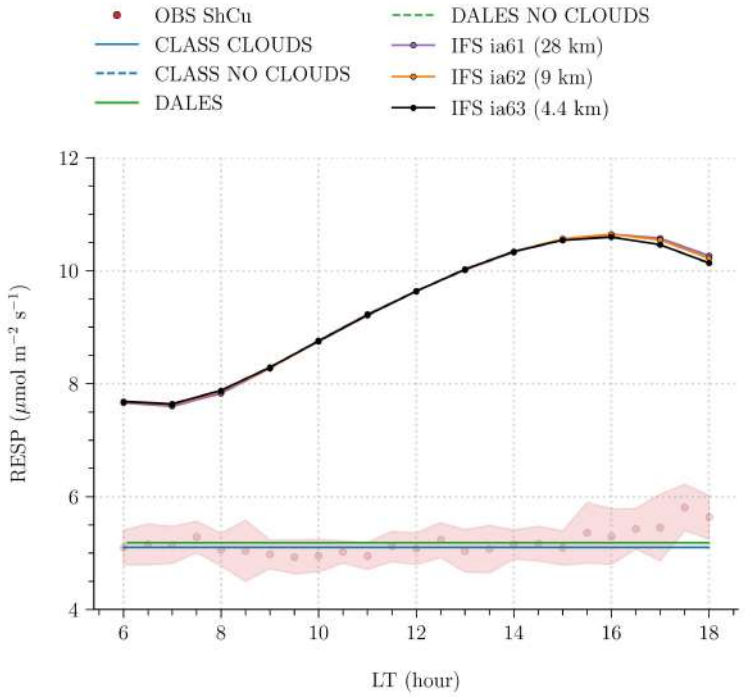



Figure 67: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (CLASS, DALES and IFS) *Resp* for the Shallow Convective (ShCu) regime.

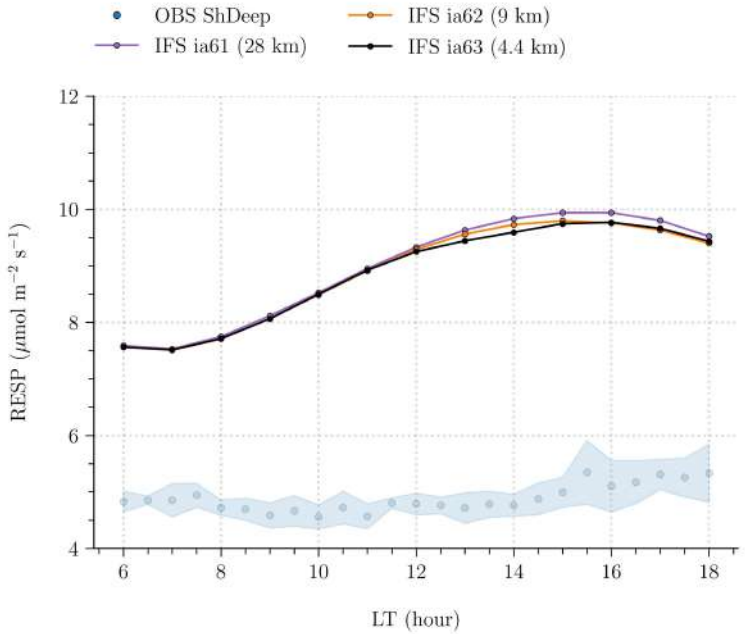



Figure 68: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (IFS) *Resp* for the Shallow-to-Deep Convective (ShDeep) regime.

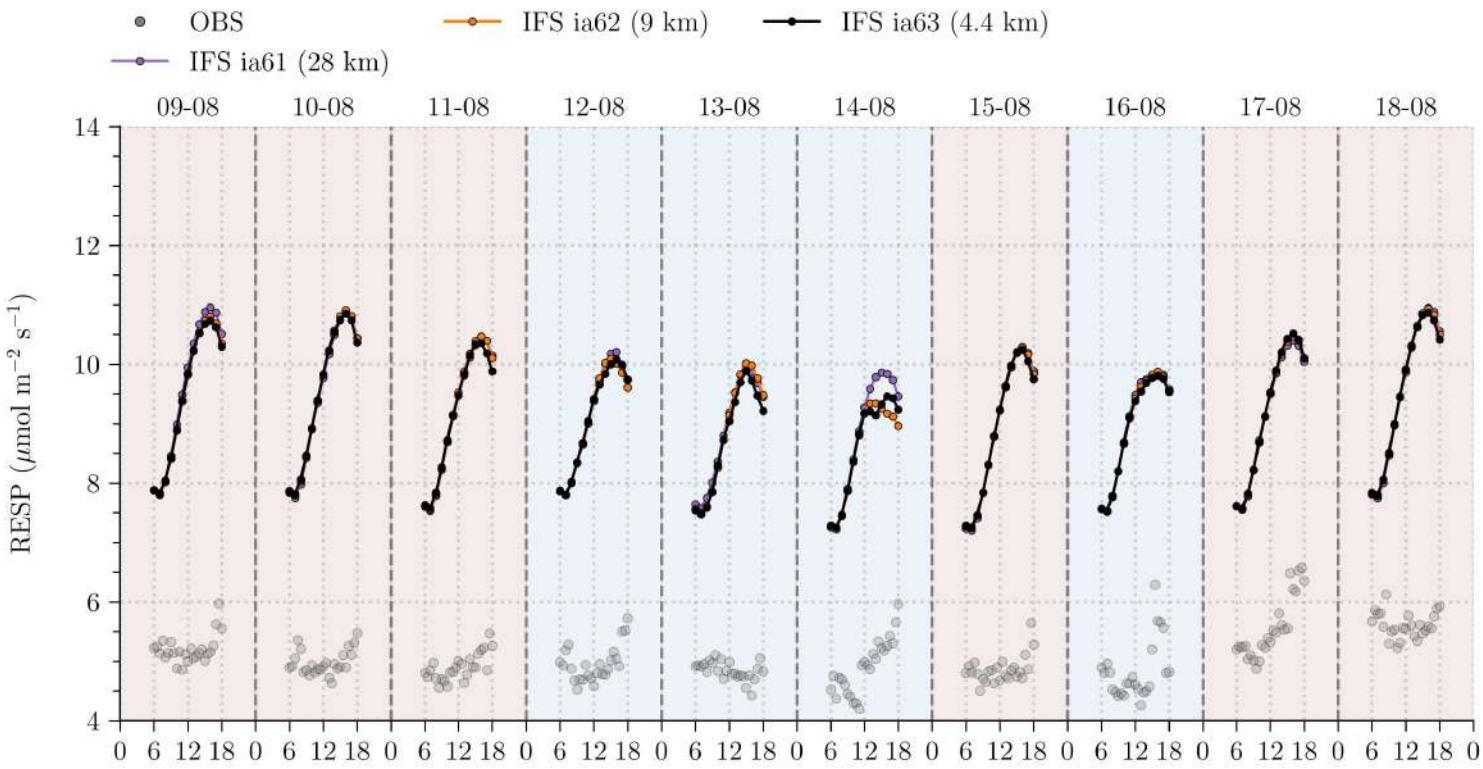



Figure 69: Diurnal cycle of observed (markers, at 50 m) and simulated (IFS) *Resp* for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).

Funded by the European Union

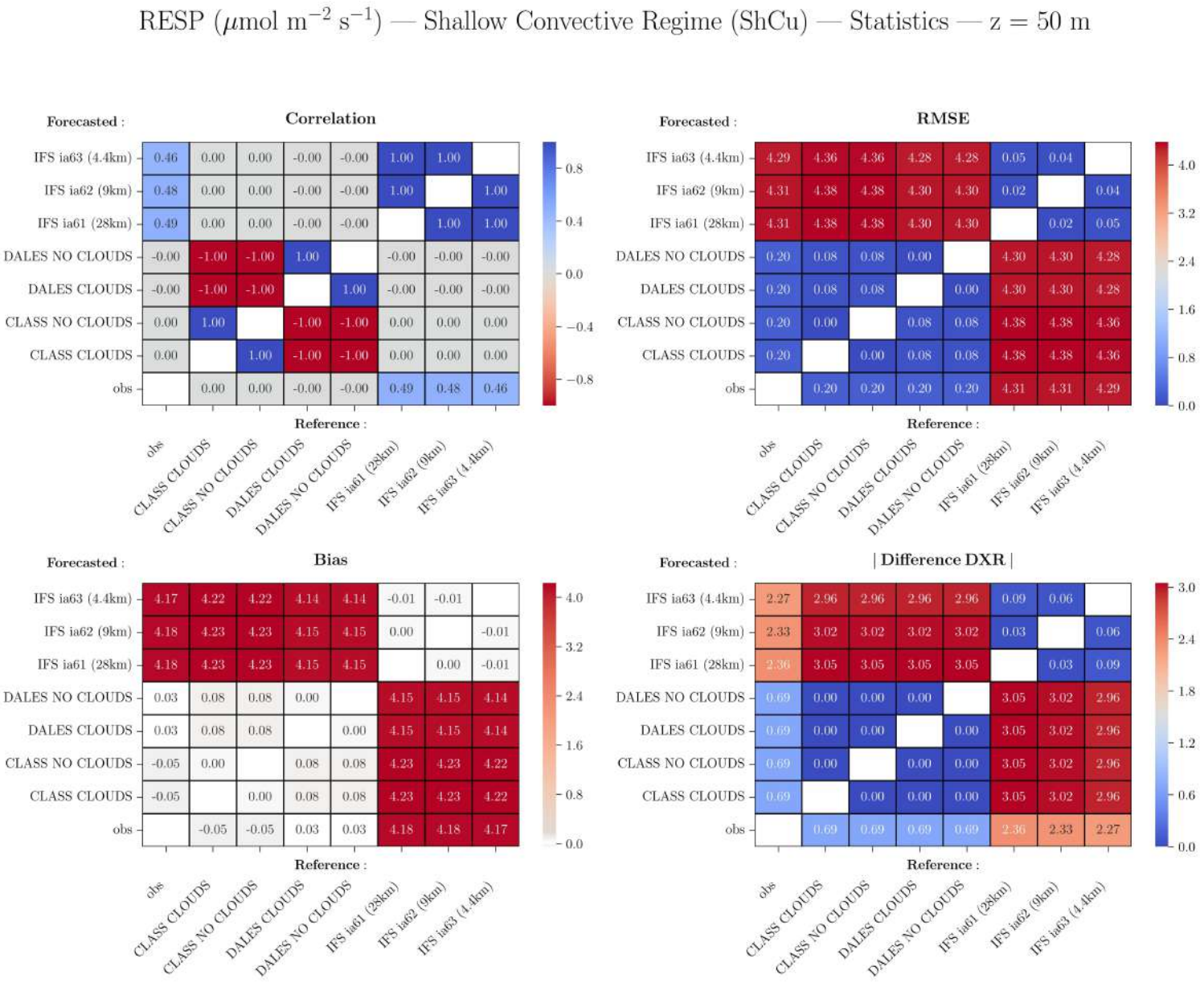



Figure 70: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (CLASS, DALES and IFS) *Resp* for the Shallow Convective (ShCu) regime.

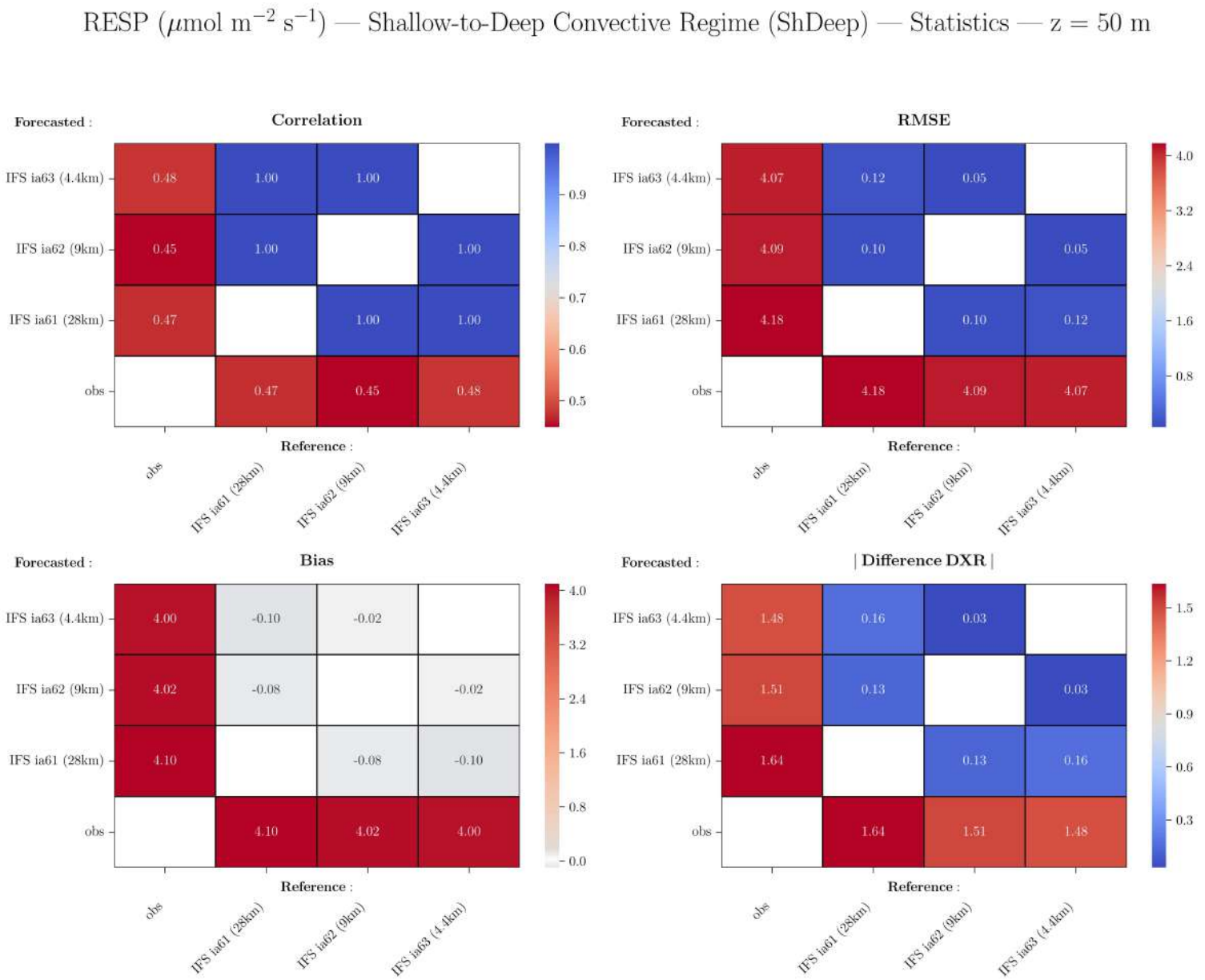



Figure 71: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) *Resp* for the Shallow-to-Deep Convective (ShDeep) regime.

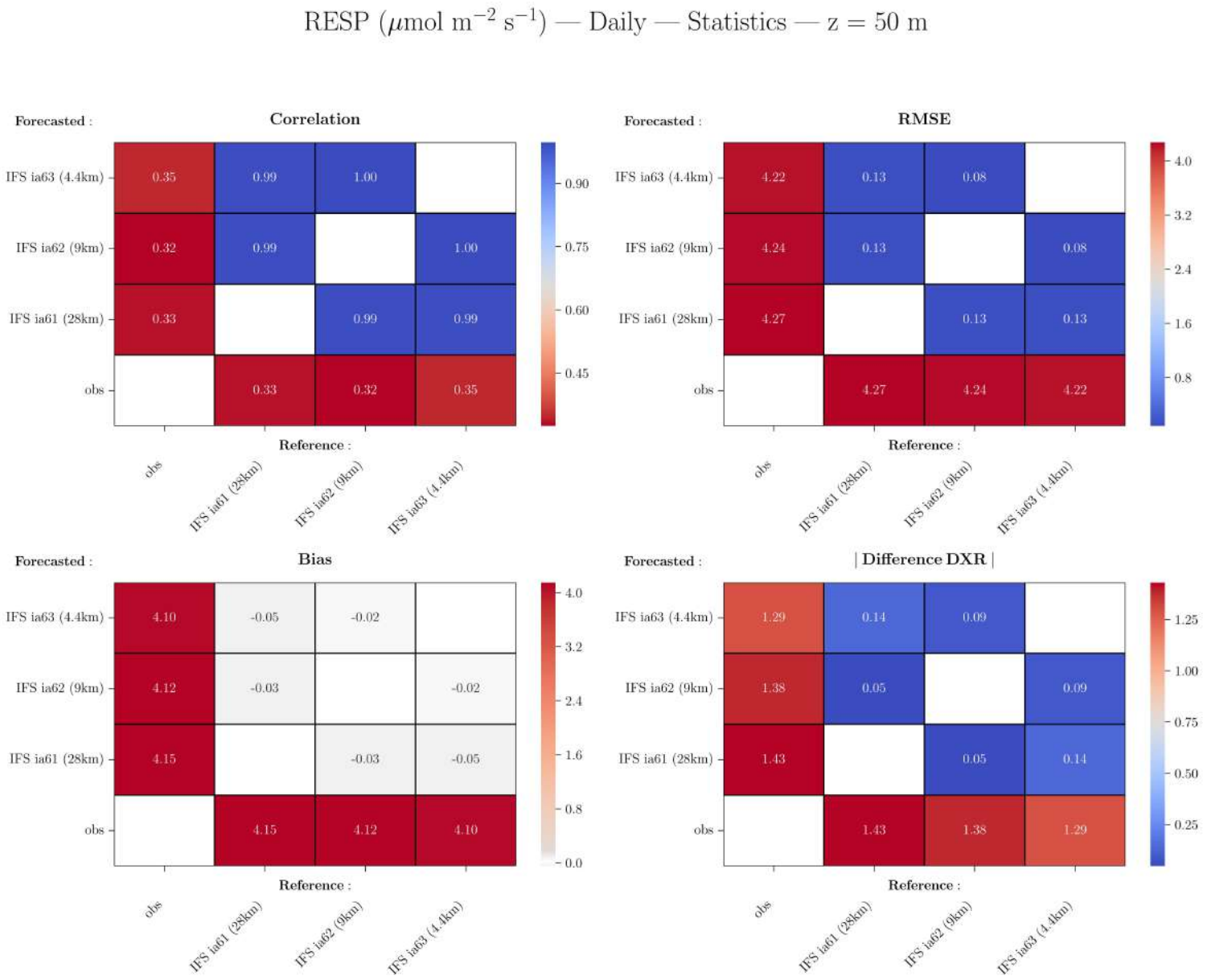



Figure 72: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) *Resp* for the full CloudRoots-Amazon22 campaign.

### 3.2.6 *GPP*

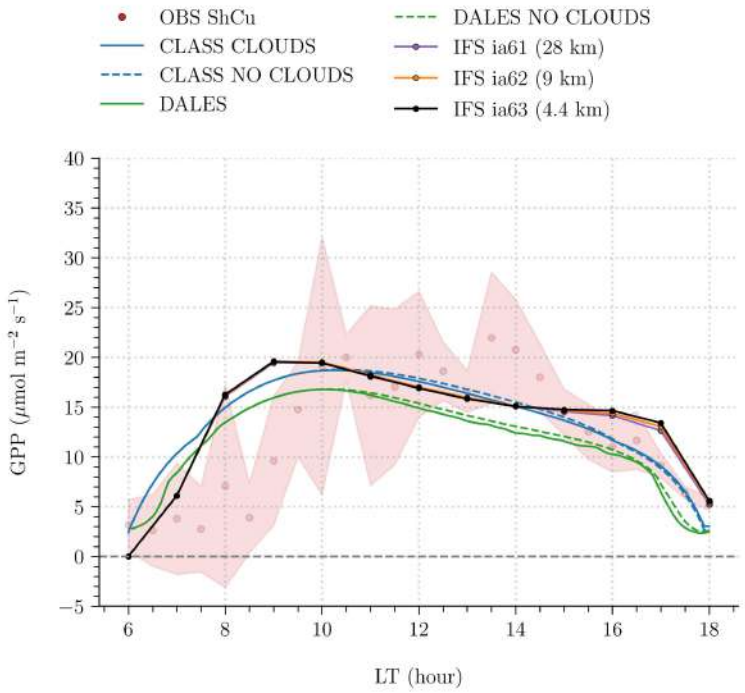



Figure 73: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (CLASS, DALES and IFS) *GPP* for the Shallow Convective (ShCu) regime.

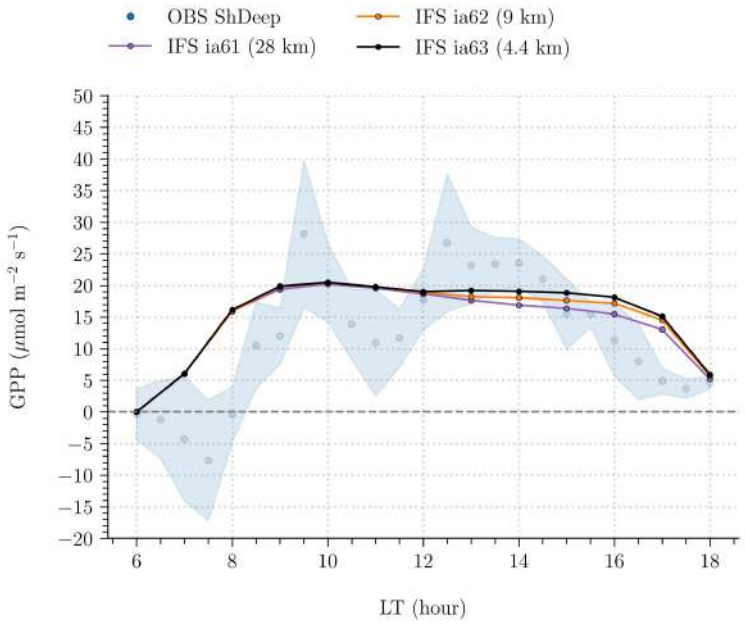



Figure 74: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (IFS) *GPP* for the Shallow-to-Deep Convective (ShDeep) regime.

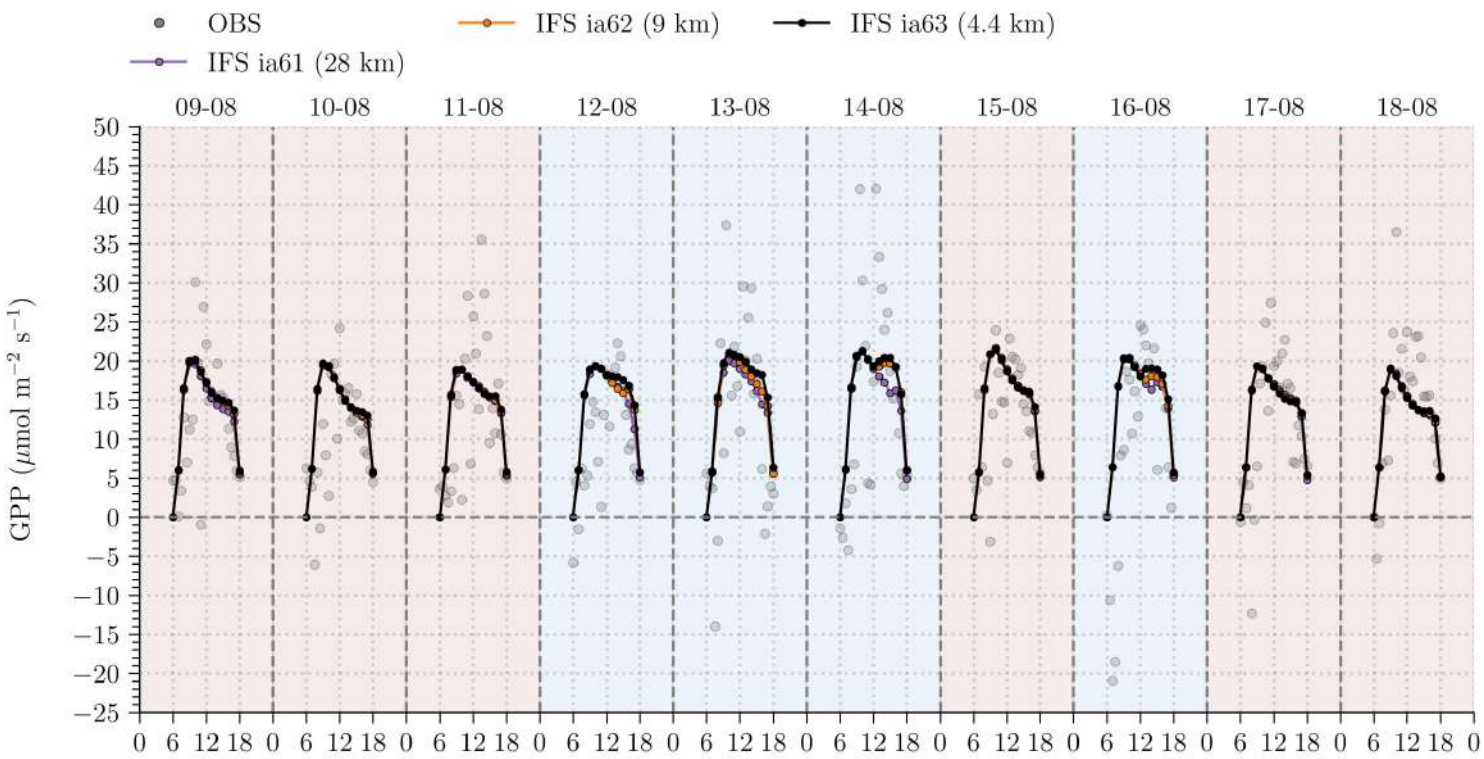



Figure 75: Diurnal cycle of observed (markers, at 50 m) and simulated (IFS) *GPP* for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).

Funded by the European Union

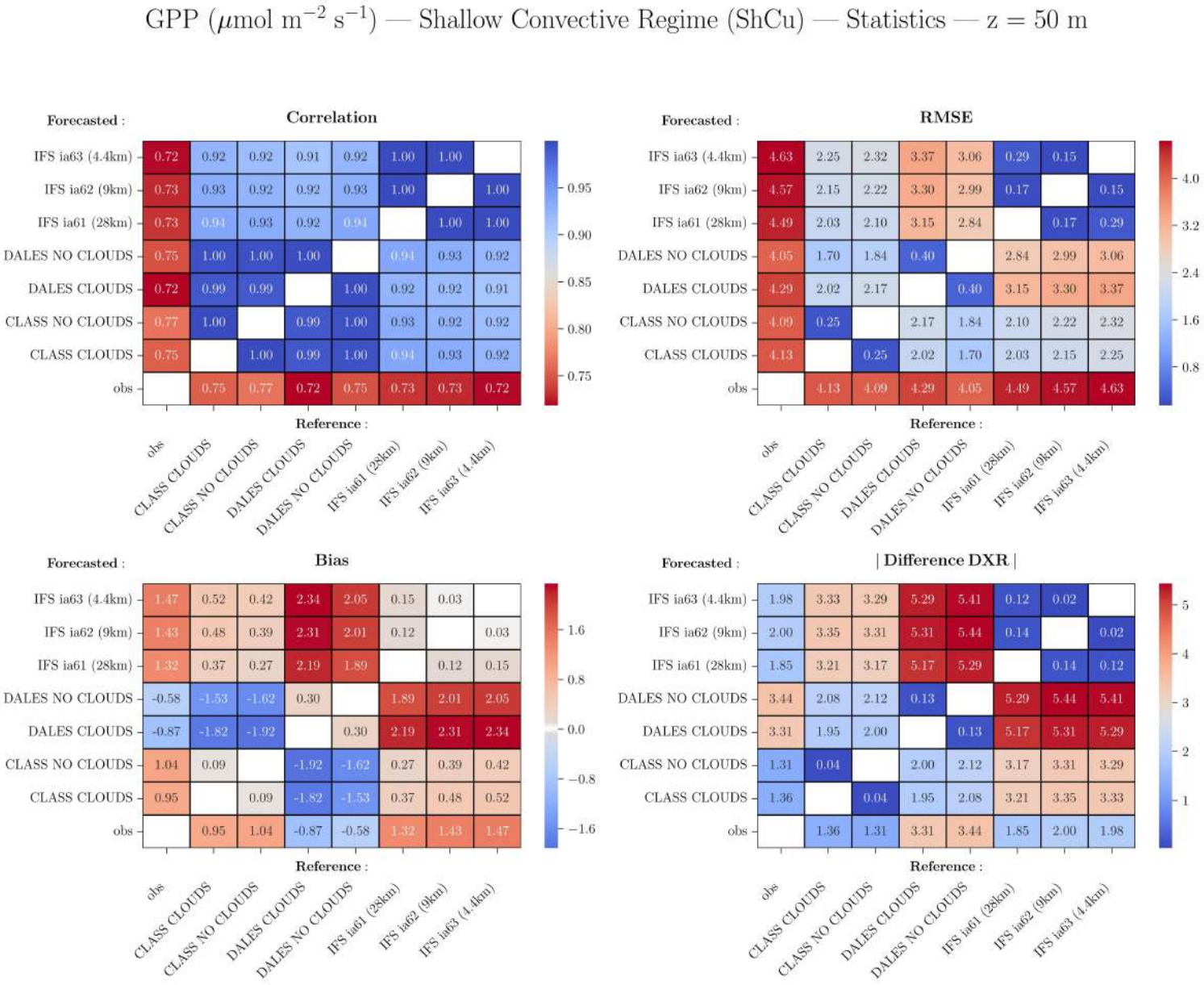



Figure 76: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (CLASS, DALES and IFS) *GPP* for the Shallow Convective (ShCu) regime.

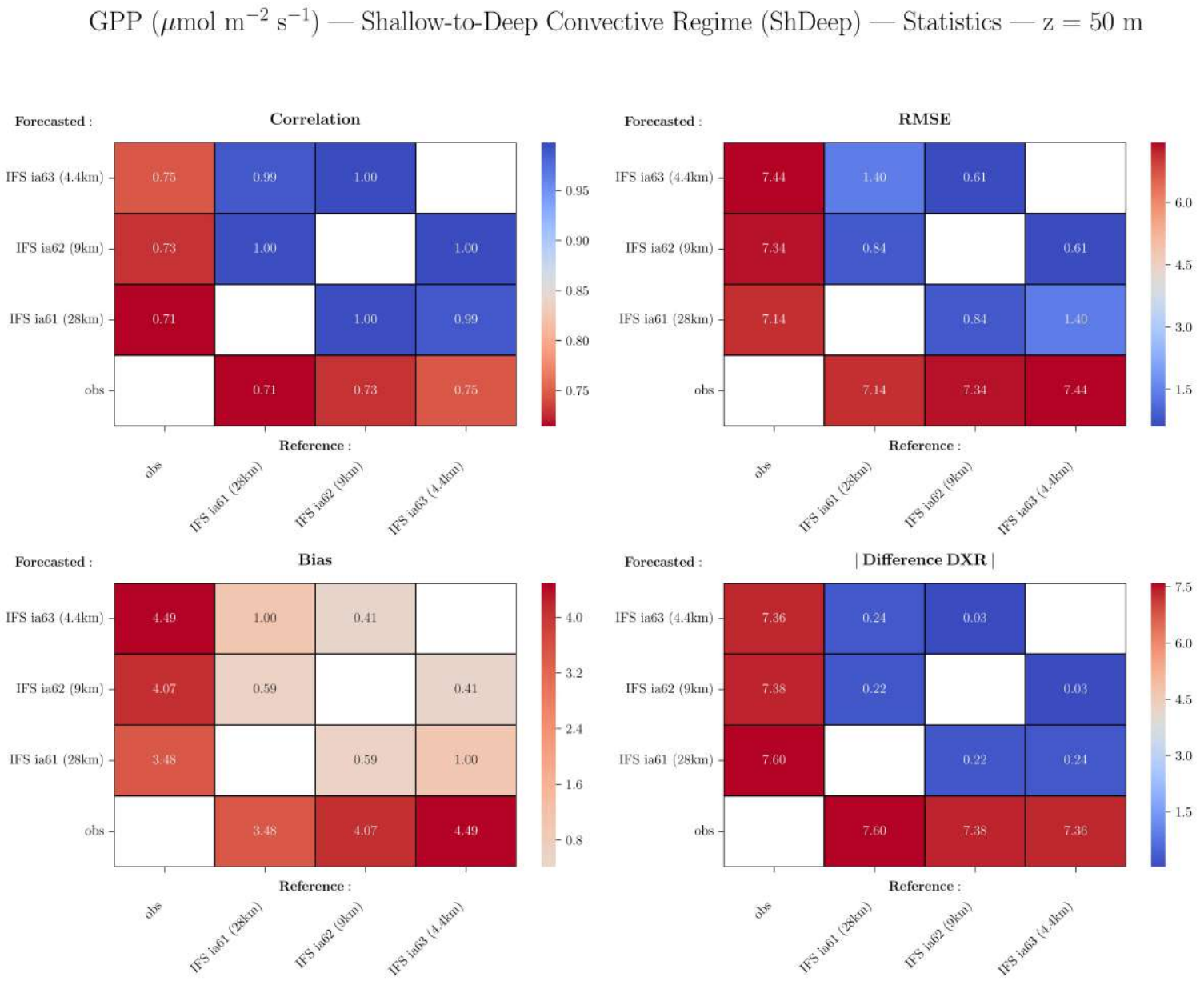



Figure 77: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) *GPP* for the Shallow-to-Deep Convective (ShDeep) regime.



Funded by the European Union

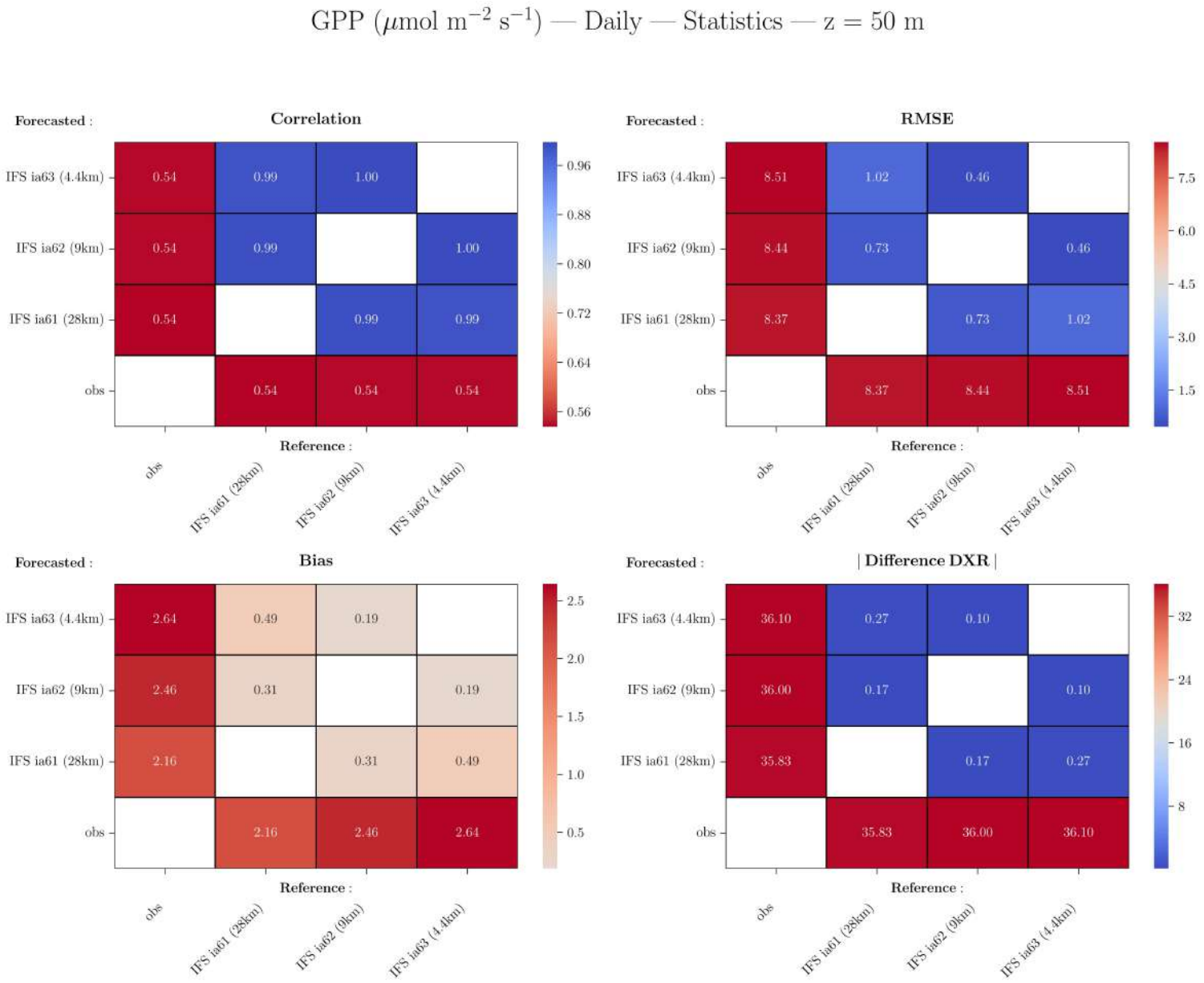



Figure 78: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) *GPP* for the full CloudRoots-Amazon22 campaign.



Funded by the European Union

# 4 Lower Troposphere

## 4.1 $z_i$

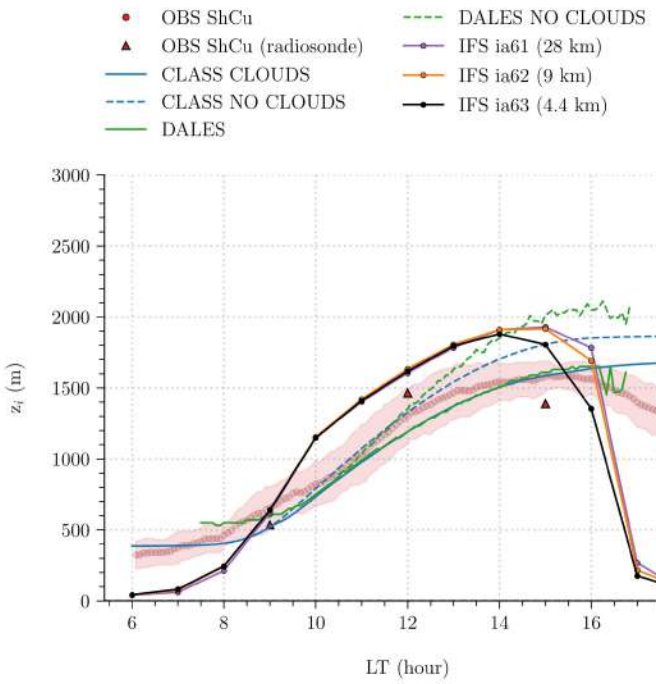



Figure 79: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), and simulated (CLASS, DALES and IFS) $z_i$ for the Shallow Convective (ShCu) regime.

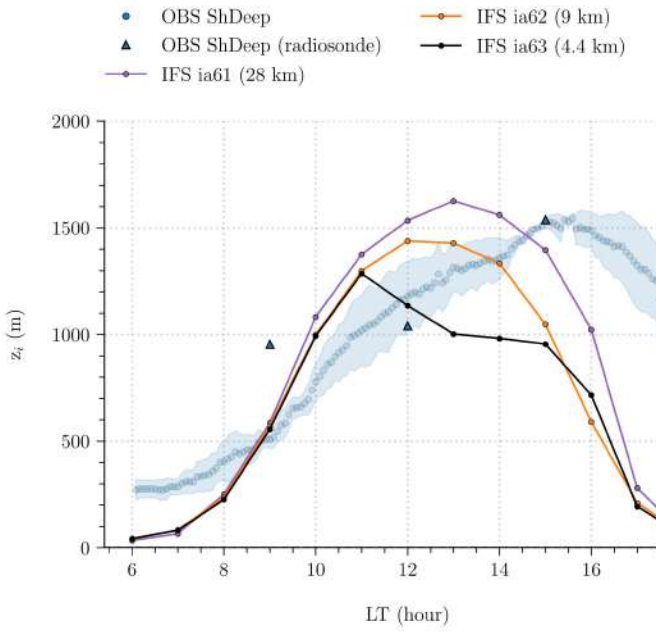



Figure 80: Diurnal cycle of observed (markers (mean) and shaded (standard deviation) and simulated (IFS) $z_i$ for the Shallow-to-Deep Convective (ShDeep) regime.

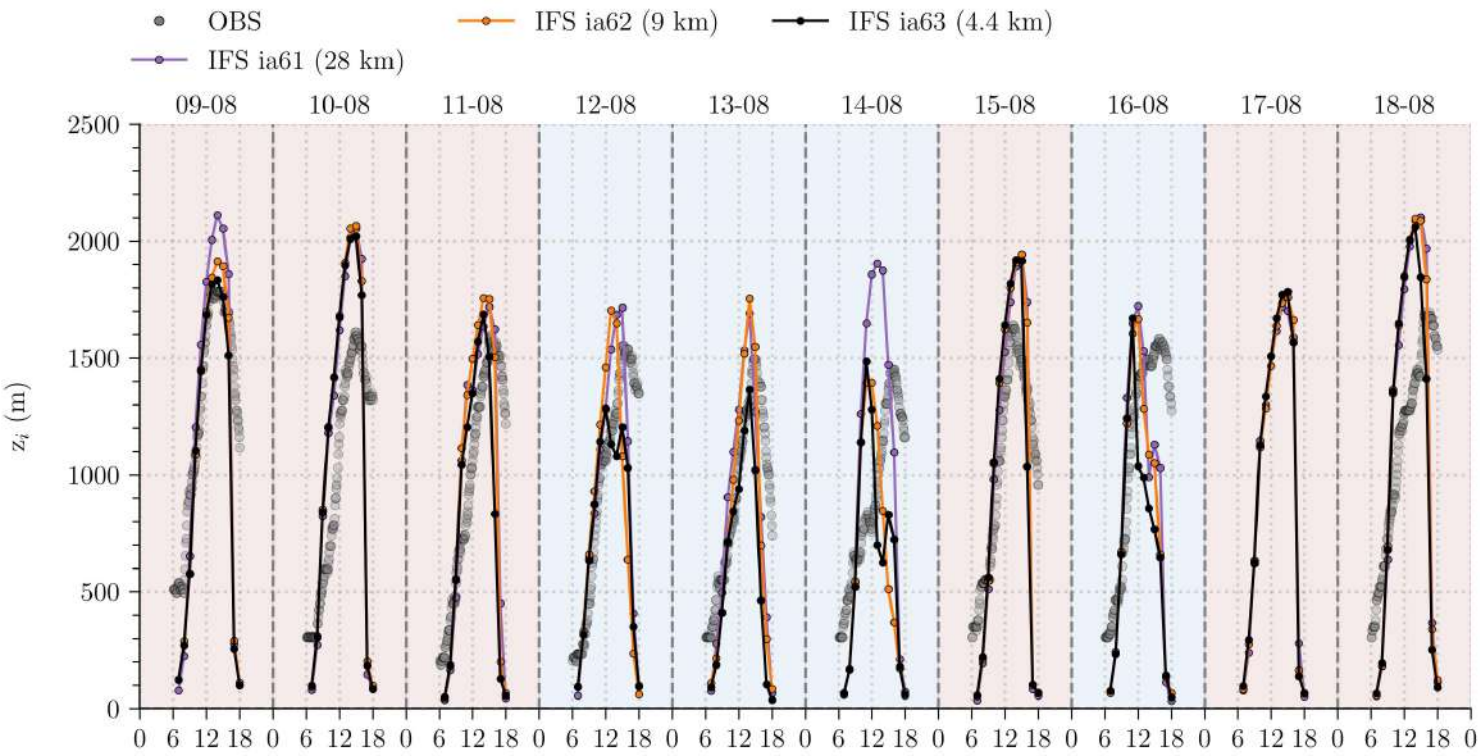



Figure 81: Diurnal cycle of observed (markers) and simulated (IFS) $z_i$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).



Funded by the European Union

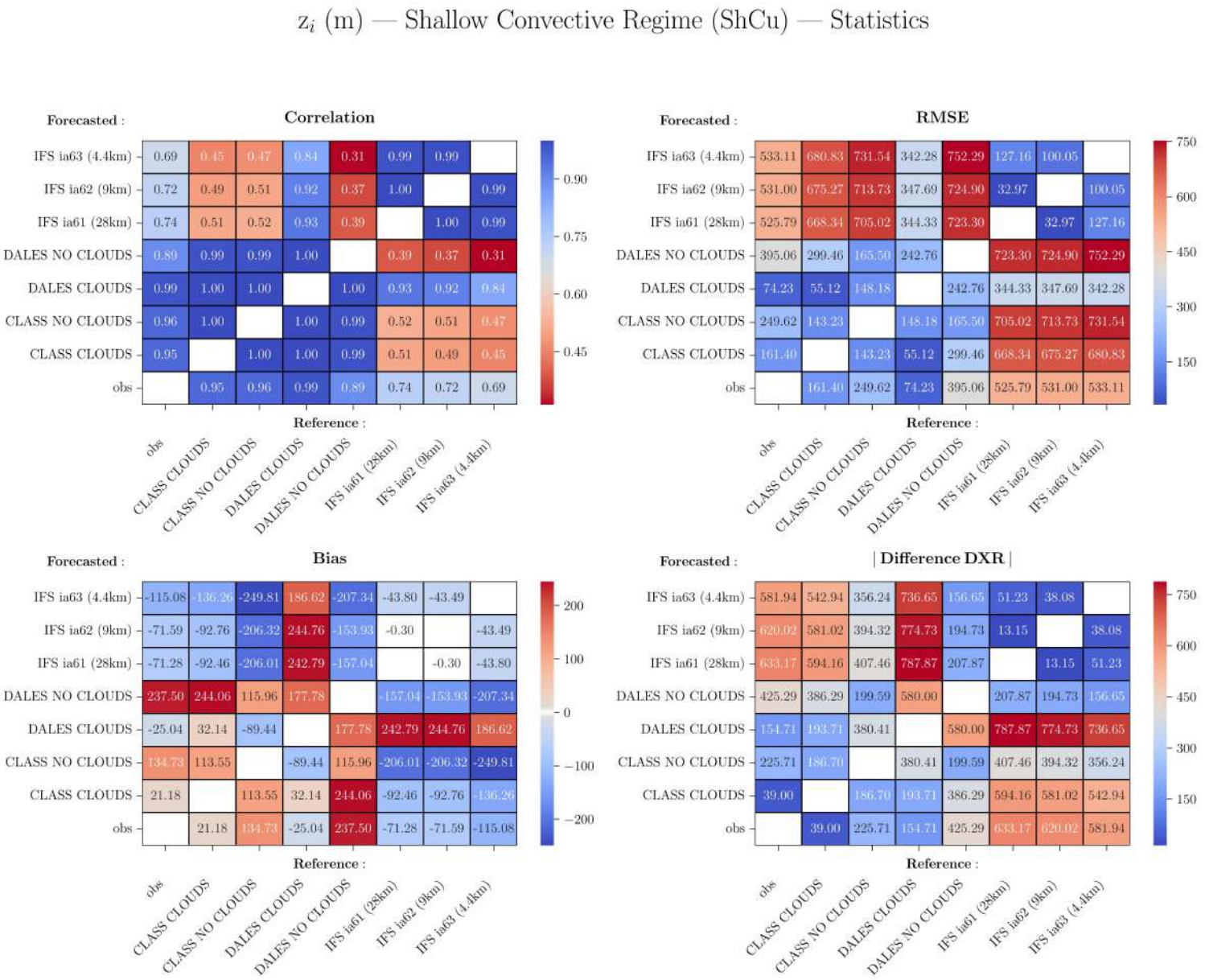



Figure 82: Statistical quantities of the diurnal cycle of observed and simulated (CLASS, DALES and IFS) $z_i$ for the Shallow Convective (ShCu) regime.

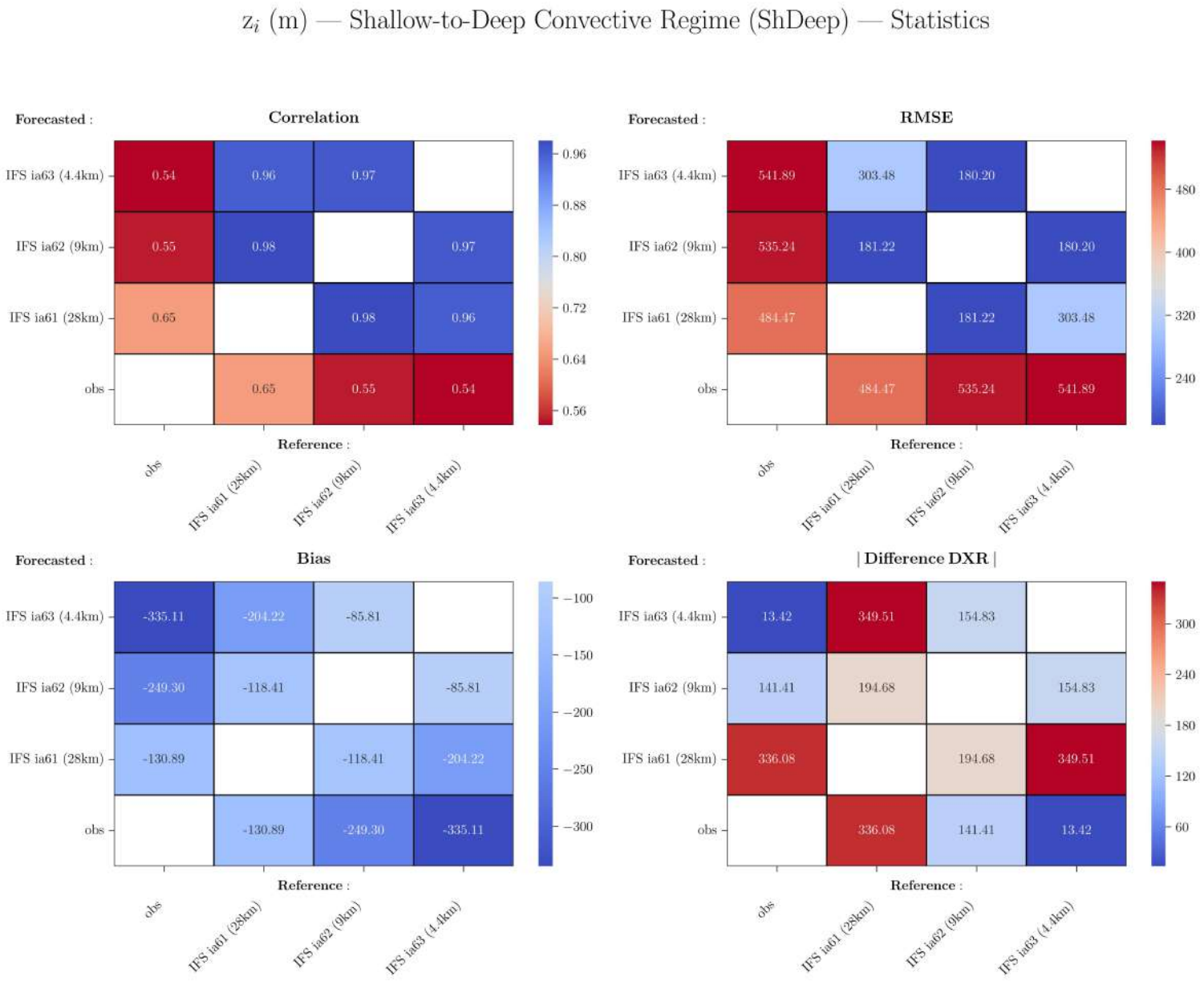



Figure 83: Statistical quantities of the diurnal cycle of observed and simulated (IFS) $z_i$ for the Shallow-to-Deep Convective (ShDeep) regime.

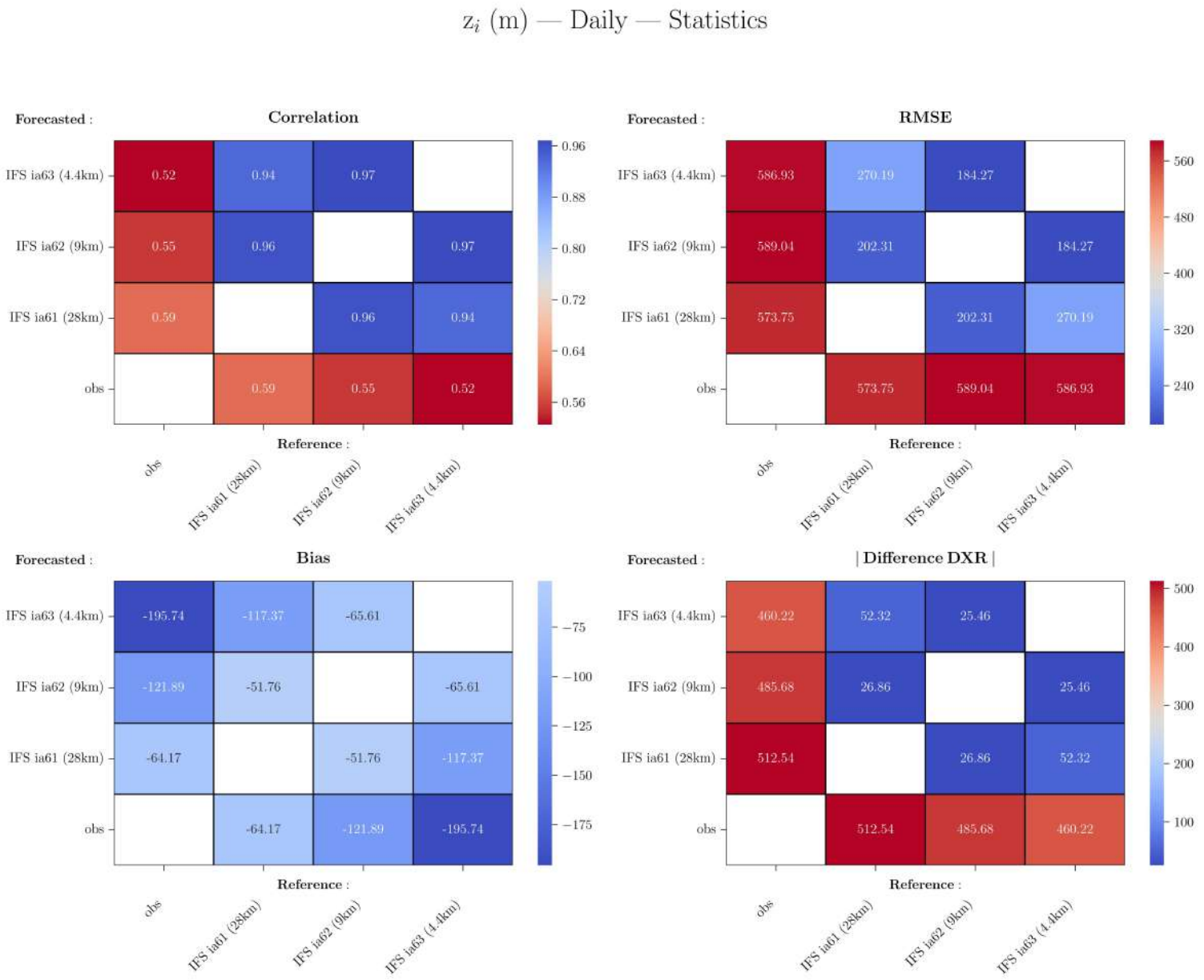



Figure 84: Statistical quantities of the diurnal cycle of observed and simulated (IFS) $z_i$ for the full CloudRoots-Amazon22 campaign.

## 4.2 $\theta$

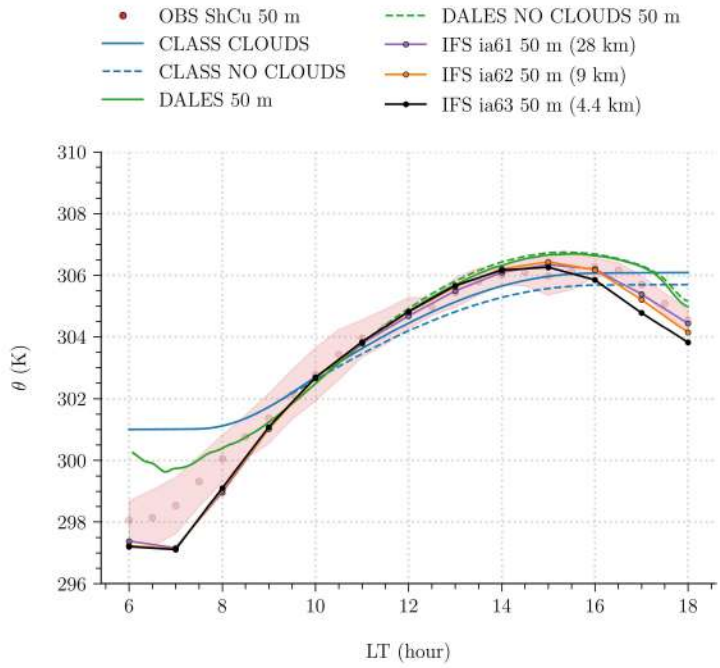



Figure 85: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (CLASS, DALES and IFS) $\theta$ for the Shallow Convective (ShCu) regime.

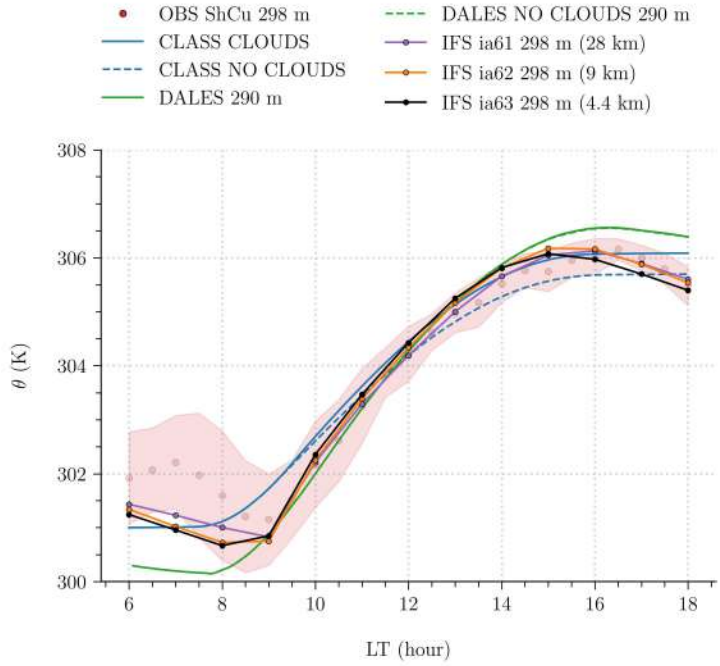



Figure 86: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 298 m) and simulated (CLASS, DALES and IFS) $\theta$ for the Shallow Convective (ShCu) regime.

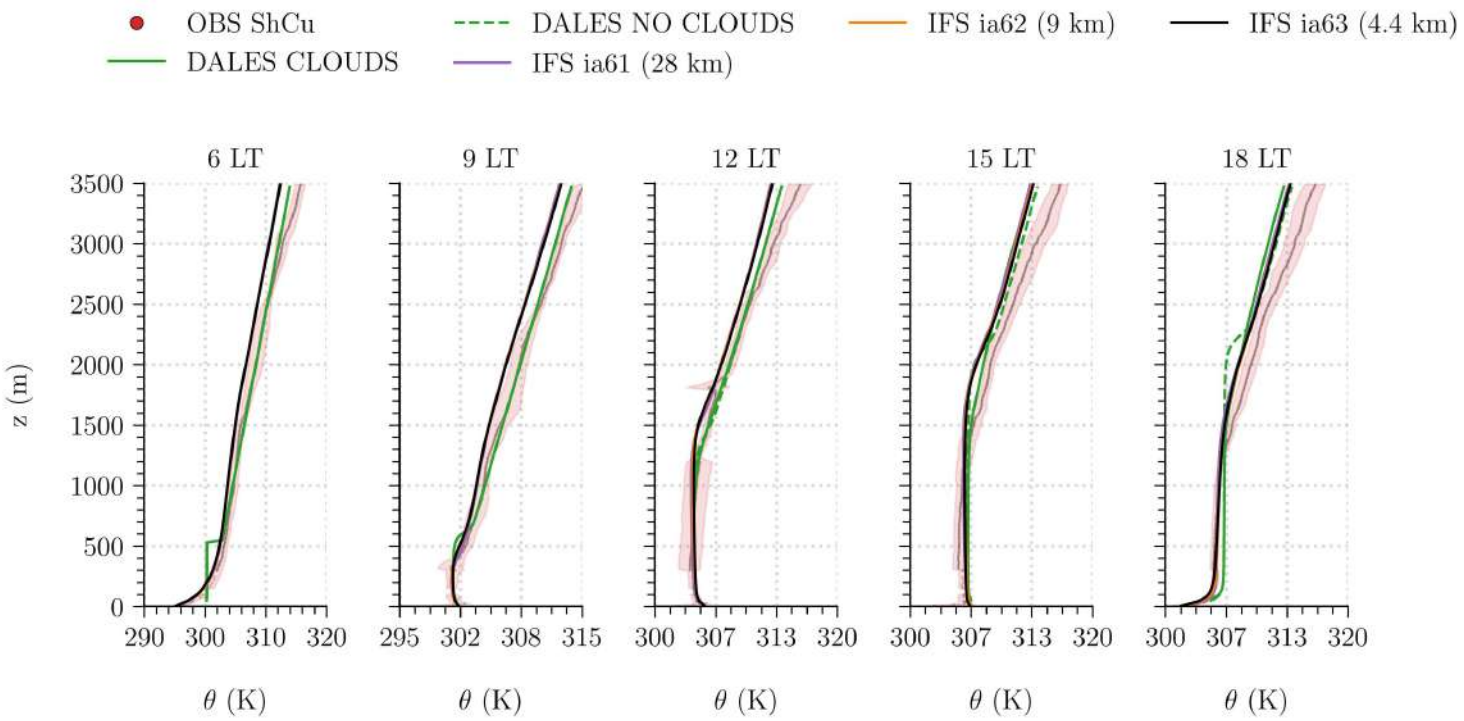



Figure 87: Vertical profiles (markers (mean) and shaded (standard deviation)) and simulated (CLASS, DALES and IFS) $\theta$ for the Shallow Convective (ShCu) regime.

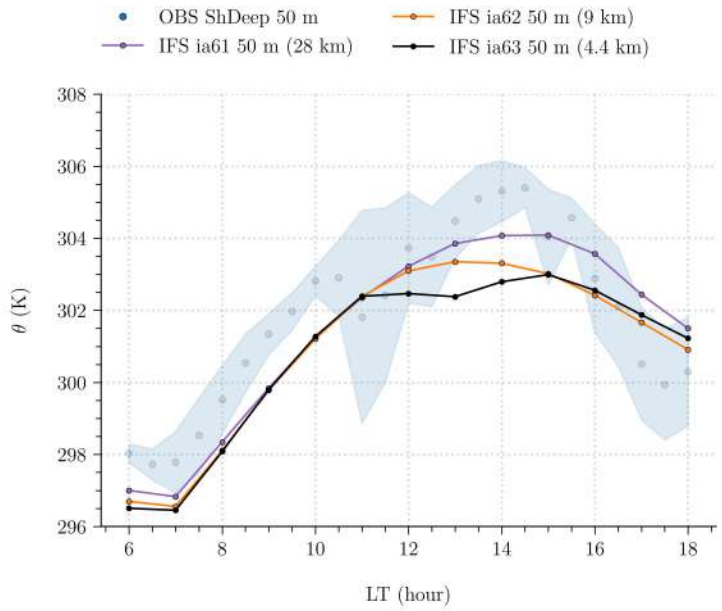



Figure 88: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (IFS) $\theta$ for the Shallow-to-Deep Convective (ShDeep) regime.

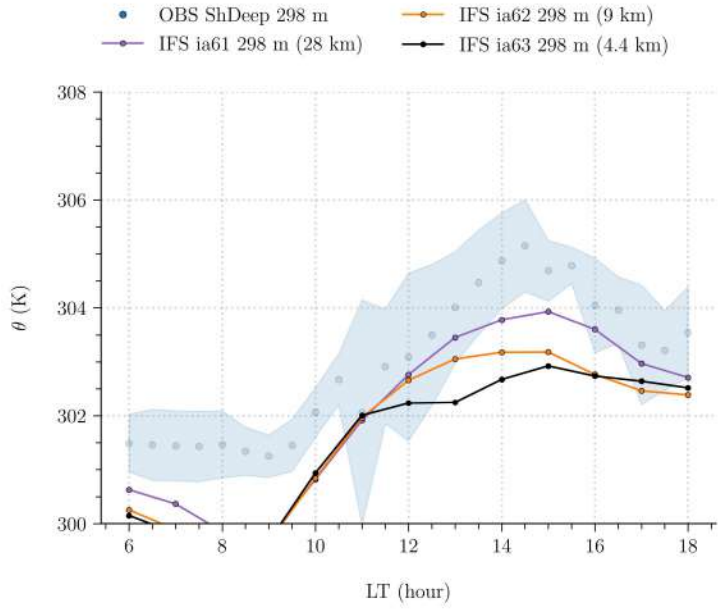



Figure 89: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 298 m) and simulated (IFS) $\theta$ for the Shallow-to-Deep Convective (ShDeep) regime.

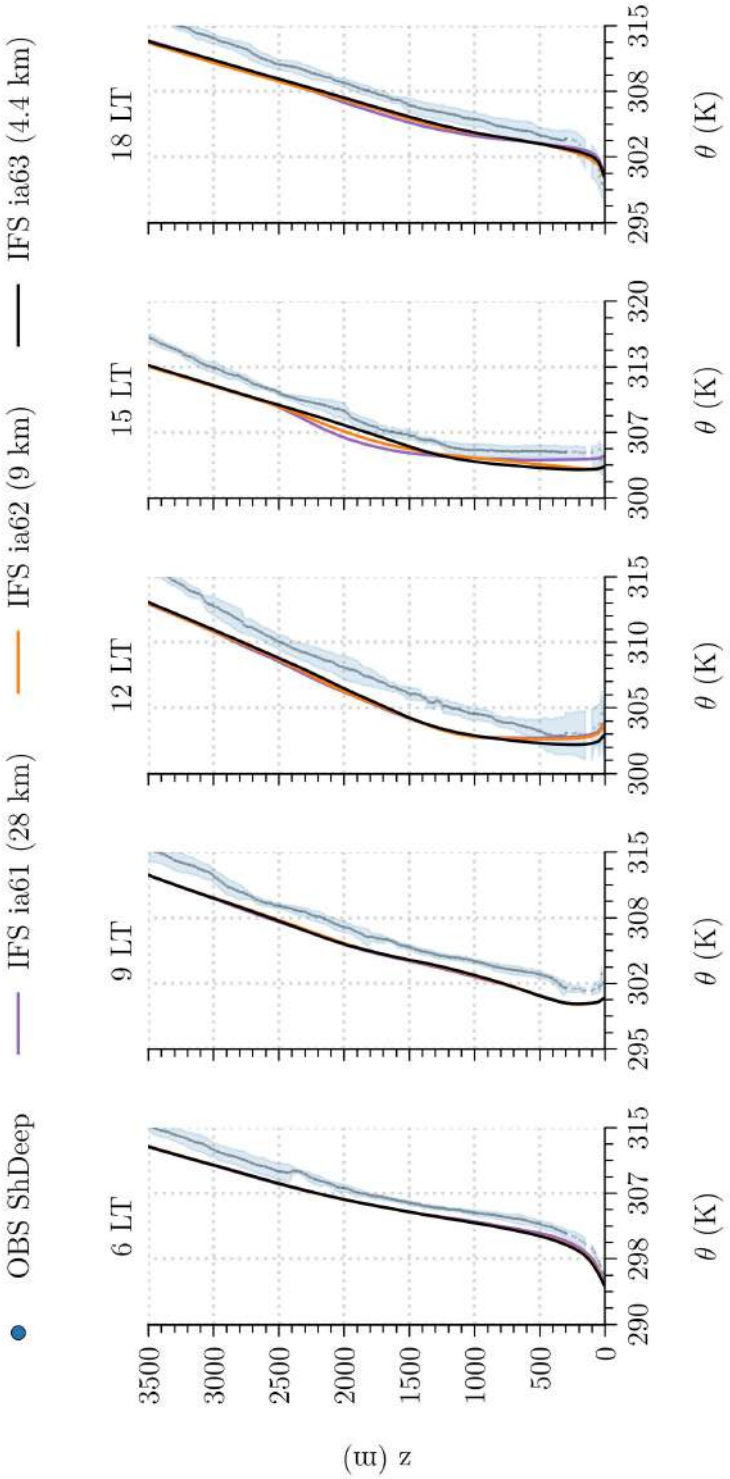



Figure 90: Vertical profiles (markers (mean) and shaded (standard deviation)) and simulated (CLASS, DALES and IFS) $\theta$ for the Shallow-to-Deep Convective (ShDeep) regime.

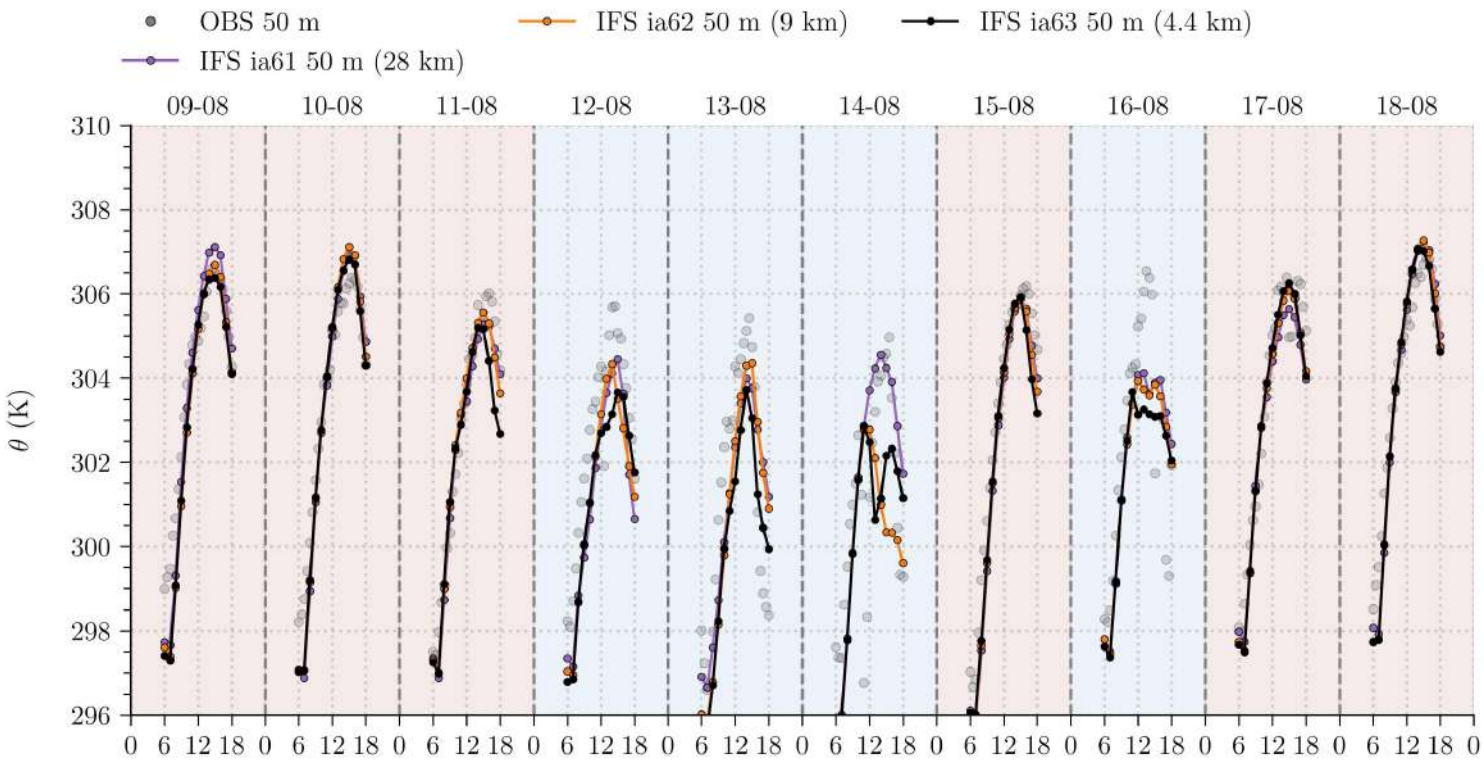



Figure 91: Diurnal cycle of observed (markers, at 50 m) and simulated (IFS) $\theta$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).

Funded by the European Union

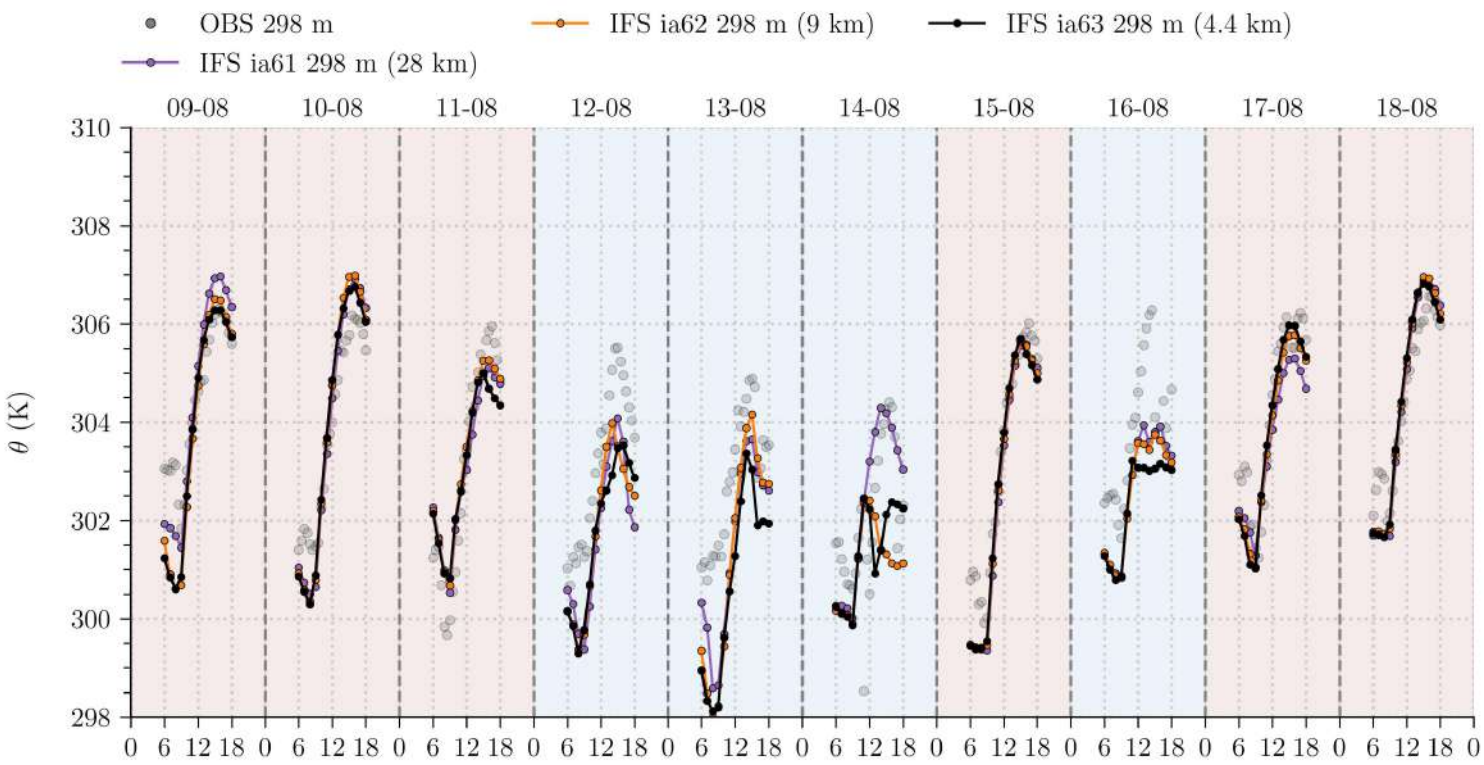



Figure 92: Diurnal cycle of observed (markers, at 298 m) and simulated (IFS) $\theta$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).



Funded by the European Union

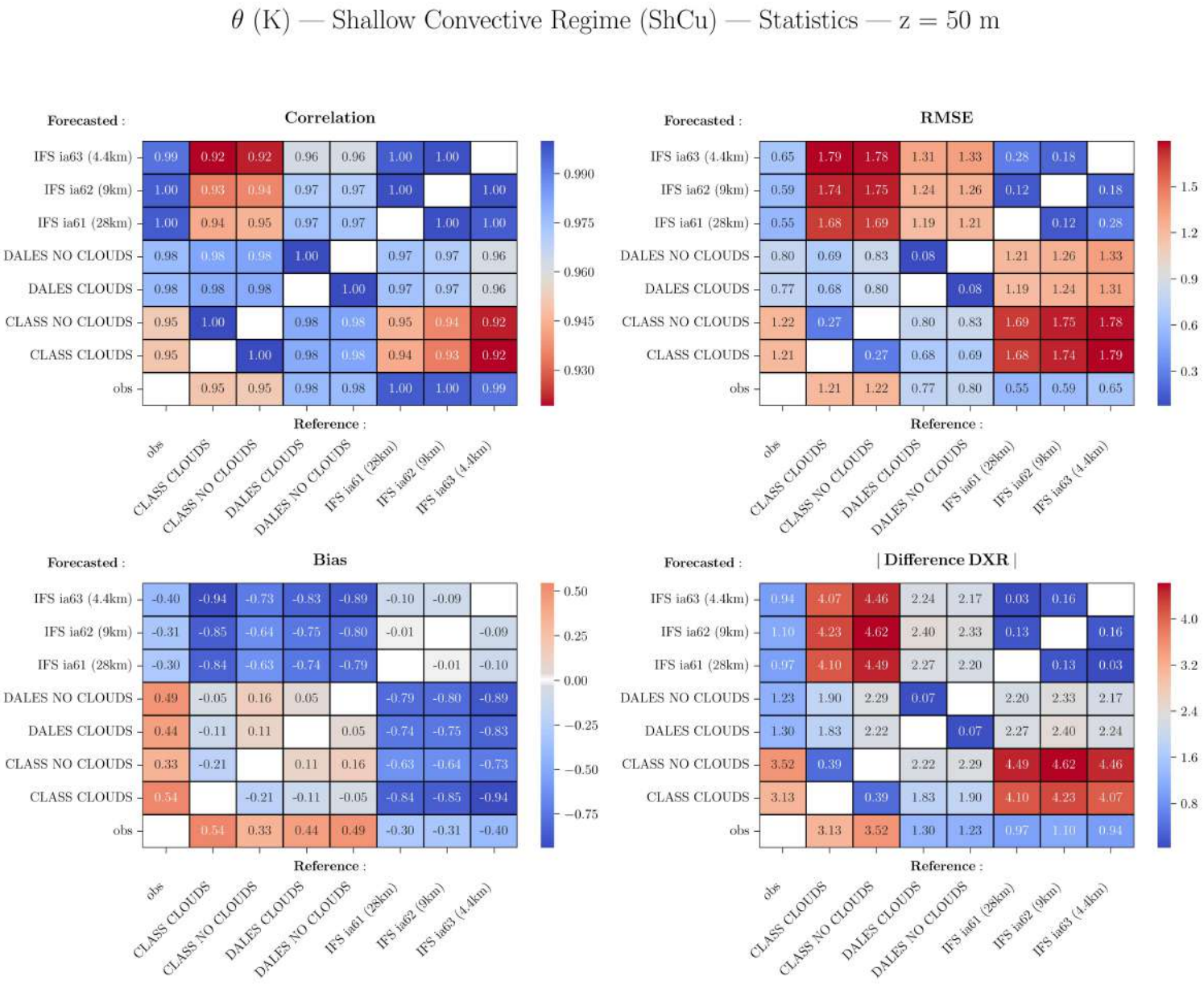



Figure 93: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (CLASS, DALES and IFS) $\theta$ for the Shallow Convective (ShCu) regime.

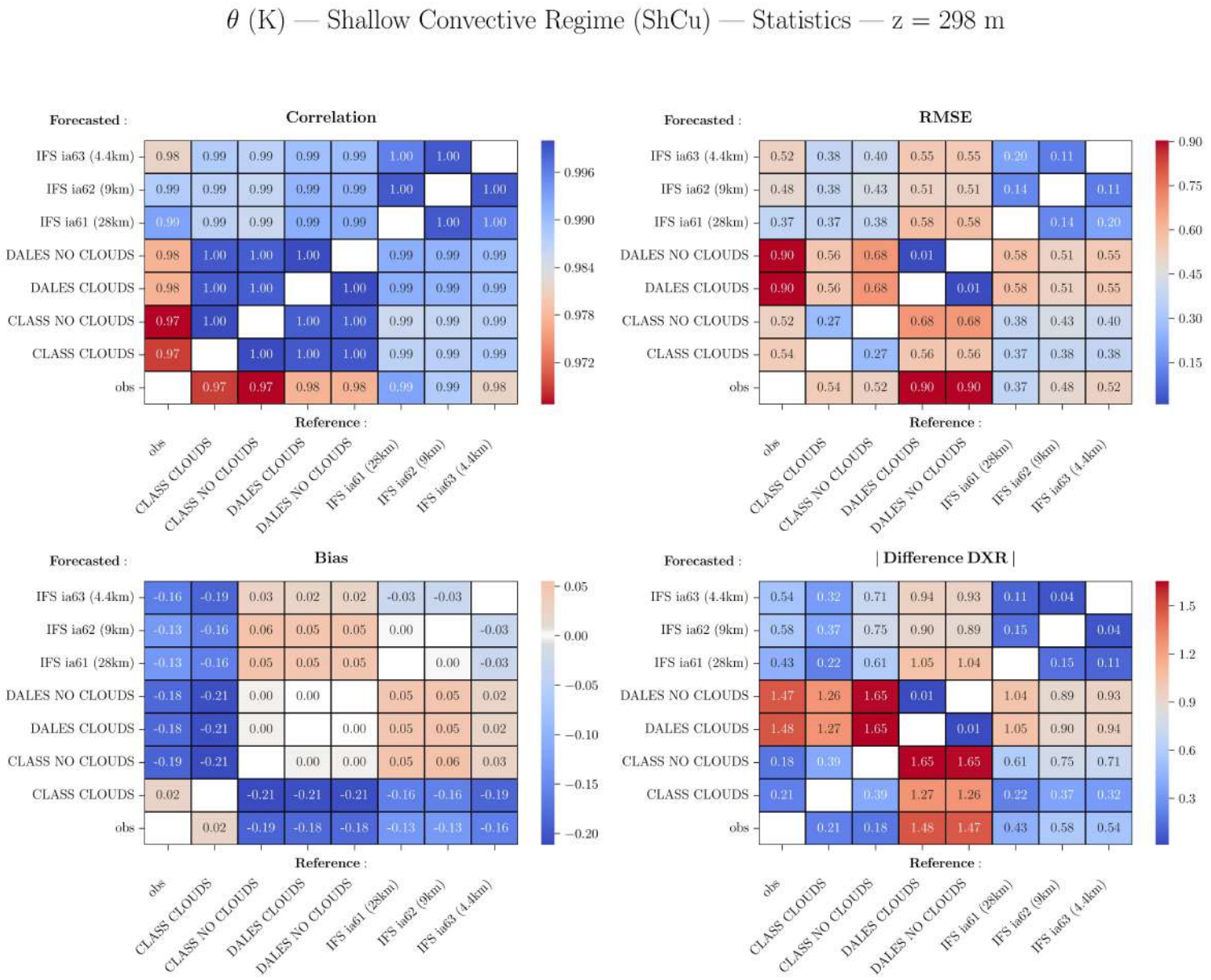



Figure 94: Statistical quantities of the diurnal cycle of observed (at 298 m) and simulated (CLASS, DALES and IFS) $\theta$ for the Shallow Convective (ShCu) regime.



Funded by the European Union

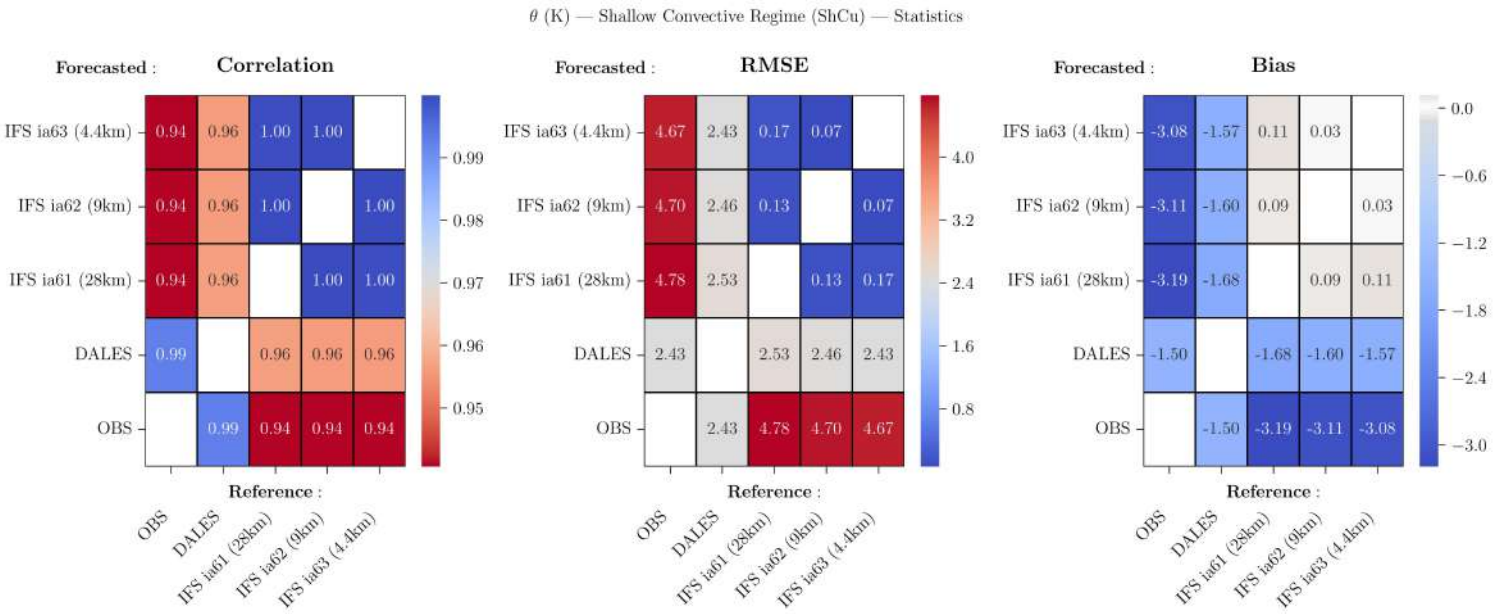



Figure 95: Statistical quantities of the vertical profiles of observed and simulated (CLASS, DALES and IFS) $\theta$ for the Shallow Convective (ShCu) regime.

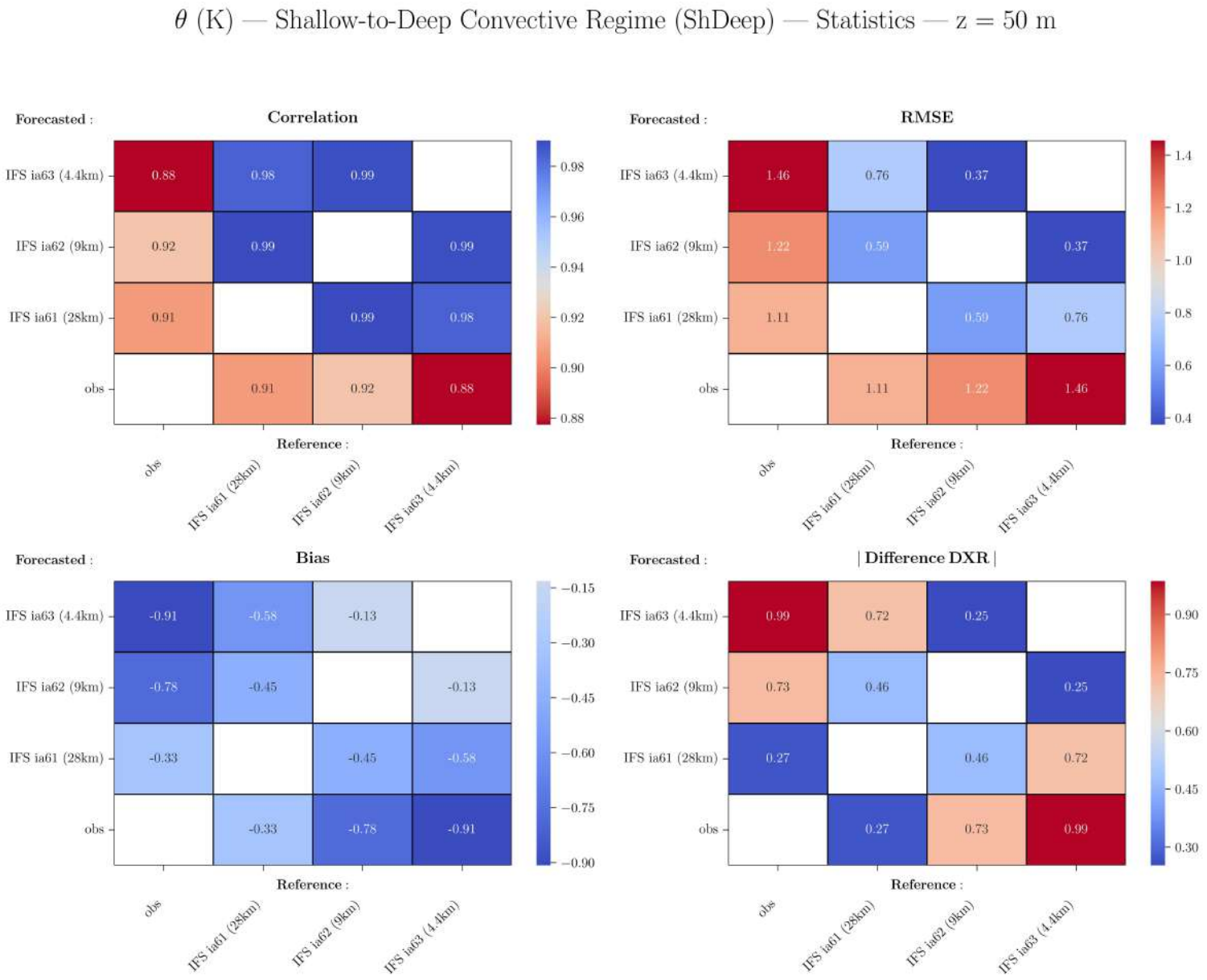



Figure 96: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) $\theta$ for the Shallow-to-Deep Convective (ShDeep) regime.

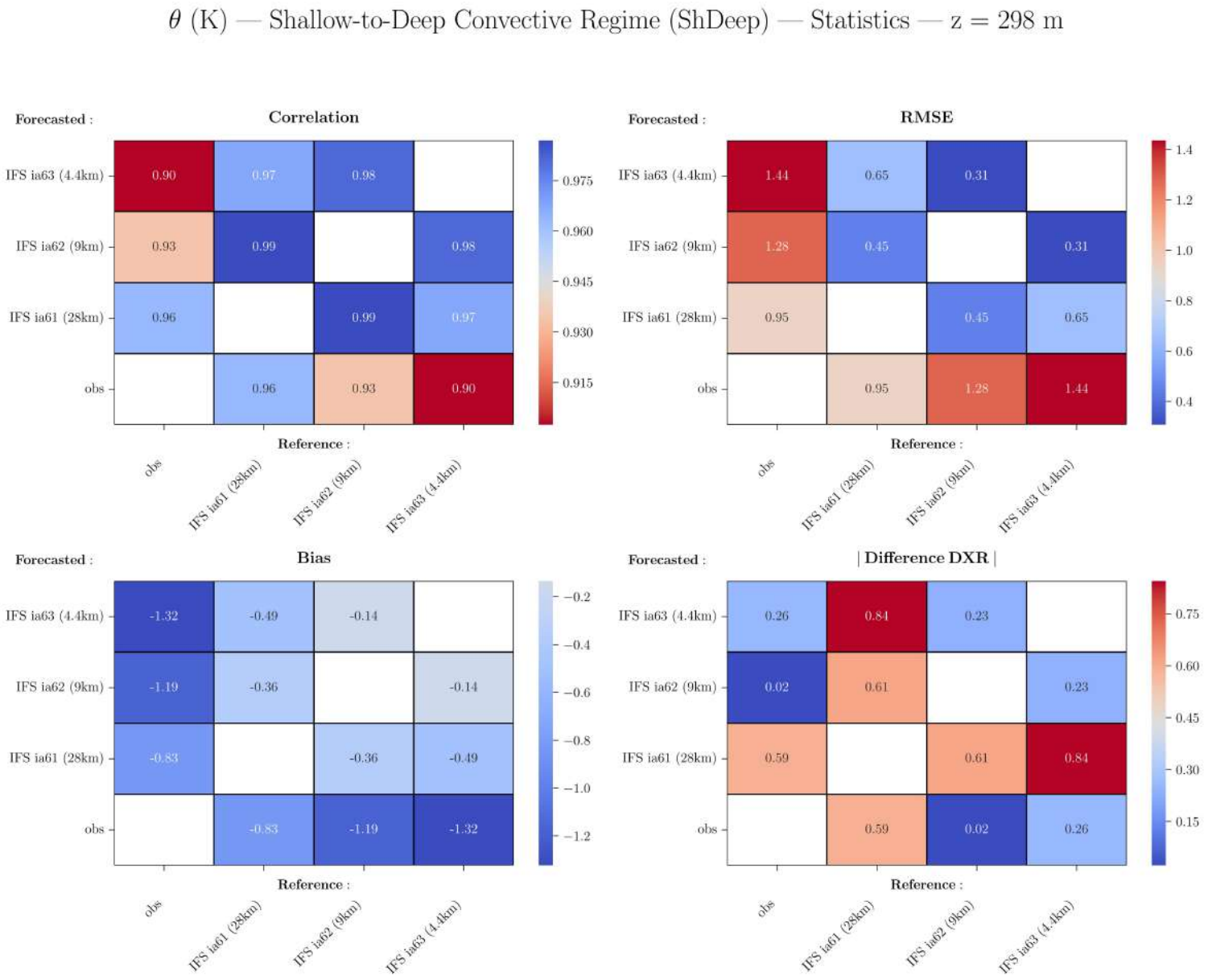



Figure 97: Statistical quantities of the diurnal cycle of observed (at 298 m) and simulated (IFS) $\theta$ for the Shallow-to-Deep Convective (ShDeep) regime.



Funded by the European Union

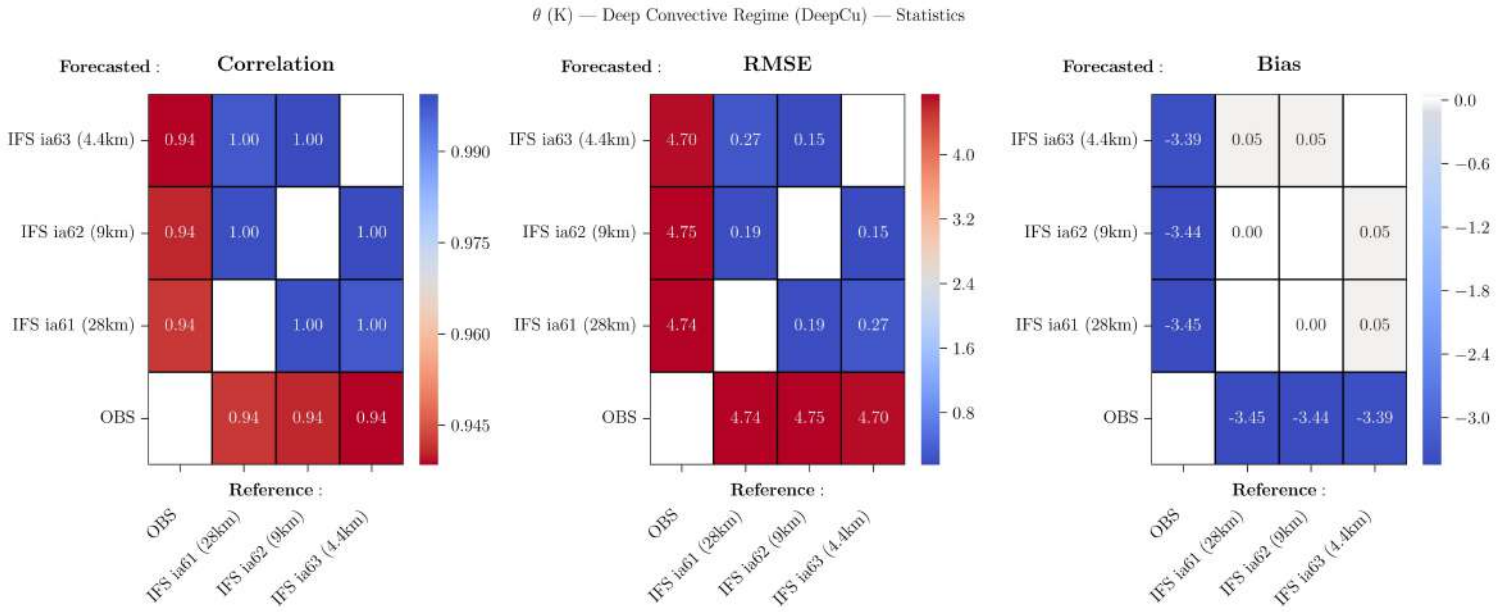



Figure 98: Statistical quantities of the vertical profiles of observed and simulated (IFS) $\theta$ for the Shallow-to-Deep Convective (ShDeep) regime.

Funded by the European Union

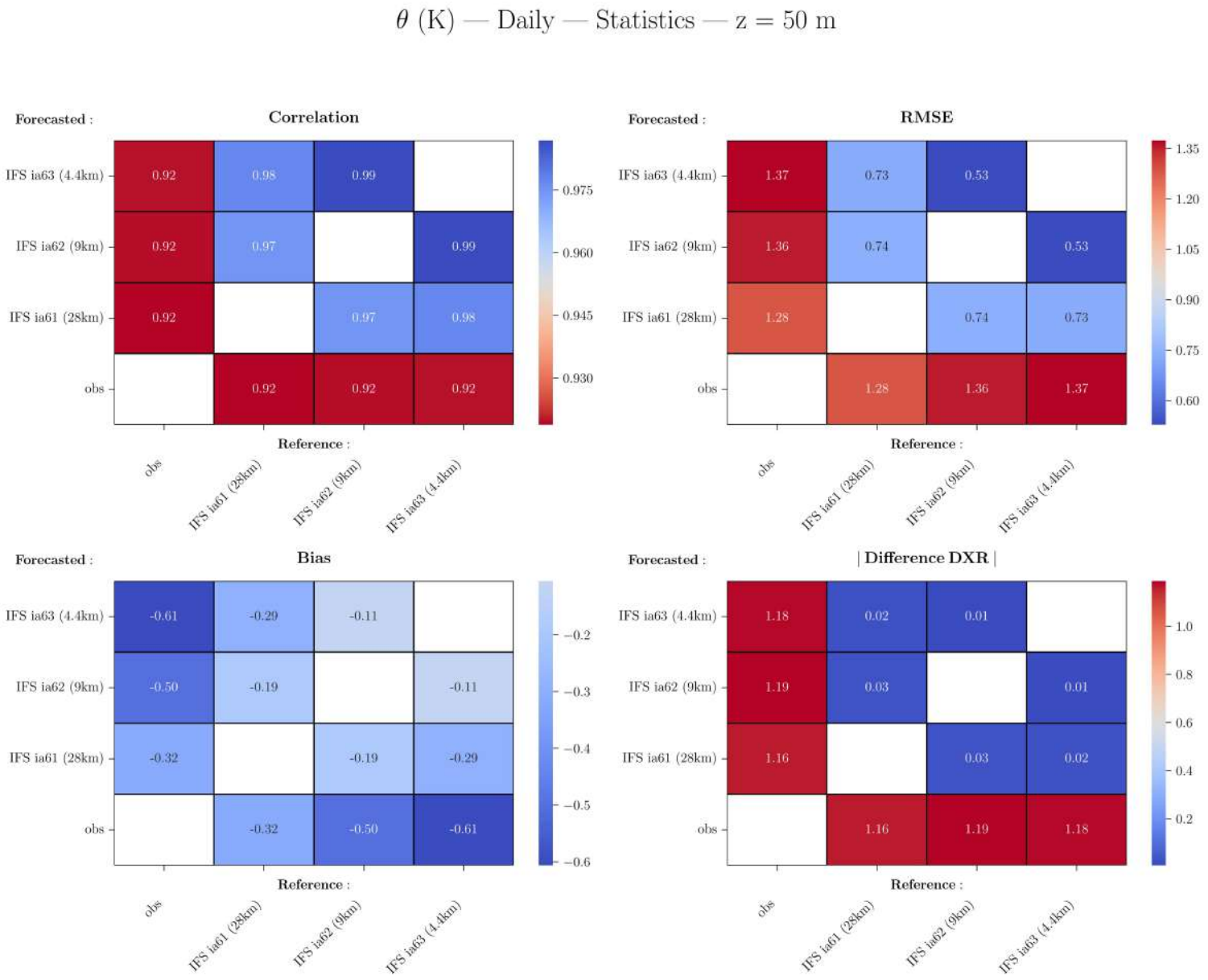



Figure 99: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) $\theta$ for the full CloudRoots-Amazon22 campaign.



Funded by the European Union

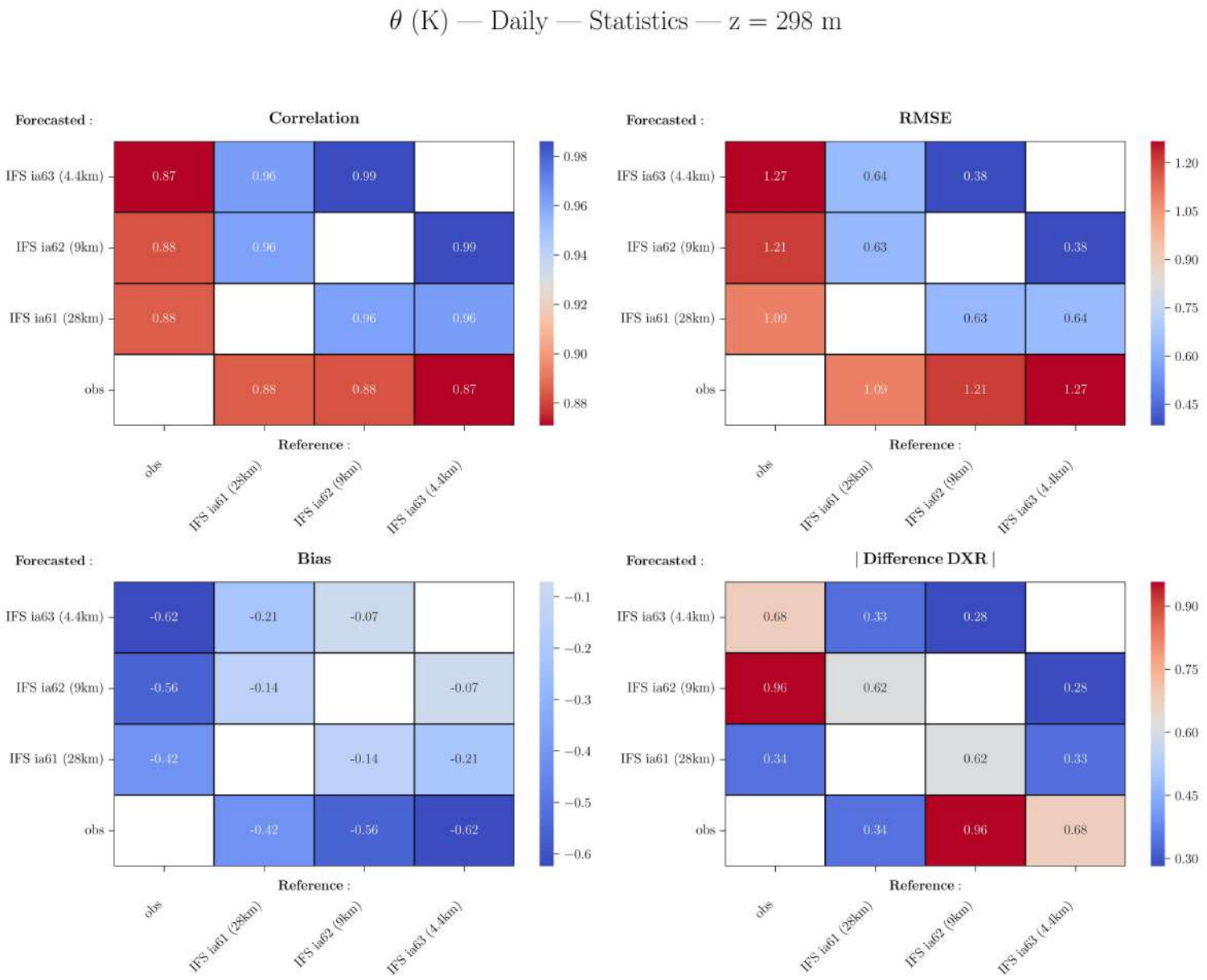



Figure 100: Statistical quantities of the diurnal cycle of observed (at 298 m) and simulated (IFS) $\theta$ for the full CloudRoots-Amazon22 campaign.



Funded by the European Union

## 4.3 $q$

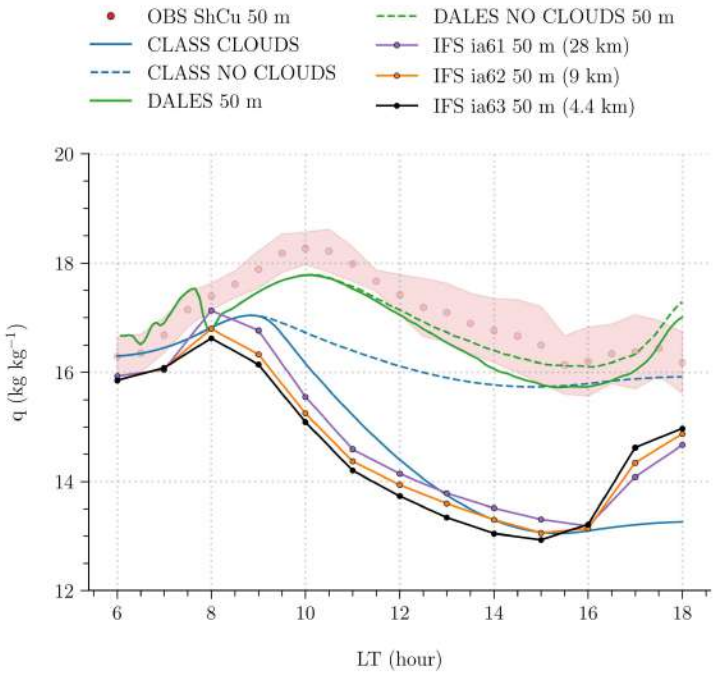



Figure 101: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (CLASS, DALES and IFS) $q$ for the Shallow Convective (ShCu) regime.

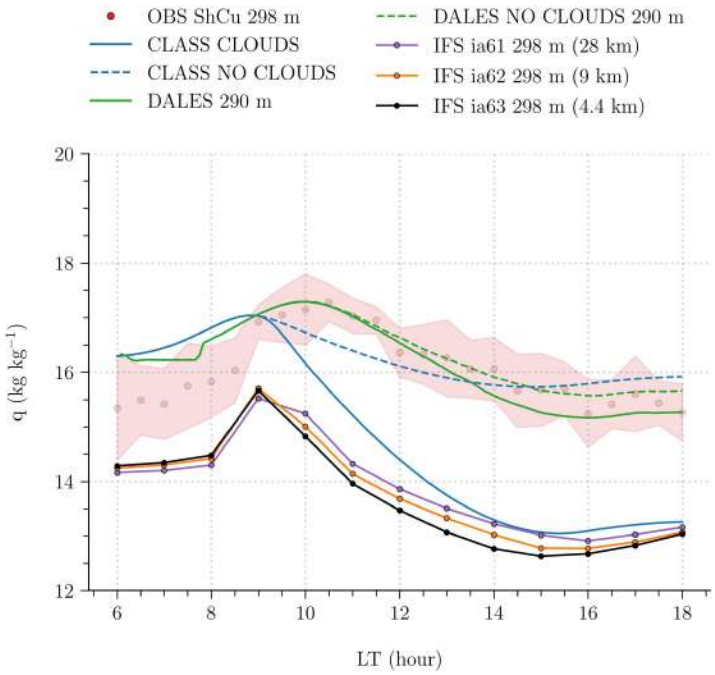



Figure 102: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 298 m) and simulated (CLASS, DALES and IFS) $q$ for the Shallow Convective (ShCu) regime.

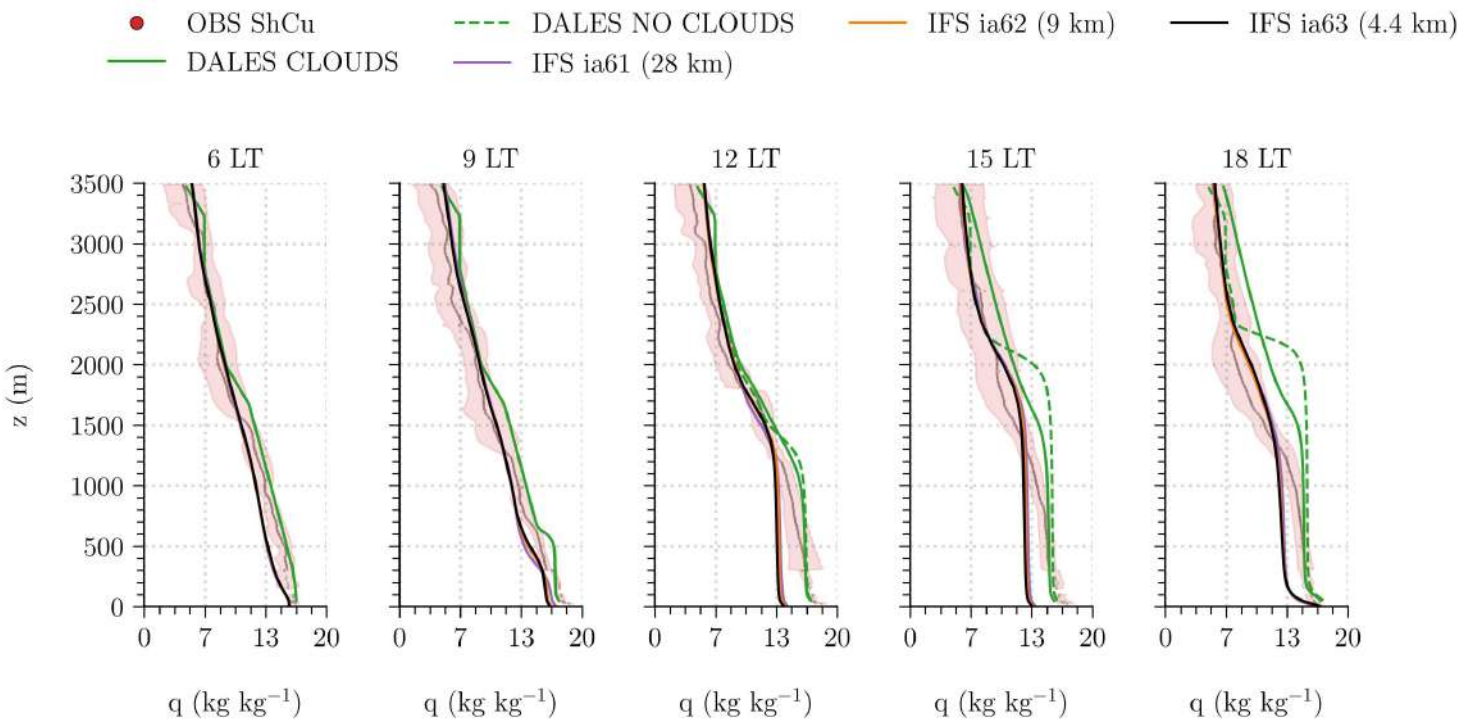



Figure 103: Vertical profiles (markers (mean) and shaded (standard deviation)) and simulated (CLASS, DALES and IFS) $q$ for the Shallow Convective (ShCu) regime.

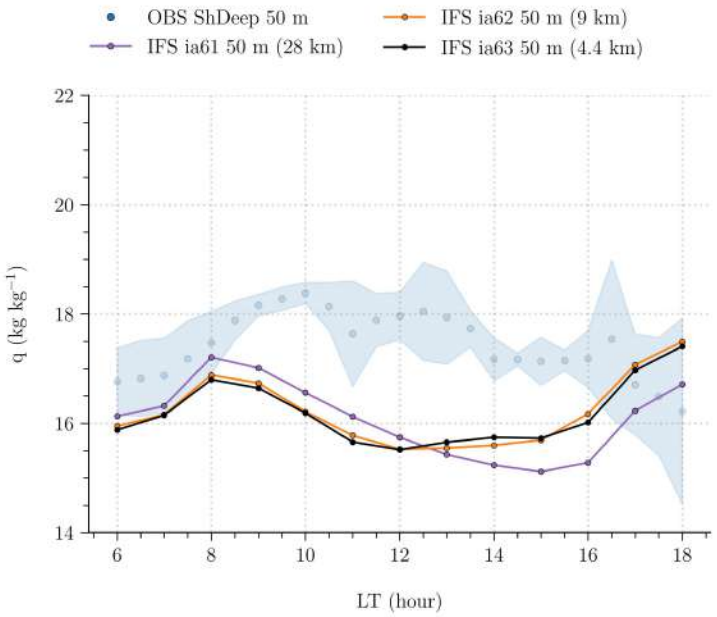



Figure 104: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (IFS) $q$ for the Shallow-to-Deep Convective (ShDeep) regime.

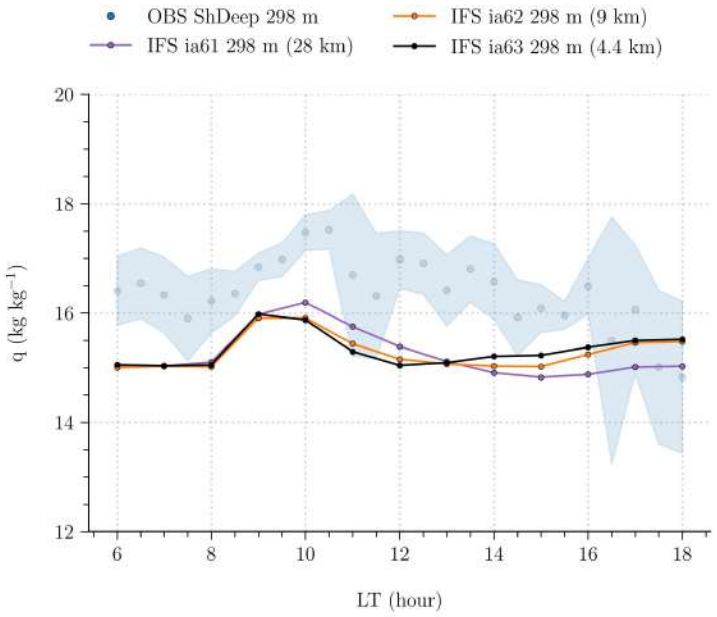



Figure 105: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 298 m) and simulated (IFS) $q$ for the Shallow-to-Deep Convective (ShDeep) regime.

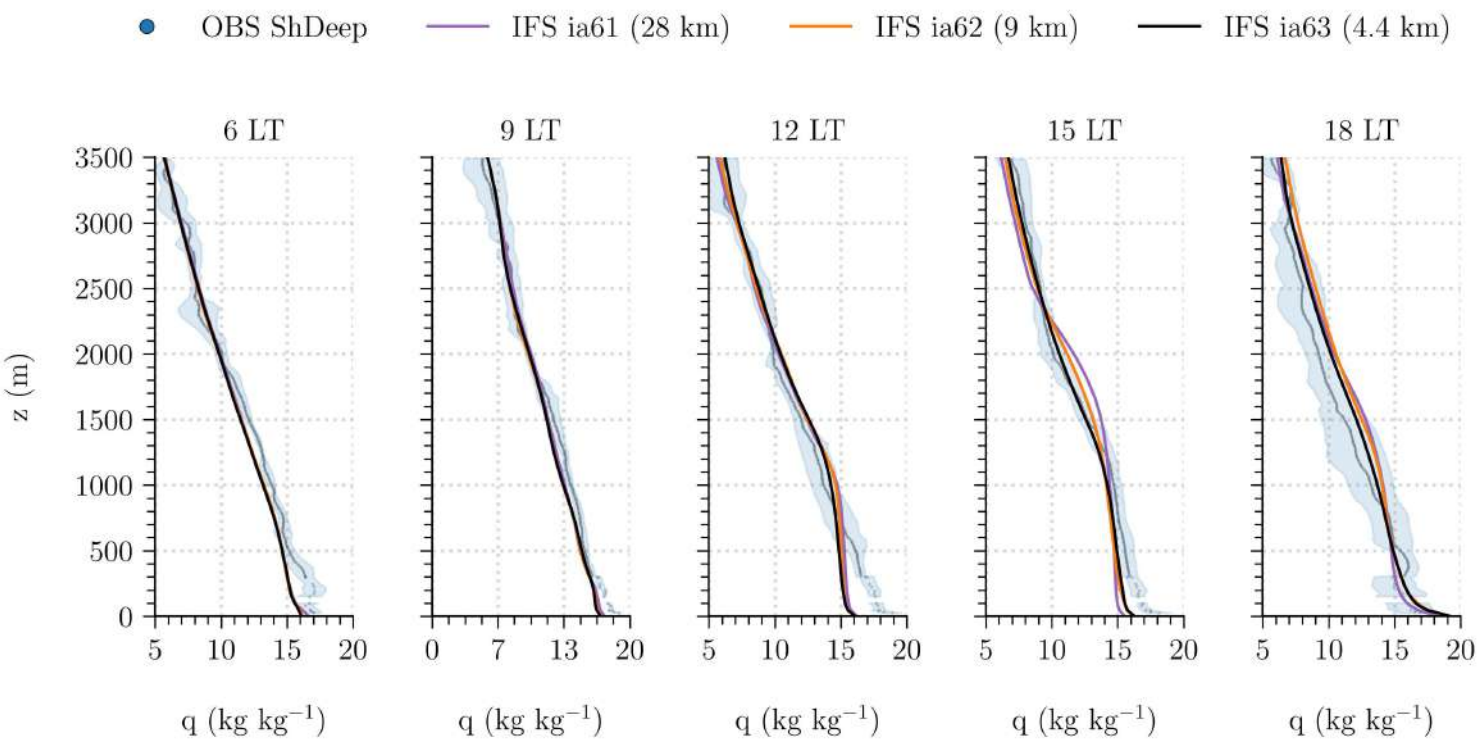



Figure 106: Vertical profiles (markers (mean) and shaded (standard deviation)) and simulated (CLASS, DALES and IFS) $q$ for the Shallow-to-Deep Convective (ShDeep) regime.

Funded by the European Union

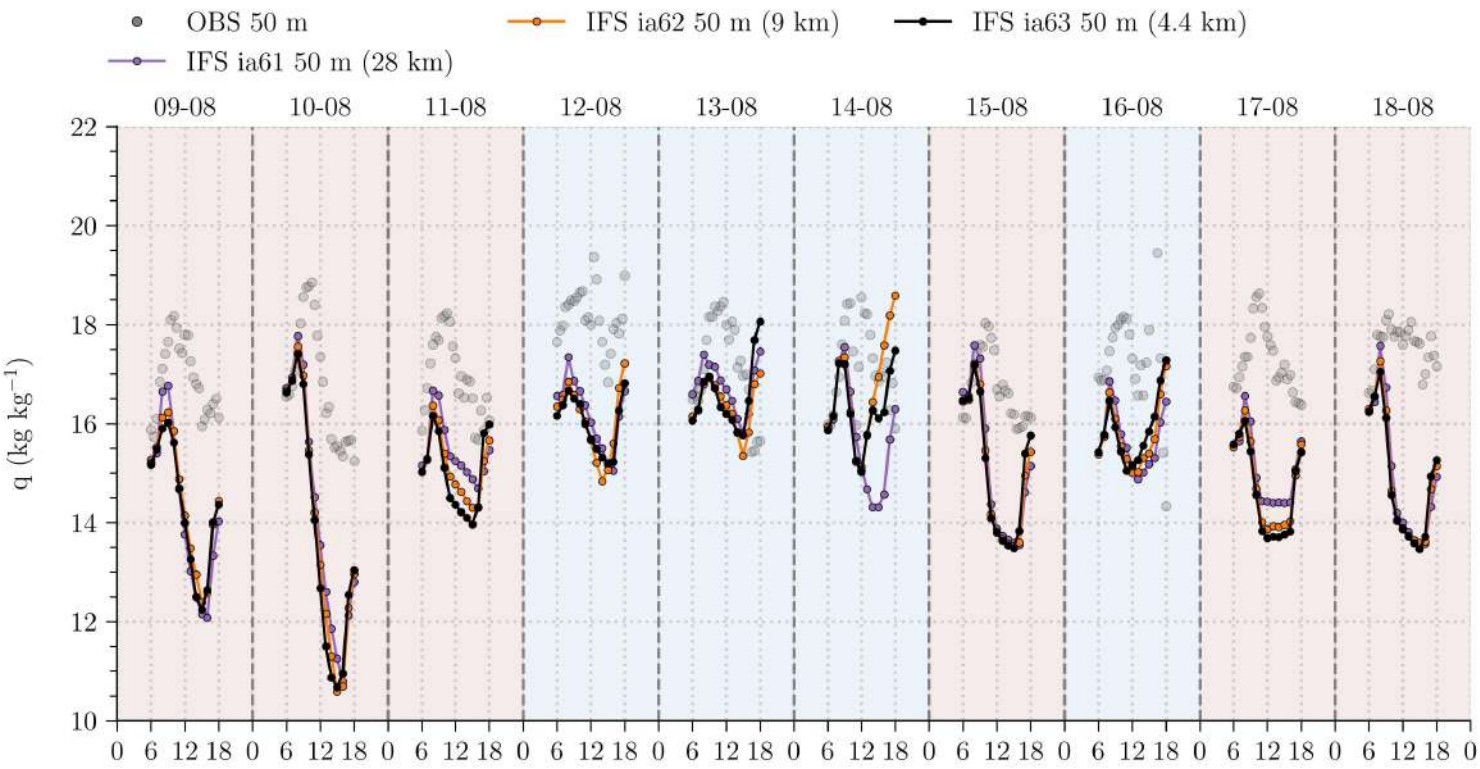



Figure 107: Diurnal cycle of observed (markers, at 50 m) and simulated (IFS) $q$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).

Funded by the European Union

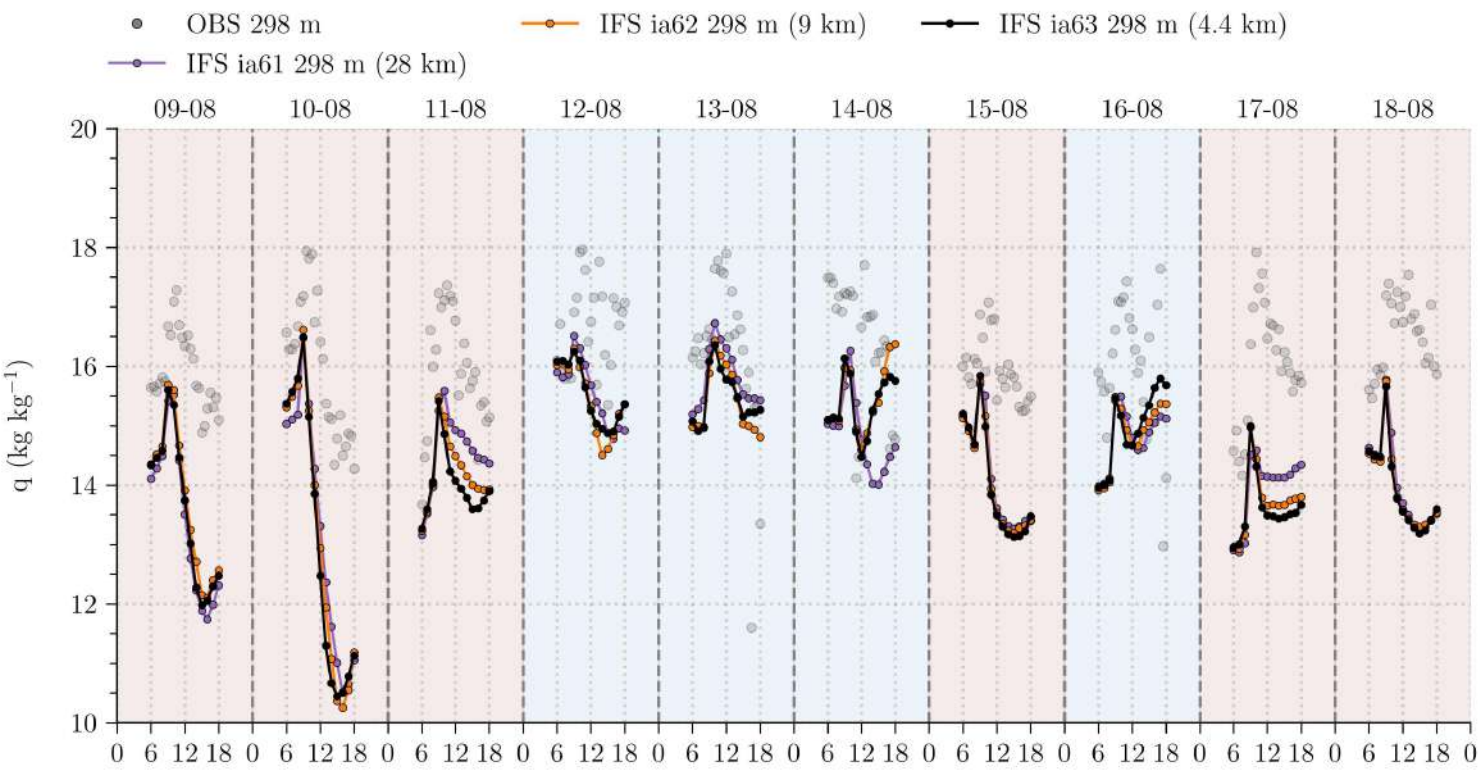



Figure 108: Diurnal cycle of observed (markers, at 298 m) and simulated (IFS) $q$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).



Funded by the European Union

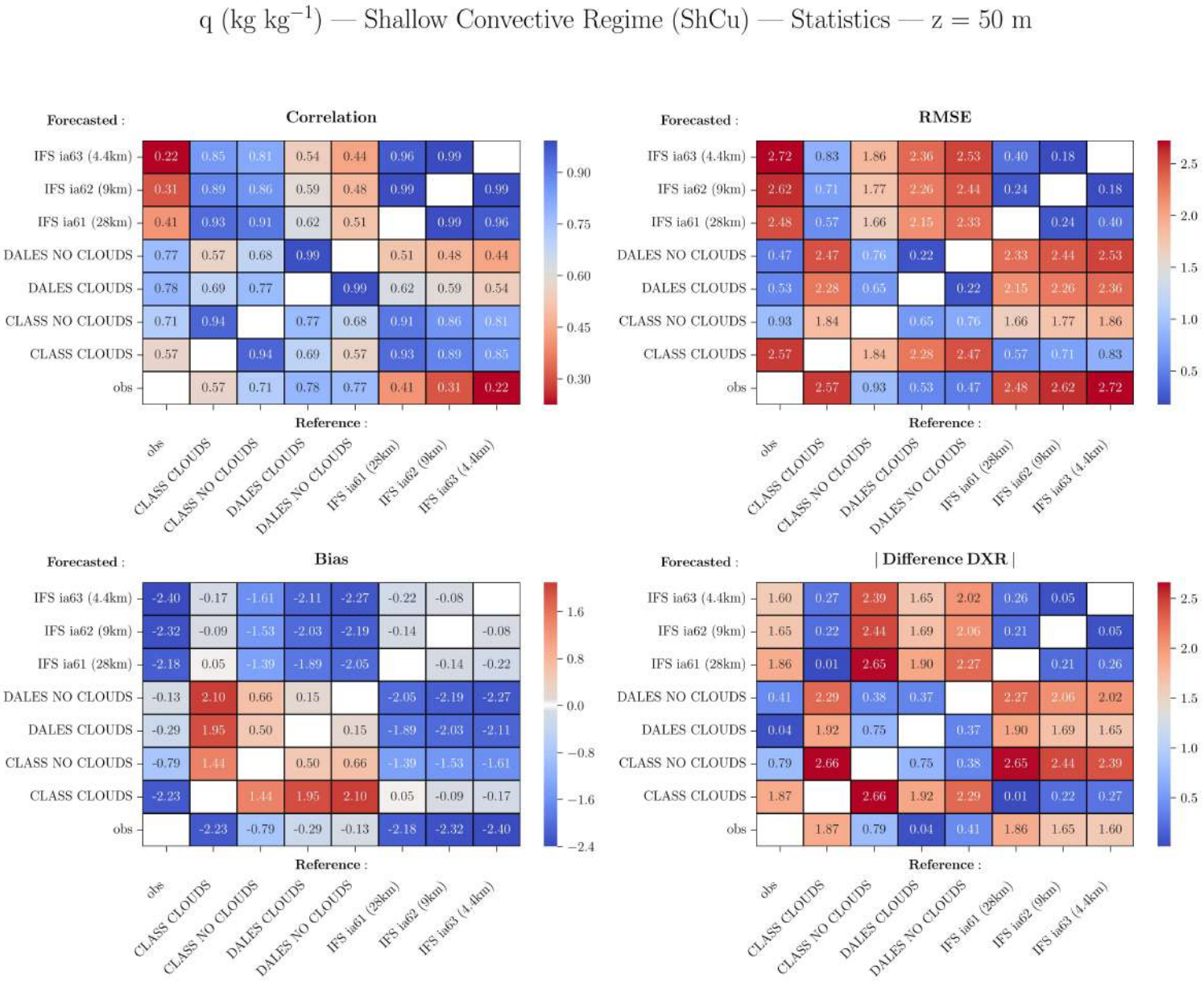



Figure 109: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (CLASS, DALES and IFS) $q$ for the Shallow Convective (ShCu) regime.

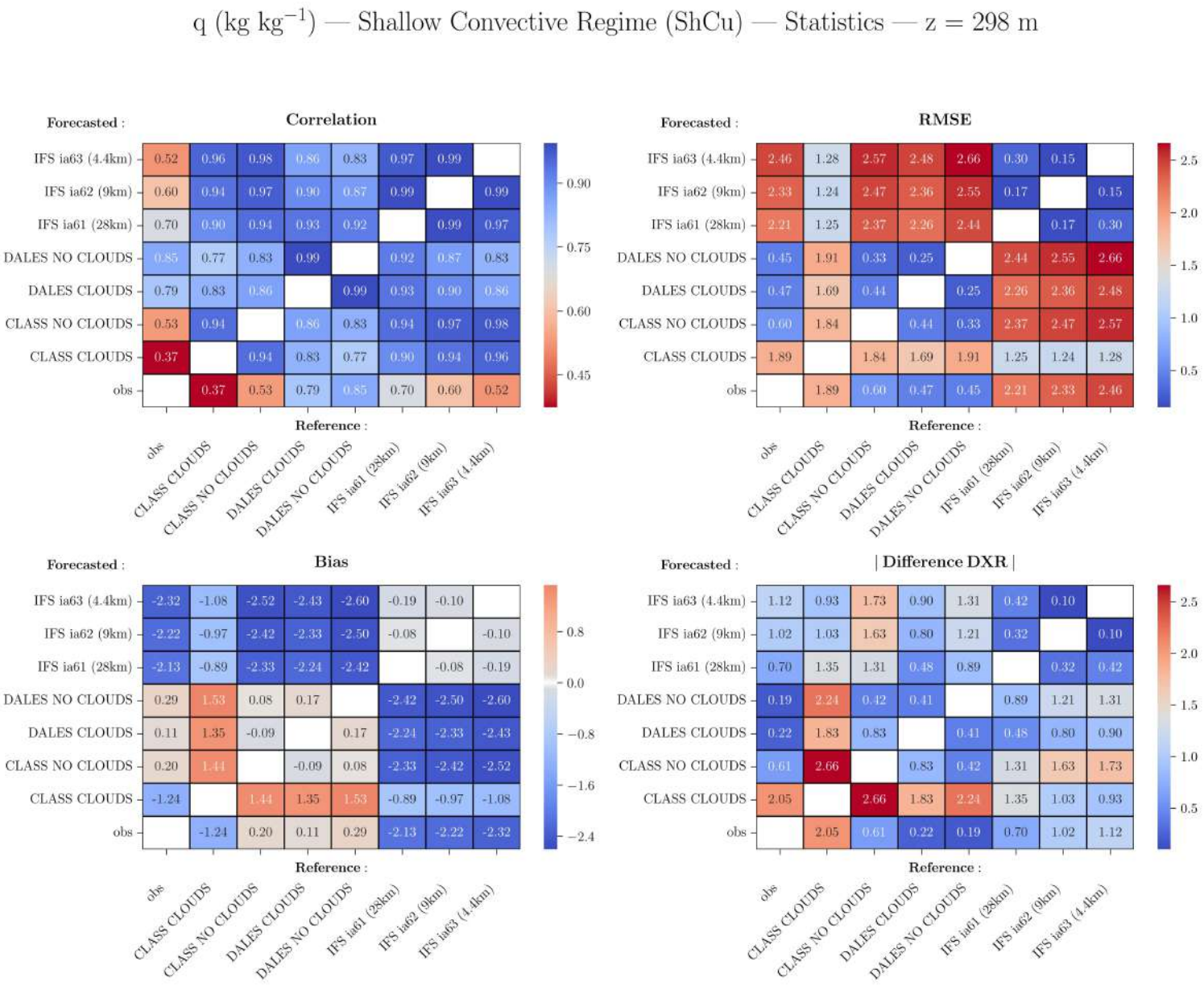



Figure 110: Statistical quantities of the diurnal cycle of observed (at 298 m) and simulated (CLASS, DALES and IFS) $q$ for the Shallow Convective (ShCu) regime.



Funded by the European Union

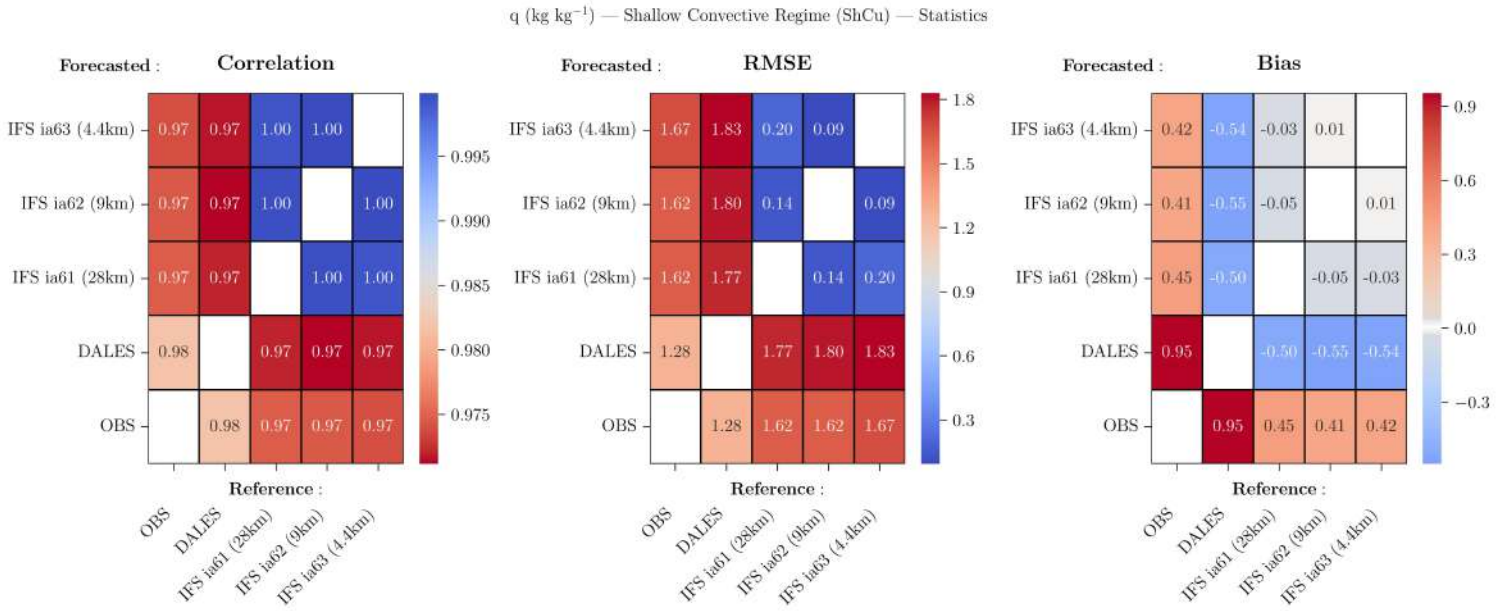



Figure 111: Statistical quantities of the vertical profiles of observed and simulated (CLASS, DALES and IFS) $q$ for the Shallow Convective (ShCu) regime.

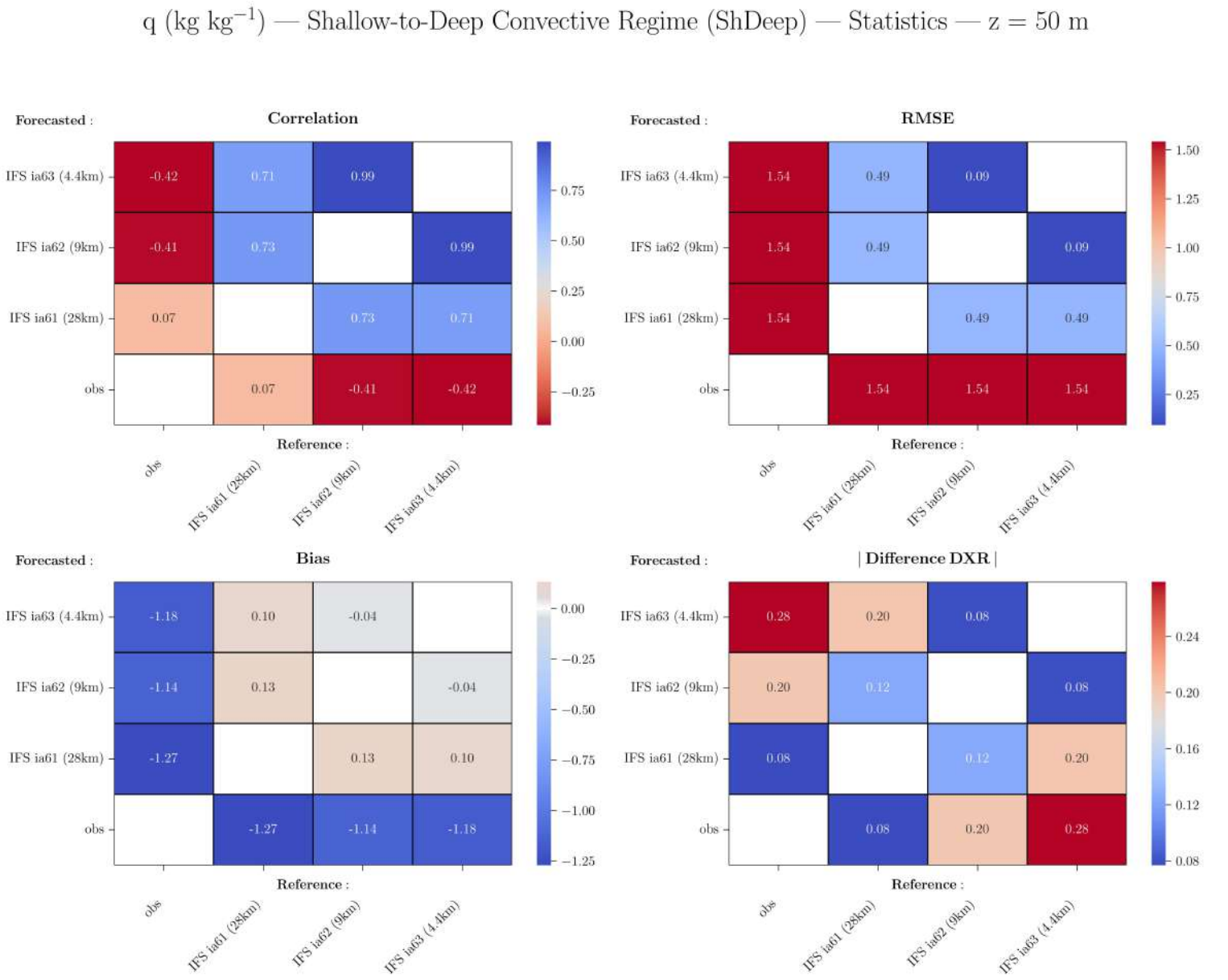



Figure 112: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) $q$ for the Shallow-to-Deep Convective (ShDeep) regime.

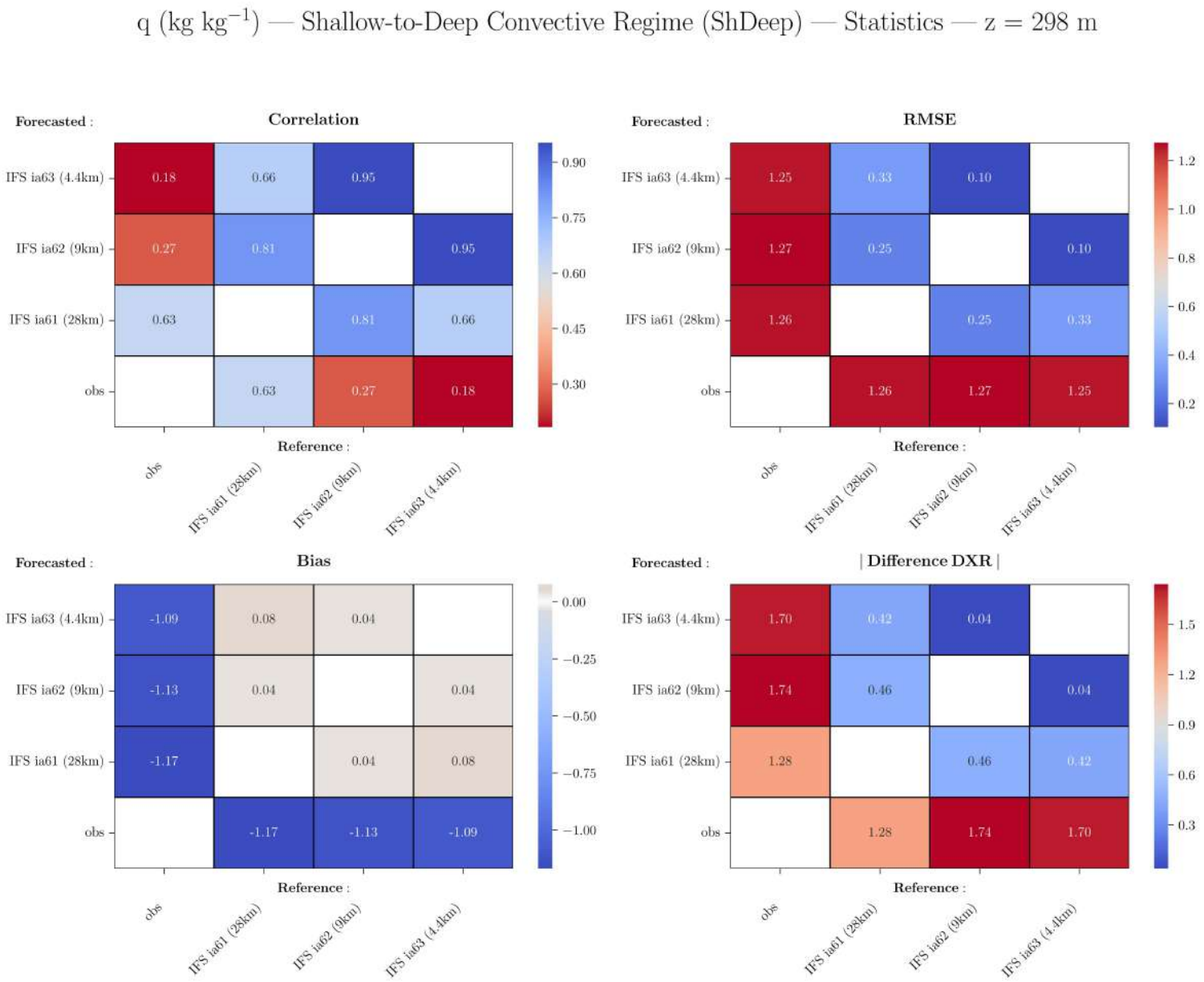



Figure 113: Statistical quantities of the diurnal cycle of observed (at 298 m) and simulated (IFS) $q$ for the Shallow-to-Deep Convective (ShDeep) regime.

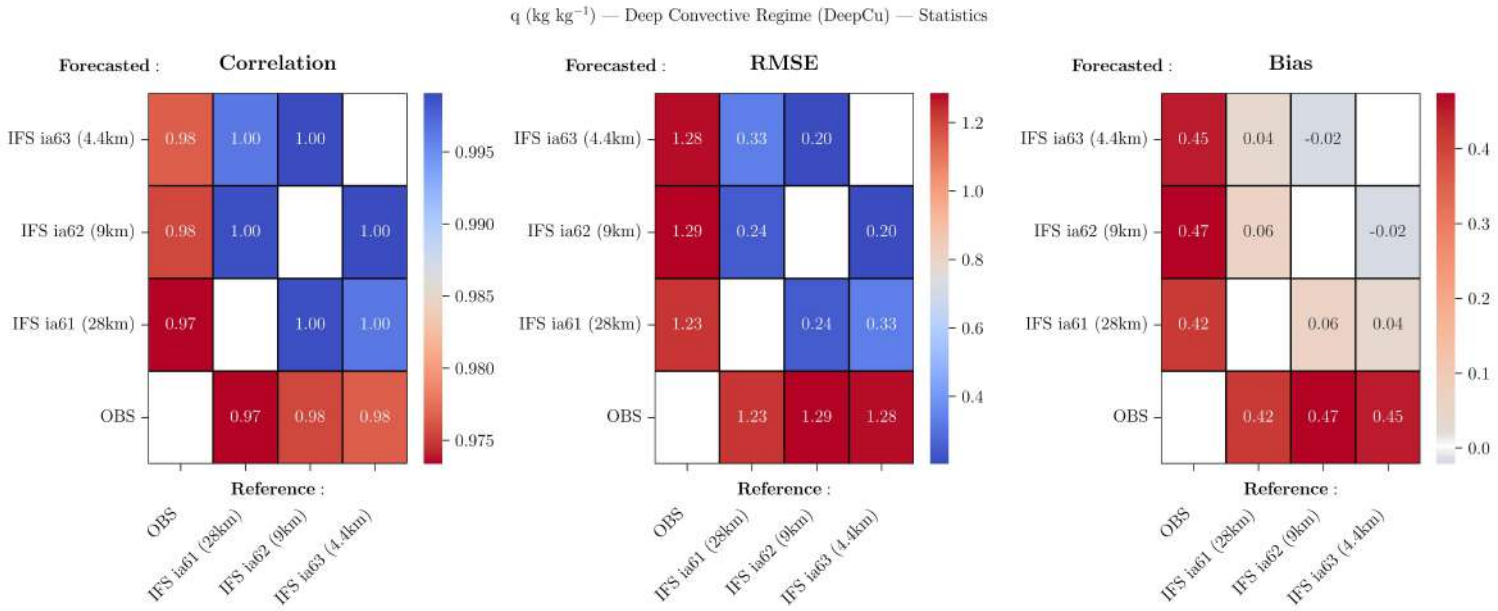



Figure 114: Statistical quantities of the vertical profiles of observed and simulated (IFS) $q$ for the Shallow-to-Deep Convective (ShDeep) regime.

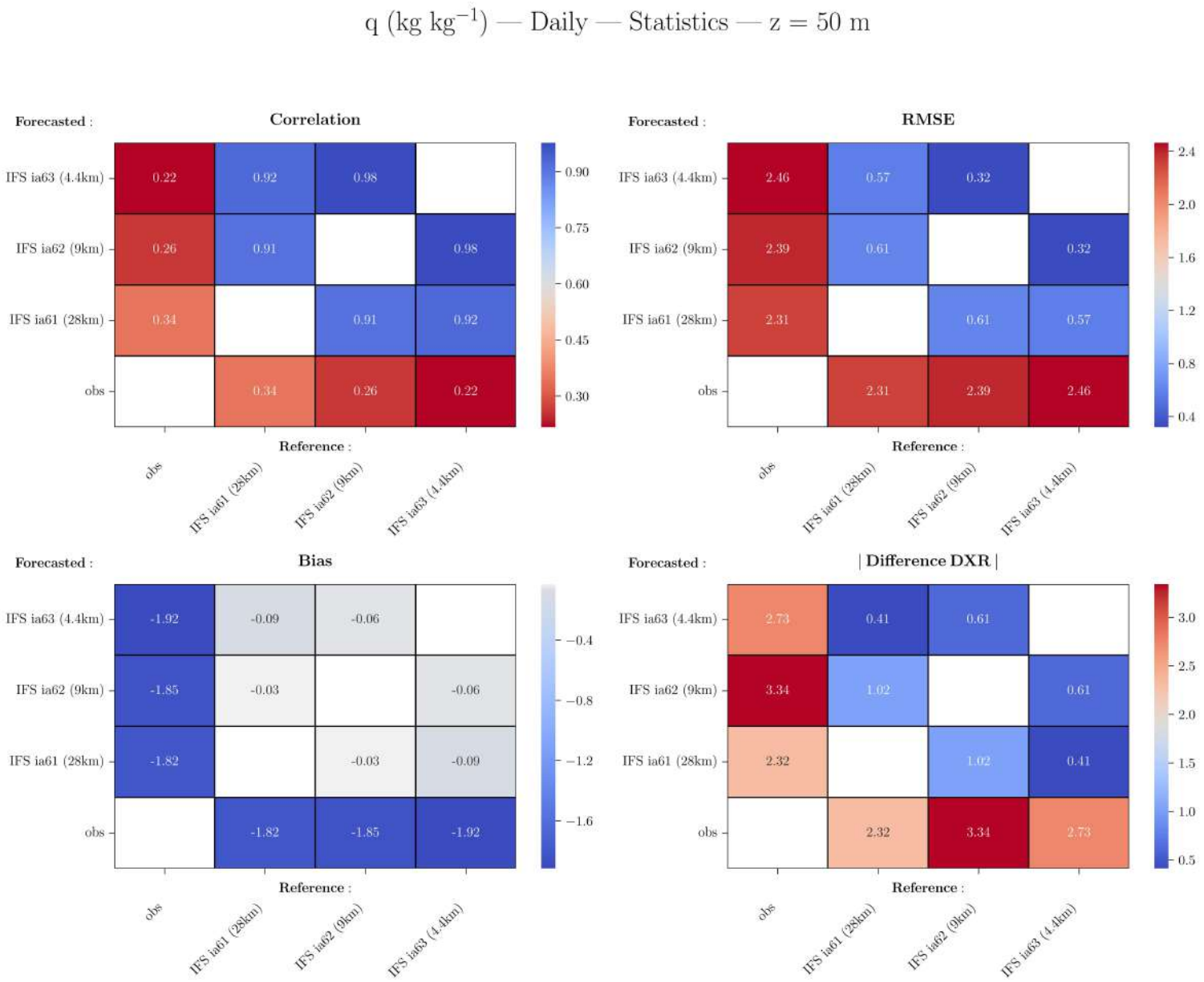



Figure 115: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) $q$ for the full CloudRoots-Amazon22 campaign.

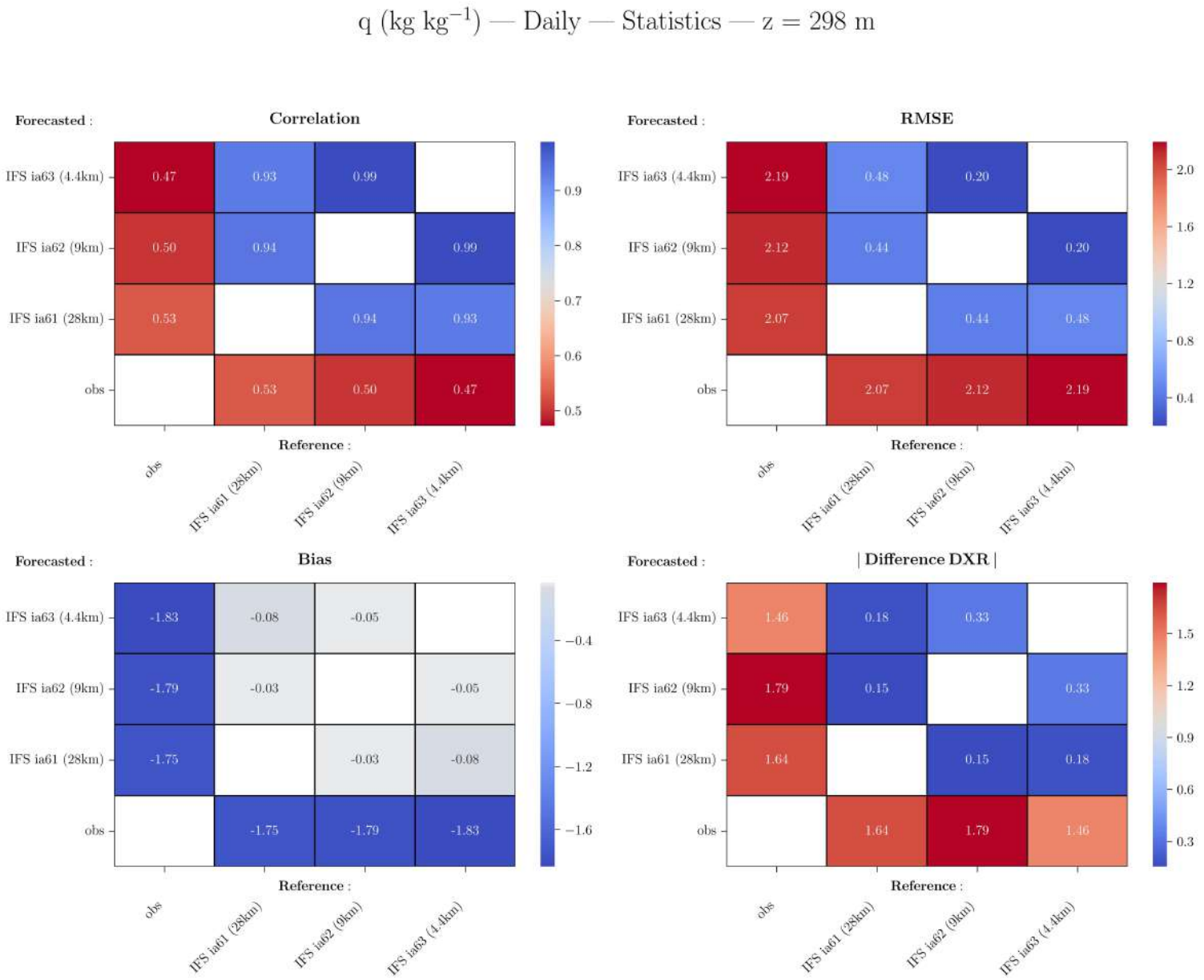



Figure 116: Statistical quantities of the diurnal cycle of observed (at 298 m) and simulated (IFS) $q$ for the full CloudRoots-Amazon22 campaign.

## 4.4 $CO_2$

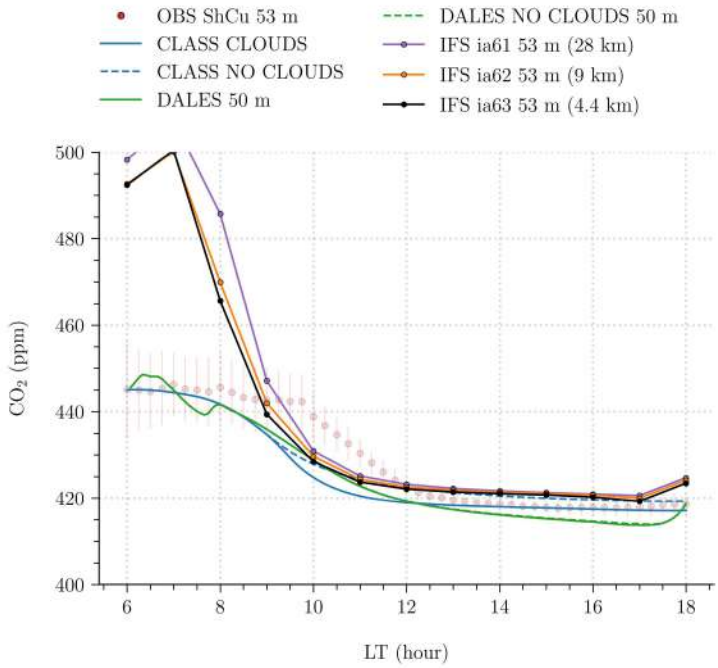



Figure 117: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (CLASS, DALES and IFS) $CO_2$ for the Shallow Convective (ShCu) regime.

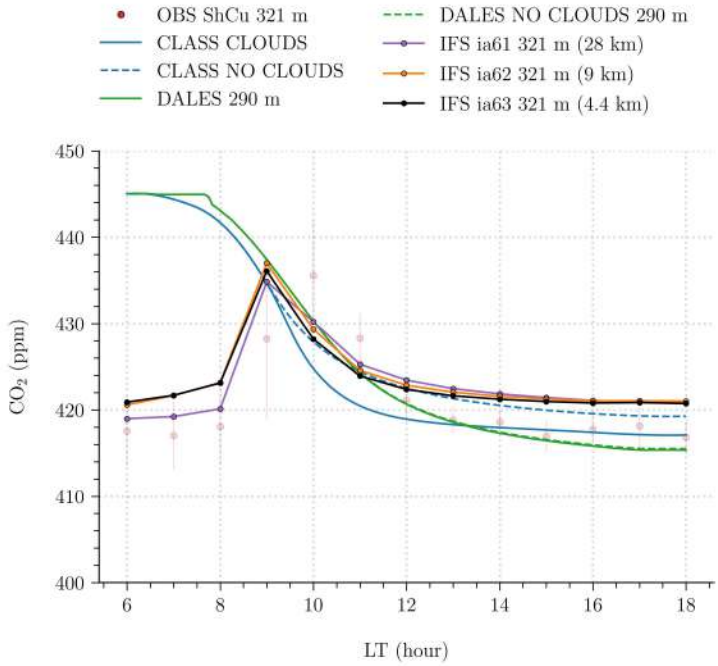



Figure 118: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 298 m) and simulated (CLASS, DALES and IFS) $CO_2$ for the Shallow Convective (ShCu) regime.

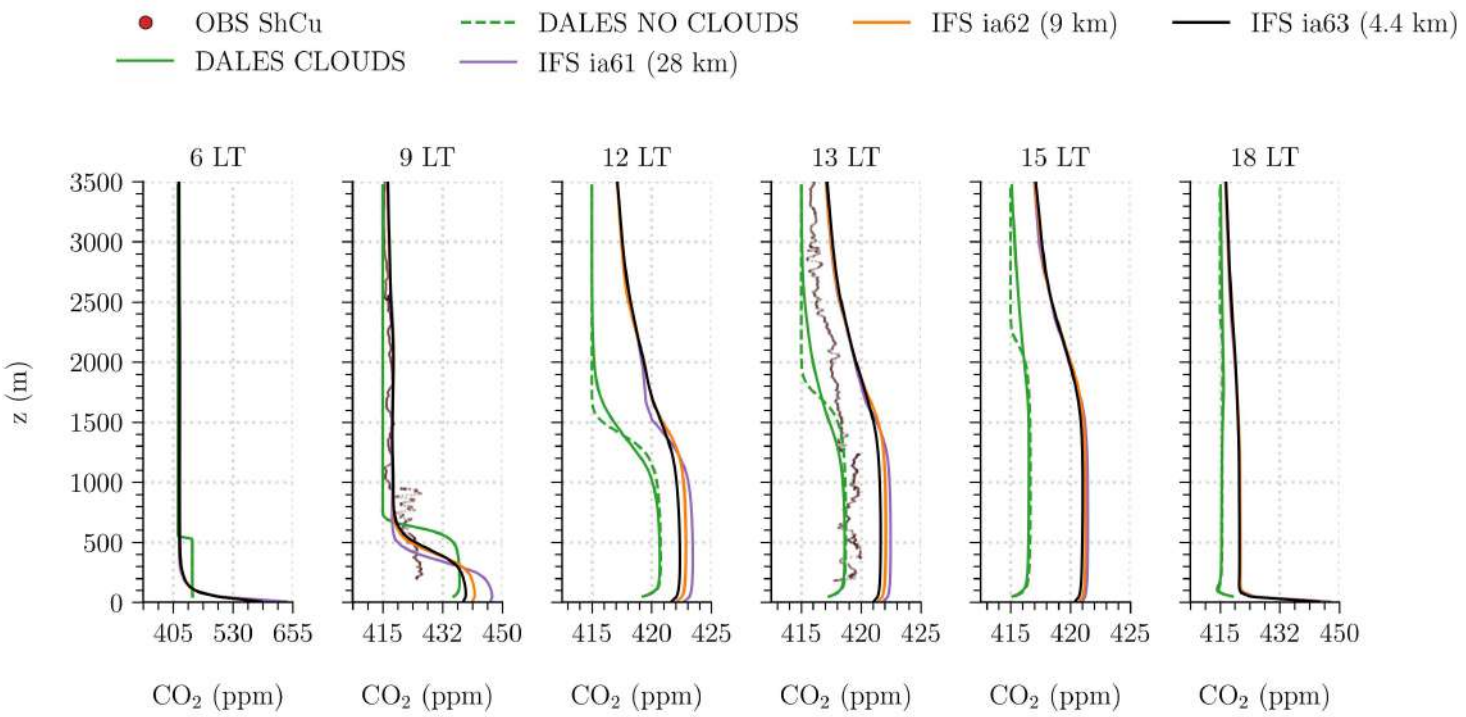



Figure 119: Vertical profiles (markers (mean) and shaded (standard deviation)) and simulated (CLASS, DALES and IFS) $CO_2$ for the Shallow Convective (ShCu) regime.

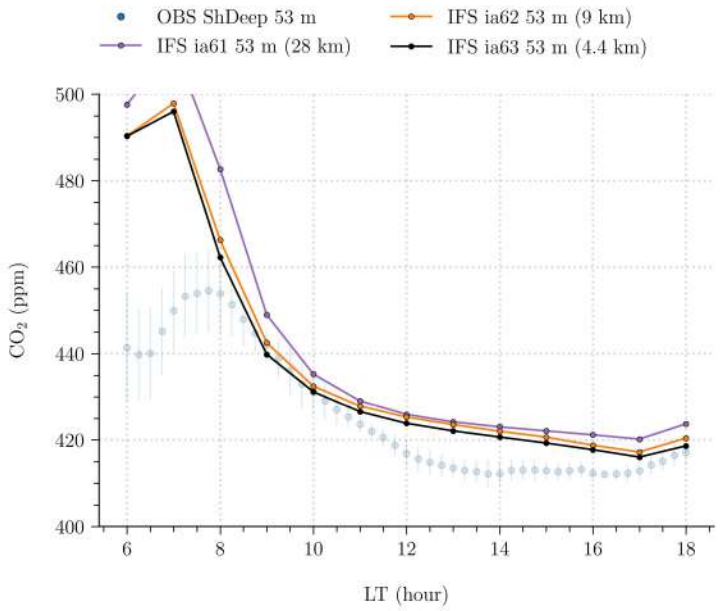



Figure 120: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (IFS) $CO_2$ for the Shallow-to-Deep Convective (ShDeep) regime.

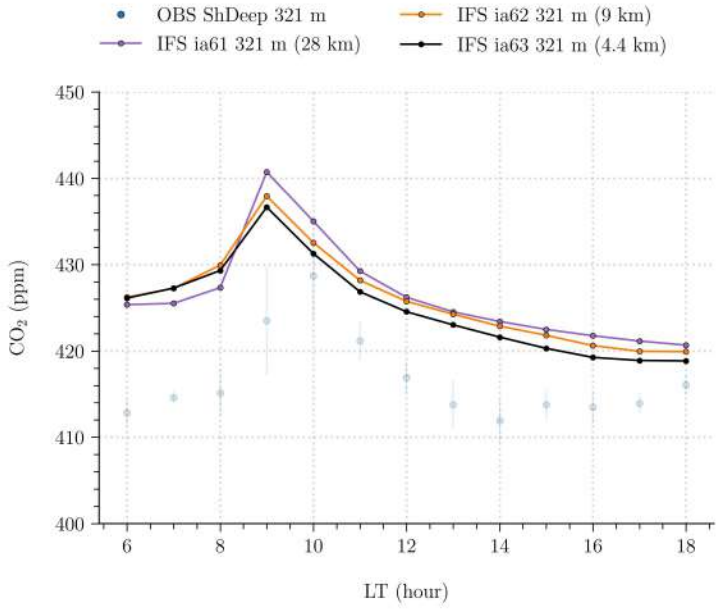



Figure 121: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 298 m) and simulated (IFS) $CO_2$ for the Shallow-to-Deep Convective (ShDeep) regime.

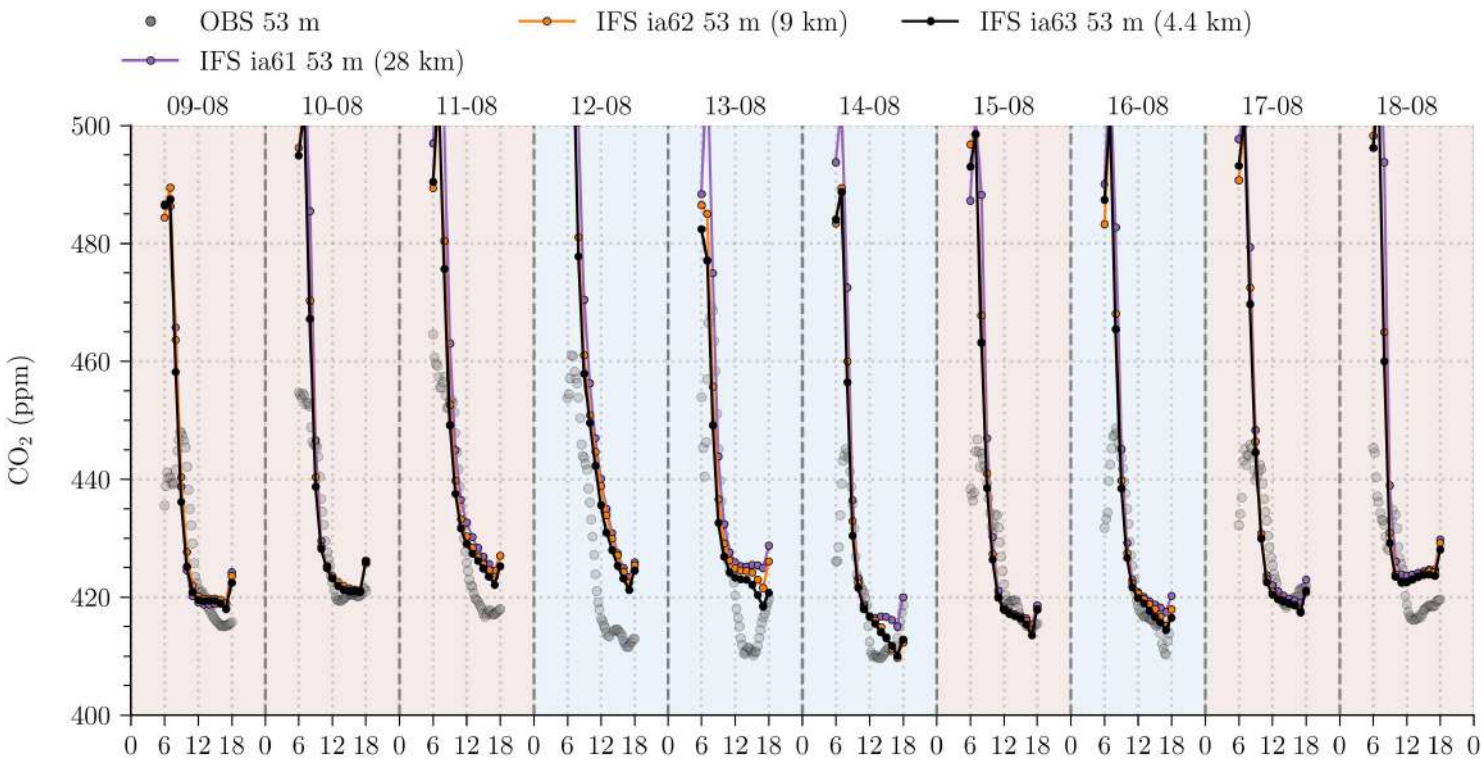



Figure 122: Diurnal cycle of observed (markers, at 50 m) and simulated (IFS) $CO_2$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).



Funded by the European Union

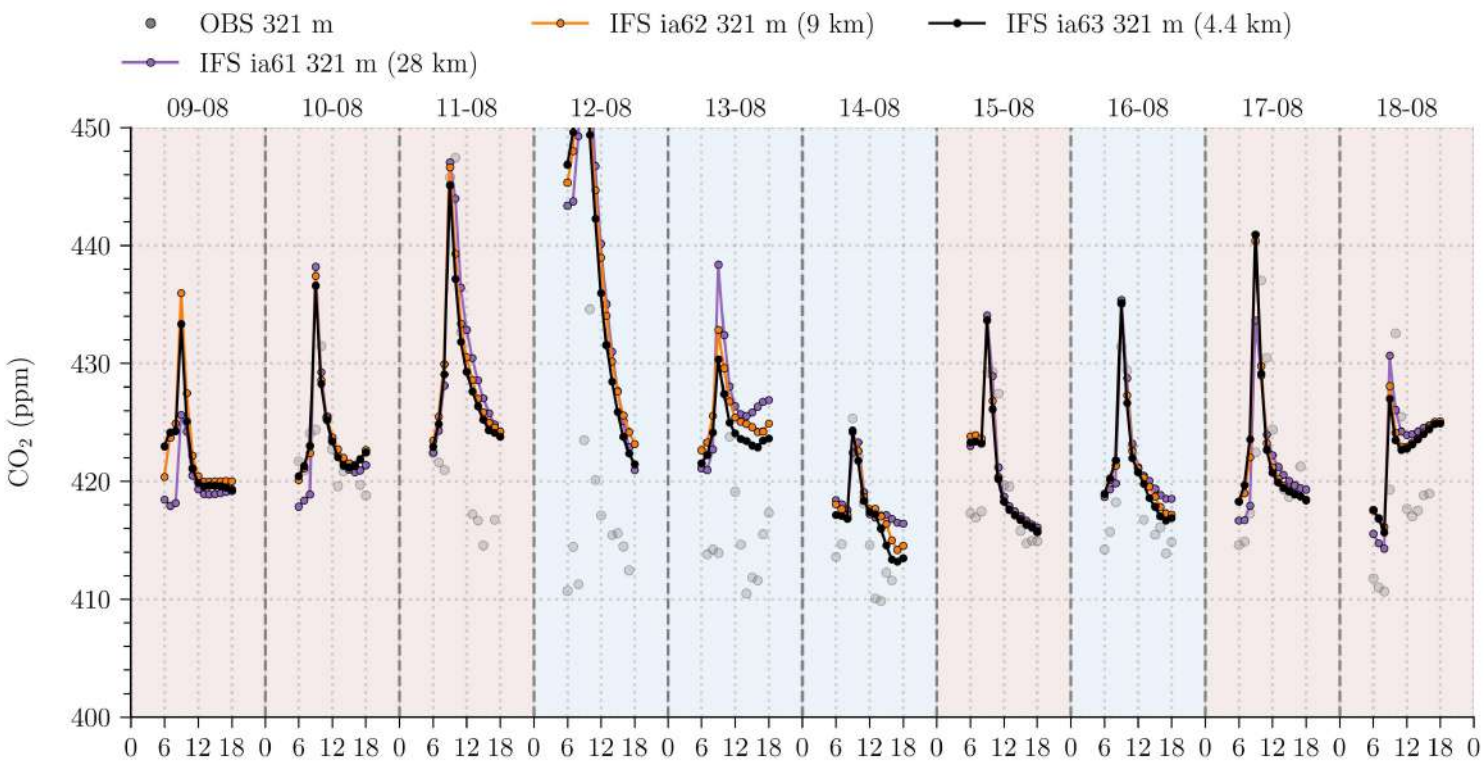



Figure 123: Diurnal cycle of observed (markers, at 298 m) and simulated (IFS) $CO_2$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).



Funded by the European Union

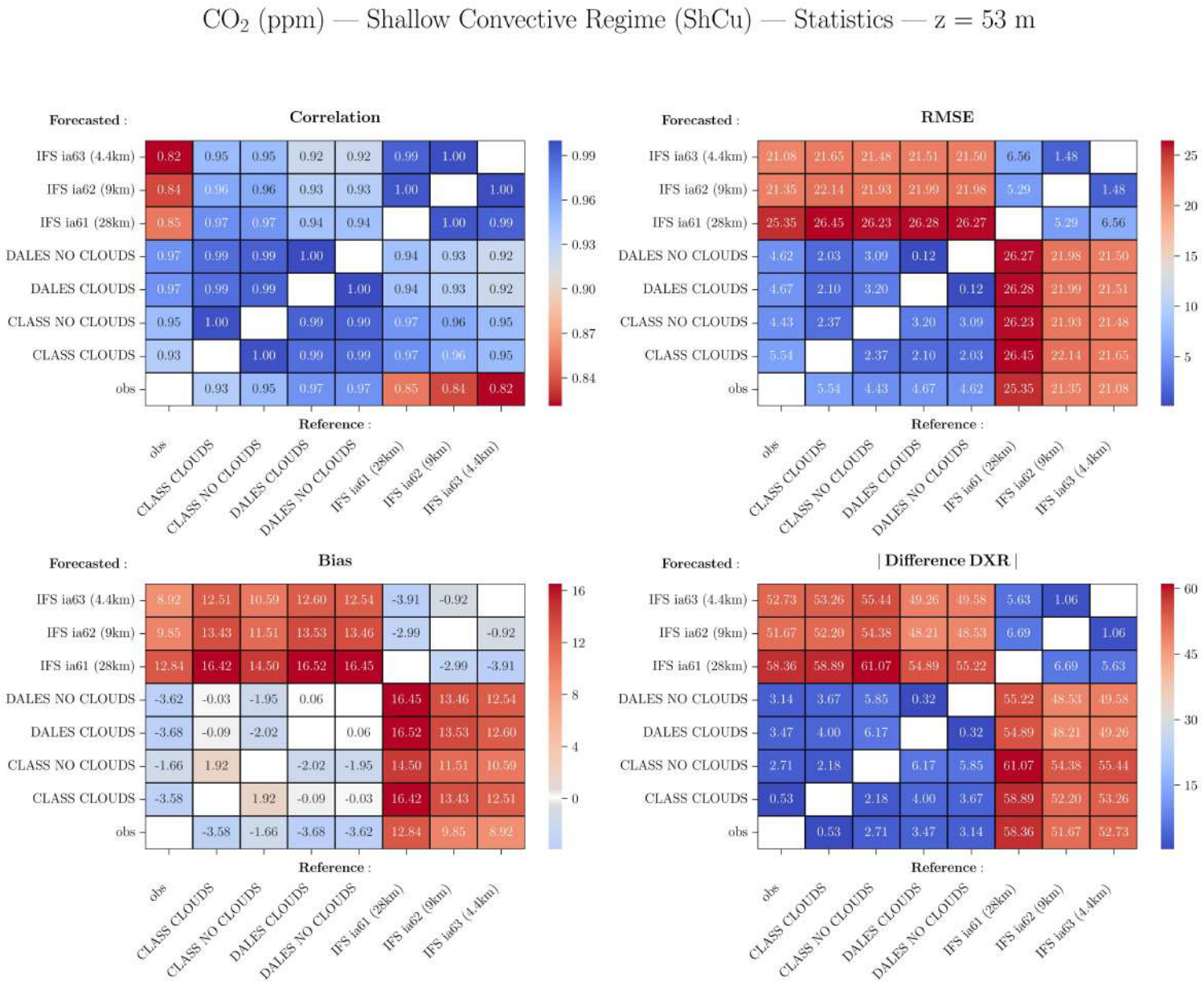



Figure 124: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (CLASS, DALES and IFS) $CO_2$ for the Shallow Convective (ShCu) regime.

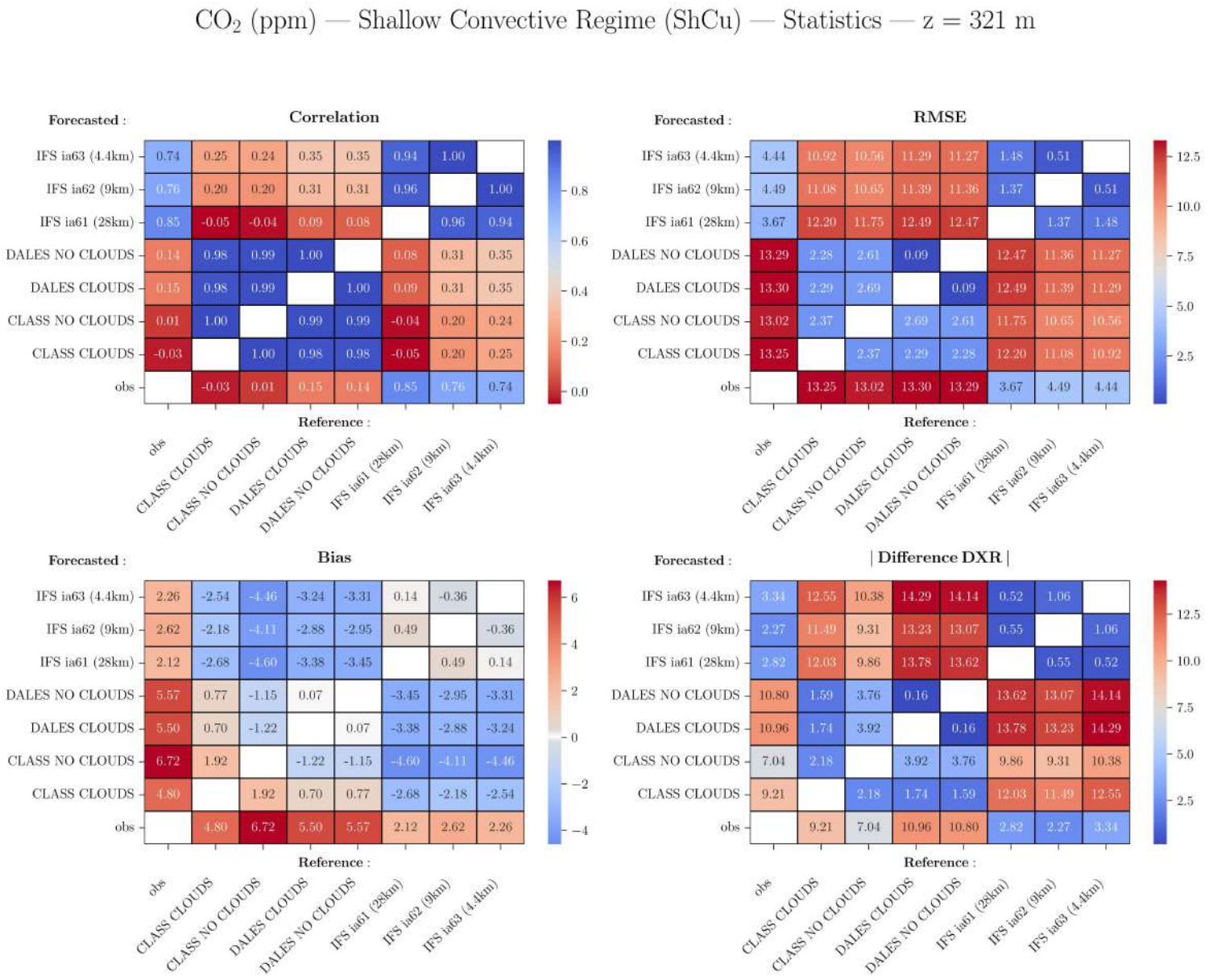



Figure 125: Statistical quantities of the diurnal cycle of observed (at 298 m) and simulated (CLASS, DALES and IFS) $CO_2$ for the Shallow Convective (ShCu) regime.

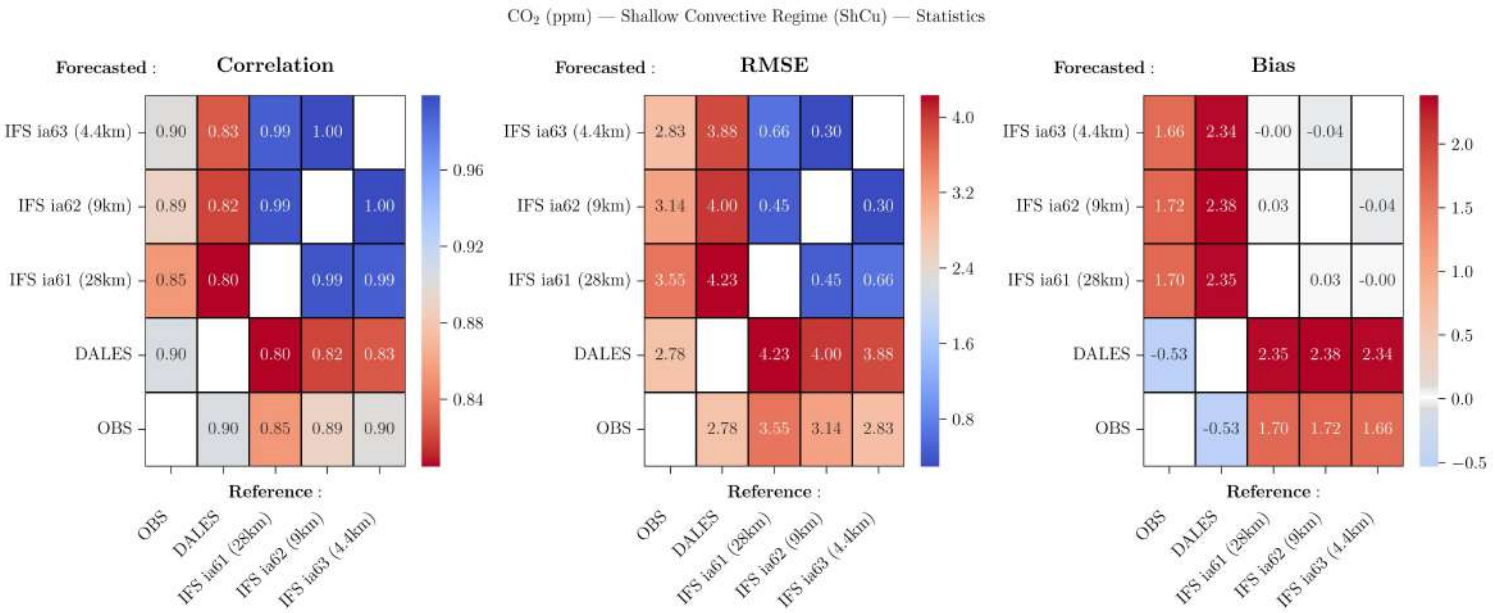



Figure 126: Statistical quantities of the vertical profiles of observed and simulated (CLASS, DALES and IFS) $CO_2$ for the Shallow Convective (ShCu) regime.

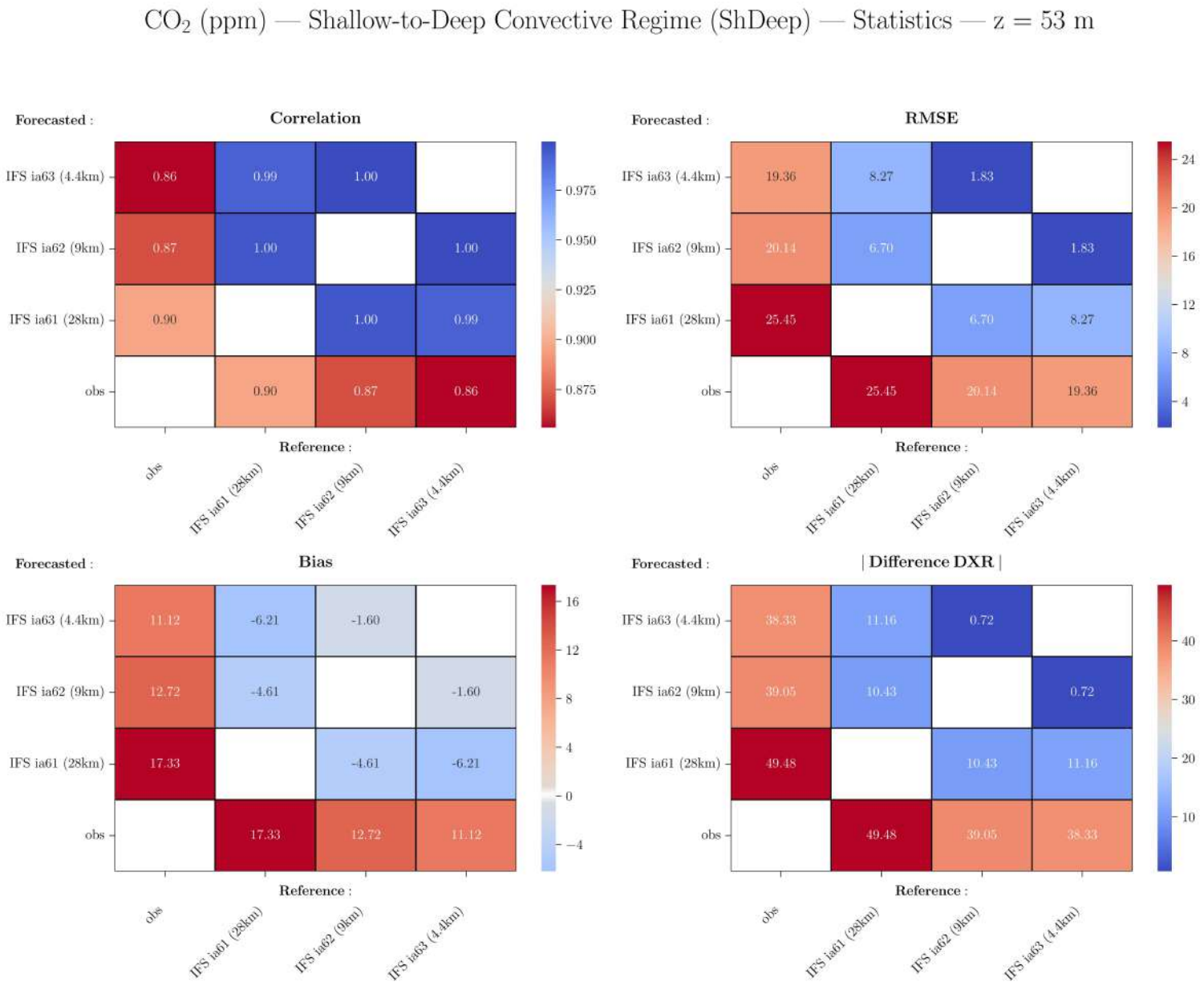



Figure 127: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) $CO_2$ for the Shallow-to-Deep Convective (ShDeep) regime.

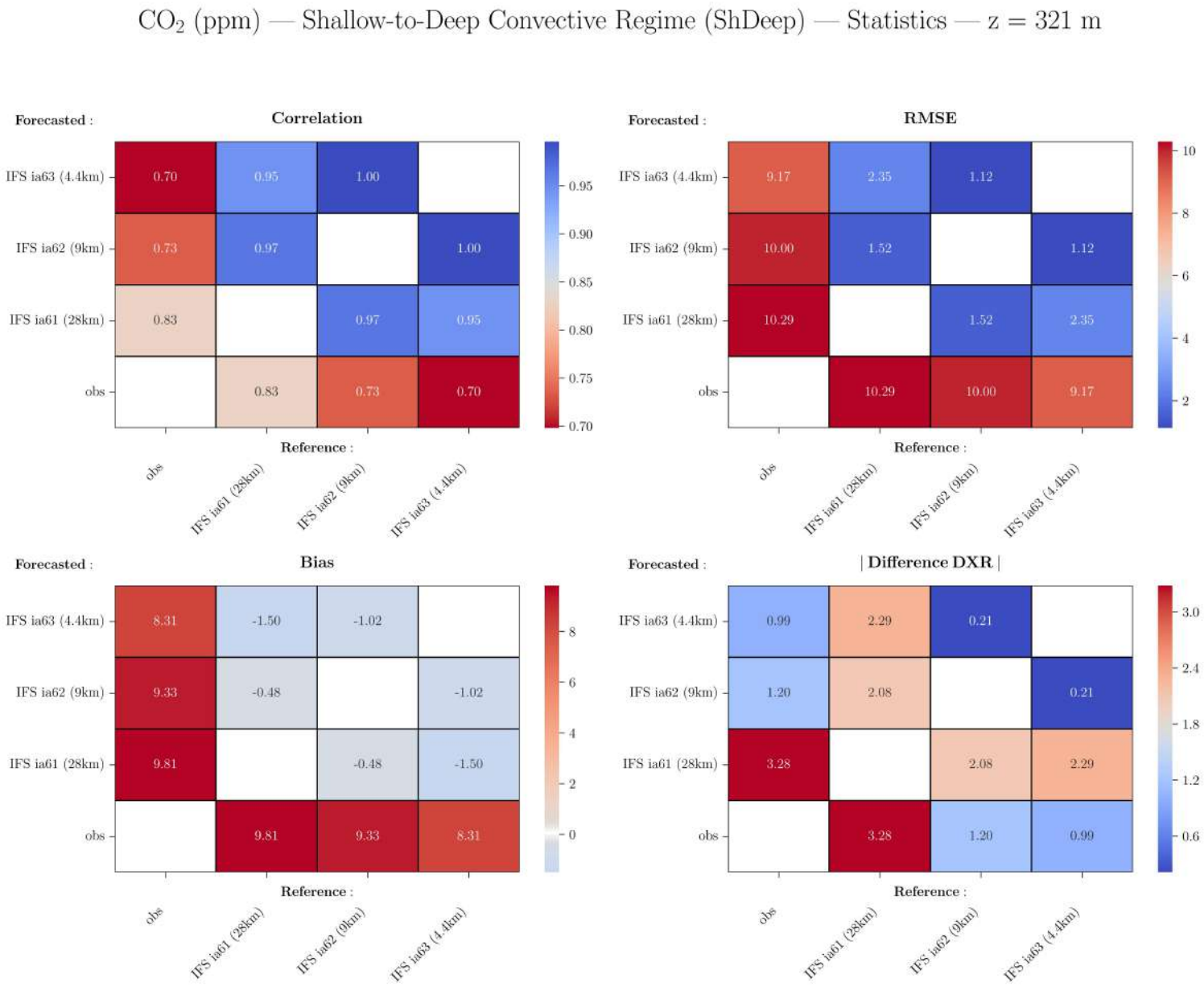



Figure 128: Statistical quantities of the diurnal cycle of observed (at 298 m) and simulated (IFS) $CO_2$ for the Shallow-to-Deep Convective (ShDeep) regime.

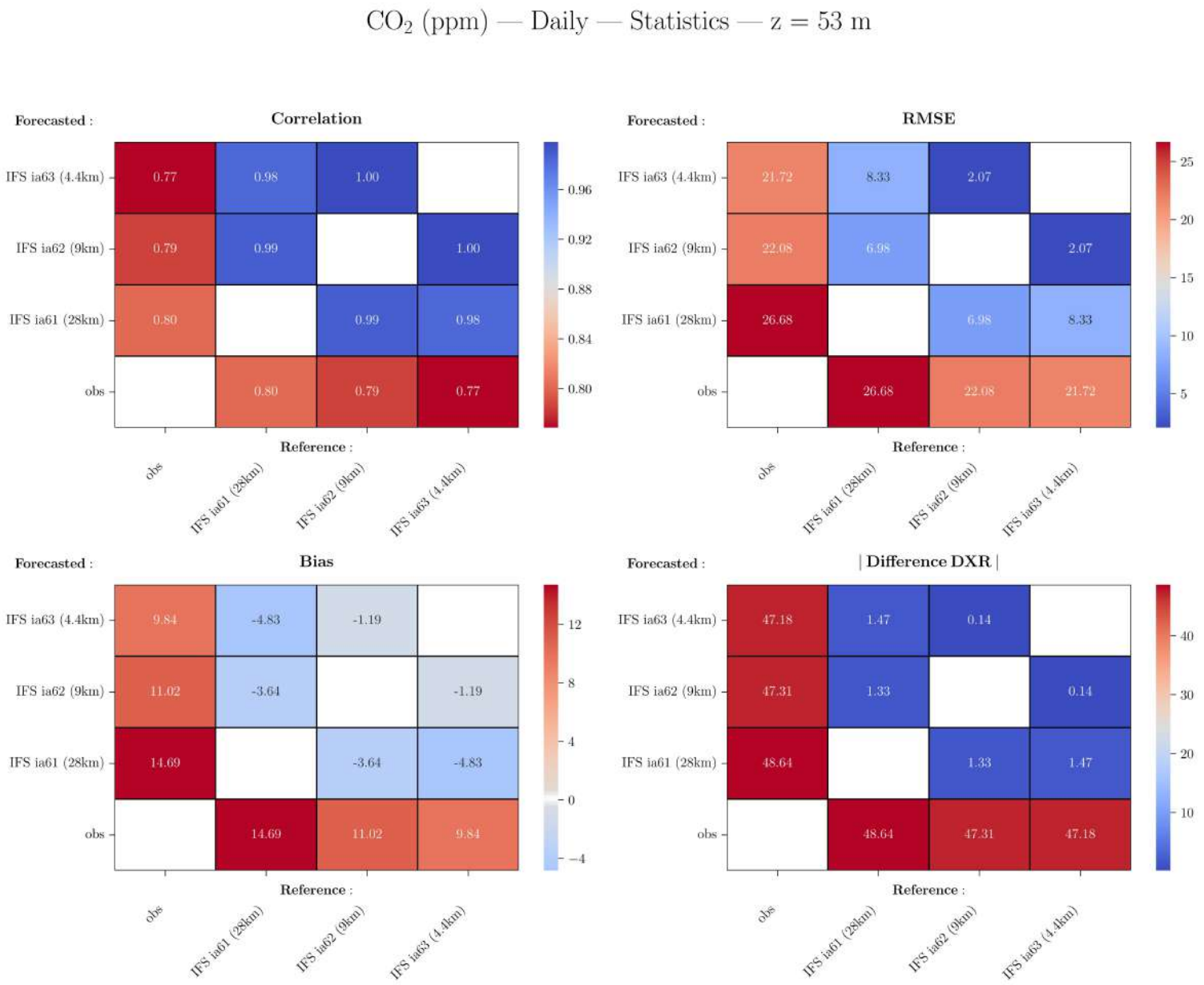



Figure 129: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) $CO_2$ for the full CloudRoots-Amazon22 campaign.

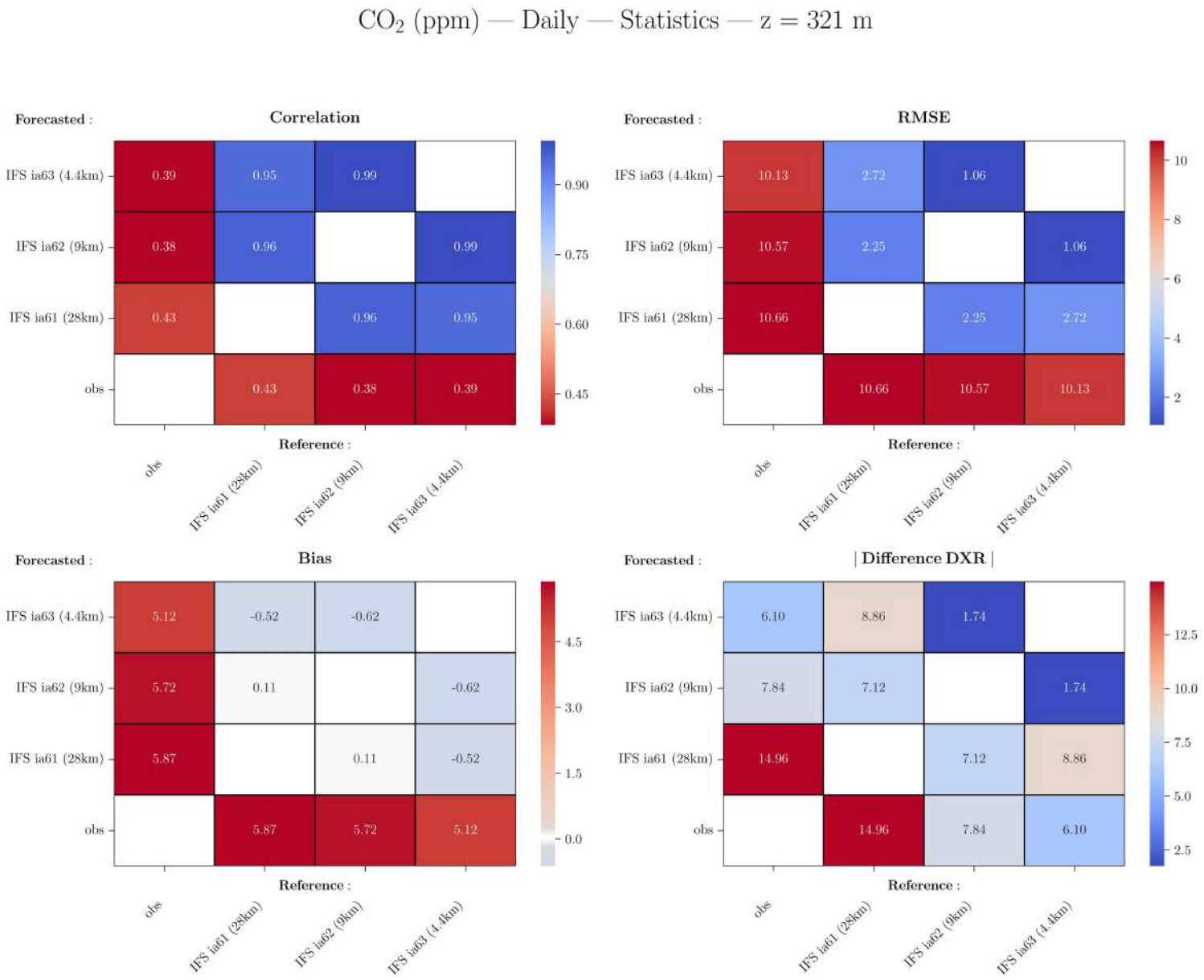



Figure 130: Statistical quantities of the diurnal cycle of observed (at 298 m) and simulated (IFS) $CO_2$ for the full CloudRoots-Amazon22 campaign.

## 4.5 $U$

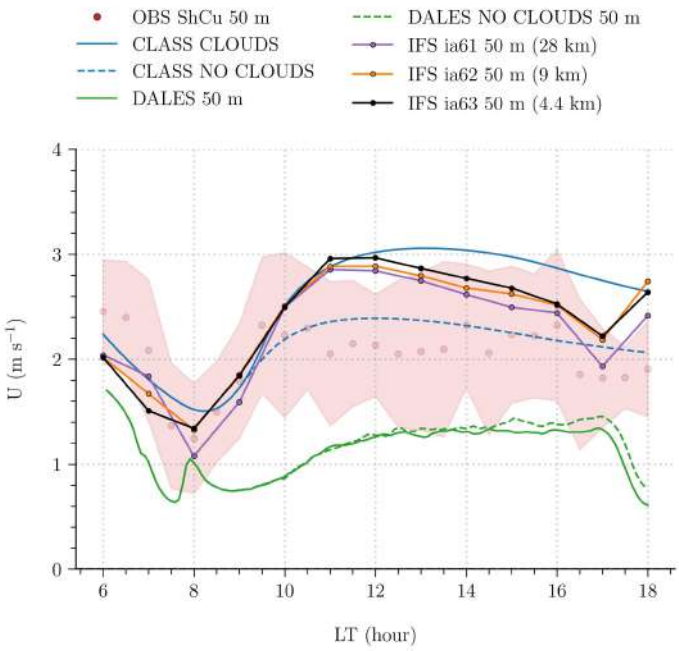



Figure 131: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (CLASS, DALES and IFS) $U$ for the Shallow Convective (ShCu) regime.

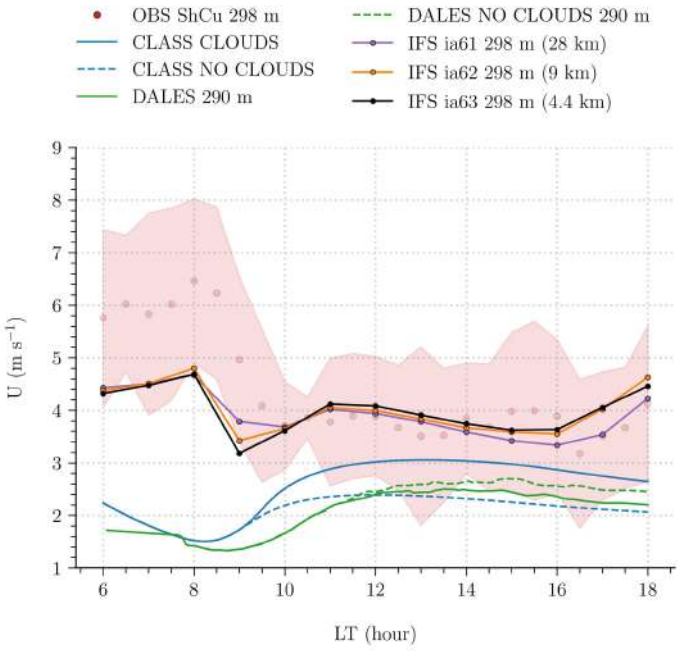



Figure 132: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 298 m) and simulated (CLASS, DALES and IFS) $U$ for the Shallow Convective (ShCu) regime.

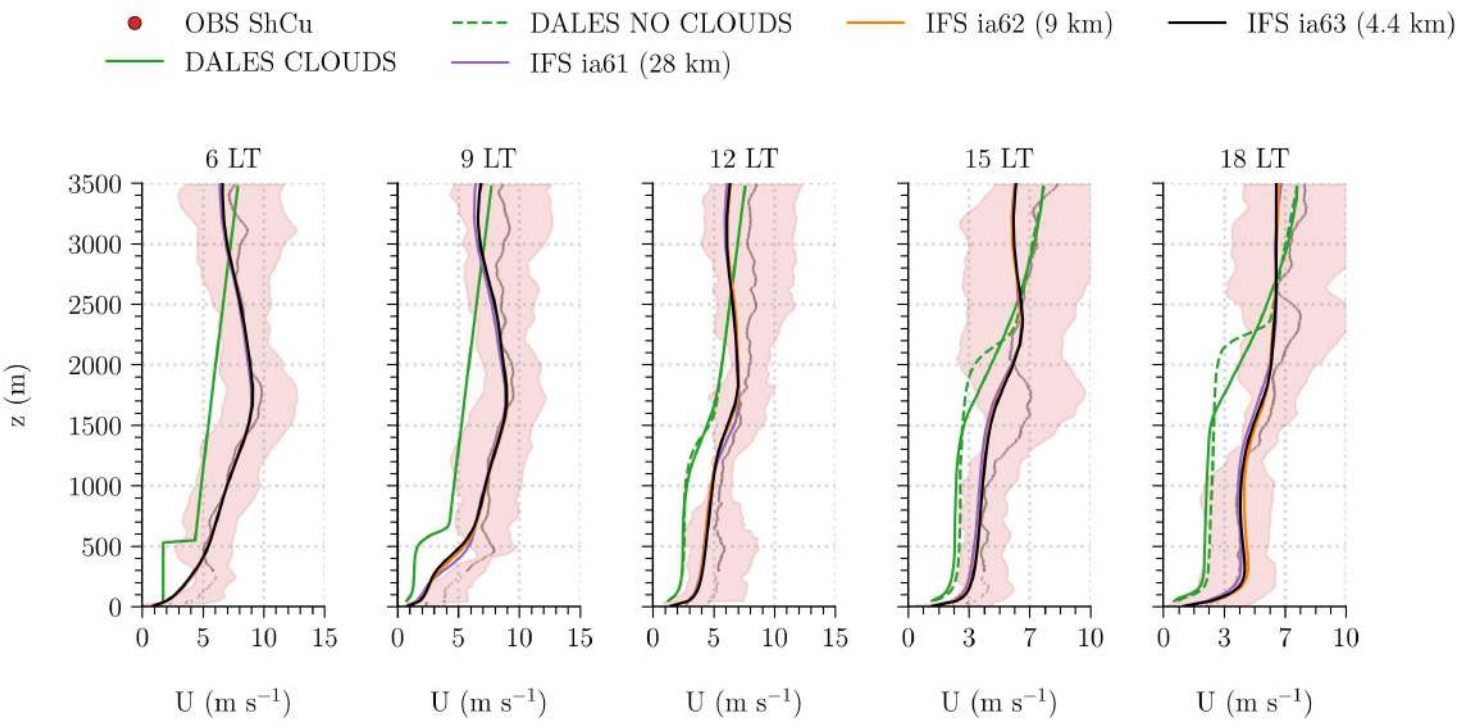



Figure 133: Vertical profiles (markers (mean) and shaded (standard deviation)) and simulated (CLASS, DALES and IFS) $U$ for the Shallow Convective (ShCu) regime.

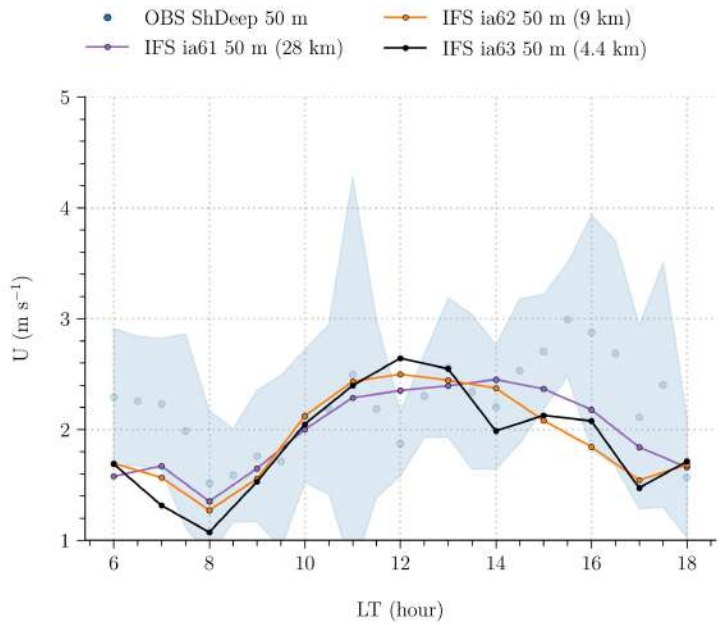



Figure 134: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (IFS) $U$ for the Shallow-to-Deep Convective (ShDeep) regime.

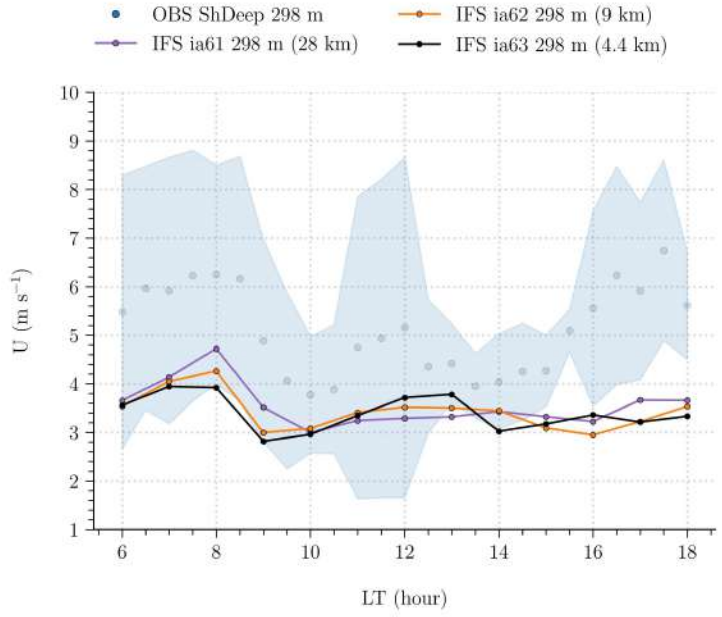



Figure 135: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 298 m) and simulated (IFS) $U$ for the Shallow-to-Deep Convective (ShDeep) regime.

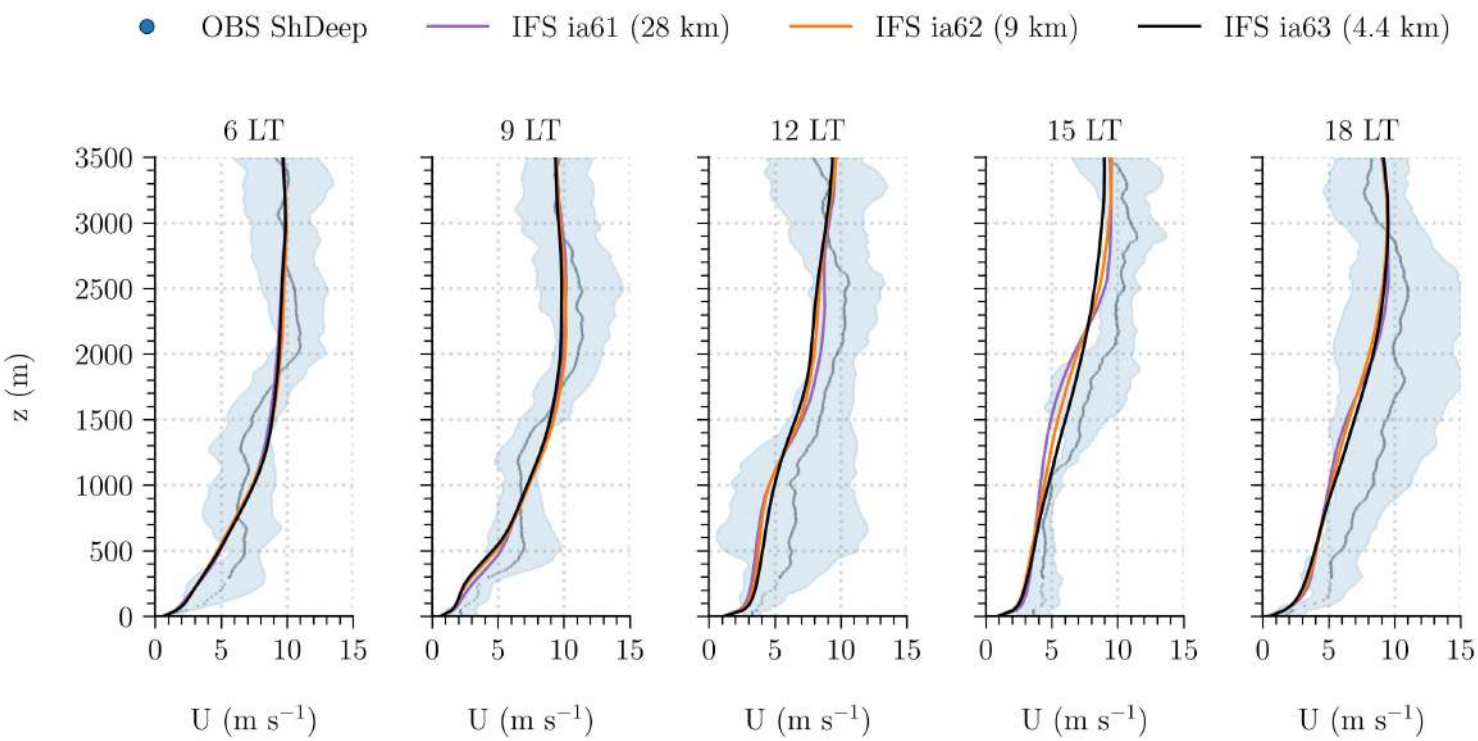



Figure 136: Vertical profiles (markers (mean) and shaded (standard deviation)) and simulated (CLASS, DALES and IFS) $U$ for the Shallow-to-Deep Convective (ShDeep) regime.

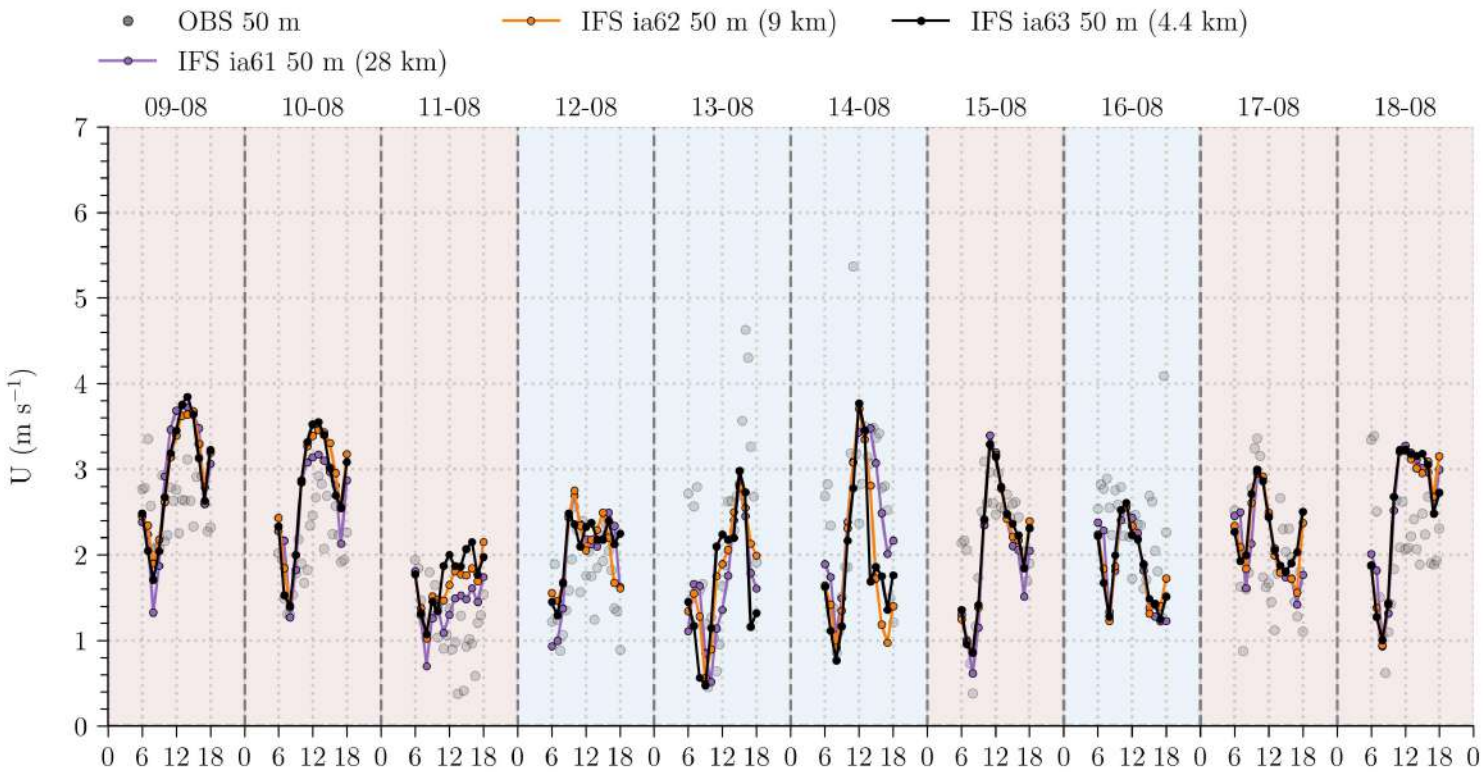



Figure 137: Diurnal cycle of observed (markers, at 50 m) and simulated (IFS) $U$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).



Funded by the European Union

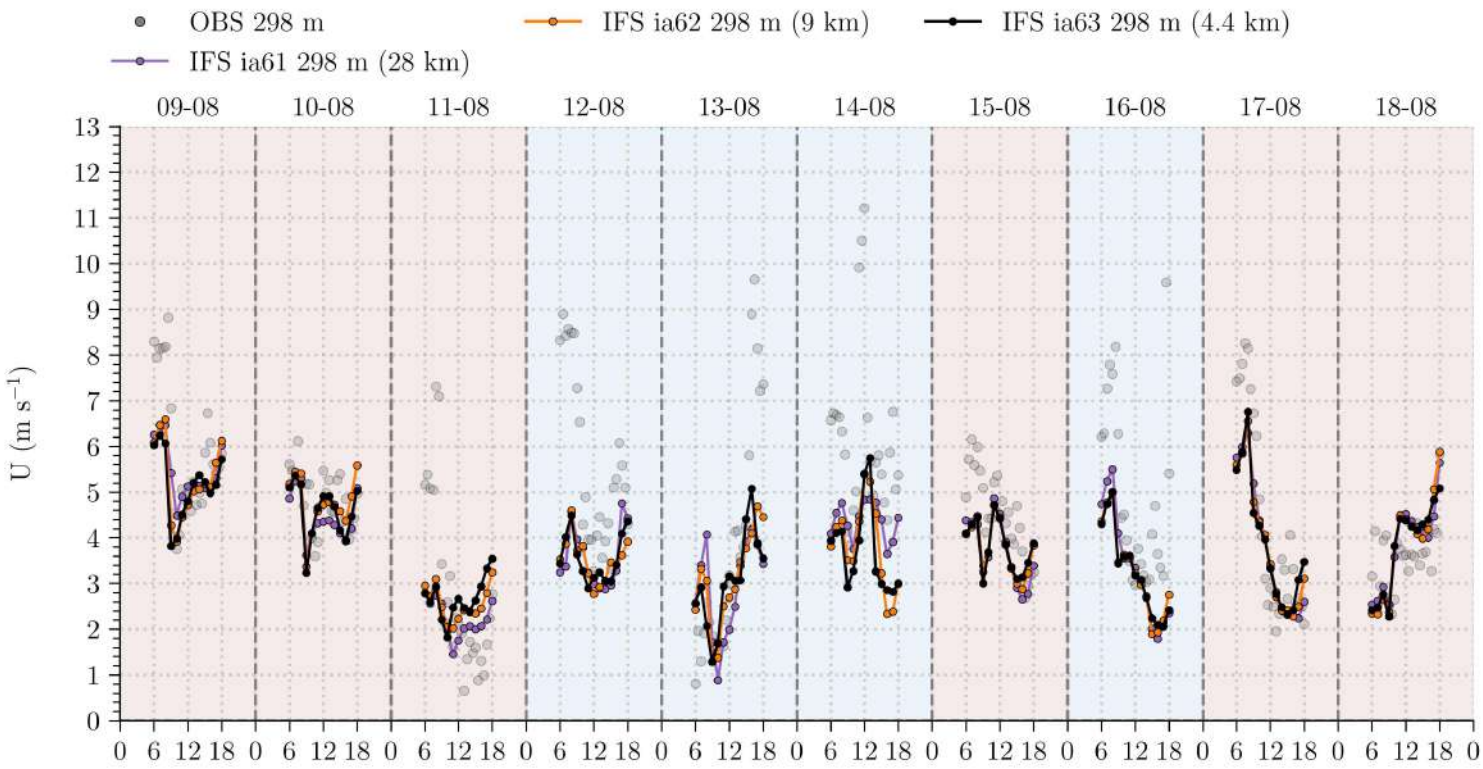



Figure 138: Diurnal cycle of observed (markers, at 298 m) and simulated (IFS) $U$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).



Funded by the European Union

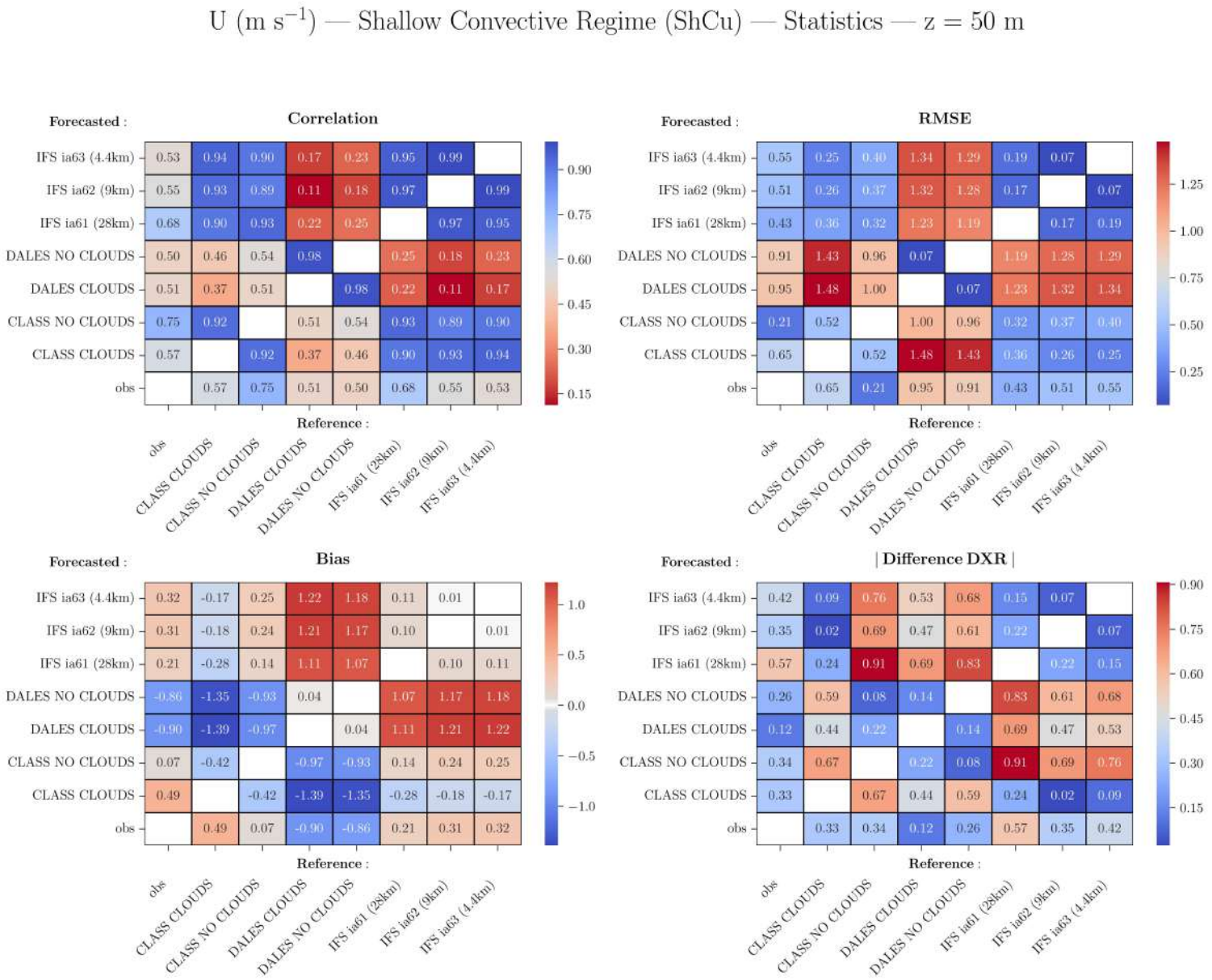



Figure 139: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (CLASS, DALES and IFS) $U$ for the Shallow Convective (ShCu) regime.

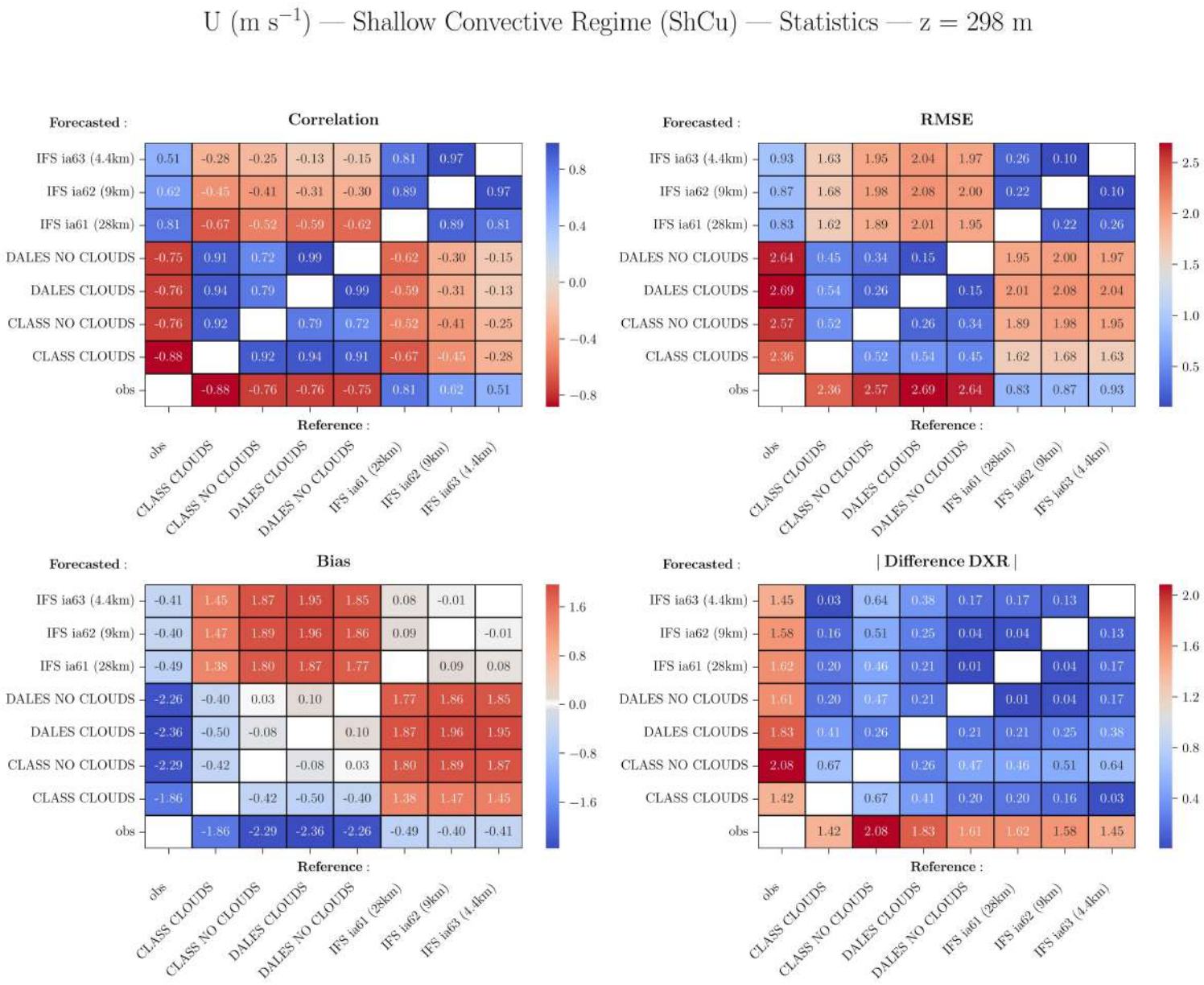



Figure 140: Statistical quantities of the diurnal cycle of observed (at 298 m) and simulated (CLASS, DALES and IFS) $U$ for the Shallow Convective (ShCu) regime.

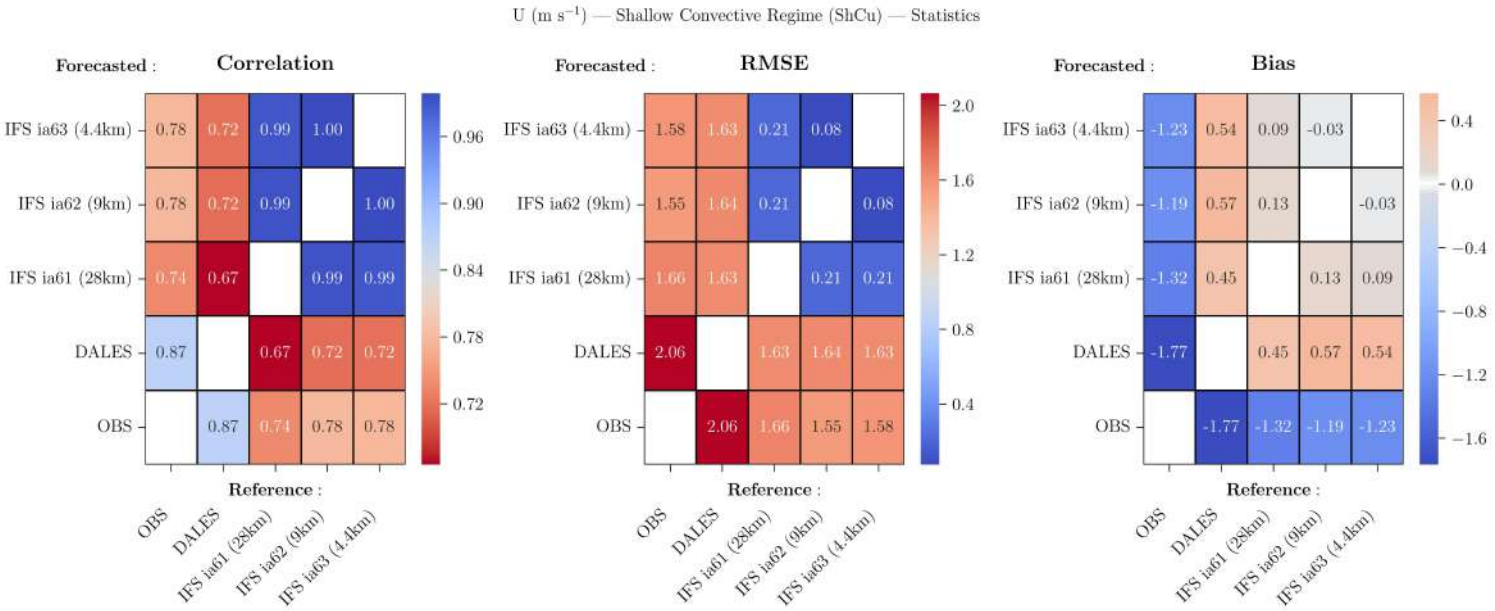



Figure 141: Statistical quantities of the vertical profiles of observed and simulated (CLASS, DALES and IFS) $U$ for the Shallow Convective (ShCu) regime.

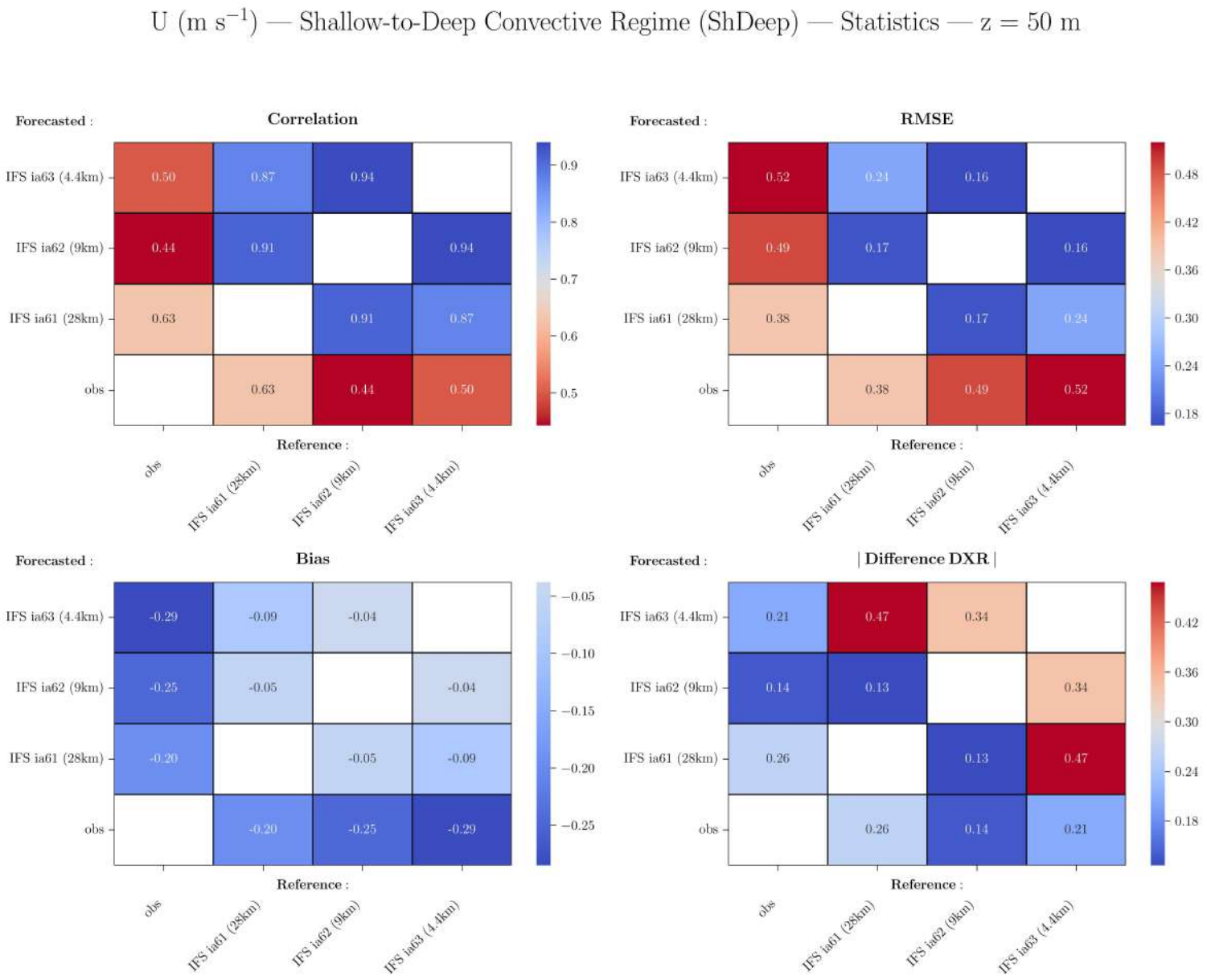



Figure 142: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) $U$ for the Shallow-to-Deep Convective (ShDeep) regime.



Funded by the European Union

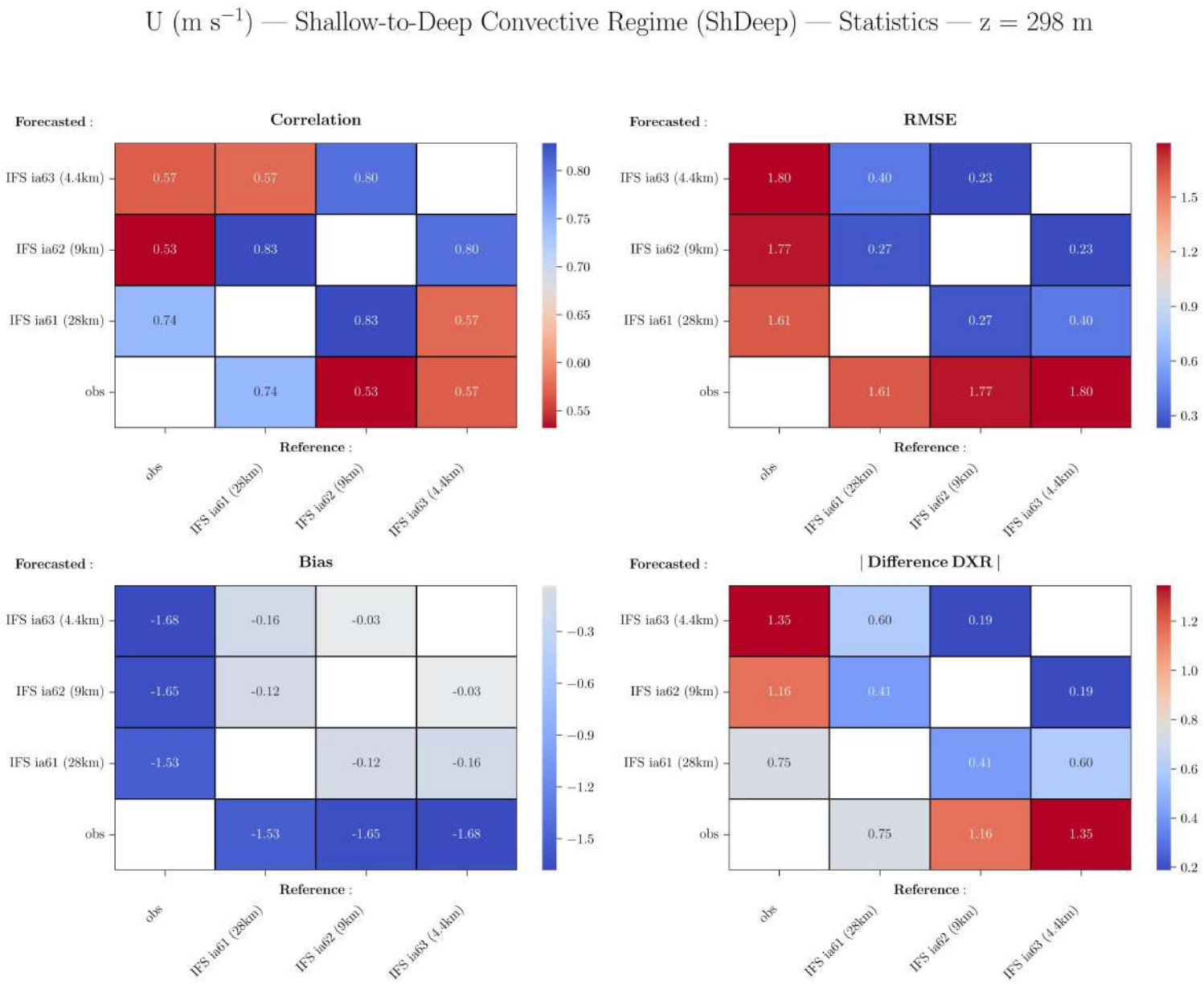



Figure 143: Statistical quantities of the diurnal cycle of observed (at 298 m) and simulated (IFS) $U$ for the Shallow-to-Deep Convective (ShDeep) regime.



Funded by the European Union

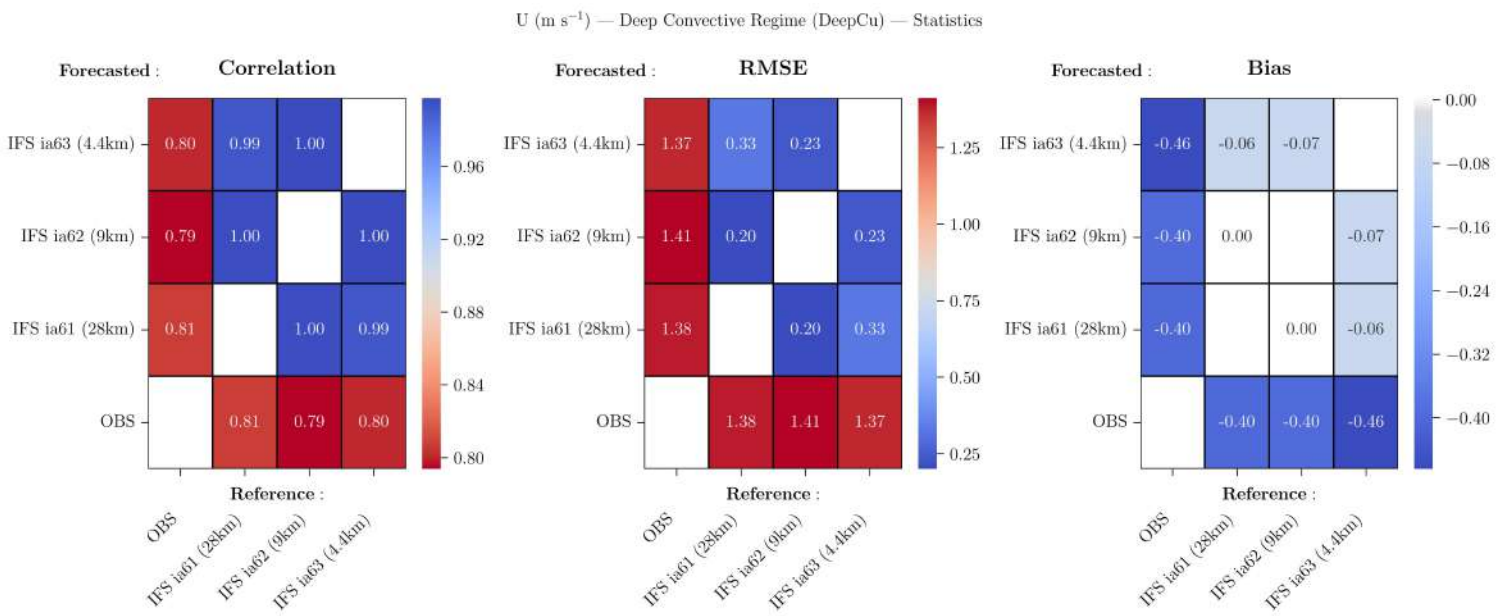



Figure 144: Statistical quantities of the vertical profiles of observed and simulated (IFS) $U$ for the Shallow-to-Deep Convective (ShDeep) regime.

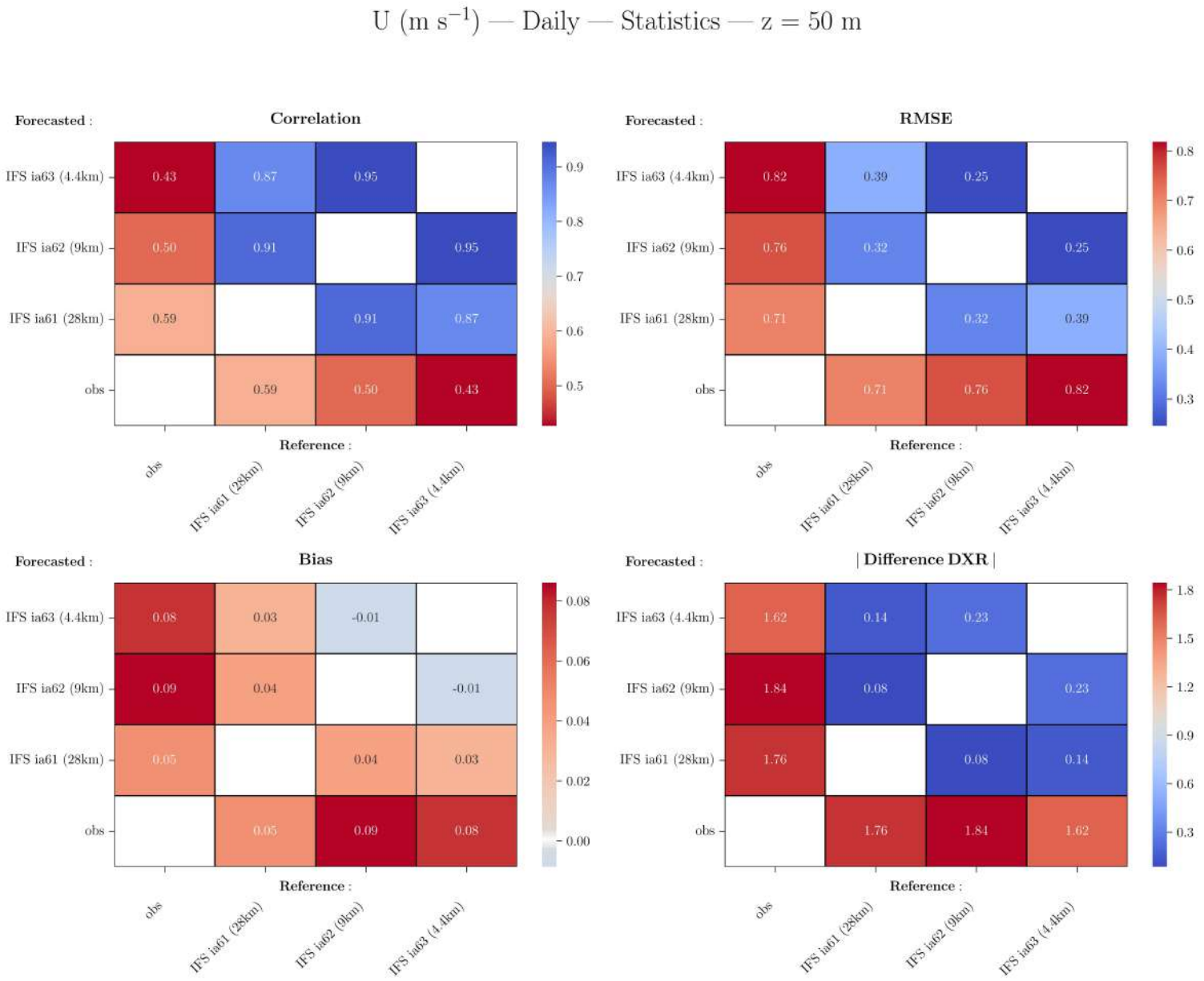



Figure 145: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) $U$ for the full CloudRoots-Amazon22 campaign.



Funded by the European Union

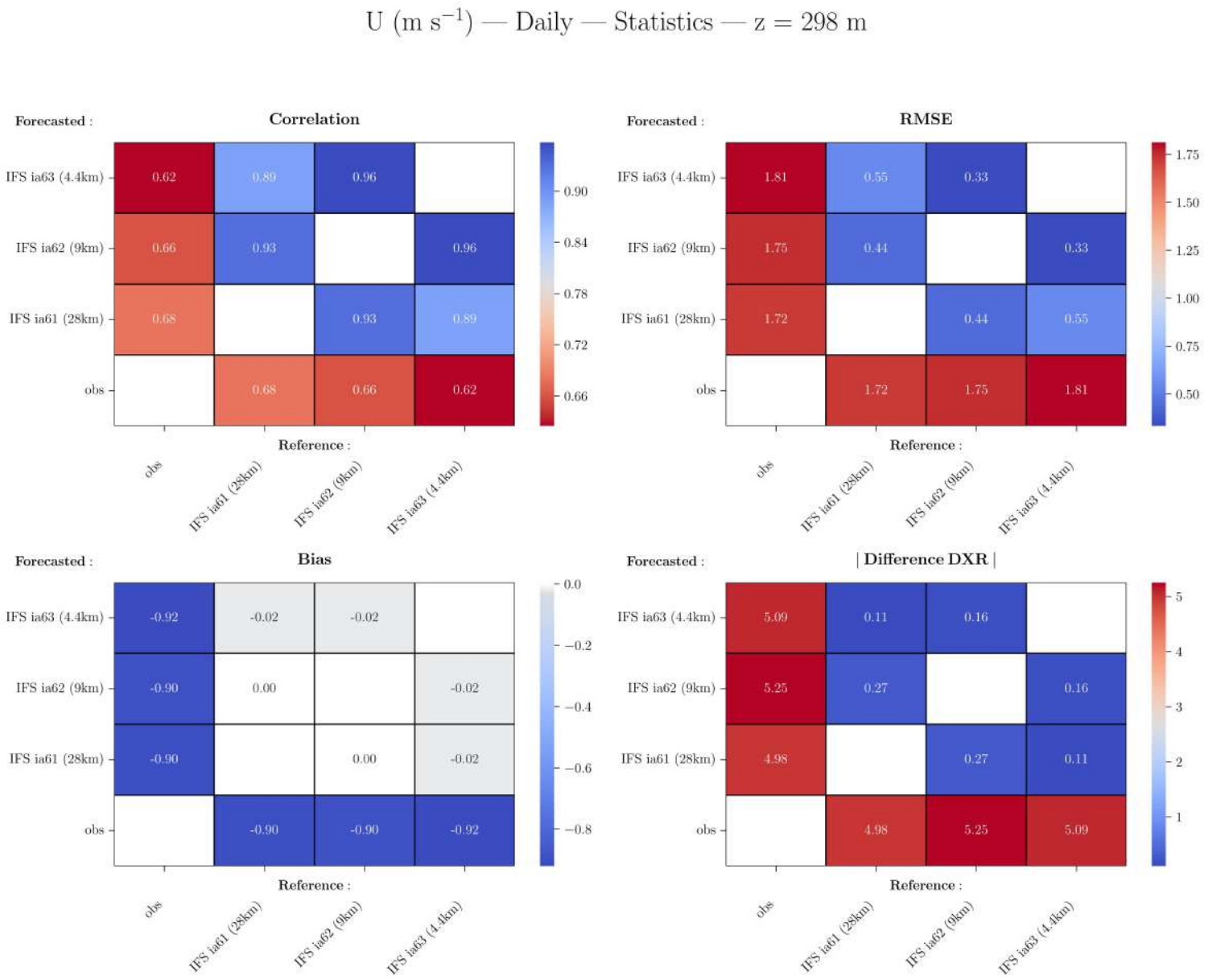



Figure 146: Statistical quantities of the diurnal cycle of observed (at 298 m) and simulated (IFS) $U$ for the full CloudRoots-Amazon22 campaign.



Funded by the European Union

## 4.6 $w_{dir}$

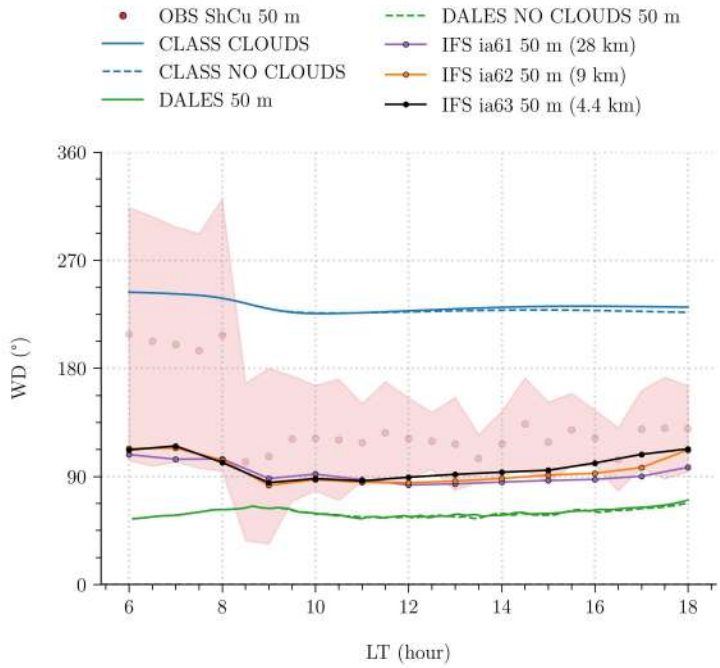



Figure 147: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (CLASS, DALES and IFS) $w_{dir}$ for the Shallow Convective (ShCu) regime.

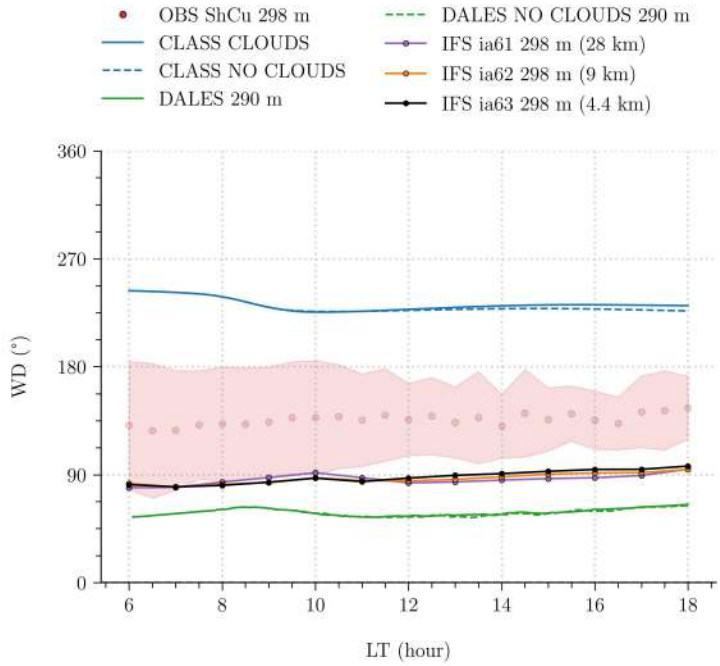



Figure 148: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 298 m) and simulated (CLASS, DALES and IFS) $w_{dir}$ for the Shallow Convective (ShCu) regime.

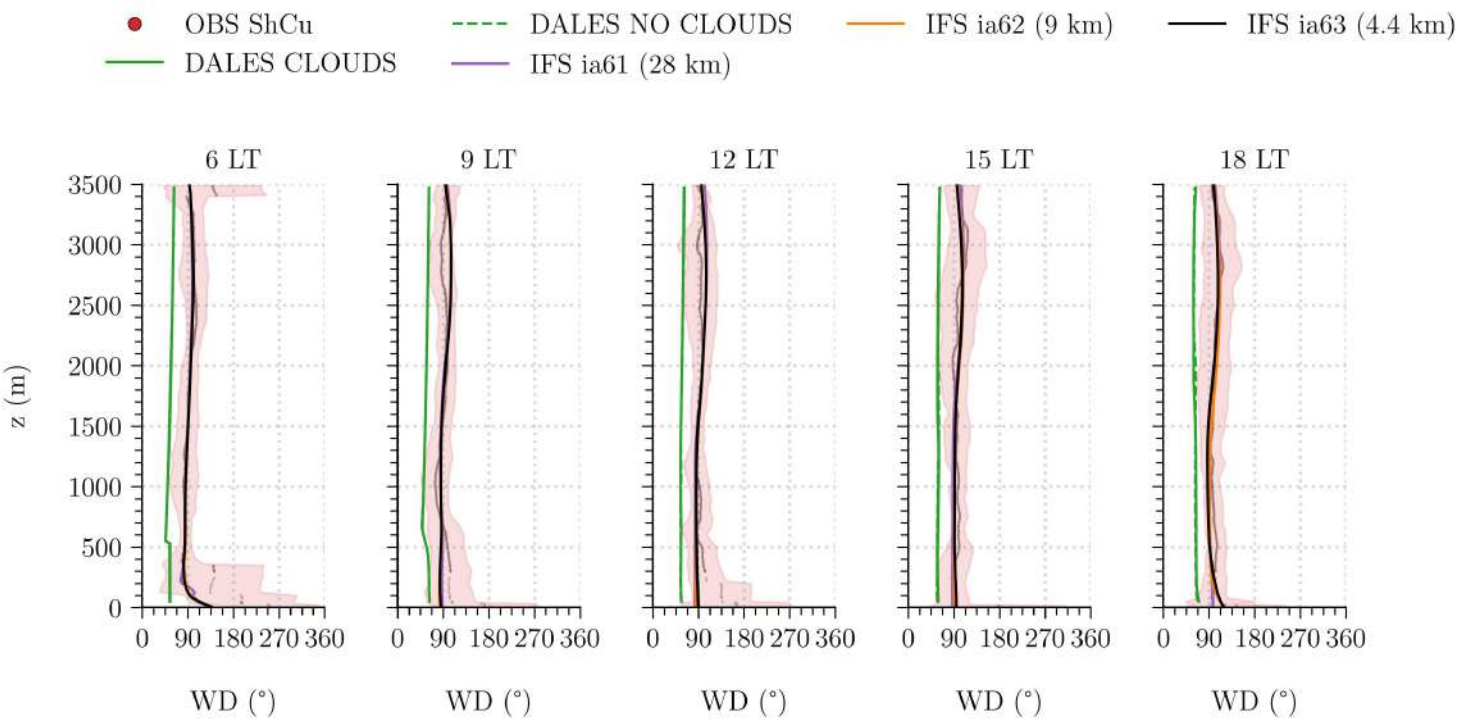



Figure 149: Vertical profiles (markers (mean) and shaded (standard deviation)) and simulated (CLASS, DALES and IFS) $w_{dir}$ for the Shallow Convective (ShCu) regime.



Funded by the European Union

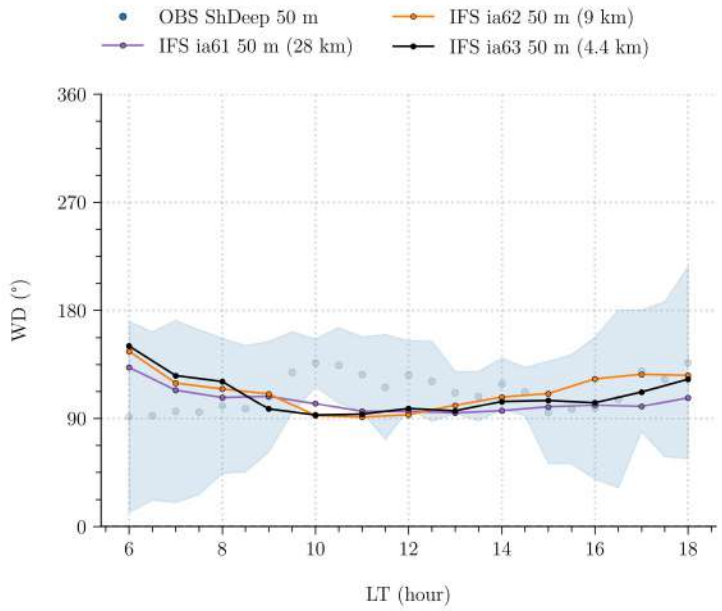



Figure 150: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 50 m) and simulated (IFS) $w_{dir}$ for the Shallow-to-Deep Convective (ShDeep) regime.

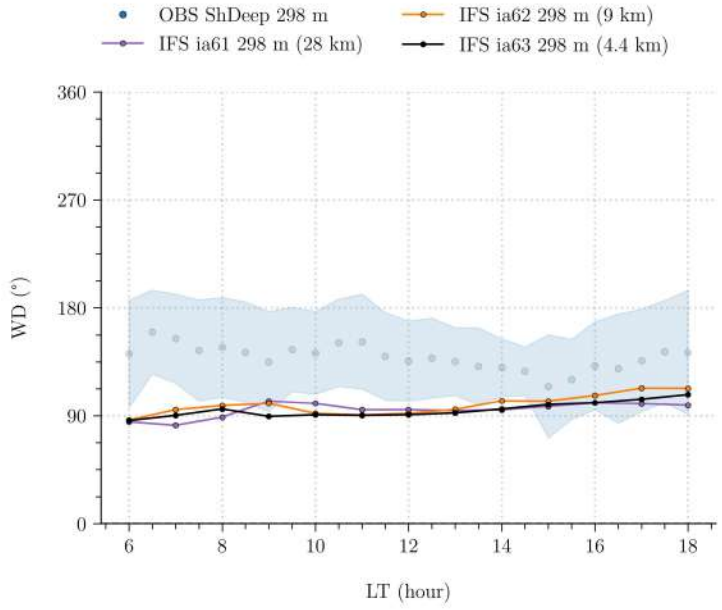



Figure 151: Diurnal cycle of observed (markers (mean) and shaded (standard deviation), at 298 m) and simulated (IFS) $w_{dir}$ for the Shallow-to-Deep Convective (ShDeep) regime.

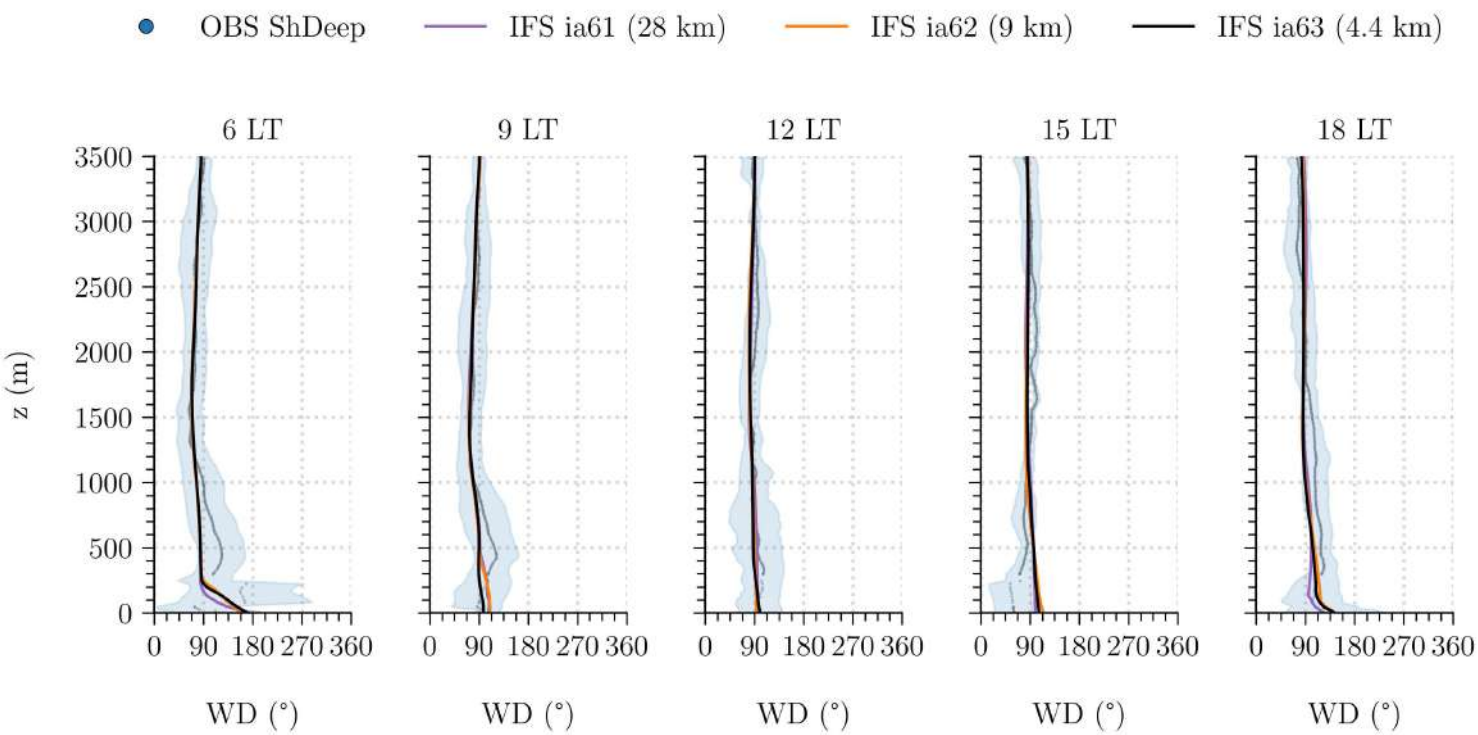



Figure 152: Vertical profiles (markers (mean) and shaded (standard deviation)) and simulated (CLASS, DALES and IFS) $w_{dir}$ for the Shallow-to-Deep Convective (ShDeep) regime.

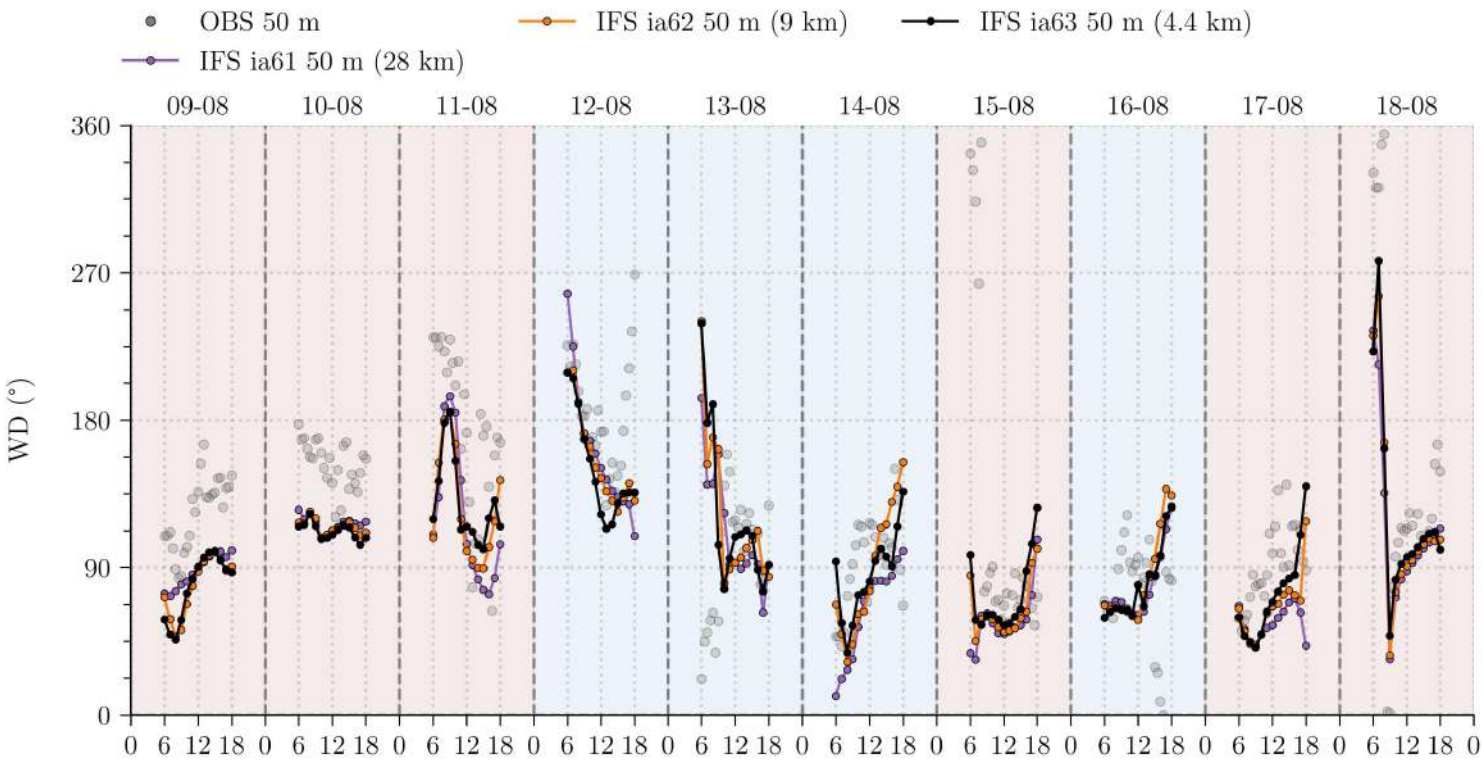



Figure 153: Diurnal cycle of observed (markers, at 50 m) and simulated (IFS) $w_{dir}$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).



Funded by the European Union

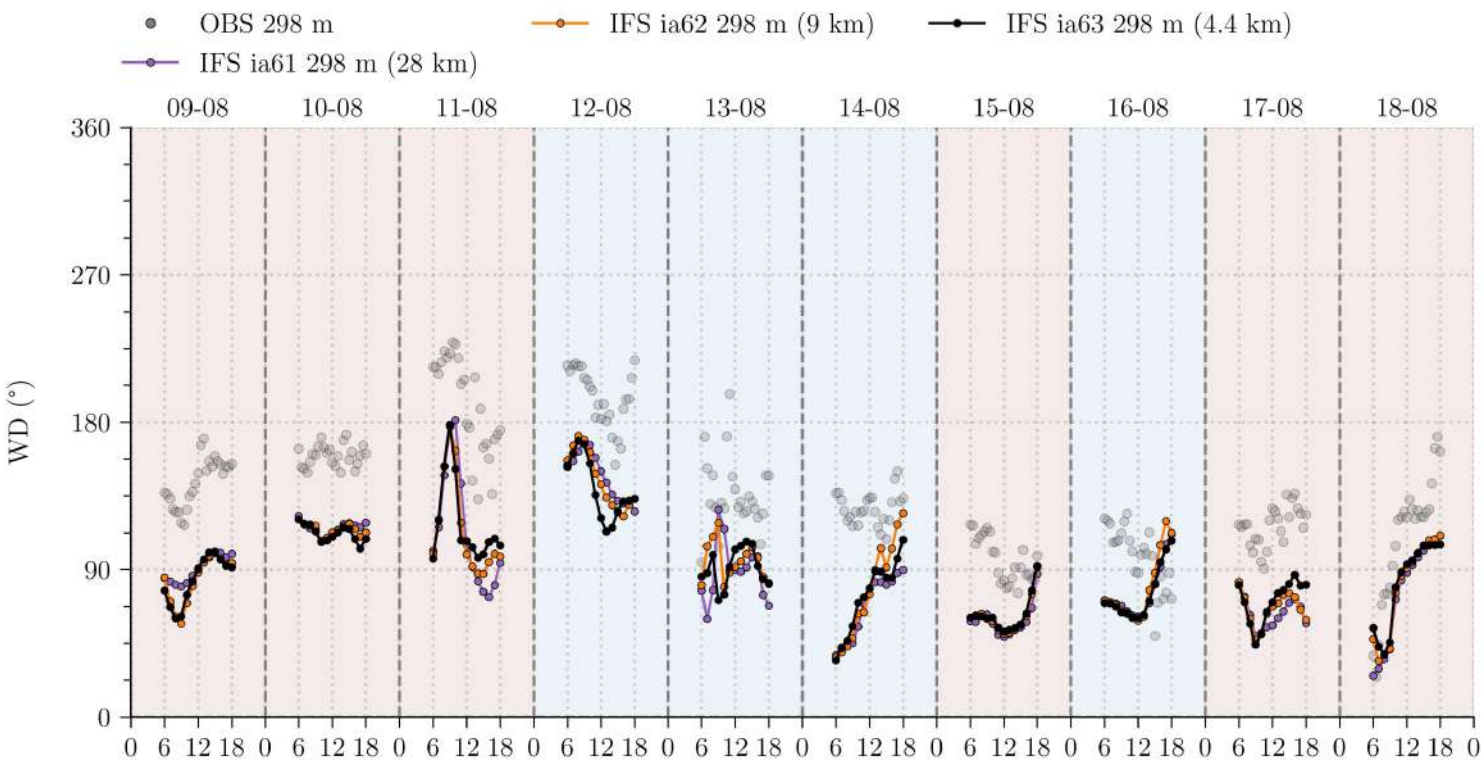



Figure 154: Diurnal cycle of observed (markers, at 298 m) and simulated (IFS) $w_{dir}$ for the full CloudRoots-Amazon22 campaign. Red shading = Shallow Convective regime (ShCu), blue shading = Shallow-to-Deep Convective regime (ShDeep).



Funded by the European Union

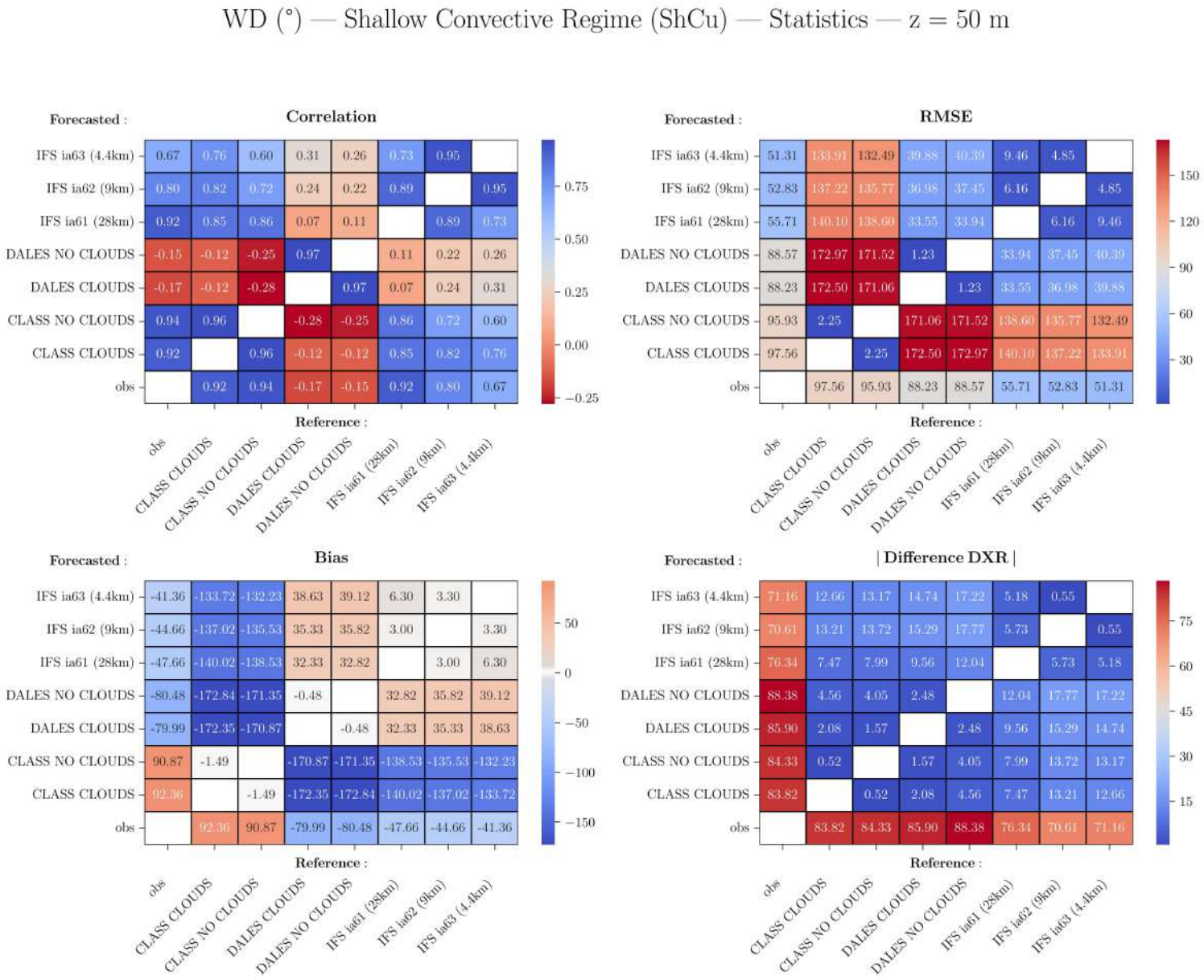



Figure 155: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (CLASS, DALES and IFS) $w_{dir}$ for the Shallow Convective (ShCu) regime.

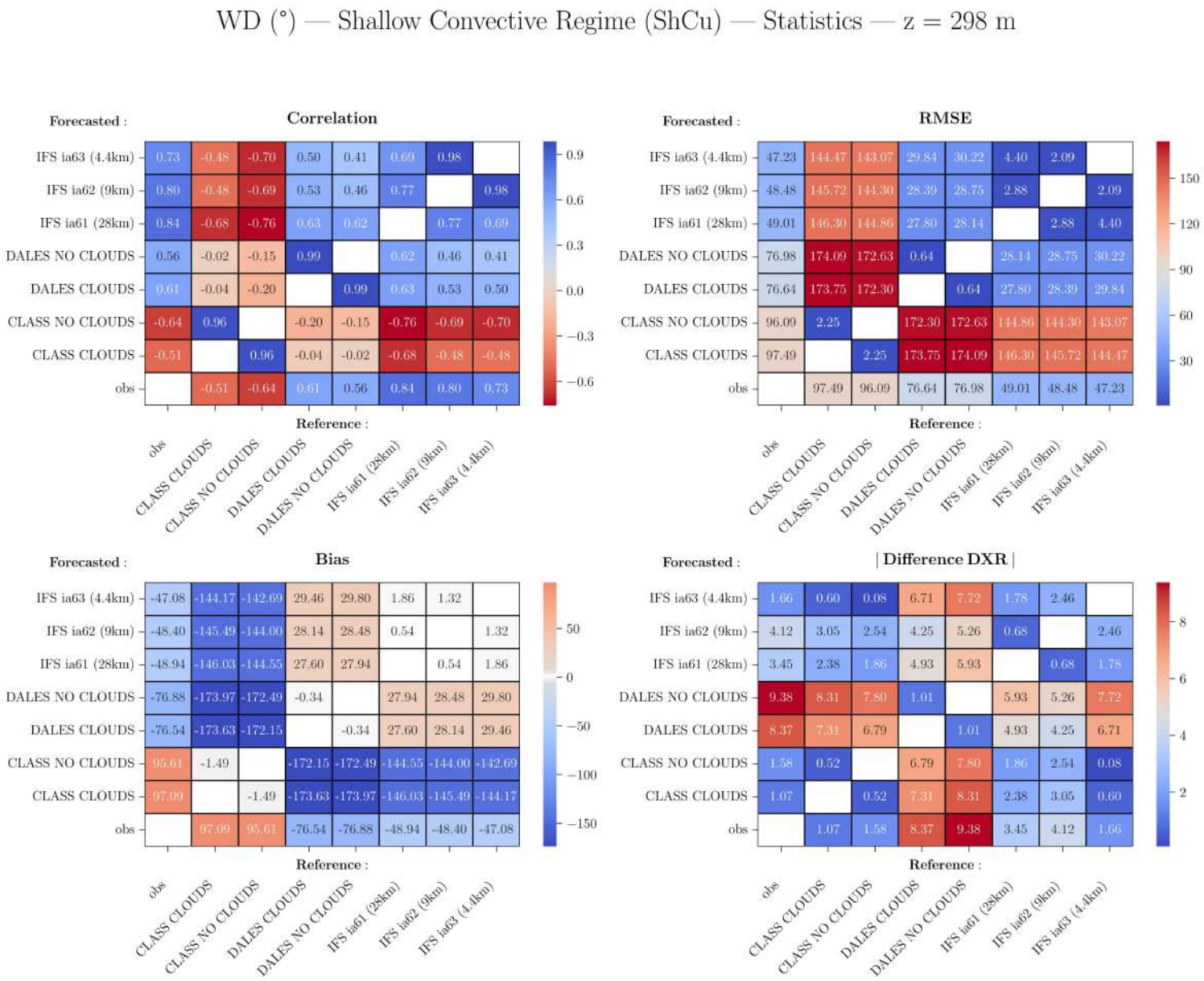



Figure 156: Statistical quantities of the diurnal cycle of observed (at 298 m) and simulated (CLASS, DALES and IFS) $w_{dir}$ for the Shallow Convective (ShCu) regime.

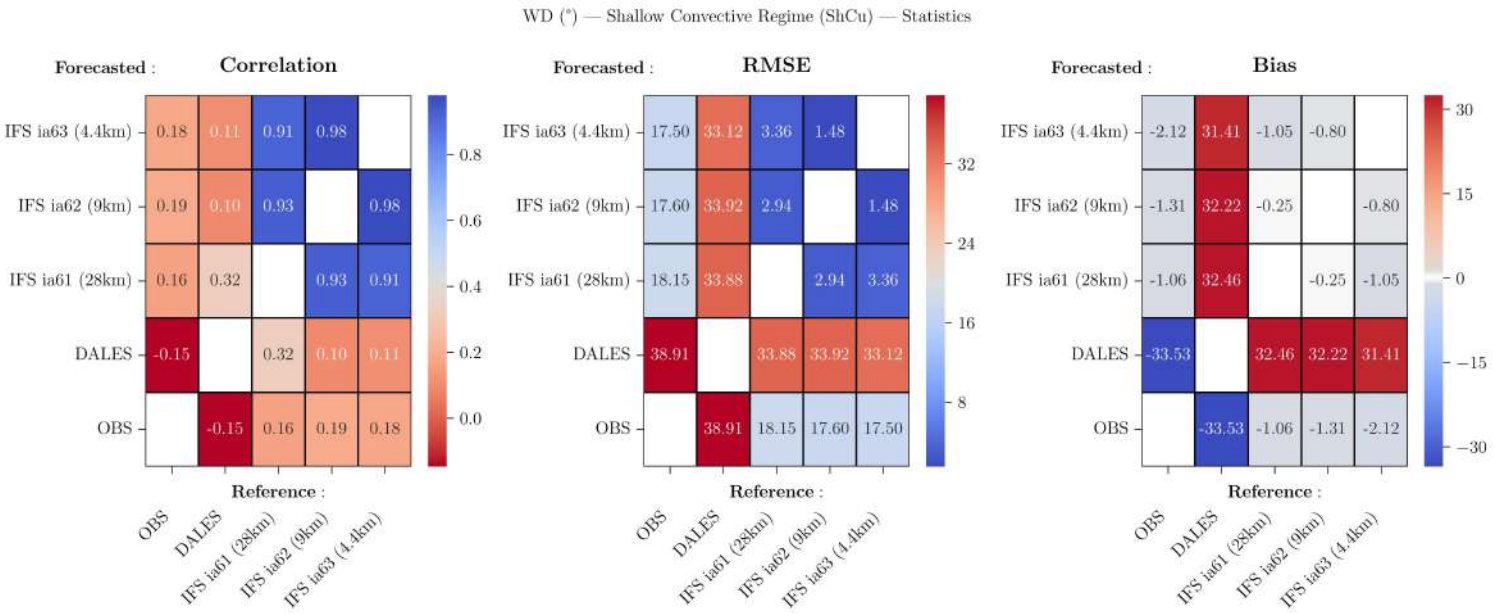



Figure 157: Statistical quantities of the vertical profiles of observed and simulated (CLASS, DALES and IFS) $w_{dir}$ for the Shallow Convective (ShCu) regime.

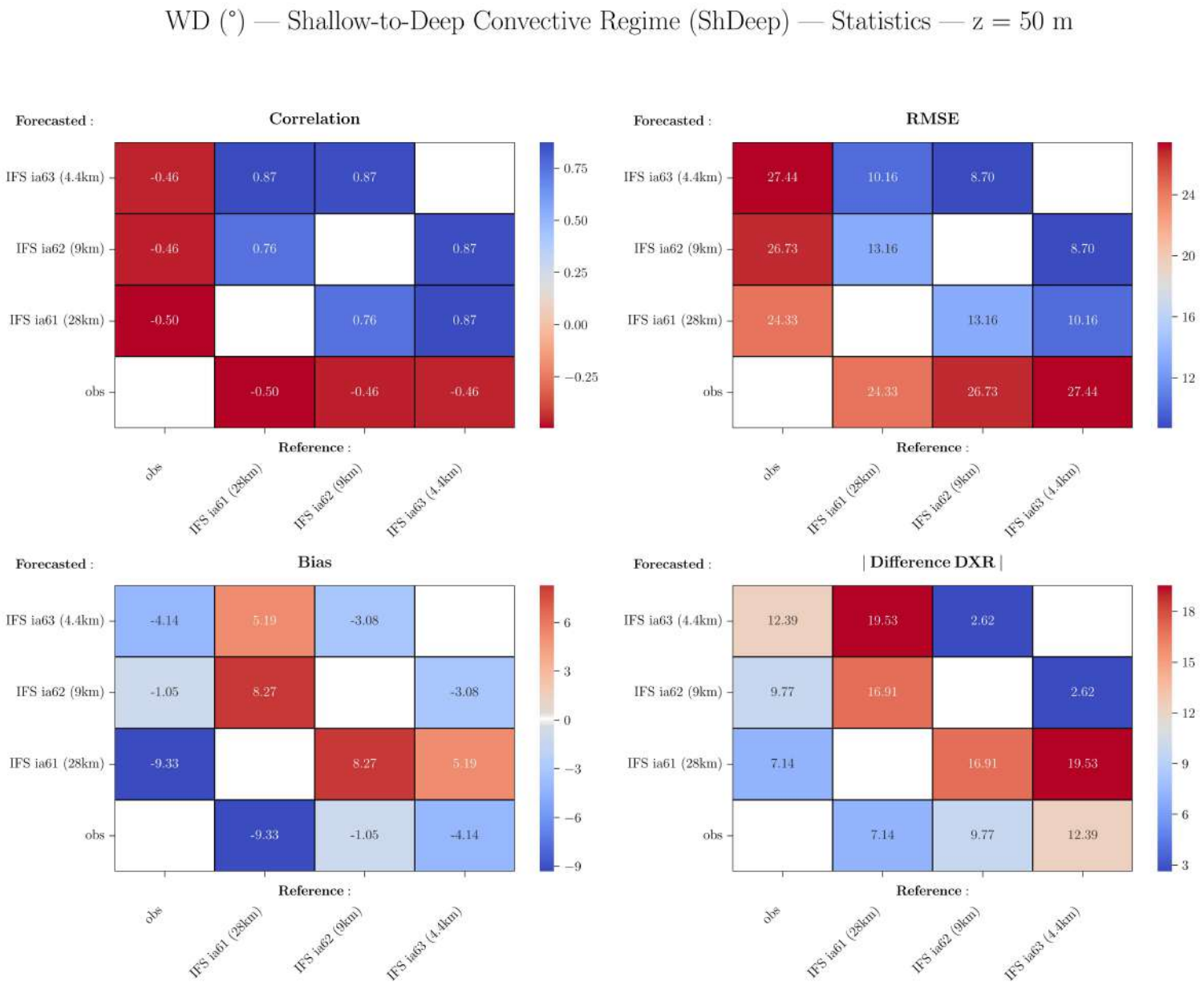



Figure 158: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) $w_{dir}$ for the Shallow-to-Deep Convective (ShDeep) regime.

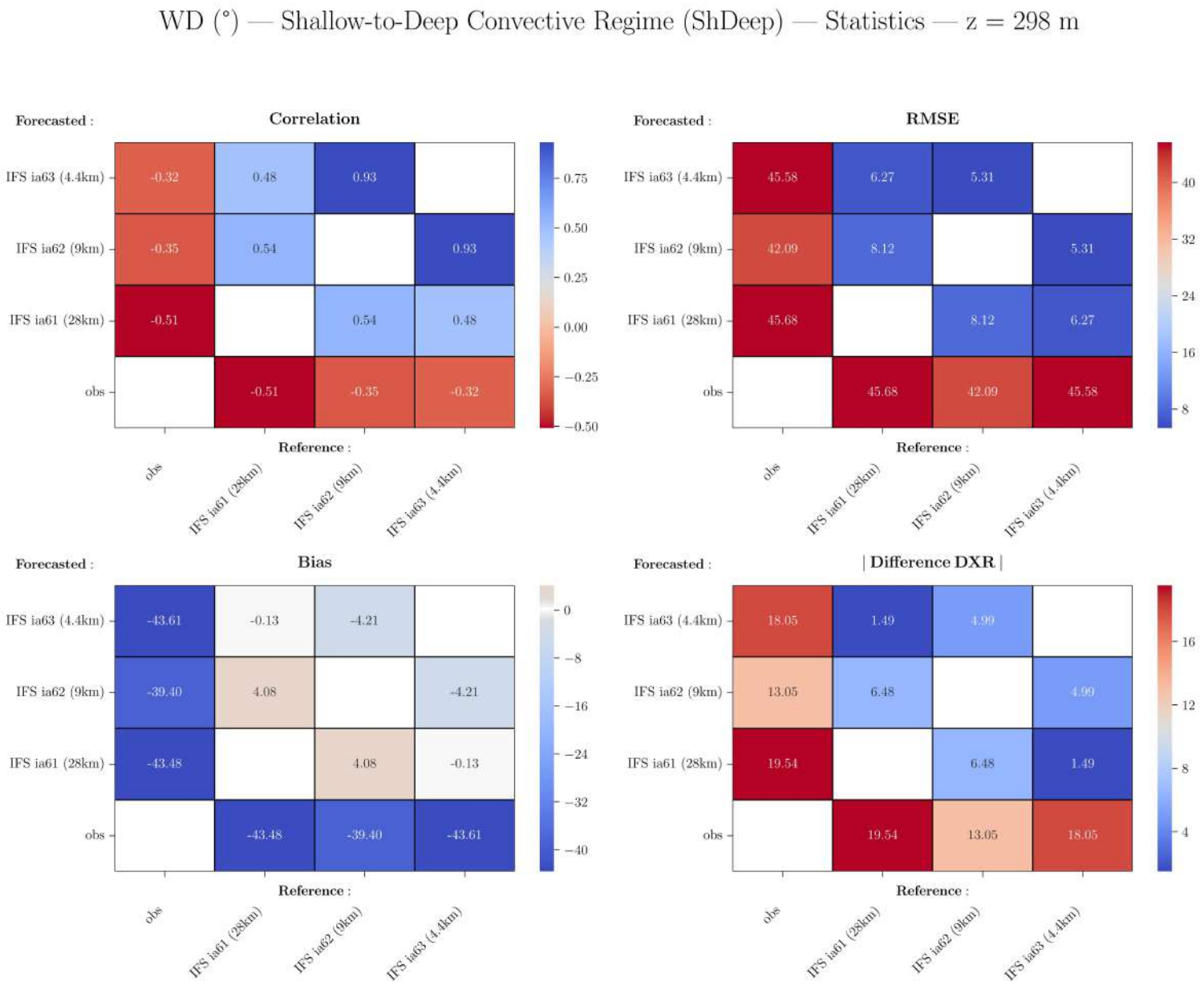



Figure 159: Statistical quantities of the diurnal cycle of observed (at 298 m) and simulated (IFS) $w_{dir}$ for the Shallow-to-Deep Convective (ShDeep) regime.

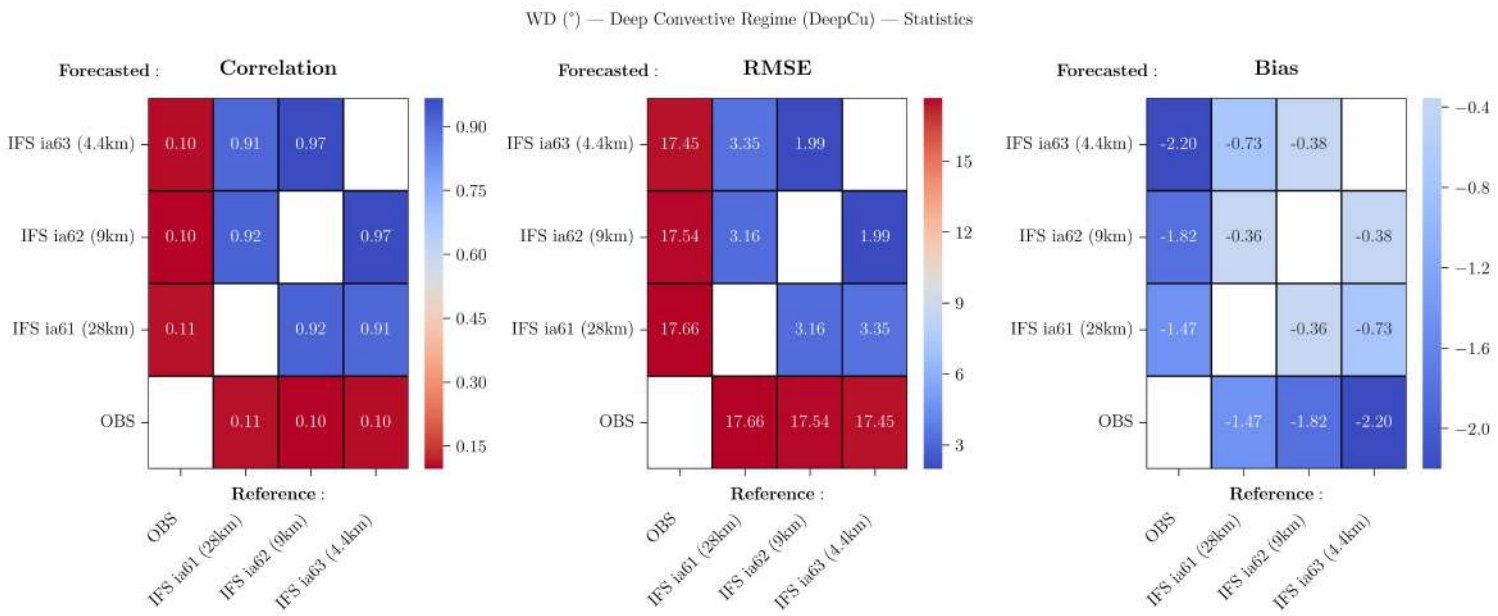



Figure 160: Statistical quantities of the vertical profiles of observed and simulated (IFS) $w_{dir}$ for the Shallow-to-Deep Convective (ShDeep) regime.

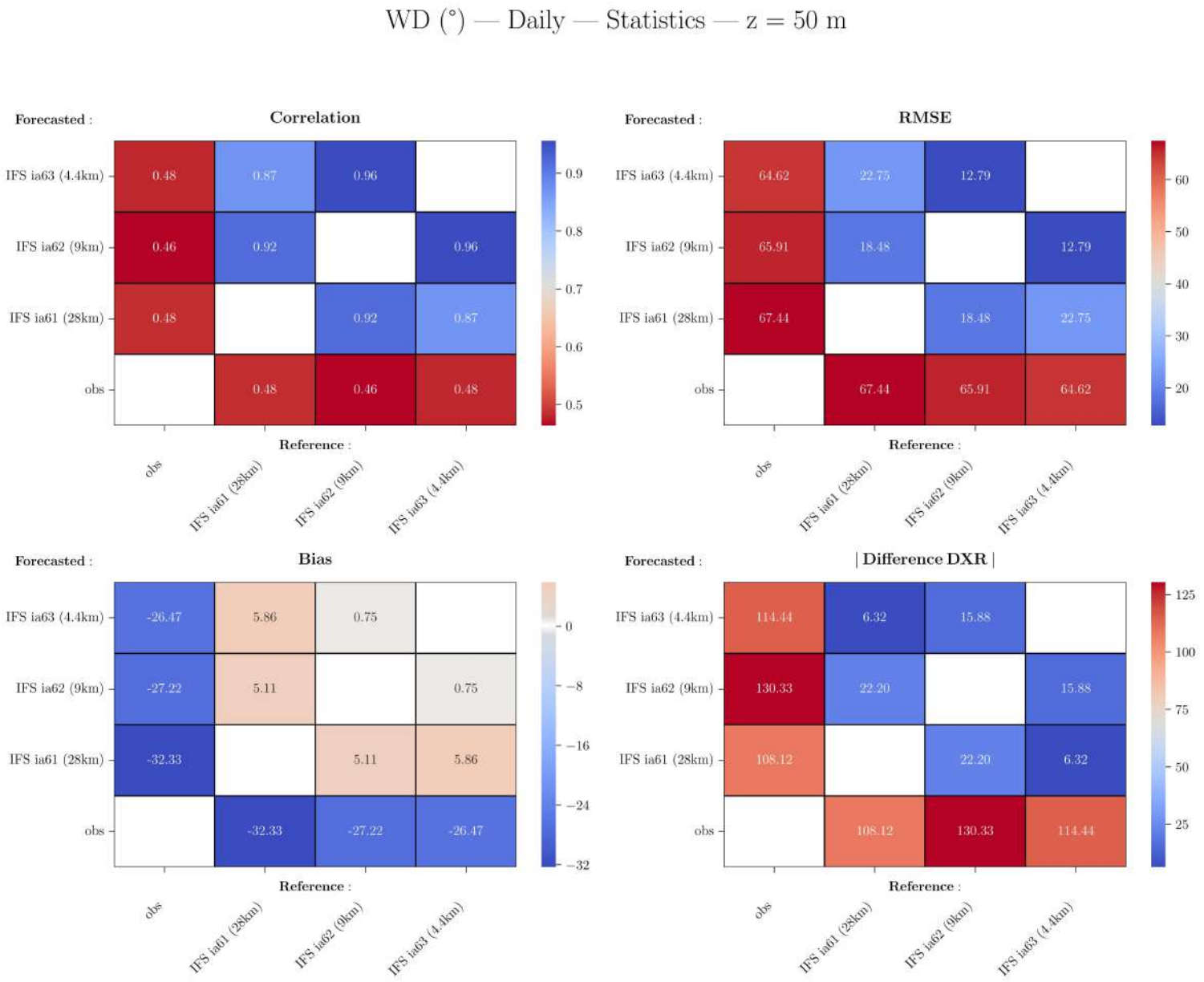



Figure 161: Statistical quantities of the diurnal cycle of observed (at 50 m) and simulated (IFS) $w_{dir}$ for the full CloudRoots-Amazon22 campaign.

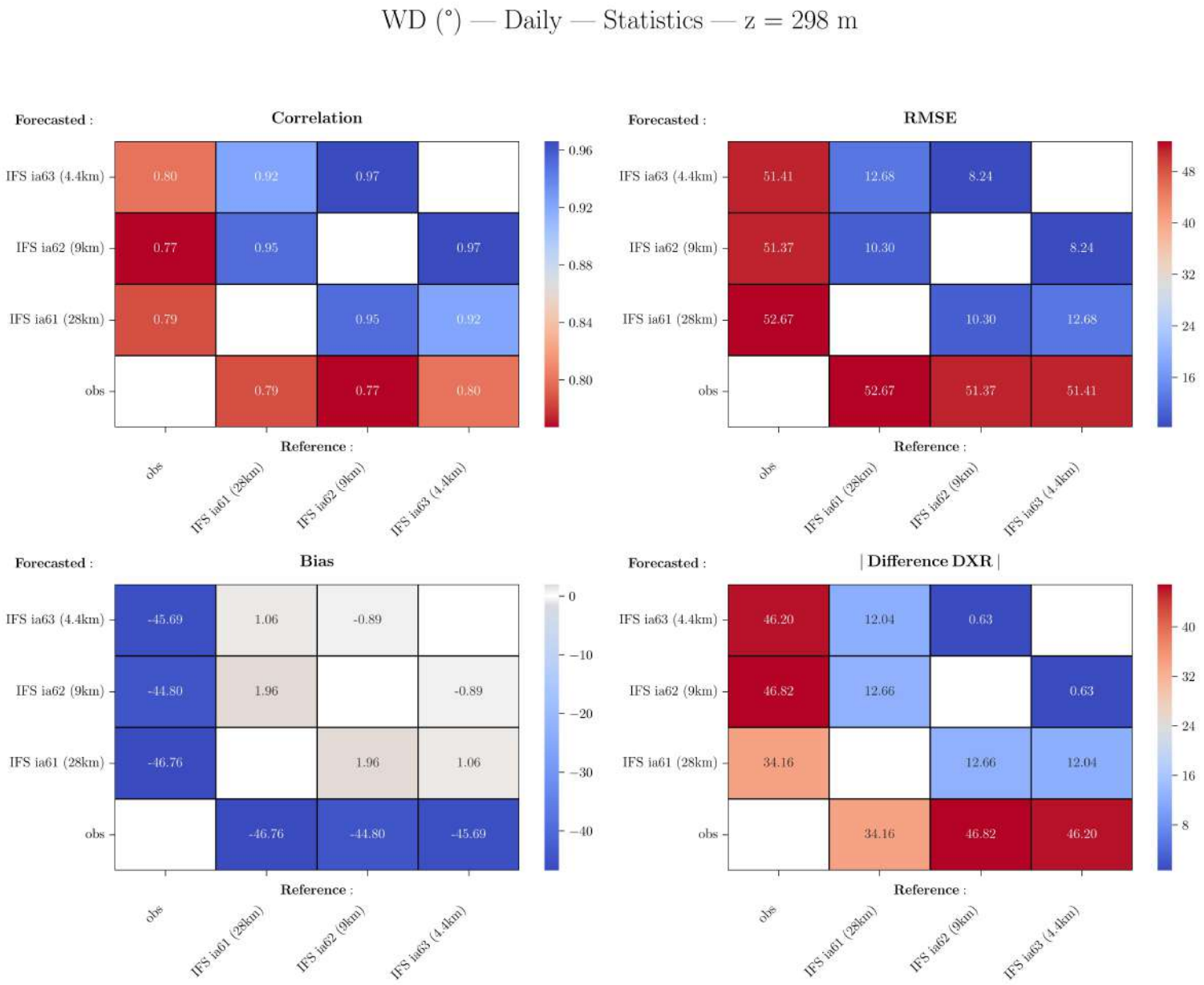



Figure 162: Statistical quantities of the diurnal cycle of observed (at 298 m) and simulated (IFS) $w_{dir}$ for the full CloudRoots-Amazon22 campaign.

## 4.7 Cloud Cover

### 4.7.1 Low Cloud Cover

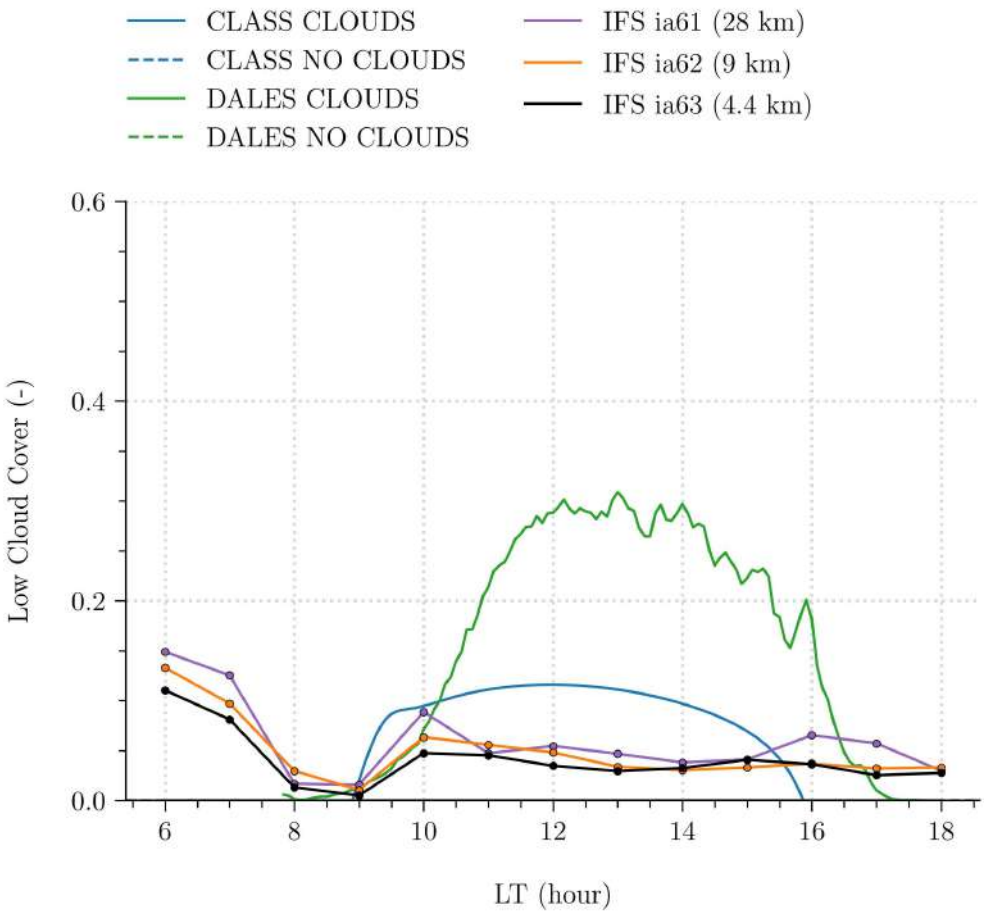



Figure 163: Diurnal cycle of simulated (CLASS, DALES and IFS) low cloud cover for the Shallow Convective (ShCu) regime.

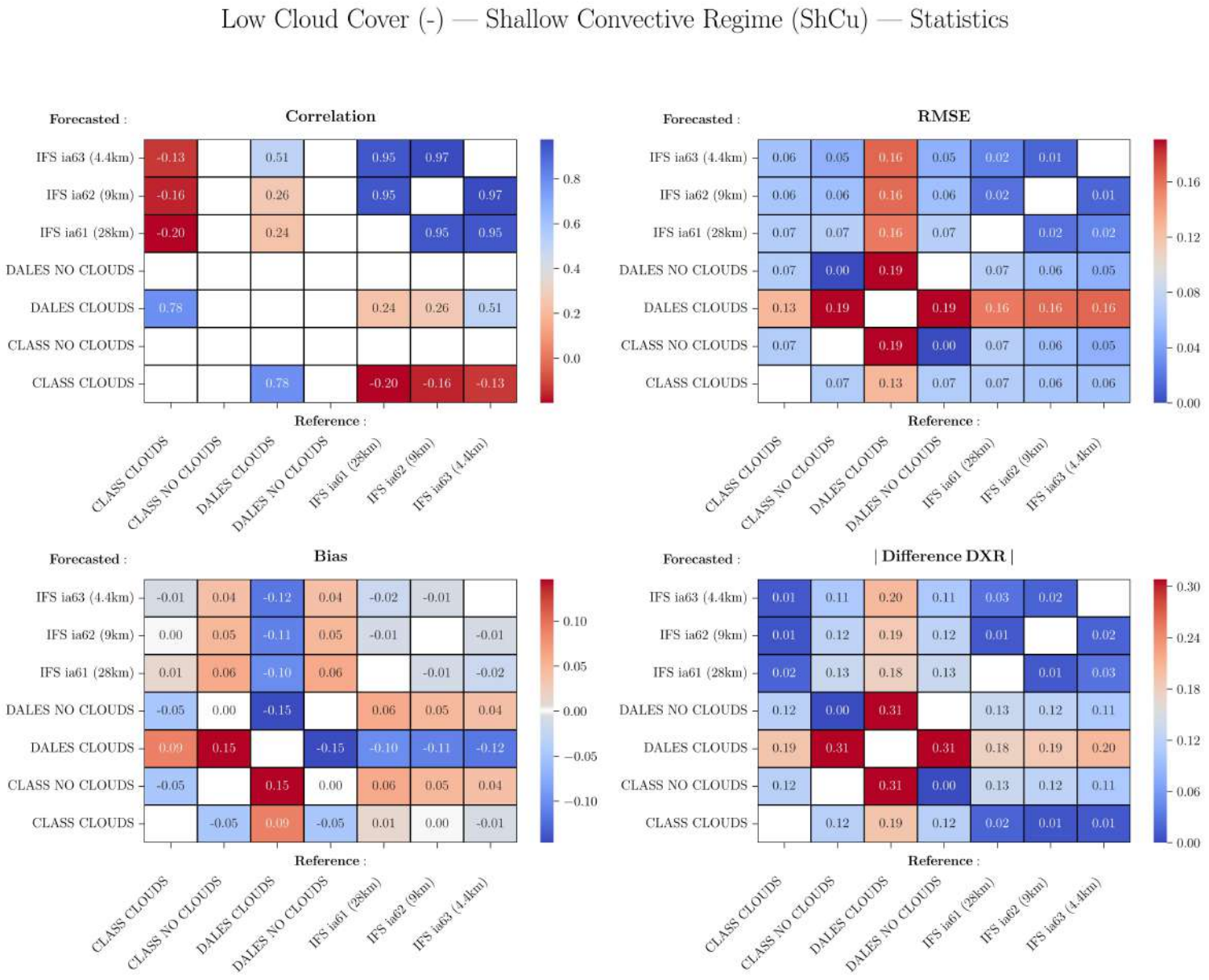



Figure 164: Statistical quantities of the diurnal cycle of simulated (CLASS, DALES and IFS) low cloud cover for the Shallow Convective (ShCu) regime.

### 4.7.2 Medium Cloud Cover

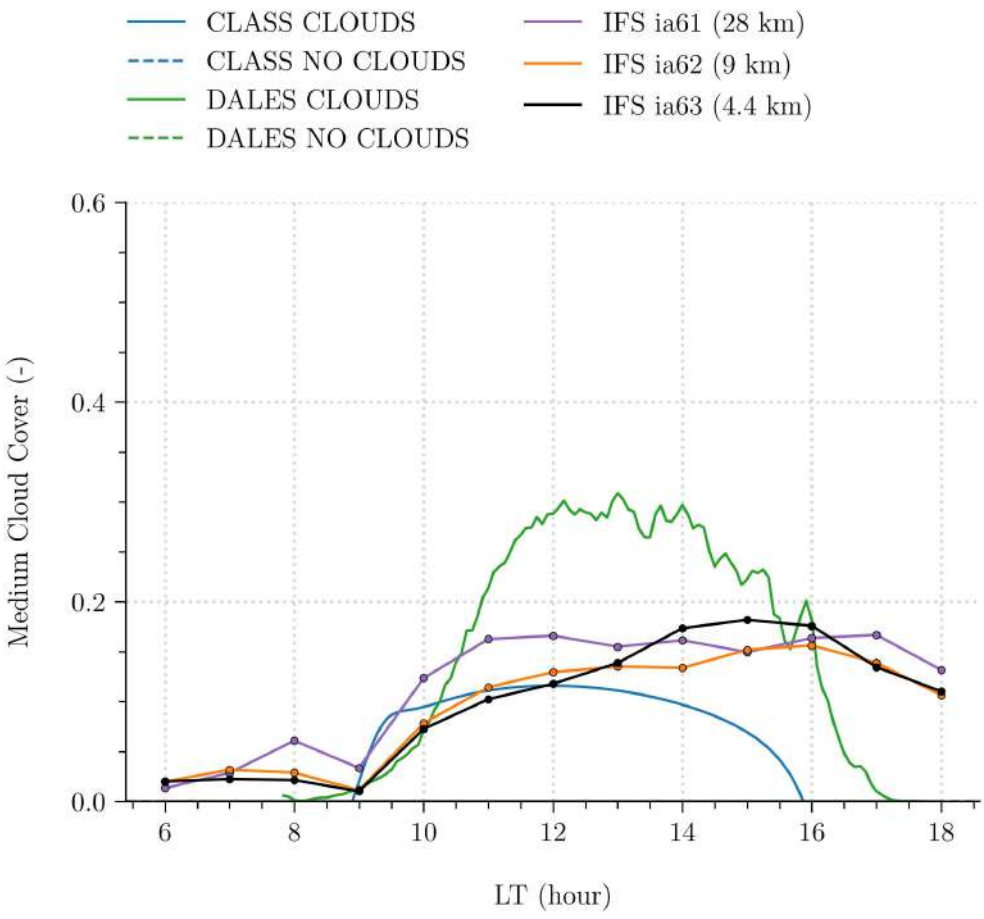



Figure 165: Diurnal cycle of simulated (CLASS, DALES and IFS) medium cloud cover for the Shallow Convective (ShCu) regime.

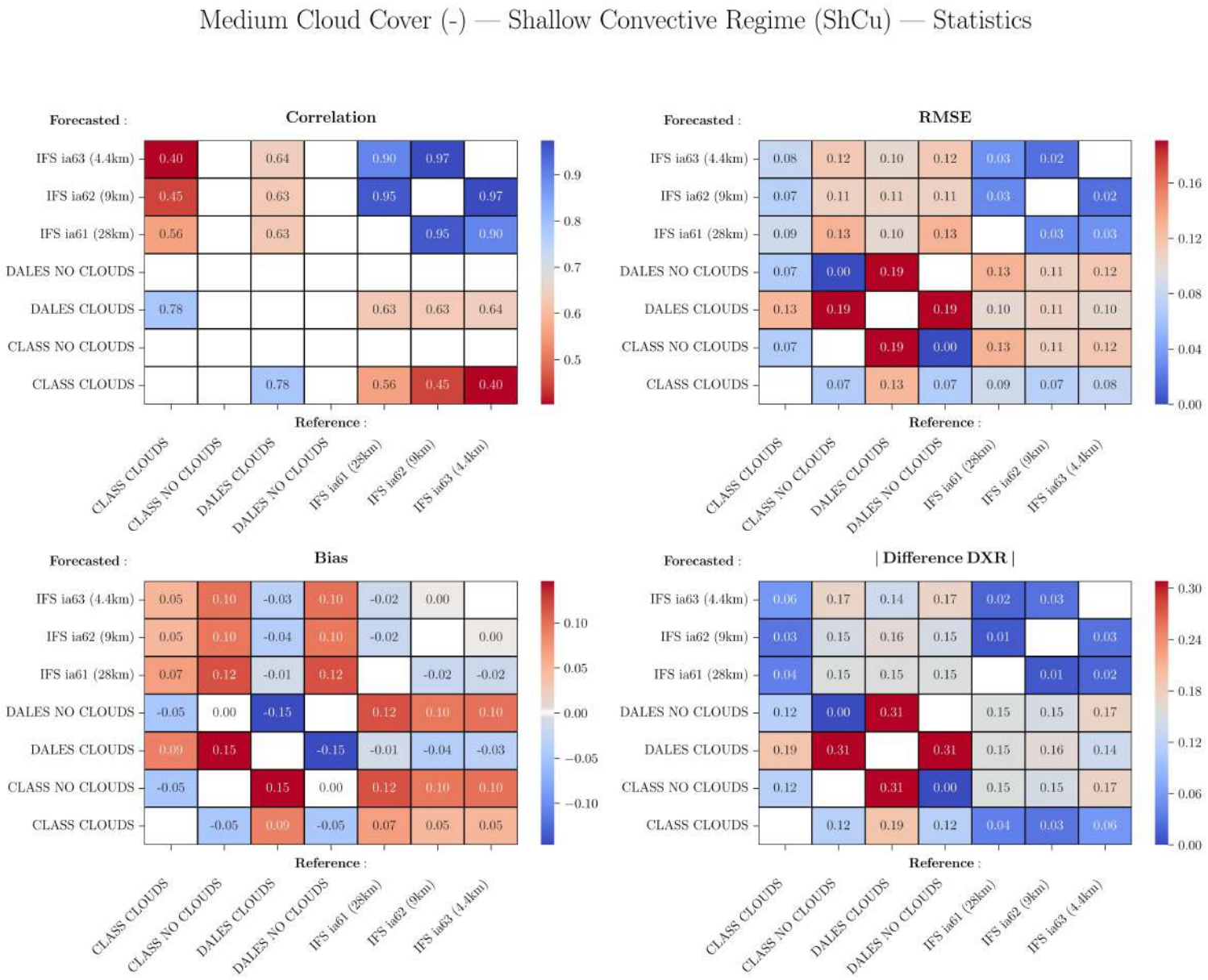



Figure 166: Statistical quantities of the diurnal cycle of simulated (CLASS, DALES and IFS) medium cloud cover for the Shallow Convective (ShCu) regime.



Funded by the European Union

### 4.7.3 High Cloud Cover

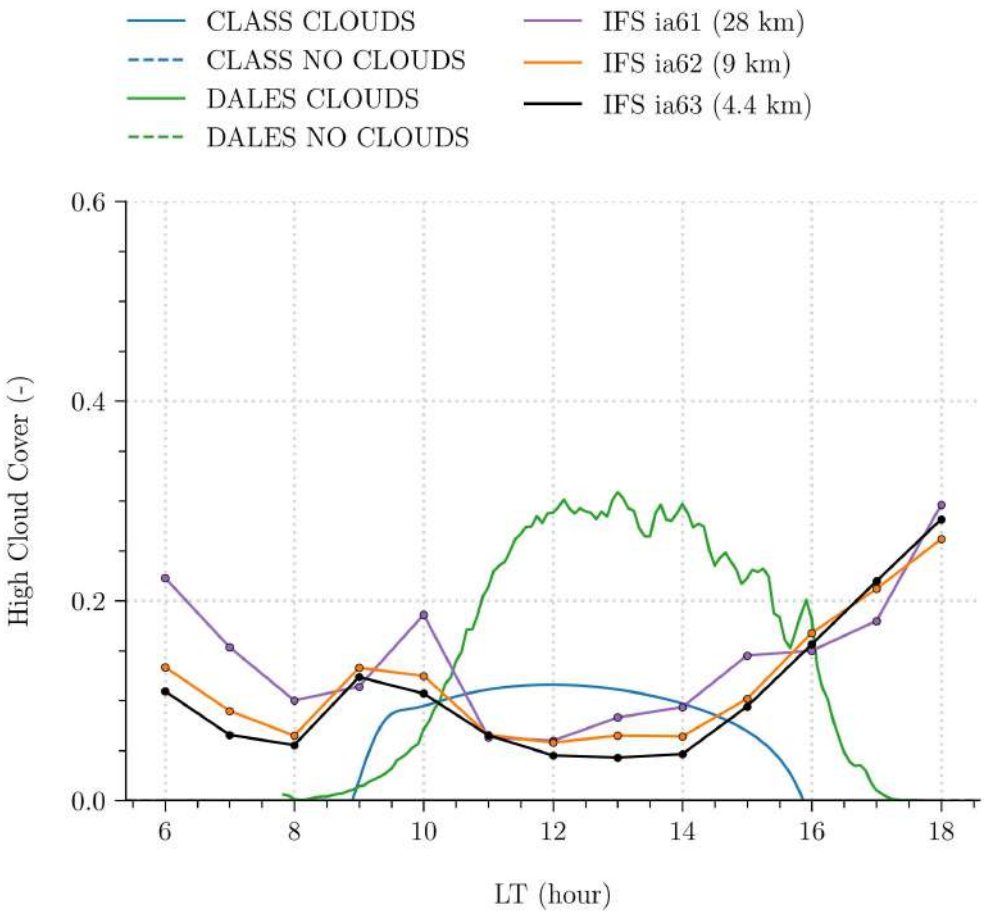



Figure 167: Diurnal cycle of simulated (CLASS, DALES and IFS) high cloud cover for the Shallow Convective (ShCu) regime.

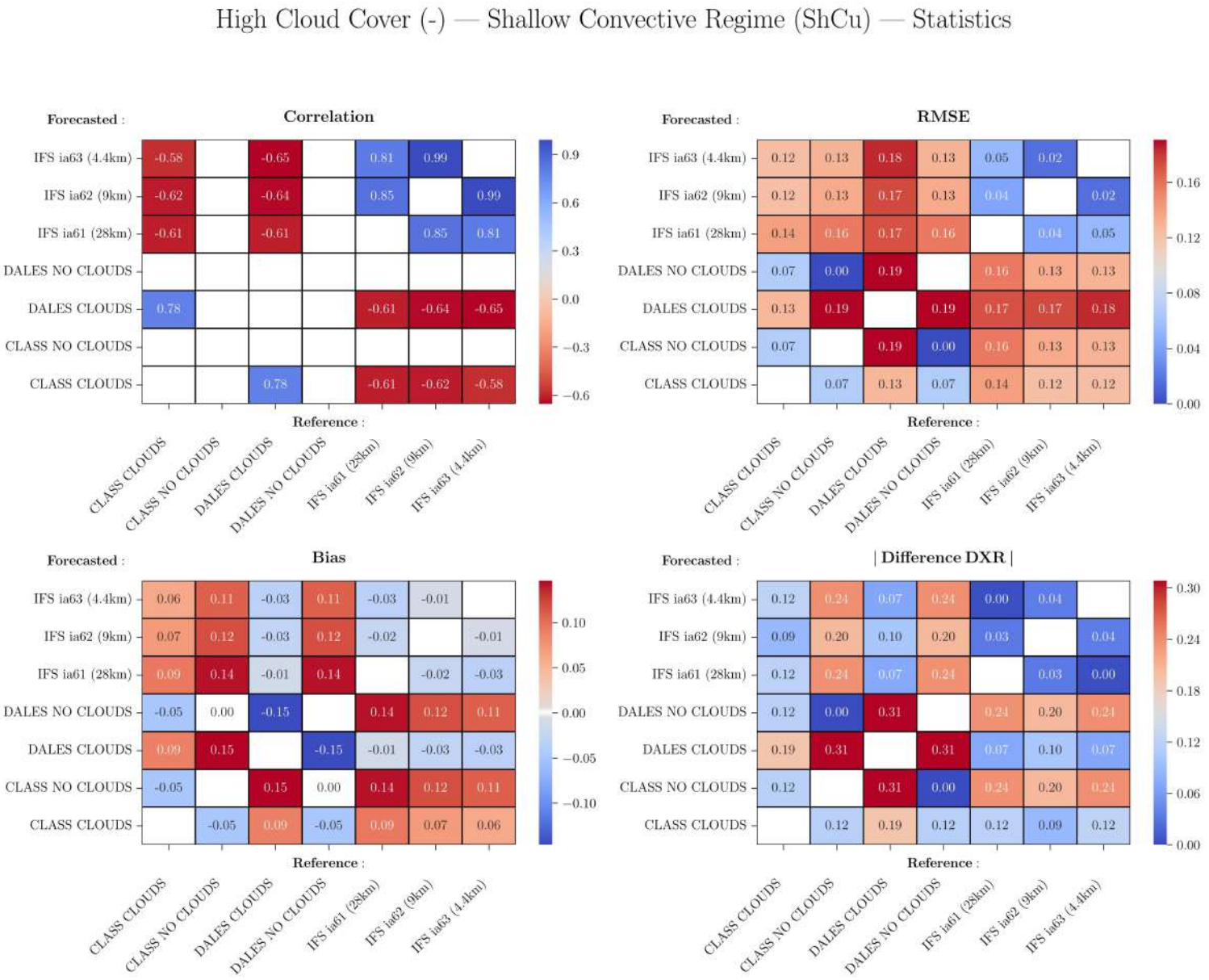



Figure 168: Statistical quantities of the diurnal cycle of simulated (CLASS, DALES and IFS) high cloud cover for the Shallow Convective (ShCu) regime.

## 4.8 Metrics

### 4.8.1 $DXR$

$\theta$

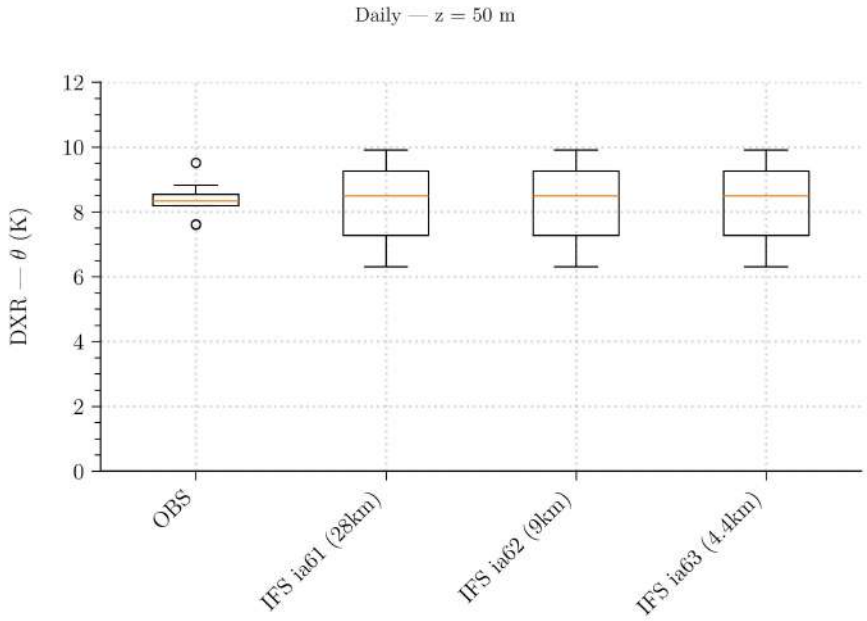



Figure 169: DXR of observed (at 50 m) and simulated (IFS) $\theta$ for the full CloudRoots-Amazon22 campaign.

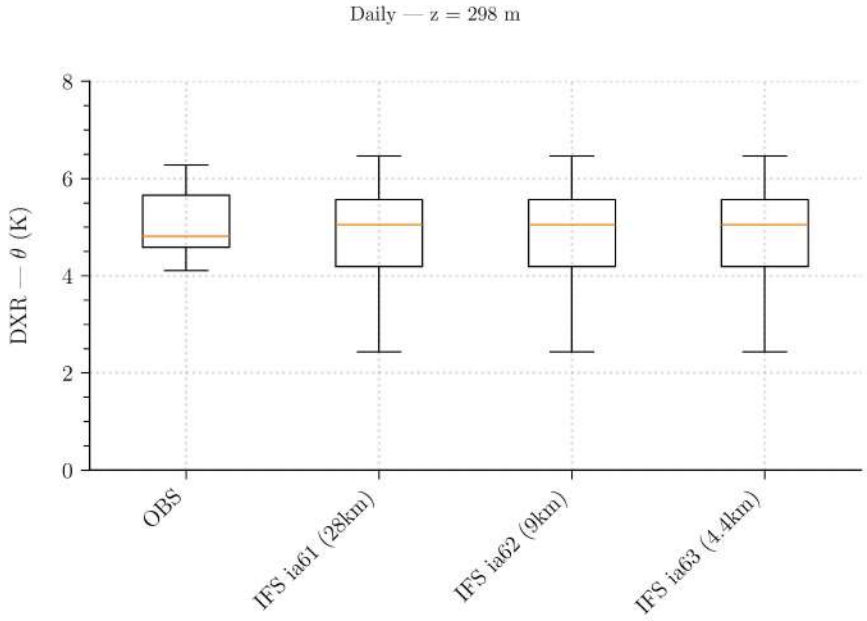



Figure 170: DXR of observed (at 298 m) and simulated (IFS) $\theta$ for the full CloudRoots-Amazon22 campaign.

$q$

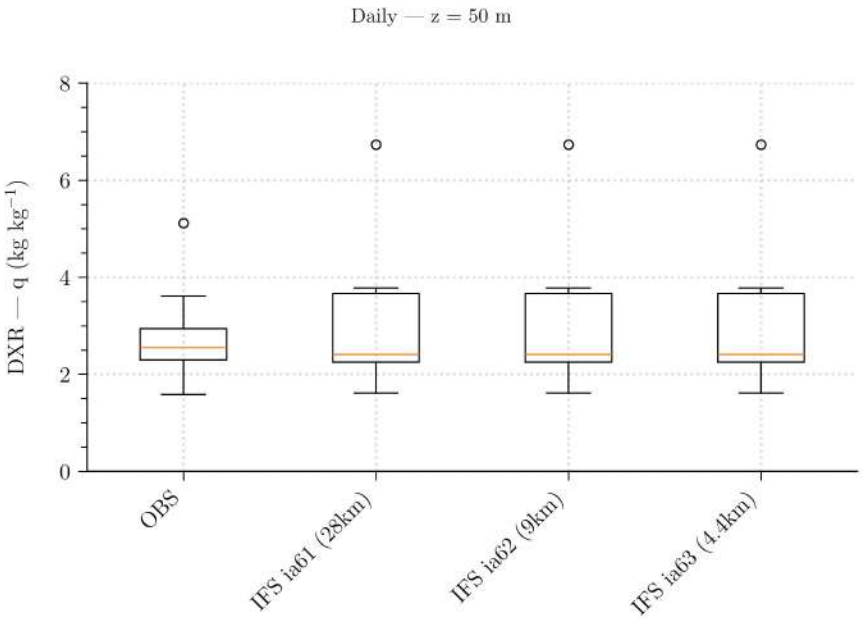



Figure 171: DXR of observed (at 50 m) and simulated (IFS) $q$ for the full CloudRoots-Amazon22 campaign.

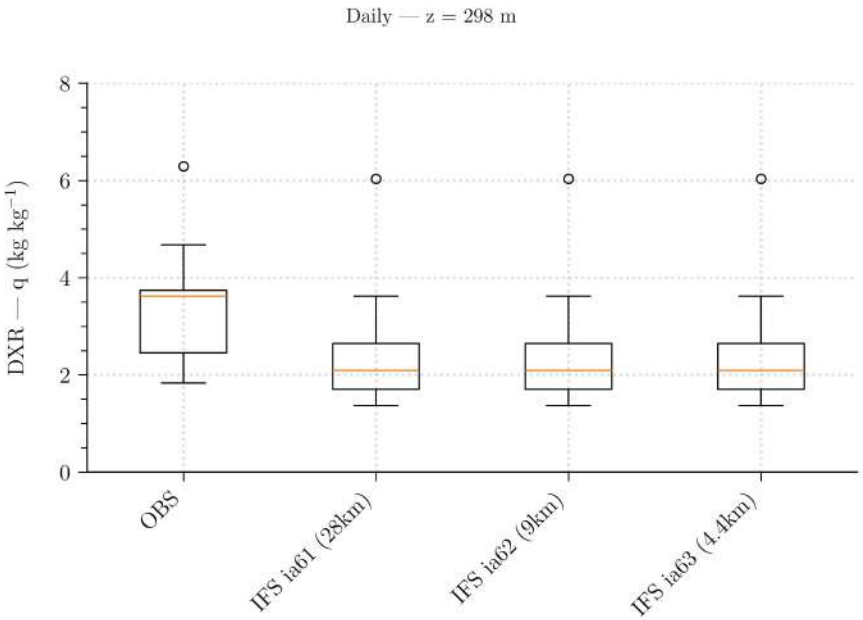



Figure 172: DXR of observed (at 298 m) and simulated (IFS) $q$ for the full CloudRoots-Amazon22 campaign.

$CO_2$

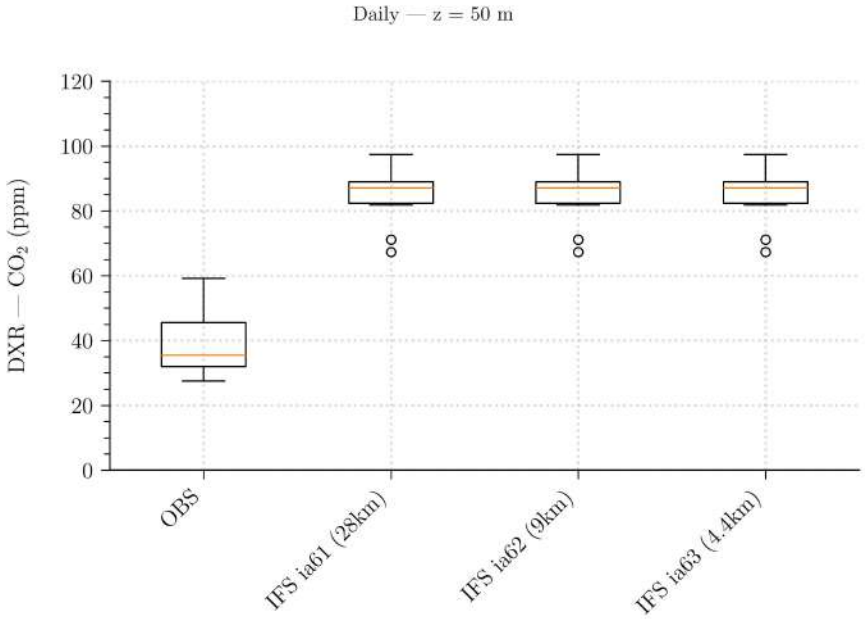



Figure 173: DXR of observed (at 50 m) and simulated (IFS) $CO_2$ for the full CloudRoots-Amazon22 campaign.

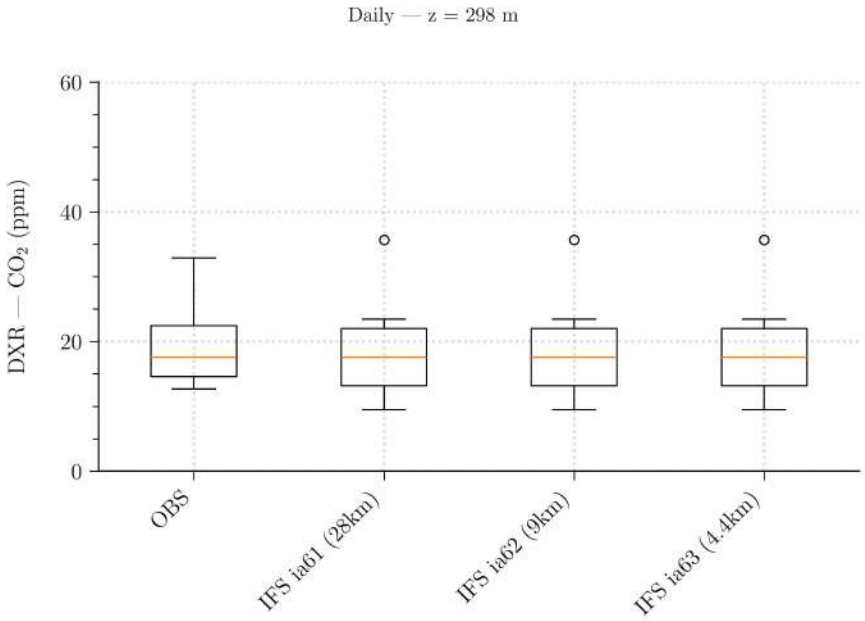



Figure 174: DXR of observed (at 298 m) and simulated (IFS) $CO_2$ for the full CloudRoots-Amazon22 campaign.

## Acknowledgments

This work was supported by the CATRINE project (grant agreement 101135000, `https://www.catrine-project.eu/`), funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the Commission. Neither the European Union nor the granting authority can be held responsible for them. The CloudRoots-Amazon22 project is funded by the Dutch Research Council (NWO under the project CloudRoots–Clouds rooted in a heterogeneous biosphere (`https://cloudroots.wur.nl`) (Dutch Research Council NWO OCENW.KLEIN.407). Additionally, this study was supported by the Amazon Tall Tower Observatory (ATTO), funded by the German Federal Ministry of Education and Research (BMBF, contracts 01LB1001A and 01LK1602A), the Brazilian Ministry of Science, Technology and Innovation (MCTI/FINEP, contract 01.11.01248.00) and the Max Planck Society (MPG). ATTO is also supported by the Fundação de Amparo à Pesquisa do Estado do Amazonas (FAPEAM), Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP), Universidade do Estado do Amazonas (UEA), Instituto Nacional de Pesquisas Amazônia (INPA), Programa de Grande Escala da Biosfera-Atmosfera na Amazônia (LBA) and the SDS/CEUC/RDS-Uatumã.

## 5 Data & Code Availability

The IFS simulations (model cycle CY49R1) used in this study are available in the Meteorological Archival and Retrieval System (MARS) of the ECMWF under the experiment identifiers: ia61 (28 km), ia62 (9 km) and ia63 (4.4 km). The download scripts, processing and visualisation code are in active development. The most recent version of the code is accessible via `https://github.com/vincentdefeiter/EUCATRINE_WP5_6.git`.

## References

Andreae, M. O., Acevedo, O. C., Araùjo, A., Artaxo, P., Barbosa, C. G. G., Barbosa, H. M. J., Brito, J., Carbone, S., Chi, X., Cintra, B. B. L., da Silva, N. F., Dias, N. L., Dias-Júnior, C. Q., Ditas, F., Ditz, R., Godoi, A. F. L., Godoi, R. H. M., Heimann, M., Hoffmann, T., Kesselmeier, J., Könemann, T., Krüger, M. L., Lavric, J. V., Manzi, A. O., Lopes, A. P., Martins, D. L., Mikhailov, E. F., Moran-Zuloaga, D., Nelson, B. W., Nölscher, A. C., Santos Nogueira, D., Piedade, M. T. F., Pöhlker, C., Pöschl, U., Quesada, C. A., Rizzo, L. V., Ro, C.-U., Ruckteschler, N., Sá, L. D. A., de Oliveira Sá, M., Sales, C. B., dos Santos, R. M. N., Saturno, J., Schöngart, J., Sörgel, M., de Souza, C. M., de Souza, R. a. F., Su, H., Targhetta, N., Tóta, J., Trebs, I., Trumbore, S., van Eijck, A., Walter, D., Wang, Z., Weber, B., Williams, J., Winderlich, J., Wittmann, F., Wolff, S., and Yáñez-Serrano, A. M. (2015). The Amazon Tall Tower Observatory (ATTO): overview of pilot measurements on ecosystem ecology, meteorology, trace gases, and aerosols. *Atmospheric Chemistry and Physics*, 15(18):10723–10776.

de Feiter, V. S., Janssens, M., de Haas, S. E. M., Hartogensis, O. K., Dias-Junior, C. Q., van Asperen, H., Martins, G., Miller, J. B., and Vilà-Guerau de Arellano, J. (2025). Turbulent Exchange of CO2 in the Lower Tropical Troposphere Across Clear-to-Cloudy Conditions. *Journal of Geophysical Research: Atmospheres*, 130(19):e2025JD044231. _eprint: https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1029/2025JD044231.

Heus, T., van Heerwaarden, C. C., Jonker, H. J. J., Pier Siebesma, A., Axelsen, S., van den Dries, K., Geoffroy, O., Moene, A. F., Pino, D., de Roode, S. R., and Vilà-Guerau de Arellano, J. (2010). Formulation of the Dutch Atmospheric Large-Eddy Simulation (DALES) and overview of its applications. *Geoscientific Model Development*, 3(2):415–444.

van Stratum, B. J. H., Vilà-Guerau de Arellano, J., van Heerwaarden, C. C., and Ouwersloot, H. G. (2014). Subcloud-Layer Feedbacks Driven by the Mass Flux of Shallow Cumulus Convection over Land. *Journal of the Atmospheric Sciences*, 71(3):881–895.

Vilà-Guerau de Arellano, J., Hartogensis, O. K., Boer, H. d., Moonen, R., González-Armas, R., Janssens, M., Adnew, G. A., Bonell-Fontás, D. J., Botía, S., Jones, S. P., Asperen, H. v., Komiya, S., Feiter, V. S. d., Rikkers, D., Haas, S. d., Machado, L. a. T., Dias-Junior, C. Q., Giovanelli-Haytzmann, G., Valenti,

W. I. D., Figueiredo, R. C., Farias, C. S., Hall, D. H., Mendonça, A. C. S., Silva, F. A. G. d., Silva, J. L. M. d., Souza, R., Martins, G., Miller, J. N., Mol, W. B., Heusinkveld, B., Heerwaarden, C. C. v., D'Oliveira, F. a. F., Ferreira, R. R., Gotuzzo, R. A., Pugliese, G., Williams, J., Ringsdorf, A., Edtbauer, A., Quesada, C. A., Portela, B. T. T., Alves, E. G., Pöhlker, C., Trumbore, S., Lelieveld, J., and Röckmann, T. (2024). CloudRoots-Amazon22: Integrating Clouds with Photosynthesis by Crossing Scales. *Bulletin of the American Meteorological Society*, 105(7):E1275–E1302.